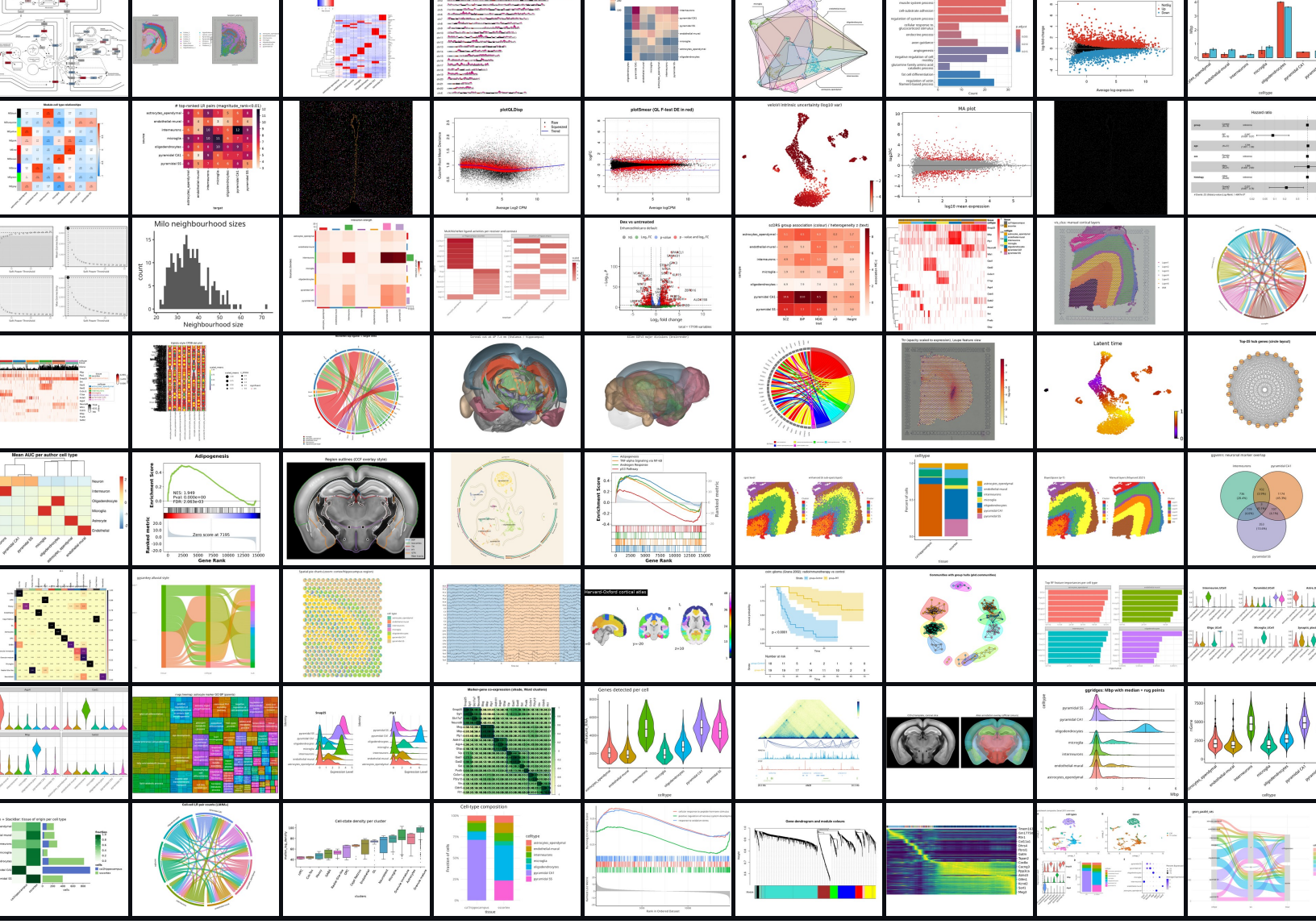


Novaflow

REFERENCE BOOK · NEURO-OMICS · EDITION 01 · OCTOBER 2026

The Figure Booklet

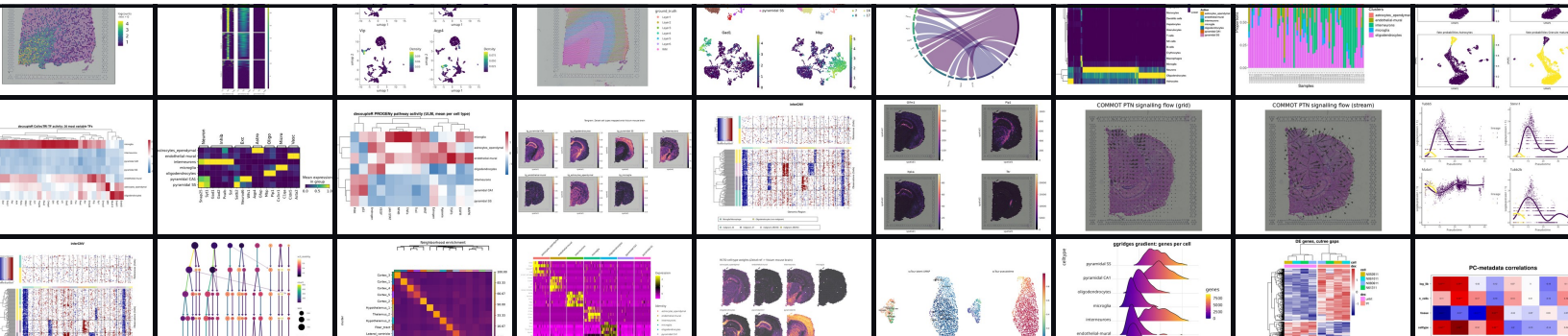
Plotting packages in neuroscience bulk, single-cell & spatial omics, 2023–26, ranked by how often papers name them and rebuilt from code.

BACKGROUND: 135 OF THE 329 RECREATED FIGURES, SORTED BY COLOUR

17,768
PAPERS MINED

98
PACKAGES

329
FIGURES RECREATED



NOVAFLOW · ANALYSIS FOR RESEARCH LABS

Your data, any question, publication-ready figures.

This booklet shows the figure types neuroscience and stem-cell omics papers rely on, each rebuilt from code on public data. Novaflow does the same work on **your** data: bulk, single-cell and spatial analyses, delivered as figures and methods text you can put in a paper or grant.

01

Send your data

Count matrices, Cell Ranger / Space Ranger outputs, h5ad or Seurat objects, or raw FASTQ.

02

Ask your question

“Which cell states change with treatment?” “Does our organoid match fetal cortex?” Scope is agreed before work starts.

03

Get figures and analysis

Publication-ready figures, statistics, reproducible code and a methods paragraph, plus comparisons with relevant public datasets.

TYPICAL REQUESTS

- QC, clustering and annotation of scRNA-seq / snRNA-seq and organoid data
- Differential expression and composition analysis with appropriate replicates
- Trajectory, RNA velocity and fate analysis for differentiation time courses
- Spatial transcriptomics: domains, deconvolution, cell–cell communication
- Mapping your samples onto public atlases (e.g. Allen, developing human brain)
- Figure revision for reviewers and resubmissions

Delivered for a fraction of the cost of a typical core-facility or contract bioinformatics service.

CONTACT
logan@novaflow.ai

BOOK A 30-MINUTE CALL
calendly.com/logan-novaflowapp/30min

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COLOPHON

About this book

A reference guide to the figures used in neuroscience omics papers. We counted how many of **17,768 open-access neuroscience papers with bulk, single-cell or spatial omics data (2023–2026)** name each plotting or analysis package in their Methods section, ranked the packages, and rebuilt **329 figure variations from 98 packages** on public datasets. Each package gets a two-page spread with the code that draws its typical figures.

What is in here

- **Part I · Mainstream workhorses**: 62 spreads. Recreated packages from the Top 100, in order of the number of papers naming them in Methods. Where several catalog entries share one recreation folder (e.g. IGV and UCSC Genome Browser → Gviz), they get one spread.
- **Part II · Niche & Cool**: 36 spreads. Newer or more specialised tools, ranked by a blend of growth in mentions and visual distinctiveness. Tools already covered in Part I are not repeated.
- **Visual index**: all 329 figures grouped into 11 figure types, each linked to its spread.
- **Appendix**: queued tools and why, plus the full Top 100 and Niche & Cool tables.

Ground rules

- Every number comes from stored Europe PMC query results and was re-derived in an independent audit (p. 9). DOIs and figure links were checked to resolve.
- Every gallery figure is a **Novaflow recreation on public data**, labelled with its dataset. None is copied from, or claims to reproduce, a cited paper.
- Code excerpts are quoted verbatim from each recreation script, with line numbers.
- All blue text is a link: DOIs, PMC figure pages, GitHub repositories and internal page references. The PDF also has bookmarks.

MAINSTREAM (TOP 100)

70/100

entries recreated (62 folders); 30 queued (Appendix A)

NICHE & COOL (73)

58/73

candidates recreated (55 folders, 19 shared with Top 100); 15 queued

REPRODUCIBILITY

98/98

recreation scripts re-run end to end without errors; all cited DOIs and figure links resolve

Reading paths

Which packages do papers report using? · Top 30 chart, then Part I in rank order.

What is rising? · Niche growth chart, then Part II.

I know the look I want · Visual index: 329 thumbnails in 11 figure types.

Spatial & tissue figures · Squidpy, spatialLIBD, BayesSpace, Banksy, RCTD, Tangram, COMMOT…

Cell–cell communication · CellChat, LIANA+, NicheNet, MultiNicheNet, pyCirclize…

Brain atlas & neuroimaging · BrainGlobe, brainrender, nilearn, MNE.

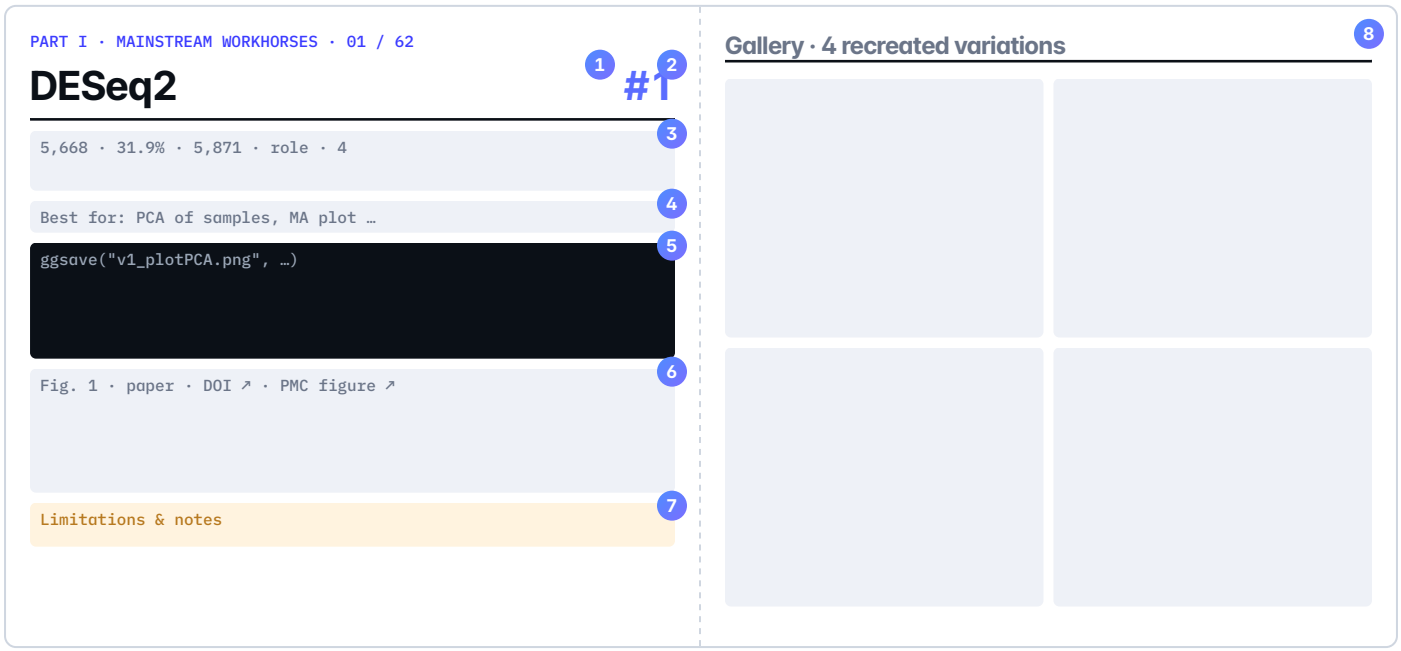
What is missing? · Appendix A: 45 queued tools and the reason each is queued.

How far can I trust a rank? · Caveats and validation.

ORIENTATION

How to use this book

Each package gets one **two-page spread**: the left page explains it, the right page shows it. Open the PDF in two-page view, with the cover shown separately, to see the spreads as facing pages. To start from a picture rather than a name, use the visual index.



1. **Name, language, category, figure types** · Chips show the language, the catalog category and the figure types drawn in the gallery. Colours match the visual index.
2. **Rank** · Part I: Top-100 rank by the number of papers naming the package in Methods. Part II: Niche & Cool rank (score = 0.5 × growth percentile + 0.5 × visual/5).
3. **Numbers strip** · Part I: papers naming it in Methods, % of the 17,768-paper corpus, papers naming it anywhere, role, variations. Part II: growth ratio († = low 2023 base), visual score, niche score, neuro-corpus papers, GitHub stars. The small bar track shows where the package sits among its peers.
4. **Best for** · The figure types the package is typically used for.
5. **Key plotting call** · A verbatim excerpt of the recreation script, ending at the line that writes the first variation (or the signature call). File and line number are given.
6. **In published papers** · Up to three figures from published papers that name the package, with DOI and PMC figure links and the sentence naming it. These are the authors' figures, linked for reference; niche tools list the most-cited papers naming them.
7. **Limitations & notes** · Package-specific caveats: simulated or non-neuro data, look-alikes of GUI tools, compatibility fixes, omitted plot types.
8. **Gallery** · Every recreated figure for the package, with the dataset it was drawn from. All are Novaflow recreations on public data.

Figure-type key

Every variation is assigned one figure type. Two catch-all types, **Composition, distributions & layouts** and **DE, QC & set overlaps**, hold figures that fit none of the others. Counts are figures, not packages.

Heatmaps & dot matrices	40	Embeddings & feature maps	26
Trajectory, pseudotime & velocity	37	Spatial & tissue maps	34
Cell–cell communication	23	Enrichment & pathway activity	32
Networks, graphs & trees	20	Genome tracks & genomic signal	19
Brain atlas, 3D & neuroimaging	13	Composition, distributions & layouts	55
DE, QC & set overlaps	30		

READ THE NUMBERS CAREFULLY

A paper naming a package in Methods is not proof that the package drew a published figure (in a hand-checked sample, only about a third of mentions tie the tool to a figure). DESeq2, limma and edgeR rank high largely because they are used for statistics. Growth ratios marked † rest on a very small 2023 base, and counts under ~10 are noisy.

Methods & corpus

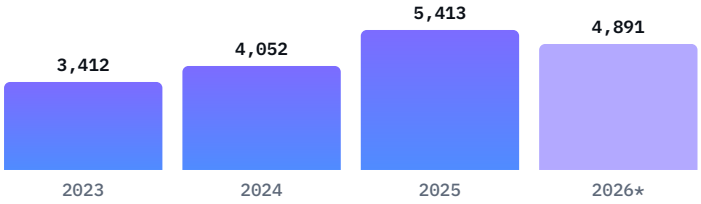
A systematic search of open-access full text in Europe PMC. Each package's count is the **number of papers whose Methods section names it**, the section where authors report the software they used. Counts were taken on 15 October 2026.

1 · CORPUS	2 · SCREEN	3 · RANK	4 · RECREATE	5 · VERIFY
17,768	**212**	**100 + 73**	**98**	**329**
OA neuro-omics papers with full text, 2023–26	candidate packages, each with its own search terms	Top 100 by papers naming them; Niche & Cool by growth + visual	folders: 70 Top-100 entries, 58 niche tools (19 shared)	figure variations; all 98 scripts re-run; links verified

CORPUS SIZE

17,768

open-access neuro-omics papers with full text (SRC:MED), published 2023–2026



*2026 is a partial year (to mid-October).

MODALITY TAGS (TERM IN ABSTRACT OR METHODS; A PAPER CAN CARRY SEVERAL)

Single-cell or multiome	6,887
Spatial	1,376
Any RNA-seq (bulk included)	16,397

TOP 10 JOURNALS IN THE CORPUS

Nat Commun	1,167
Sci Rep	545
Int J Mol Sci	539
Adv Sci (Weinh)	433
J Neuroinflammation	405
iScience	396
Sci Adv	341
Front Immunol	280
Proc Natl Acad Sci U S A	268
Elife	263

SEARCH (EUROPE PMC REST /WEBSERVICES/REST/SEARCH)

Neuro terms in the title or abstract, **AND** omics terms in the abstract or Methods, **AND** `PUB_YEAR:[2023 TO 2026] OPEN_ACCESS:Y HAS_FT:Y SRC:MED`. Full query:

```
(TITLE:(neuron OR neurons OR neuronal OR brain OR cortex OR cortical OR hippocampus OR hippocampal OR neurodevelopment OR neurodevelopmental OR "neural progenitor" OR "neural stem" OR neurodegeneration OR neurodegenerative OR cerebellum OR "spinal cord" OR retina OR glia OR glial OR astrocyte OR astrocytes OR microglia OR oligodendrocyte OR "cerebral organoid" OR "brain organoid" OR Alzheimer OR Parkinson OR "nervous system") OR ABSTRACT:(neuron OR neurons OR neuronal OR brain OR cortex OR cortical OR hippocampus OR hippocampal OR neurodevelopment OR neurodevelopmental OR "neural progenitor" OR "neural stem" OR neurodegeneration OR neurodegenerative OR cerebellum OR "spinal cord" OR retina OR glia OR glial OR astrocyte OR astrocytes OR microglia OR oligodendrocyte OR "cerebral organoid" OR "brain organoid" OR Alzheimer OR Parkinson OR "nervous system")) AND (ABSTRACT:("single-cell RNA" OR "single-nucleus RNA" OR "single-cell sequencing" OR "scRNA-seq" OR "snRNA-seq" OR "RNA-seq" OR "RNA sequencing" OR "spatial transcriptomics" OR Visium OR MERFISH OR Xenium OR CosMx OR "Slide-seq" OR "Stereo-seq" OR CODEX OR "snATAC-seq" OR "scATAC-seq" OR multiome) OR METHODS:("single-cell RNA" OR "single-nucleus RNA" OR "single-cell sequencing" OR "scRNA-seq" OR "snRNA-seq" OR "RNA-seq" OR "RNA sequencing" OR "spatial transcriptomics" OR Visium OR MERFISH OR Xenium OR CosMx OR "Slide-seq" OR "Stereo-seq" OR CODEX OR "snATAC-seq" OR "scATAC-seq" OR multiome)) AND PUB_YEAR:[2023 TO 2026] AND OPEN_ACCESS:Y AND HAS_FT:Y AND SRC:MED
```

Screening & counting

1. **212 candidate packages** are screened. Each has its own search phrases. Ambiguous names are searched only as specific phrases or with context words: "STRING database", "GSEA software"/ GSEApy rather than the bare words; *patchwork* with ggplot2, "R package" or CRAN; *MAST* with hurdle wording; *PROGENy* with pathway-activity wording.
2. **Paper count** = number of corpus papers whose Methods section names the package (Europe PMC section search). **Full-text count** = papers naming it anywhere, given for comparison.
3. Only **figure-making tools** are ranked. 26 analysis-only, QC and data packages (Harmony, DoubletFinder, GSVA, DAVID, anndata…) go to the *Excluded* sheet, and general GUI tools (Prism, ImageJ, Illustrator, BioRender) to *GUI_tools*.
4. **Example figures.** For each Top-100 package, up to 10 of the most-cited papers mentioning it are parsed (fullTextXML). Figures are scored +6 if the caption names the package, +3 if a citing paragraph names it, +1–3 for figure-type keywords and −2 if supplementary. Up to 3 are kept. Each PMC figure link was checked against the article's full-text XML, and each DOI against the DOI registry (19 Oct 2026); figures not hosted on PMC are given as article links.
5. **Niche & Cool** covers 73 newer or specialised tools (`niche_candidates.py`). The growth ratio is (broad mentions per 1k papers in 2025–26 + 0.01) ÷ (per 1k in 2023 + 0.01), measured in a broad OA omics corpus of 28,714 / 33,012 / 42,965 / 35,775 papers. Score = 0.5 × percentile(growth) + 0.5 × visual/5. Ratios built on fewer than 5 mentions in 2023 are marked †: they mostly reflect a recent release.

Recreations

Each package has one script that draws 1–7 figure variations and runs on its own (Rscript, python3 or bash). All figures are **Novaflow recreations on public datasets**, listed below and on every gallery page. They show what each package's figures look like; they are not reproductions of the cited papers' figures and carry no new biological claims.

Three recreations call **live web APIs** (Enrichr, g:Profiler, STRING). Two use **synthetic signal**: deepTools (simulated bigWigs) and the Gviz data track (jittered peak counts); both are labelled. Four use non-neuro example data shipped with the package (airway bulk RNA-seq, Signac PBMC, ggcoverage, plotgardener IMR90).

CAVEATS

1. **Open-access bias.** Only papers with Europe PMC full text are included, so closed-access Cell, Neuron and Nature Neuroscience papers are missing unless an OA or author-manuscript copy exists.
2. **A mention is not figure use.** Counts measure how often papers report using a package (29/30 hand-checked hits were genuine use), not how many figures it drew (10/30 tied it to a figure). DESeq2, limma and edgeR rank high largely for statistics.
3. **Under-counting.** Tools named only in Key Resources tables or code-availability statements outside Methods, or not named at all (matplotlib and ggplot2 often go unnamed), are under-counted. Compare with the full-text count.
4. **Name collisions** are handled with specific phrases and context terms (p. 9). None appeared in the sample, but common words (Harmony, Giotto, CARD) can still collide.
5. **Topic noise.** The neuro filter works on title or abstract, so a few off-topic papers are included.
6. **Not recreated:** web, GUI and commercial tools (Cytoscape, Metascape, IPA, QuPath…) and database-heavy tools (SCENIC). See Appendix A.
7. **Niche growth.** Ratios marked † (fewer than 5 mentions in 2023) mostly reflect a recent release. Counts under ~10 are noisy, the visual score is curator judgement, and 2026 is partial (to mid-October).
8. **Snapshot.** Europe PMC keeps indexing; re-running queries later gives slightly higher counts.

Public datasets behind the recreations

Zeisel 2015 · mouse S1 cortex / CA1 scRNA-seq	3,005 cells. Most single-cell figures; C1 chip used as replicate in composition tests.
La Manno 2018 · dentate gyrus P12/P35	scVelo, CellRank, Palantir, Mellon, veloVI, scTour, moscot (P12→P35 optimal transport), CytoTRACE 2
10x Visium · mouse coronal brain	Squidpy, Tangram, COMMOT, Loupe-style
Moffitt 2018 · MERFISH hypothalamus	Squidpy, Datashader, plotly 3D
Maynard 2021 · human DLPFC 151673	spatialLIBD, ggspavis, BayesSpace, Banksy (STexampleData)
Allen CCFv3 25 µm (BrainGlobe)	brainrender, atlas overlays
NeuroVault / Harvard-Oxford · PhysioNet EEGBCI S001	nilearn · MNE-Python
Tirosh 2016 oligodendroglioma · scDRS MAGMA sets	inferCNV · scDRS
Bulk & other	airway bulk RNA-seq (Himes 2014; not neuro) · coin::glioma · package example data (Signac, ggcoverage, plotgardener) · simulated (deepTools)

Recreations use current R/Bioconductor and Python releases (October 2026). Where a package no longer runs as published on current dependencies, a minimal compatibility fix is applied and noted on its spread. Scripts are available to clients on request.

VALIDATION · AUDIT OF 19 OCTOBER 2026

How far the numbers can be trusted

Every count, percentage, rank, growth ratio and star count in this booklet was re-derived from the stored query results, and every external link was re-checked. A random sample of hits was read by hand to estimate how often a counted mention is real.

1 · Hand-checked sample of 30 hits

One hit drawn at random (seed 2026) from each of 30 randomly chosen Top-100 packages. For each, the Europe PMC full text was retrieved and the sentence naming the package was read.

Right software (not a homonym)	**30/30 (100%; 95% CI 88–100%)**
Authors used it in their own work	**29/30 (97%; 95% CI 83–100%)**
Text ties it to a figure or plot	**10/30 (33%; 95% CI 17–53%)**

So the counts measure **reported use of a package** reliably, but most mentions do not say which figure a package drew. The one non-use hit was a passing mention ("tools like CellChat or NicheNet can…"). Read ranks as "how often papers report using this tool", not "how many published figures it made".

2 · Name collisions

Ambiguous names are searched only as specific phrases or with context words: STRING only as "STRING database" or "STRING v11/v12"; GSEA only as "GSEA software", "Gene Set Enrichment Analysis (GSEA)", "GSEA v4" or GSEApy, so generic use of the word "GSEA" for any enrichment analysis is **not** counted; *slingshot* needs trajectory/pseudotime; *Signac* needs ATAC/chromatin; *patchwork* needs ggplot2/CRAN; MAST and PROGENy need method wording. Unique names (ggplot2, Seurat, DESeq2) are searched directly. No homonyms appeared in the sample, but rare collisions can remain for common words.

3 · Re-derivation and links

- All Top-100 counts and percentages match the stored Europe PMC hit counts and per-package paper lists exactly (0 mismatches); all 73 Niche & Cool growth ratios, scores, ranks and GitHub star counts re-computed with 0 mismatches.
- Counts are a snapshot taken on 15 October 2026. Re-running six queries on 19 October gave 0.1–0.2% more hits as Europe PMC indexes new papers; the six packages kept their relative order.
- 141 DOIs resolve in the DOI registry, and all 168 PMC figure links point to a figure that exists in the article's full-text XML (checked 19 Oct 2026). No links were invented; links that could not be verified are given as article links only.

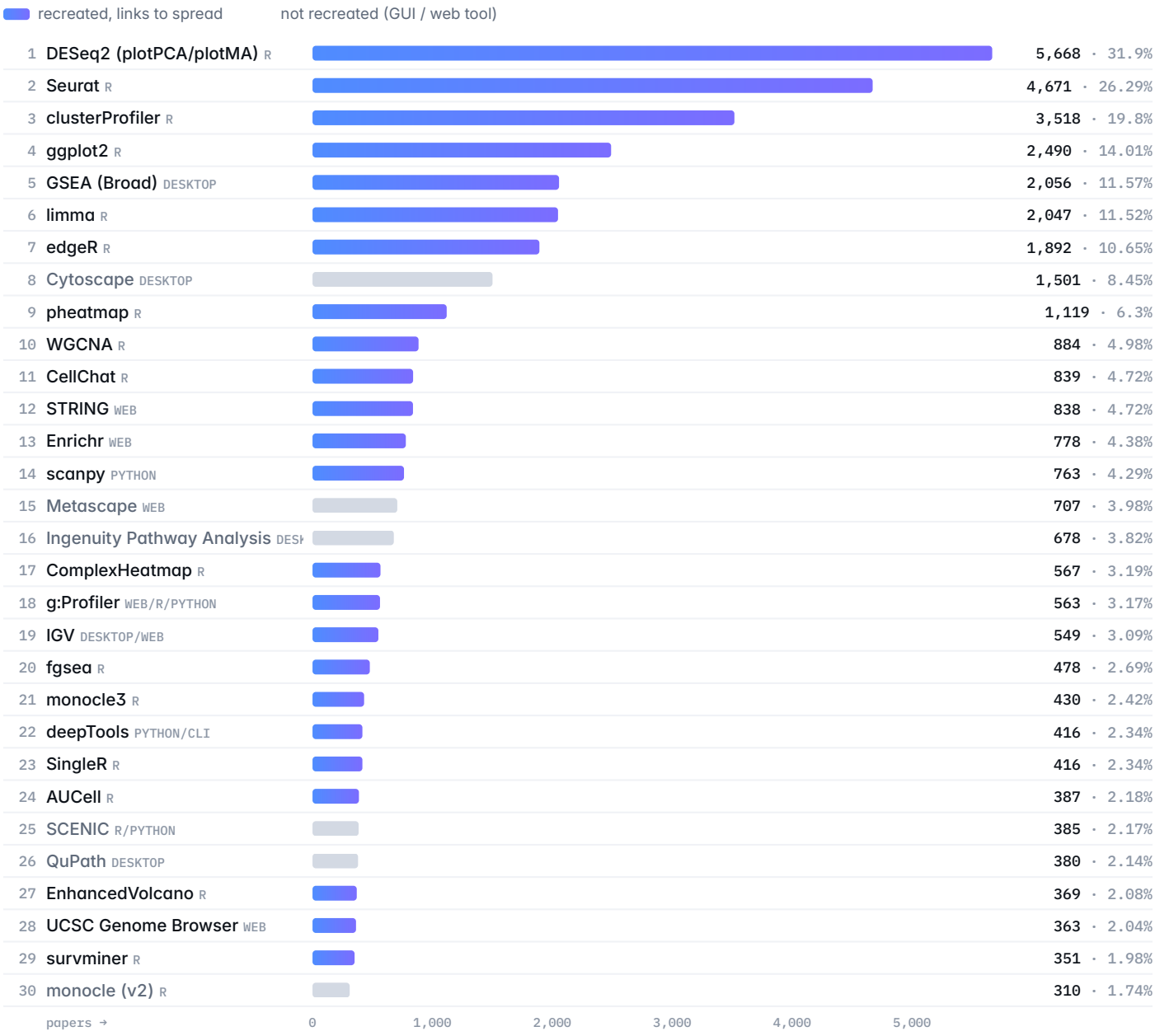
Sample (correct software · used · figure)

Ingenuity Pathway Analysis	PMC12684901	√	√	√
CellPhoneDB	PMC9905408	√	√	–
cellxgene	PMC12528736	√	√	–
NicheNet	PMC12436503	√	✗	–
Squidpy	PMC10709140	√	√	–
scanpy	PMC12043673	√	√	–
survminer	PMC13505878	√	√	–
FactoMineR	PMC10751163	√	√	–
EnrichmentMap	PMC11151750	√	√	√
variancePartition	PMC10639075	√	√	–
circlize	PMC10234549	√	√	–
scCustomize	PMC12450660	√	√	√
GOplot	PMC11466412	√	√	√
ABC atlas / Allen Brain Map	PMC13365245	√	√	–
RColorBrewer	PMC12447325	√	√	√
UpSetR	PMC9957872	√	√	√
Space Ranger/Loupe	PMC13558159	√	√	–
DESeq2 (plotPCA/plotMA)	PMC10336585	√	√	–
patchwork	PMC10771486	√	√	√
CellChat	PMC12103954	√	√	–
Metascape	PMC11947734	√	√	–
Signac	PMC11488568	√	√	–
Enrichr	PMC11046969	√	√	–
ArchR	PMC11166570	√	√	–
Seurat	PMC10436266	√	√	–
Tangram	PMC12994606	√	√	–
Nebulosa	PMC10703680	√	√	√
EnhancedVolcano	PMC10545437	√	√	√
ClueGO	PMC10015208	√	√	–
ggpubr	PMC12847059	√	√	√

Full sentences: audit/precision_review.csv. 95% intervals are exact (Clopper–Pearson).

Top 30 by papers naming them

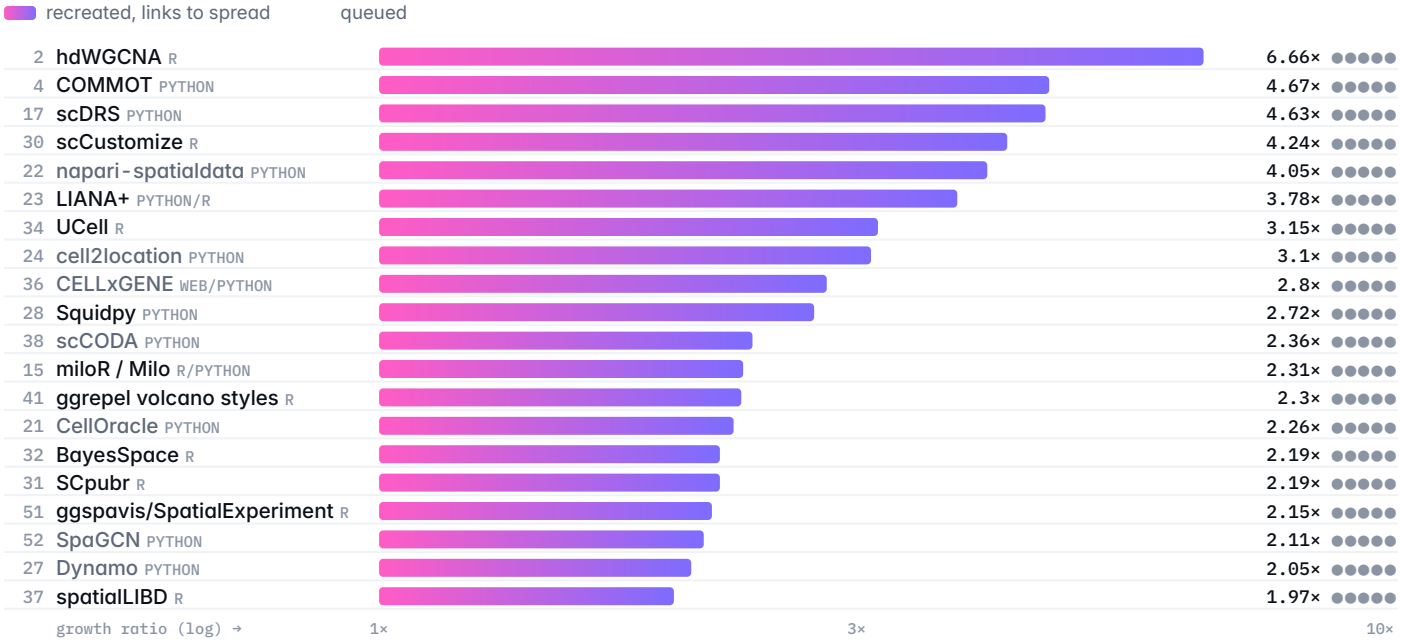
Number of the 17,768 corpus papers that name each package in their **Methods** section, with the share of the corpus. Bars link to spreads; hatched bars are not recreated (Appendix A).



DESeq2, limma and edgeR are named mostly for statistics, not figures. IGV (#19) and UCSC Genome Browser (#28) are recreated as Gviz look-alikes.

Niche & Cool: fastest growing

The 20 fastest-growing Niche & Cool candidates with a measurable 2023 base (at least 5 mentions). Growth ratio = papers per 1,000 naming the tool in 2025–26 ÷ in 2023, in a broad open-access omics corpus (log scale). Dots show the curator visual score (1–5); the left number is the niche rank.



† 25 candidates had fewer than 5 mentions in 2023 (e.g. CytoTRACE2, pyCirclize, moscot); their ratios (up to 324.85×) mostly reflect a recent release and are not charted. The niche rank blends growth percentile with visual score.

Niche & Cool top 20 · by score

#	TOOL	GROWTH	VISUAL	NEURO PAPERS 2023–26	GITHUB STARS	RECREATED	PAGE
1	**pyCirclize**	20.05†	5	0	1127	done (3)	146
2	**hdWGCNA**	6.66	5	147	494	done (5)	94
3	**brainrender**	4.81†	5	4	680	done (3)	148
4	**COMMOT**	4.67	5	15	149	done (3)	150
5	**CytoTRACE2**	324.85†	4	24	269	done (3)	104
6	**MultiNicheNet**	89.9†	4	13	202	done (3)	152
7	**moscot**	42.91†	4	7	217	done (3)	154
8	**scplotter**	17.51†	4	1	333	done (4)	156
9	SCENIC+	14.32†	4	32	267	queued (0)	
10	**Banksy**	14.16†	4	21	167	done (2)	158
11	**ggsankey**	10.71†	4	8	338	done (3)	142
12	**scTour**	6.52†	4	7	89	done (3)	160
13	**plot1cell**	2.7†	5	6	131	done (3)	162
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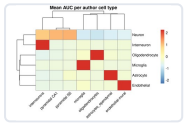

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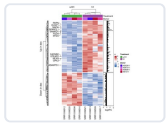

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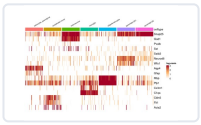

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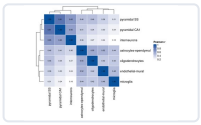

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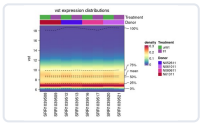

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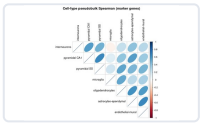

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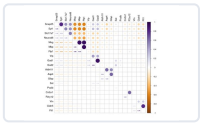

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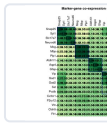

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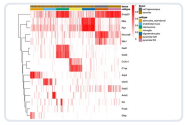

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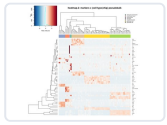

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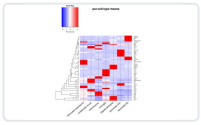

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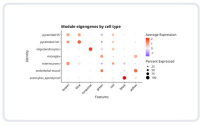

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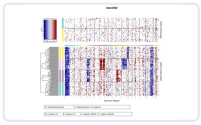

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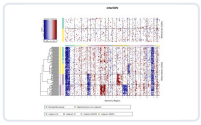

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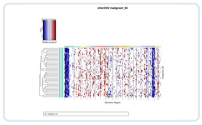

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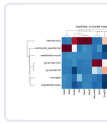

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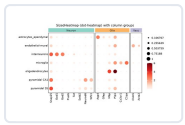

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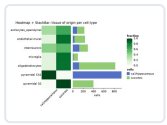

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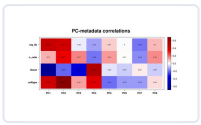

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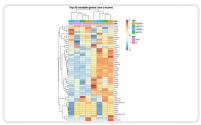

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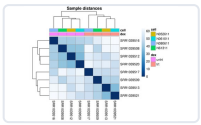

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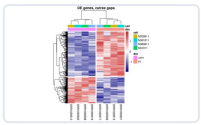

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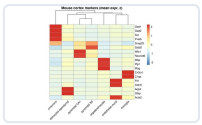

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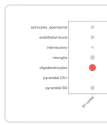

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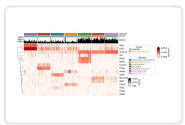

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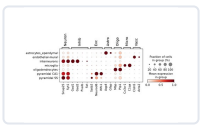

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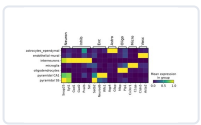

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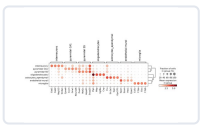

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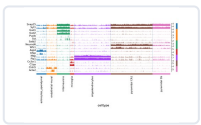

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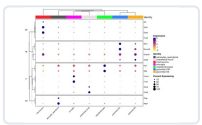

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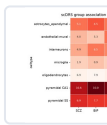

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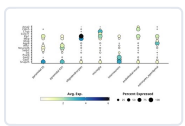

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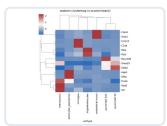

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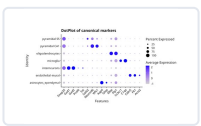

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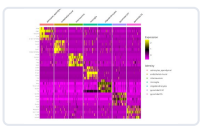

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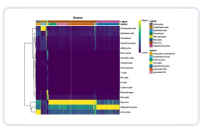

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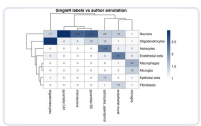

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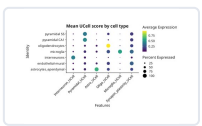

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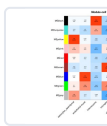

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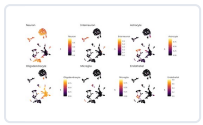

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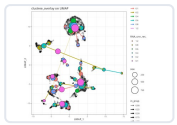

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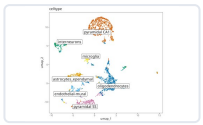

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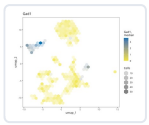

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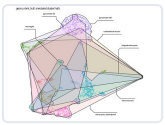

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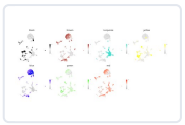

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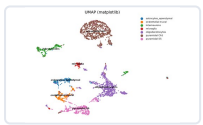

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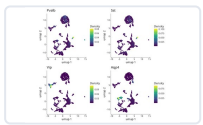

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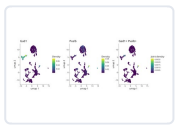

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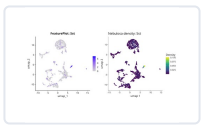

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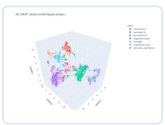

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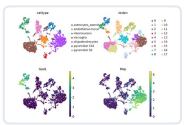

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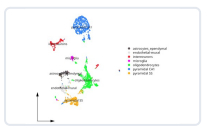

scCustomize
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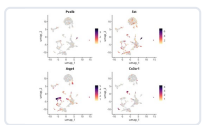
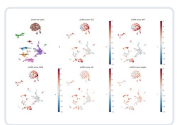


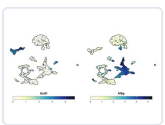
scCustomize
v2_FeaturePlot_scCusto



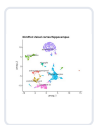
scDRS
v1_umap_scores

scplotter
v1_celldimplot

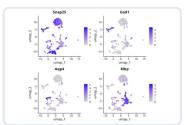
SCpubr
v1_do_DimPlot



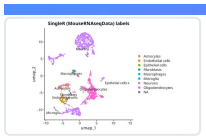
SCpubr
v2_do_NebulosaPlot



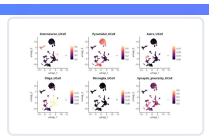
Seurat
v1_DimPlot



Seurat
v2_FeaturePlot



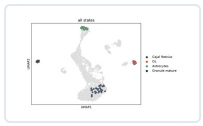
SingleR
v4_umap_labels



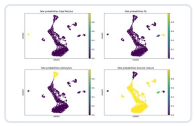
UCell
v1_UCell_featureplots

Trajectory, pseudotime & velocity

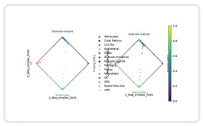
37 FIGURES · 12 PACKAGES



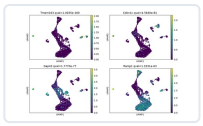
CellRank
v1_macrostates



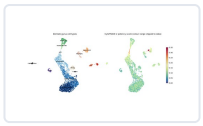
CellRank
v2_fate_probabilities



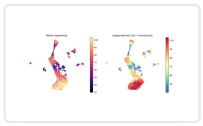
CellRank
v3_circular_projection



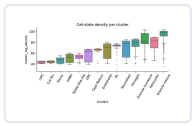
CellRank
v4_lineage_drivers



CytoTRACE 2
v1_potency_umap

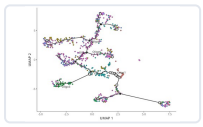


Mellon
v1_density_umap

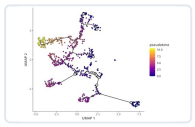


Mellon
v2_density_by_cluster

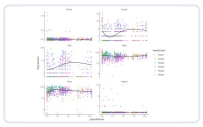
Mellon
v3_densi



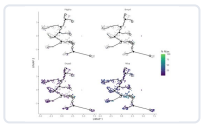
monocle3
v1_trajectory_subclass



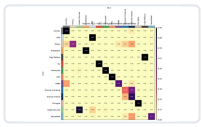
monocle3
v2_pseudotime



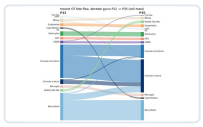
monocle3
v3_genes_in_pseudotime



monocle3
v4_genes_on_trajectory

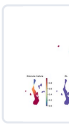


moscot
v1_cell_transition

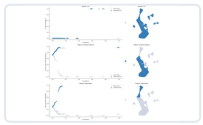


moscot
v2_sankey

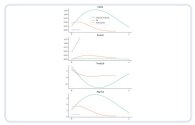
moscot
v3_push_umap



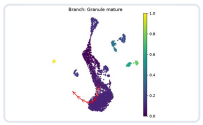
Palantir
v1_palar



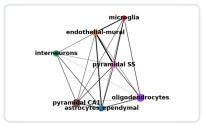
Palantir
v2_branch_selection



Palantir
v3_gene_trends

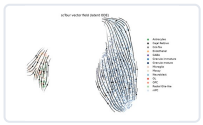


Palantir
v4_trajectory_arrows

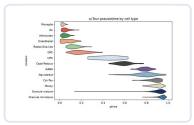


Scanpy
v7_paga

scTour
v1_pseudotime_umap

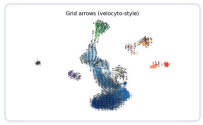


scTour
v2_vector_field

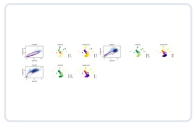


scTour
v3_ptime_by_celltype

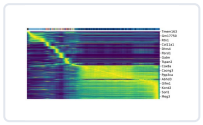
scVelo
v1_stred



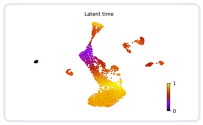
scVelo
v2_grid_arrows



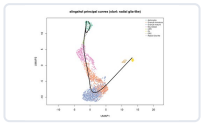
scVelo
v3_phase_portraits



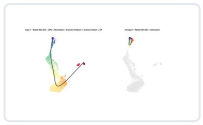
scVelo
v4_latent_time_heatmap



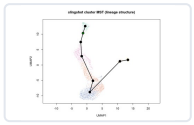
scVelo
v5_latent_time



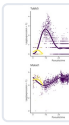
slingshot
v1_curves_on_umap



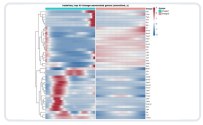
slingshot
v2_pseudotime_per_line



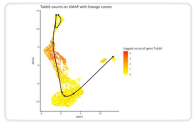
slingshot
v3_cluster_MST



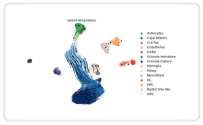
tradeSeq
v1_plotS



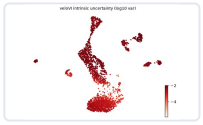
tradeSeq
v2_dynamic_gene_heatma



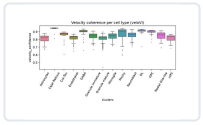
tradeSeq
v3_plotGeneCount



veloVI
v1_velovi_stream



veloVI
v2_uncertainty



veloVI
v3_coherence_by_cluste

Spatial & tissue maps

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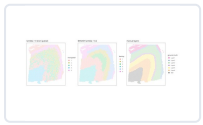
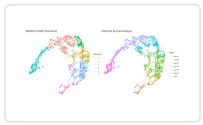
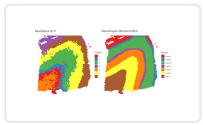


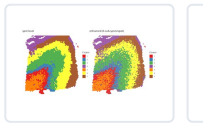
Banksy
v1_banksy_vs_nonspatia



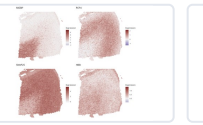
Banksy
v2_banksy_umap



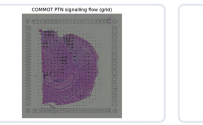
BayesSpace
v1_clusterPlot_vs_trut



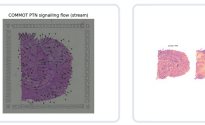
BayesSpace
v2_enhanced_subspot



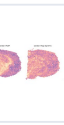
BayesSpace
v3_enhanced_genes



COMMOT
v1_flow_grid

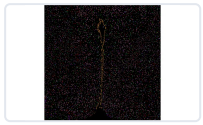


COMMOT
v2_flow_stream



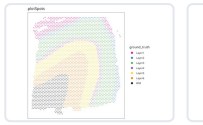
COMMOT
v3_sender_maps

Datashader
v1_density_fire

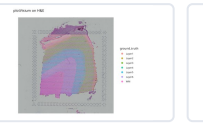


Datashader
v2_categorical_shading

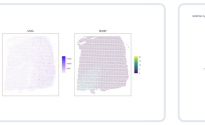
Datashader
v3_all_slices_strip



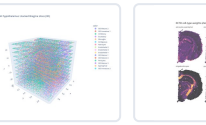
ggspavis
v1_plotSpots_layers



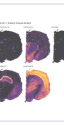
ggspavis
v2_plotVisium_HE



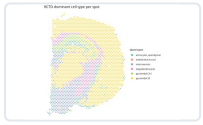
ggspavis
v3_continuous_qc_gene



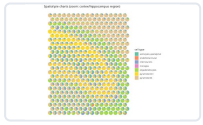
plotly (3D)
v2_merfish3d



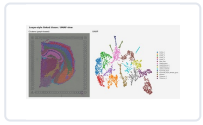
RCTD/spacexr
v1_weight_maps



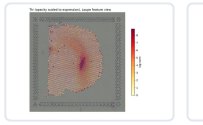
RCTD/spacexr
v2_dominant_type



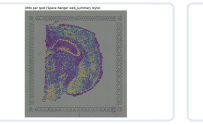
RCTD/spacexr
v3_scatterpie_zoom



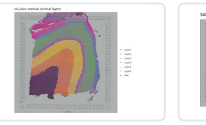
Space Ranger / Lou
v1_tissue_clusters_uma



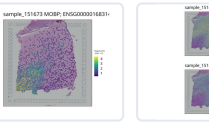
Space Ranger / Lou
v2_feature_opacity



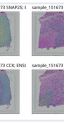
Space Ranger / Lou
v3_umi_per_spot



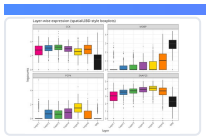
spatialLIBD
v1_vis_clus_layers



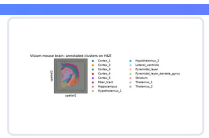
spatialLIBD
v2_vis_gene_MOBP



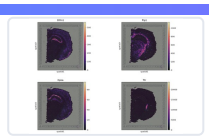
spatialLIBD
v3_vis_grid_gene



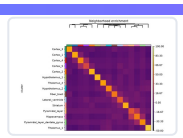
spatialLIBD
v4_layer_boxplots



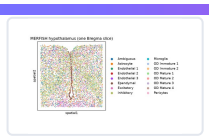
Squidpy
v1_clusters_on_HE



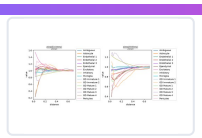
Squidpy
v2_genes_on_HE



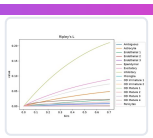
Squidpy
v3_nhood_enrichment



Squidpy
v4_merfish_cell_classe

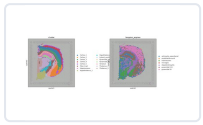


Squidpy
v5_co_occurrence

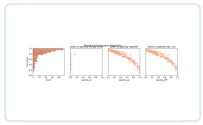


Squidpy
v6_ripley_L

Tangram
v1_celltype_proj



Tangram
v2_argmax_vs_clusters



Tangram
v3_training_scores

Cell–cell communication

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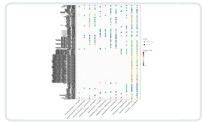
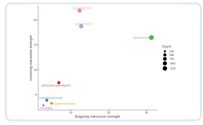
CellChat
v1_circle_counts

CellChat
v2_heatmap

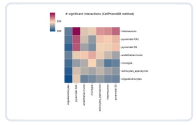


CellChat
v3_bubble

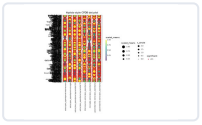


CellChat
v4_signalingRole_scatt

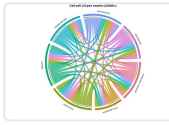
CellChat
v5_chord_pathway



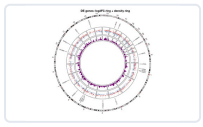
CellPhoneDB
v1_cpdb_heatmap



CellPhoneDB
v2_cpdb_dotplot



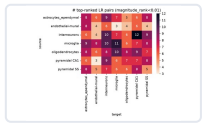
circlize
v1_chord_direction



circlize
v2_circos_genome_rings

LIANA+
v1_dotplot

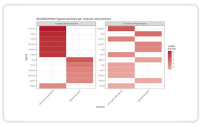
LIANA+
v2_circle_plot



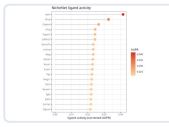
LIANA+
v3_interaction_count_h

MultiNicheNet
v1_circos_group_compar

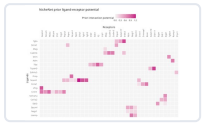
MultiNicheNet
v2_sample_lr_activity



MultiNicheNet
v3_ligand_activity_til

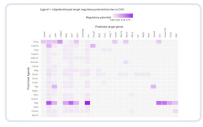


NicheNet
v1_ligand_activity



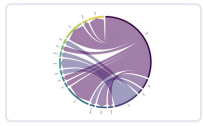
NicheNet
v2_ligand_receptor_hea

NicheNet
v3_circos



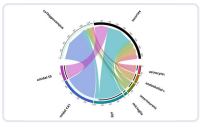
NicheNet
v4_ligand_target_heatm

pyCirclize
v1_chord_interactions



pyCirclize
v2_ligand_receptor_cho

pyCirclize
v3_circos_tracks



SCpubr
v4_do_ChordDiagram

Enrichment & pathway activity

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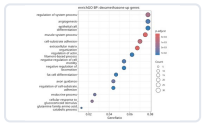
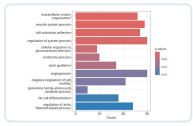
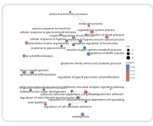


clusterProfiler
v1_dotplot

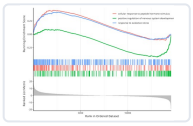


clusterProfiler
v2_barplot

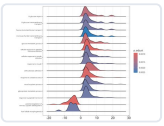
clusterProfiler
v3_cnetplot



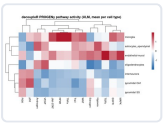
clusterProfiler
v4_emapplot



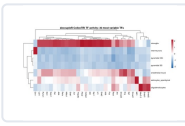
clusterProfiler
v5_gseaplot2



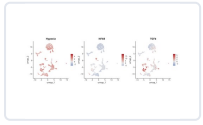
clusterProfiler
v6_ridgeplot



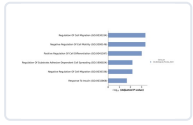
decoupleR
v1_progeny_heatmap



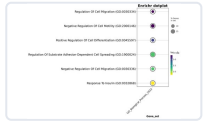
decoupleR
v2_collectri_tf_heatm



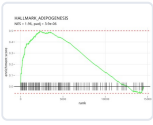
decoupleR
v3_pathway_umaps



Enrichr
v1_barplot_grouped

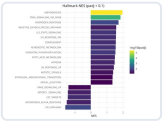


Enrichr
v2_dotplot_by_library



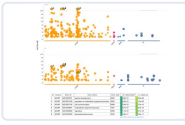
fgsea
v1_plotEnrichment

fgsea
v2_plotGseaTable

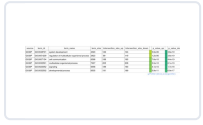


fgsea
v3_NES_barplot

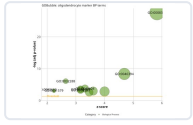
g:Profiler
v1_manhattan



g:Profiler
v2_publish_highlight

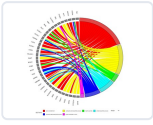


g:Profiler
v3_gosttable

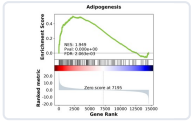


GOplot
v1_GOBubble

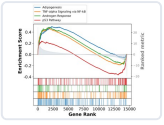
GOplot
v2_GOCircle



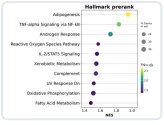
GOplot
v3_GOChord



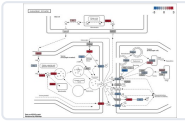
GSEA (Broad) via G
v1_classic_enrichment



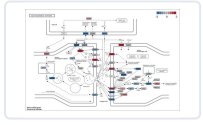
GSEA (Broad) via G
v2_multi_term



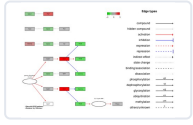
GSEA (Broad) via G
v3_dotplot



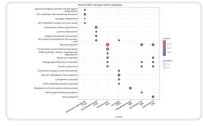
pathview
v1_GABAergic_synapse_



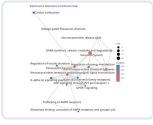
pathview
v2_glutamatergic_synap



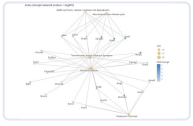
pathview
v3_GABAergic_graphviz



ReactomePA
v1_compareCluster_dotp

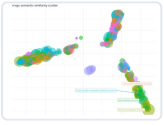


ReactomePA
v2_emapplot

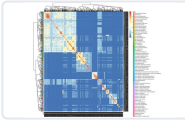


ReactomePA
v3_cnetplot

rrvgo
v1_treemap



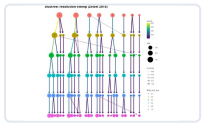
rrvgo
v2_scatter



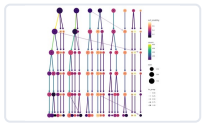
rrvgo
v3_heatmap

Networks, graphs & trees

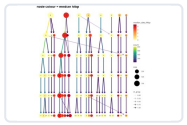
20 FIGURES · 7 PACKAGES



clustree
v1_clustree_tree



clustree
v2_clustree_sc3stabili

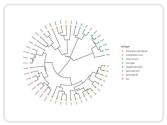


clustree
v3_clustree_gene_Mbp

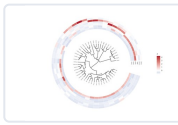
ggraph
v1_coexpression_networ

ggraph
v2_circular_dendrogram

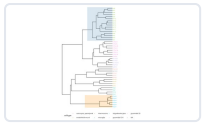
ggraph
v3_circlepack



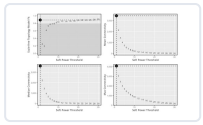
ggtree / ggtreeExt
v1_circular_tree



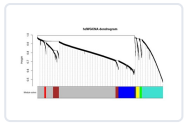
ggtree / ggtreeExt
v2_fan_heatmap_bar_



ggtree / ggtreeExt
v3_rect_tree_clade_hig



hdWGCNA
v1_soft_powers



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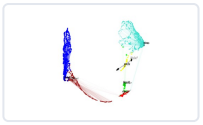
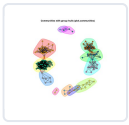
v2_dendrogram

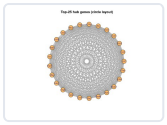


hdWGCNA
v5_ModuleUMAPPlot

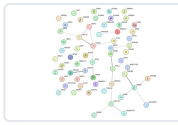
igraph
v1_fr_communities



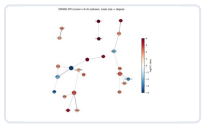
igraph
v2_mark_groups



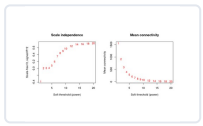
igraph
v3_circle_hubs



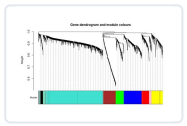
STRING
v1_string_api_networ



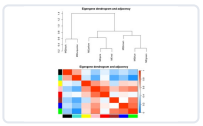
STRING
v2_networkx_fc



WGCNA
v1_soft_threshold



WGCNA
v2_dendro_modules



WGCNA
v4_eigengene_network

Genome tracks & genomic signal

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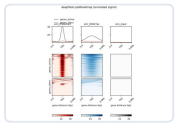
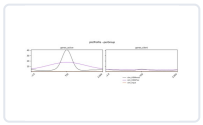
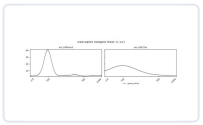


deepTools
v1_plotHeatmap

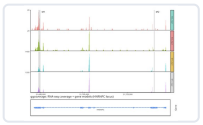


deepTools
v2_plotProfile_perGrou

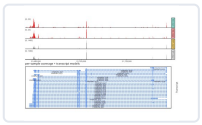
deepTools
v3_kmeans3



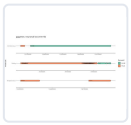
deepTools
v4_scale_regions_profi



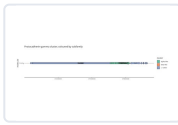
ggcoverage
v1_coverage_gene



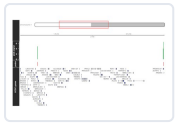
ggcoverage
v2_samples_transcripts



gggenes
v1_gene_arrows



gggenes
v2_pcdhg_cluster

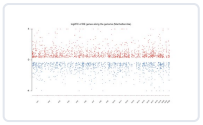


IGV-style tracks (
v1_igv_style_tracks

IGV-style tracks (
v2_zoom_gene_model

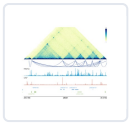
IGV-style tracks (
v3_ucsc_style_dense

karyoploteR
v1_karyotype_density



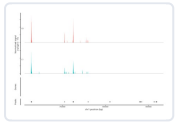
karyoploteR
v2_lfc_scatter_ideogra

karyoploteR
v3_markers_chr

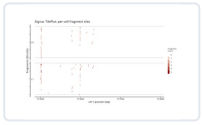


plotgardener
v1_page_layout

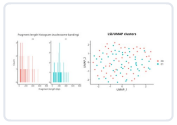
plotgardener
v2_gene_and_ideogram



Signac
v1_coverage_plot



Signac
v2_tile_plot



Signac
v3_fraghist_umap

Brain atlas, 3D & neuroimaging

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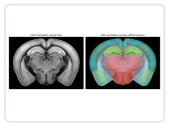
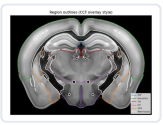
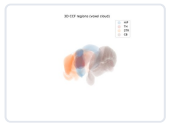


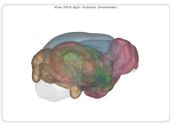
Allen CCF atlas (B
v1_coronal_overlay



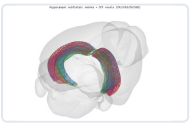
Allen CCF atlas (B
v2_region_outlines



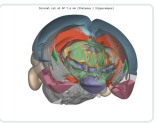
Allen CCF atlas (B
v3_ccf_3d_regions



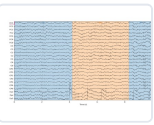
brainrender
v1_regions_3quarter



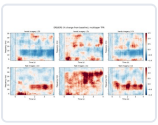
brainrender
v2_hippocampus_points



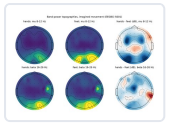
brainrender
v3_coronal_cut



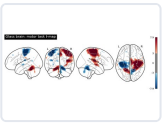
MNE-Python
v1_raw_browser



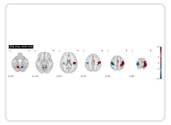
MNE-Python
v2_erds_tfr



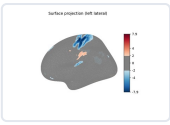
MNE-Python
v3_band_topomaps



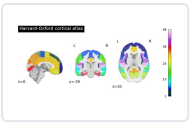
nilearn
v1_glass_brain



nilearn
v2_stat_map_cuts



nilearn
v3_surface_stat_map



nilearn
v4_atlas_roi

Composition, distributions & layouts

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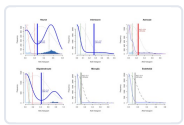
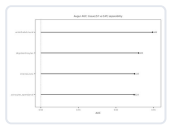
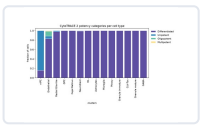


AUCell
v1_exploreThresholds

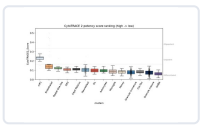


Augur
v1_lollipop

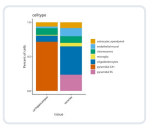
Augur
v3_feature_importance



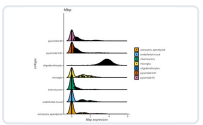
CytoTRACE 2
v2_potency_composition



CytoTRACE 2
v3_score_ranking



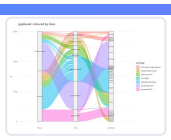
dittoSeq
v2_dittoBarPlot



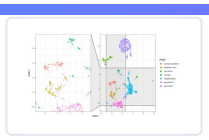
dittoSeq
v3_dittoPlot_ridgejitt

ggalluvial / ggsan
v1_ggsankey

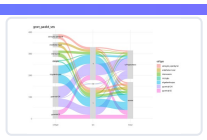
ggalluvial / ggsan
v2_ggsankey_alluvial



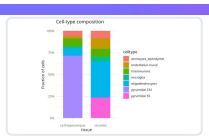
ggalluvial / ggsan
v3_ggalluvial



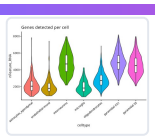
ggforce
v2_facet_zoom



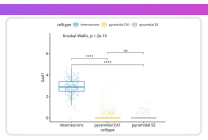
ggforce
v3_parallel_sets



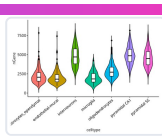
ggplot2
v3_composition_bar



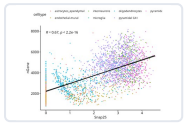
ggplot2
v4_violin_box



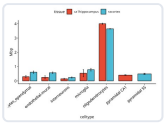
ggpubr
v1_boxplot_pvalues



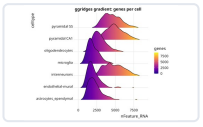
ggpubr
v2_violin



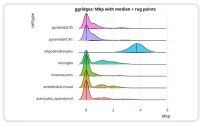
ggpubr
v3_scatter_cor



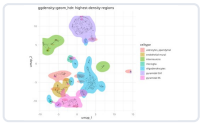
ggpubr
v4_barplot_mean_se



ggridges
v1_ridges_gradient

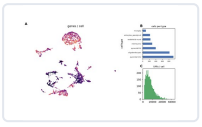


ggridges
v2_ridges_points

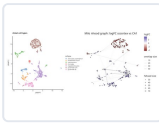


ggridges
v3_hdr_umap

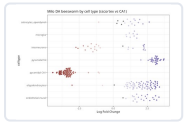
ggridges
v4_hdr_lines



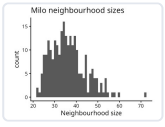
matplotlib
v4_multipanel_gridspec



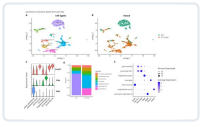
miloR
v1_nhood_graph_DA



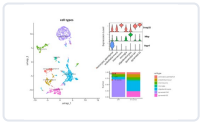
miloR
v2_DA_beeswarm



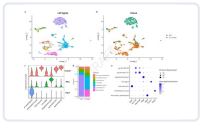
miloR
v3_nhood_size_hist



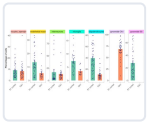
patchwork
v1_patchwork_composite



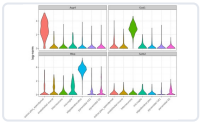
patchwork
v2_patchwork_design_in



patchwork
v3_cowplot_grid_waterm

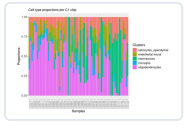


plot1cell
v2_cell_fraction

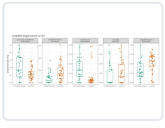


plotnine
v2_faceted_violins

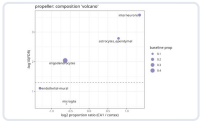
plotnine
v3_bin2d_facets



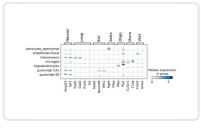
propeller (speckle
v1_plotCellTypeProps



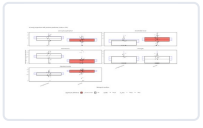
propeller (speckle
v2_props_by_group



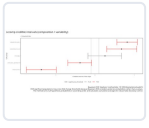
propeller (speckle
v3_composition_volcano



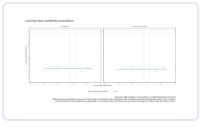
Scanpy
v4_stacked_violin



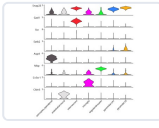
sccomp
v1_posterior_boxplot



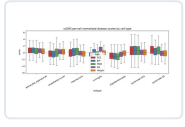
sccomp
v2_1D_intervals



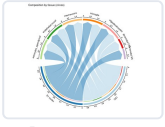
sccomp
v3_2D_mean_variability



scCustomize
v4_Stacked_VlnPlot

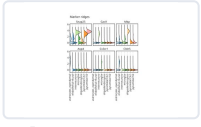


scDRS
v3_score_boxplots

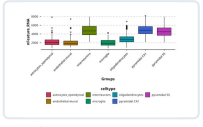


scplotter
v2_cellstat_circos

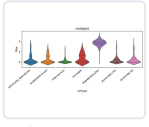
scplotter
v3_cellstat_radar_ring



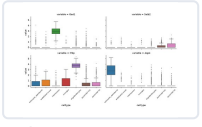
scplotter
v4_featurestat_ridge



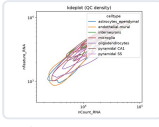
SCpubr
v5_do_BoxPlot



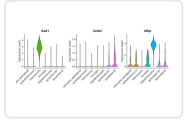
seaborn
v2_violin



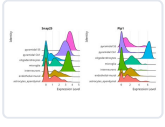
seaborn
v3_catplot_facets



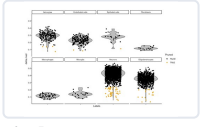
seaborn
v4_kde_qc



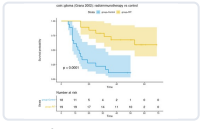
Seurat
v3_VlnPlot



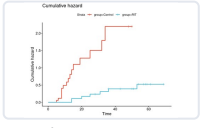
Seurat
v6_RidgePlot



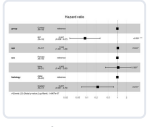
SingleR
v2_plotDeltaDistributi



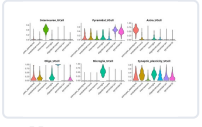
survminer
v1_km_risktable



survminer
v2_cumhaz



survminer
v3_forest



UCell
v2_UCell_violins

DE, QC & set overlaps

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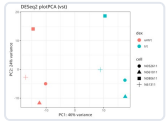
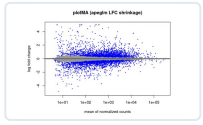
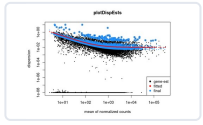


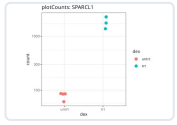
DESeq2
v1_plotPCA



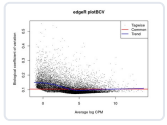
DESeq2
v2_plotMA_shrunken



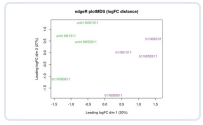
DESeq2
v3_plotDispEsts



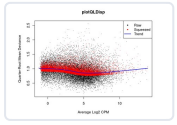
DESeq2
v4_plotCounts



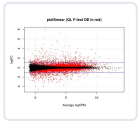
edgeR
v1_plotBCV



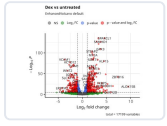
edgeR
v2_plotMDS



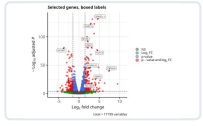
edgeR
v3_plotQLDisp



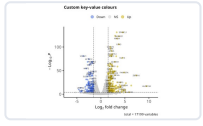
edgeR
v4_plotSmear



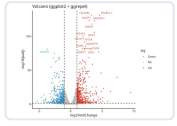
EnhancedVolcano
v1_default



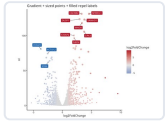
EnhancedVolcano
v2_selected_boxed



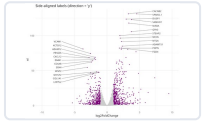
EnhancedVolcano
v3_custom_colours



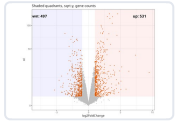
ggplot2
v1_volcano



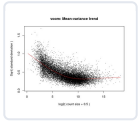
ggrepel volcano st
v1_gradient_labels



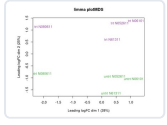
ggrepel volcano st
v2_side_aligned



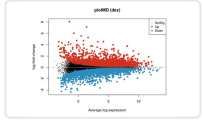
ggrepel volcano st
v3_quadrants_counts



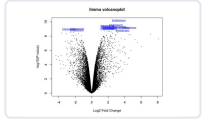
limma
v1_voom_meanvar



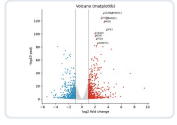
limma
v2_plotMDS



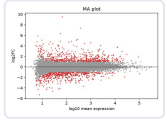
limma
v3_plotMD



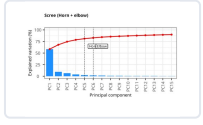
limma
v4_volcanoplot



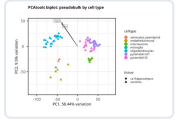
matplotlib
v1_volcano



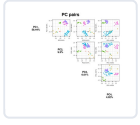
matplotlib
v2_MA



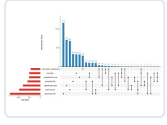
PCAtools
v1_screeplot



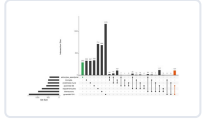
PCAtools
v2_biplot



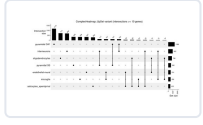
PCAtools
v4_pairsplot



UpSetR
v1_upset_frequency

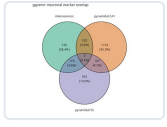


UpSetR
v2_upset_degree_querie

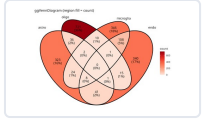


UpSetR
v3_complexheatmap_UpSe

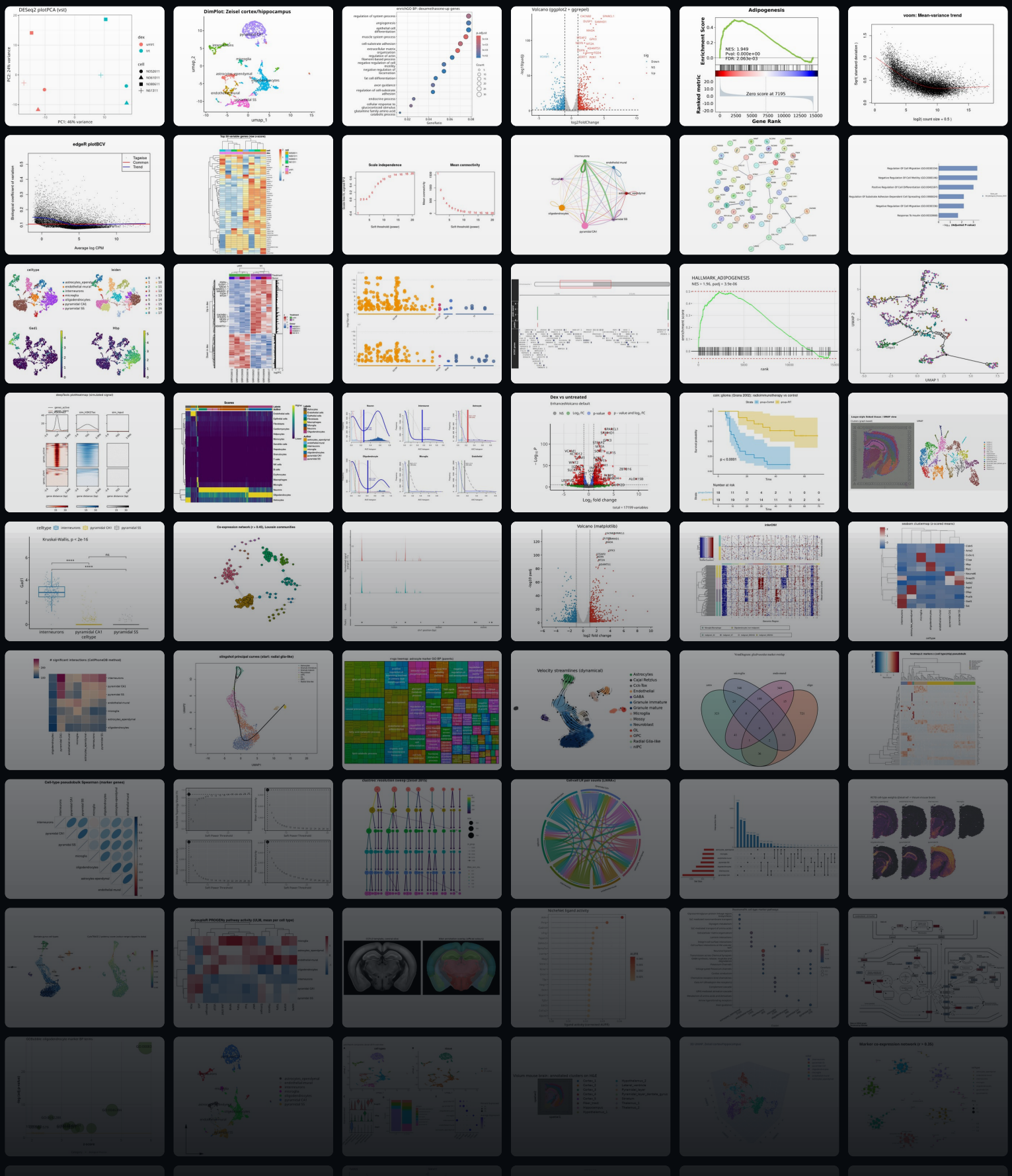
VennDiagram / ggve
v1_VennDiagram_4set



VennDiagram / ggve
v2_ggvenn_3set



VennDiagram / ggve
v3_ggVennDiagram_gradi



PART I · 62 PACKAGES · 217 RECREATED FIGURES

novaflow

Mainstream workhorses

The packages that neuroscience omics papers name most often in their Methods sections, in Top-100 rank order. They cover bulk DE diagnostics, single-cell embeddings, enrichment, communication, trajectories and spatial maps. **70 of the Top 100 entries are recreated in 62 folders**; the other 30 are GUI, web or commercial tools, or not yet recreated (Appendix A).

62
spreads

217
figures

#1–#100
rank span

5,668
papers for #1 (DESeq2)

IN THIS PART · RANK · PAGE

1	DESeq2	20	27	EnhancedVolcano	62	60	CytoTRACE 2	104
2	Seurat	22	29	survminer	64	61	decoupleR	106
3	clusterProfiler	24	31	Space Ranger / Loupe-sty	66	63	Allen CCF atlas (BrainGl	108
4	ggplot2	26	33	ggpubr	68	66	NicheNet	110
5	GSEA (Broad) via GSEApy	28	36	igraph	70	68	ReactomePA	112
6	limma	30	37	Signac	72	70	pathview	114
7	edgeR	32	38	matplotlib	74	71	GOplot	116
9	pheatmap	34	39	inferCNV	76	74	scCustomize	118
10	WGCNA	36	40	seaborn	78	79	patchwork	120
11	CellChat	38	41	CellPhoneDB	80	83	Squidpy	122
12	STRING	40	42	slingshot	82	84	plotly (3D)	124
13	Enrichr	42	44	rrvgo	84	86	ggraph	126
14	Scanpy	44	45	scVelo	86	87	Tangram	128
17	ComplexHeatmap	46	46	VennDiagram / ggvenn	88	88	PCAtools	130
18	g:Profiler	48	47	gplots (heatmap.2)	90	91	tradeSeq	132
19	IGV-style tracks (Gviz)	50	48	corrplot	92	92	LIANA+	134
20	fgsea	52	50	hdWGCNA	94	97	BayesSpace	136
21	monocle3	54	52	clustree	96	98	CellRank	138
22	deepTools	56	54	circlize	98	99	Nebulosa	140
23	SingleR	58	57	UpSetR	100	100	ggalluvial / ggsankey	142
24	AUCell	60	59	RCTD/spacexr	102			

DESeq2

#1

OF 100 · PAPERS NAMING IT

R | BULK QC/DE | DE, QC & SET OVERLAPS

5,668	31.9%	5,871	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

PCA of samples (plotPCA), MA plot, dispersion plot, per-gene counts; results feed volcano/heatmaps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

DESeq2 built-in diagnostic plots (canonical bulk RNA-seq QC/DE figures) Data: airway (dexamethasone vs untreated)

KEY PLOTTING CALL

RECREATIONS/DESEQ2/DESEQ2_RECREATE.R — EXCERPT · ENDS LINE 11

```
x <- readRDS("../_data/airway_deseq2.rds"); dds <- x$dds; res <- x$res
vsd <- vst(dds, blind = FALSE)
# V1: plotPCA (the most-reused DESeq2 figure)
p <- plotPCA(vsd, intgroup = c("dex","cell"), returnData = TRUE); pv <- round(100*attr(p,"percentVar"))
g <- ggplot(p, aes(PC1, PC2, color = dex, shape = cell)) + geom_point(size = 4) +
  xlab(paste0("PC1: ", pv[1], "% variance")) + ylab(paste0("PC2: ", pv[2], "% variance")) + coord_fixed() + theme_bw() +
ggtitle("DESeq2 plotPCA (vst)")
ggsave("v1_plotPCA.png", g, width = 6, height = 4.5, dpi = 150)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 FIGURE CAPTION

GDF15 promotes weight loss by enhancing energy expenditure in muscle. *Nature, 2023*

"...P values were calculated using two-sided unpaired Mann–Whitney U-tests. i, PCA of liver samples from vehicle-treated and GDF15-treated (5 nmol per kg) mice and pair-fed controls using VST data from DESeq2. n = ..."

DOI ↗ Fig. 1 on PMC ↗

Fig. 3 FIGURE CAPTION

APOE4/4 is linked to damaging lipid droplets in Alzheimer's disease microglia. *Nature, 2024*

"...st; per condition, mean ± s.e.m). d, Normalized gene expression counts for significant differentially expressed genes in APOE4/4 iMG ± fAβ (n = 3 replicate wells per condition, P values determined by DEseq2 ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 2 FIGURE CAPTION

Microglia ferroptosis is regulated by SEC24B and contributes to neurodegeneration. *Nat Neurosci, 2023*

"...s and astrocytes in iron + RSL3 versus vehicle. g, Violin plots for FTH1 expression in microglia (red), neurons (green) and astrocytes (yellow) under vehicle or iron + RSL3 conditions. Wald test with DESeq2 ..."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

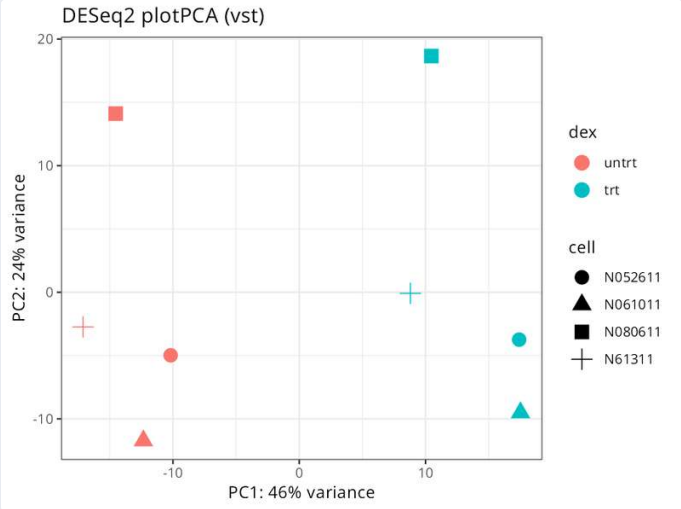
Reproduce: `Rscript recreations/DESeq2/DESeq2_recreate.R`

√ re-run end to end · 16 s

DESeq2 · 4 variations

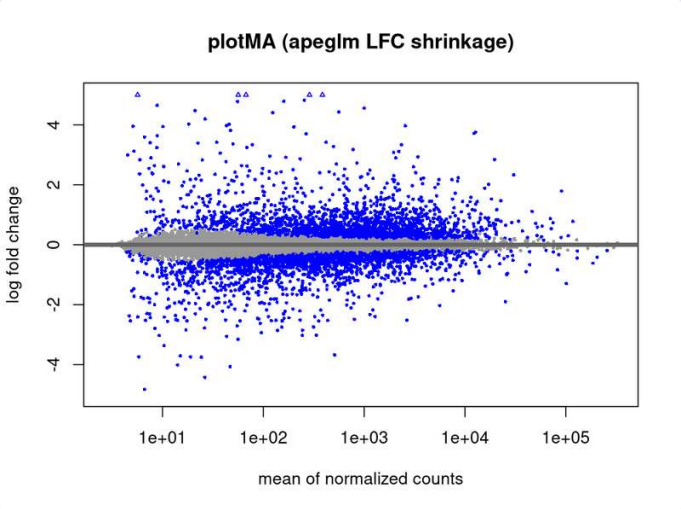
NOVAFLOW RECREATION

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



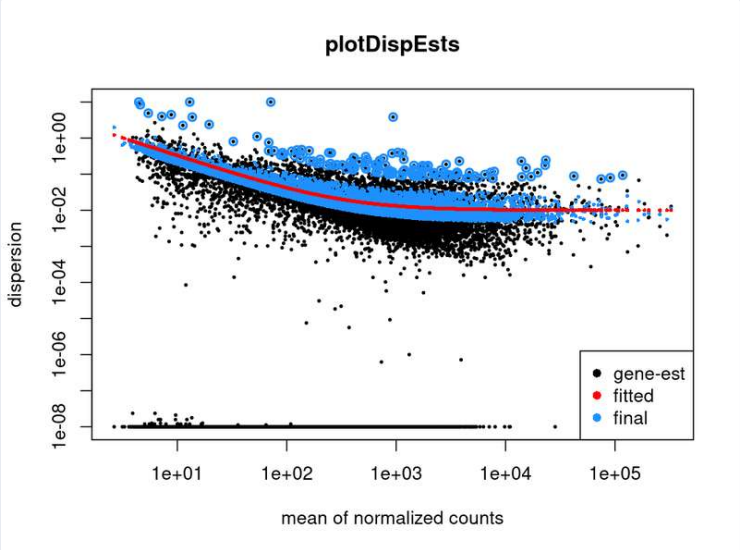
v1 plotPCA

DE



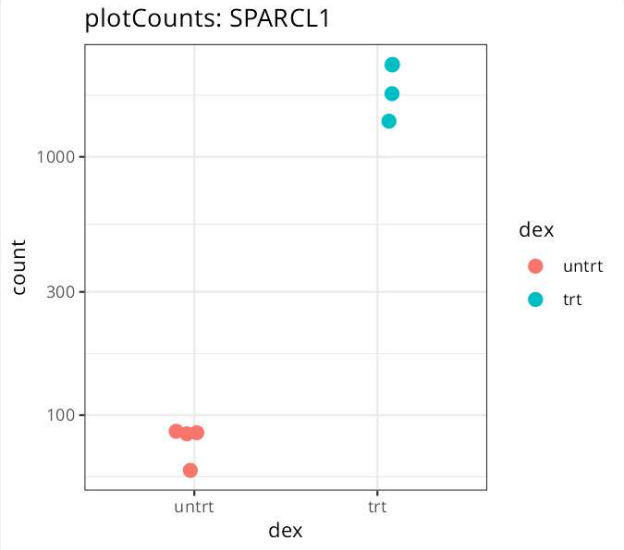
v2 plotMA_shrunken

DE



v3 plotDispEsts

DE



v4 plotCounts

DE

Seurat

#2

OF 100 · PAPERS NAMING IT

R · SINGLE-CELL (EMBEDDING/DOT/VIOLIN/FEATURE) · EMBEDDINGS & FEATURE MAPS · COMPOSITION, DISTRIBUTIONS & LAYOUTS · HEATMAPS & DOT MATRICES

4,671	26.29%	4,846	analysis+plotting	6
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

UMAP/t-SNE DimPlot, FeaturePlot, VlnPlot, DotPlot, DoHeatmap, RidgePlot, spatial feature plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Seurat plotting functions: DimPlot, FeaturePlot, VlnPlot, DotPlot, DoHeatmap, RidgePlot

KEY PLOTTING CALL

RECREATIONS/SEURAT/SEURAT_RECREATE.R — EXCERPT · ENDS LINE 5

```
so <- readRDS("../_data/zeisel_seurat.rds")
ggsave("v1_DimPlot.png", DimPlot(so, label = TRUE, repel = TRUE) + NoLegend() + ggtitle("DimPlot: Zeisel cortex/hippocampus"), w
idth = 6, height = 5.5, dpi = 150, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 — FIGURE CAPTION

Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST. *Nat Commun, 2023*

"...ote the upper and lower quartiles, and whiskers denote the 1.5× interquartile range. B H&E image and manual annotation from the original study. C Clustering results by nonspatial and spatial methods, Seurat ..."

DOI ↗ Fig. 2 on PMC ↗

Fig. 4 — TEXT

cGAS-STING drives ageing-related inflammation and neurodegeneration. *Nature, 2023*

"...For each dataset, we filtered out low-quality and potential doublet cells on the basis of violin plot distributions of genes, UMI counts and mitochondrial gene expression (as determined using the Seurat ..."

DOI ↗ Fig. 4 on PMC ↗

Figure 3 — TEXT

hdWGCNA identifies co-expression networks in high-dimensional transcriptomics data. *Cell Rep Methods, 2023*

"...•hdWGCNA constructs co-expression networks in high-dimensional transcriptomics data•hdWGCNA provides tools for statistics, visualization, and downstream interpretation•hdWGCNA is an open-source R package that ..."

DOI ↗ Figure 3 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

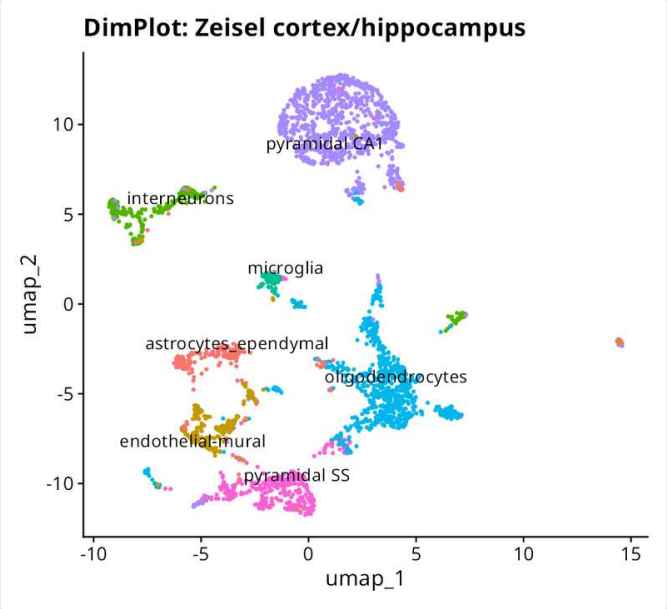
Reproduce: `Rscript recreations/Seurat/Seurat_recreate.R`

√ re-run end to end · 13 s

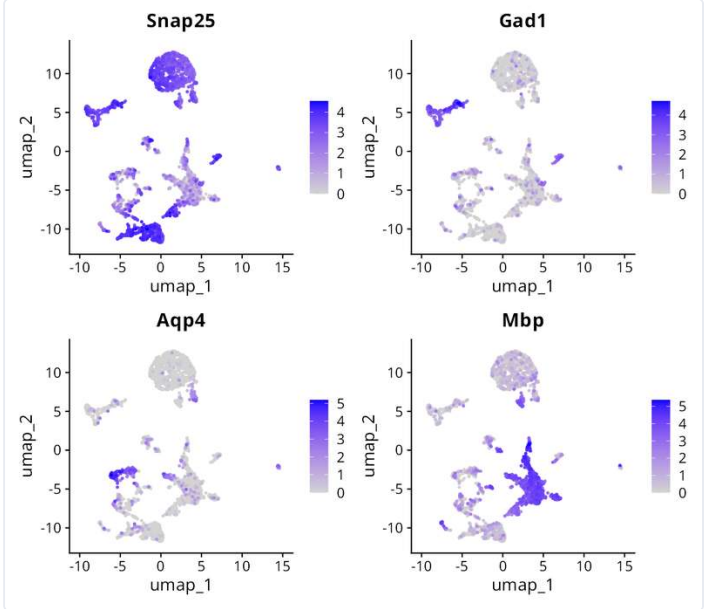
Seurat · 6 variations

NOVAFLOW RECREATION

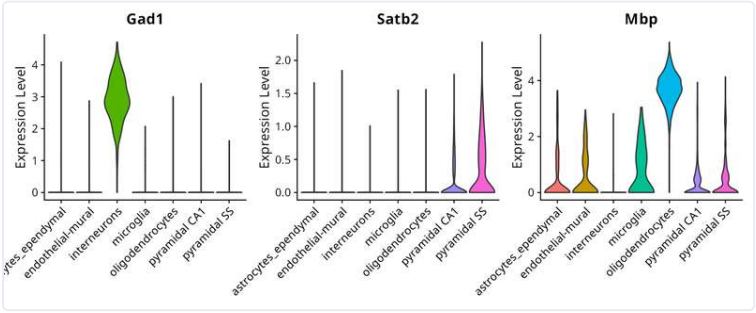
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



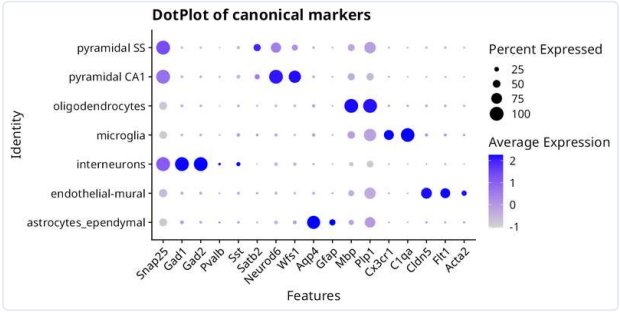
v1 DimPlot — EMBEDDINGS



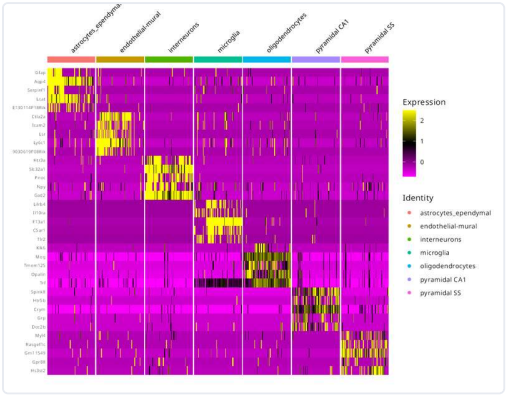
v2 FeaturePlot — EMBEDDINGS



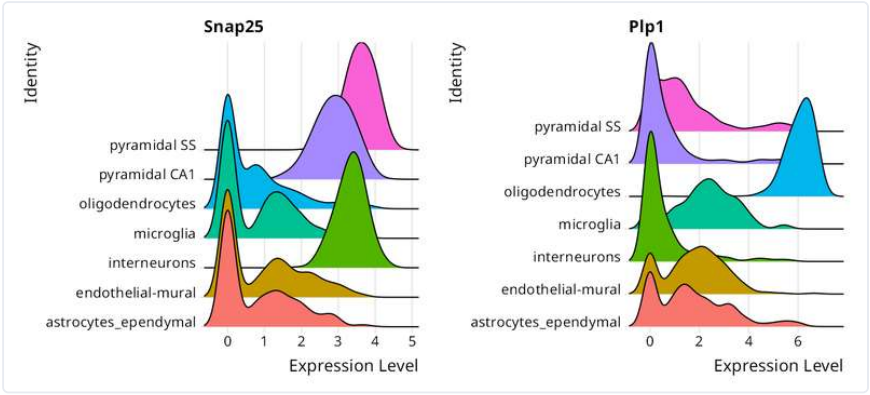
v3 VlnPlot — COMPOSITION



v4 DotPlot — HEATMAPS



v5 DoHeatmap — HEATMAPS



v6 RidgePlot — COMPOSITION

clusterProfiler

#3

OF 100 · PAPERS NAMING IT

R | ENRICHMENT | ENRICHMENT & PATHWAY ACTIVITY

3,518	19.8%	3,609	analysis+plotting	6
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

GO/KEGG ORA dot & bar plots, cnetplot, emapplot, GSEA running-score (via enrichplot)

Also catalogued as: **enrichplot** (#32, 280 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

clusterProfiler + enrichplot: ORA and GSEA visualisations

KEY PLOTTING CALL

RECREATIONS/CLUSTERPROFILER/CLUSTERPROFILER_RECREATE.R — EXCERPT · ENDS LINE 7

```
res <- as.data.frame(readRDS("../_data/airway_deseq2.rds")$res); res <- res[!is.na(res$padj) & !is.na(res$entrez),]
up <- unique(res$entrez[res$padj < .01 & res$log2FoldChange > 1]); uni <- unique(res$entrez)
ego <- enrichGO(up, OrgDb = org.Hs.eg.db, ont = "BP", universe = uni, readable = TRUE); ego <- simplify(ego)
ggsave("v1_dotplot.png", dotplot(ego, showCategory = 15) + ggtitle("enrichGO BP: dexamethasone-up genes"), width = 7.5, height
= 6.5, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 7 — FIGURE CAPTION

Cell-type-specific CAG repeat expansions and toxicity of mutant Huntingtin in human striatum and cerebellum. *Nat Genet, 2024*

"...es are omitted from the plot. For panels c and f, the significance threshold for enrichment analysis was: q value < 0.05, Padj < 0.05, adjusted for multiple comparisons after hypergeometric test with ..."

DOI ↗ Fig. 7 on PMC ↗

Extended Data Fig. 7 — FIGURE CAPTION

Blood-brain barrier disruption and sustained systemic inflammation in individuals with long COVID-associated cognitive impairment. *Nat Neurosci, 2024*

"...ed Gene Ontology biological processes in each cohort. Bubble plots show top 20 enriched terms with absLog2FC cut-off < 0.59 and adjusted p-value < 0.05. Pathway enrichment p values were calculated in ..."

DOI ↗ Extended Data Fig. 7 on PMC ↗

Fig. 1 — TEXT

Transcriptional and epigenetic decoding of the microglial aging process. *Nat Aging, 2023*

"...categories were simplified using the 'simplify' function with a cutoff value = 0.7, and visualization was performed using the dotplot and emapplot functions in clusterProfiler 3.14.3."

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

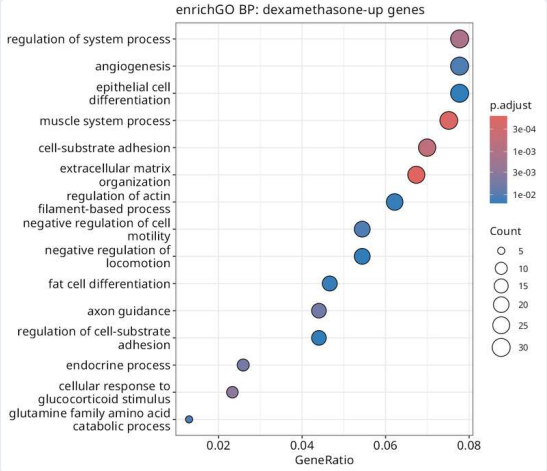
- enrichplot (#32): Shares clusterProfiler recreations (all plots are enrichplot functions)

Reproduce: `Rscript recreations/clusterProfiler/clusterProfiler_recreate.R`

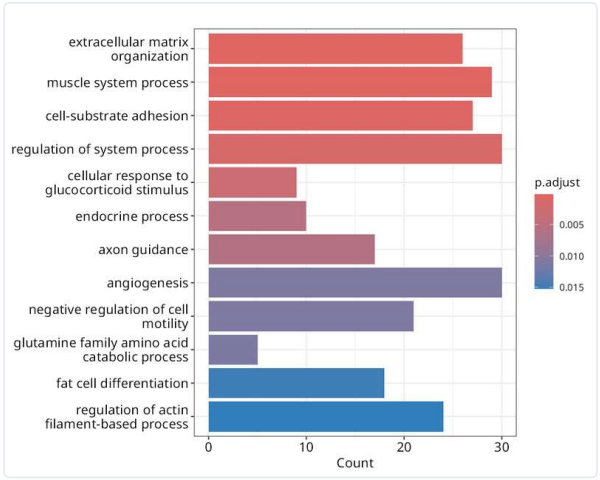
✓ re-run end to end · 57 s

clusterProfiler · 6 variations

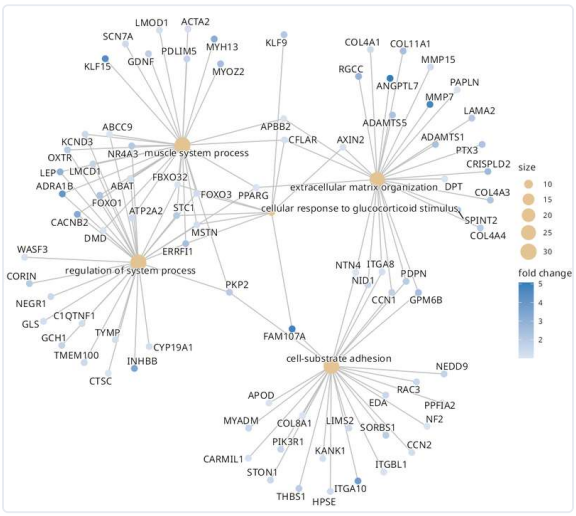
Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



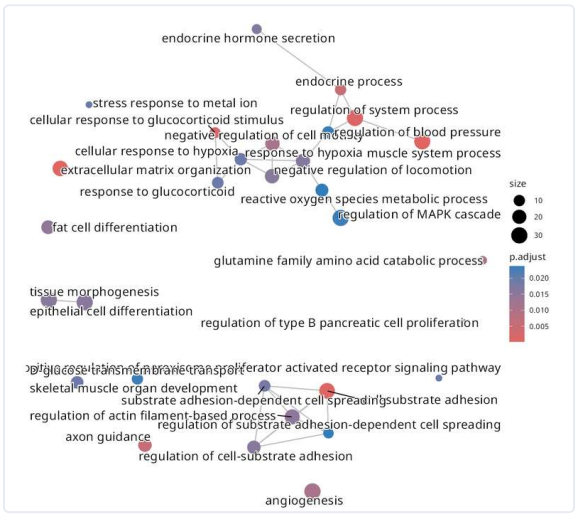
v1 dotplot ENRICHMENT



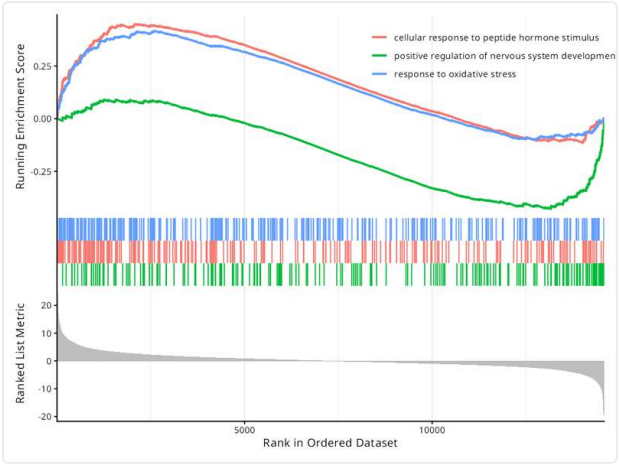
v2 barplot ENRICHMENT



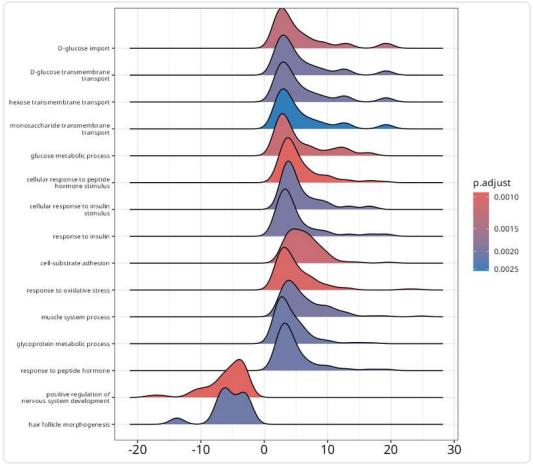
v3 cnetplot ENRICHMENT



v4 emapplot ENRICHMENT



v5 gseaplot2 ENRICHMENT



v6 ridgeplot ENRICHMENT

PART I · MAINSTREAM WORKHORSES · 04 / 62

ggplot2

#4

OF 100 · PAPERS NAMING IT

R · GENERAL · DE, QC & SET OVERLAPS · EMBEDDINGS & FEATURE MAPS · COMPOSITION, DISTRIBUTIONS & LAYOUTS

2,490	14.01%	2,546	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Everything custom: volcano, bar/box/violin, UMAP scatter, composition stacked bars, multi-panel

Also catalogued as: **ggrepel** (#64, 92 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

ggplot2: the grammar-of-graphics backbone of most omics figures

KEY PLOTTING CALL

RECREATIONS/GGPLOT2/GGPLOT2_RECREATE.R — EXCERPT · ENDS LINE 11

```
res <- res |> filter(!is.na(padj)) |> mutate(sig = case_when(padj < .05 & log2FoldChange > 1 ~ "Up", padj < .05 &
log2FoldChange < -1 ~ "Down", TRUE ~ "NS"))
lab <- res |> filter(sig != "NS") |> arrange(padj) |> head(15)
# V1: volcano plot
g <- ggplot(res, aes(log2FoldChange, -log10(padj), color = sig)) + geom_point(size = .8, alpha = .7) +
  scale_color_manual(values = c(Up = "#D7301F", Down = "#2B8CBE", NS = "grey75")) + geom_vline(xintercept = c(-1,1), lty = 2) +
geom_hline(yintercept = -log10(.05), lty = 2) +
  geom_text_repel(data = lab, aes(label = symbol), size = 3, max.overlaps = 30, show.legend = FALSE) + theme_classic() + labs(t
itle = "Volcano (ggplot2 + ggrepel)")
ggsave("v1_volcano.png", g, width = 6, height = 5, dpi = 150)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 TEXT
Microglia ferroptosis is regulated by SEC24B and contributes to neurodegeneration. *Nat Neurosci, 2023*
"...UMAP plots and gene expression plots were generated using built-in Seurat/ggplot2 plotting functions or scCustomize version 0.7.0 unless otherwise described."
DOI ↗ Fig. 1 on PMC ↗

Figure 7 TEXT
Macrophage-mediated myelin recycling fuels brain cancer malignancy. *Cell, 2024*
"...Plots were generated using the package ggplot (v.3.3.6) or ggpubr (v.0.4.0), except where mentioned."
DOI ↗ Figure 7 on PMC ↗

Fig. 6 TEXT
Microglia regulate central nervous system myelin growth and integrity. *Nature, 2023*
"...ggplot2 v.3.3.5 was used to perform custom plots, here v.1.0.1 was used to ensure reproducible paths, and Matrix v.1.3.4 was used for handling sparse matrices."
DOI ↗ Fig. 6 on PMC ↗

LIMITATIONS & NOTES

- ggrepel (#64): See ggplot2 v1_volcano.png (ggrepel labels)

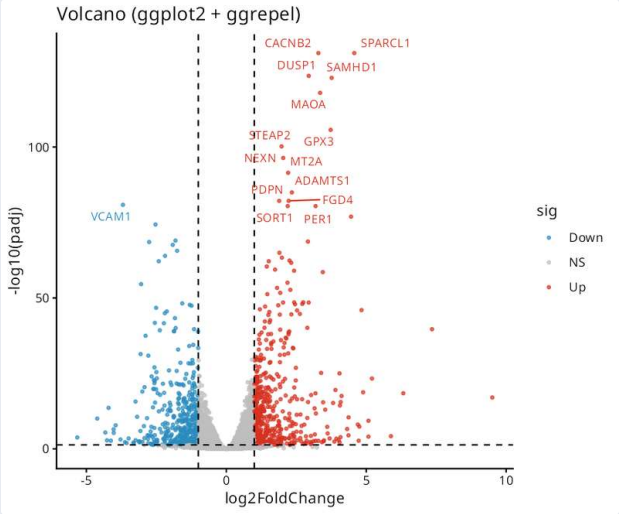
Reproduce: `Rscript recreations/ggplot2/ggplot2_recreate.R`

✓ re-run end to end · 2 s

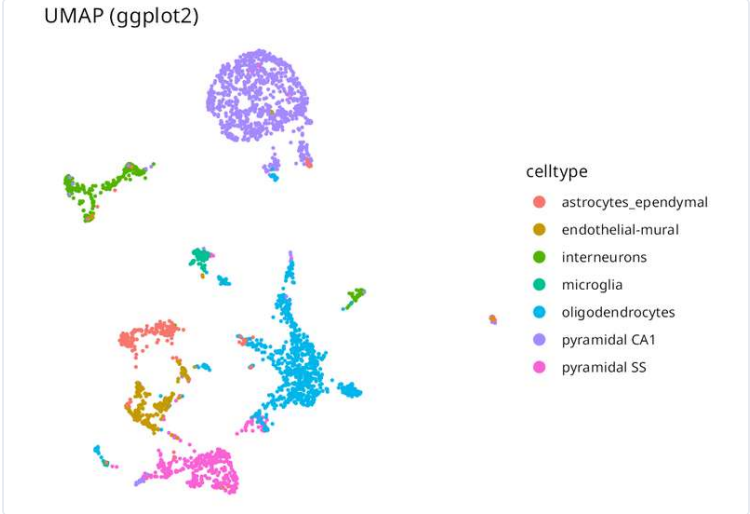
ggplot2 · 4 variations

NOVAFLOW RECREATION

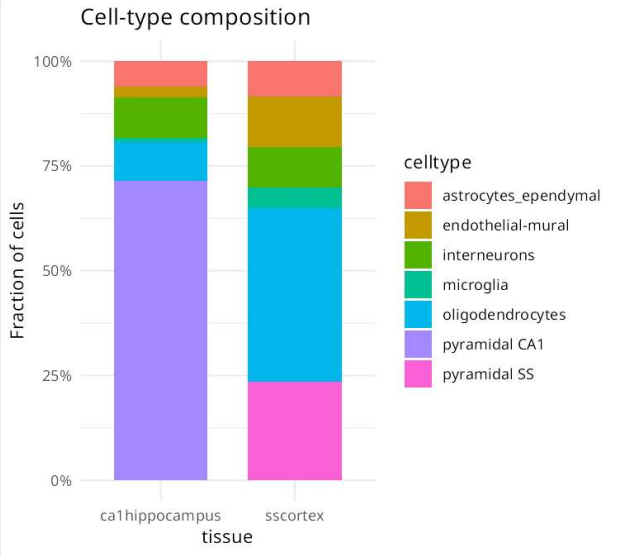
Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*); Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



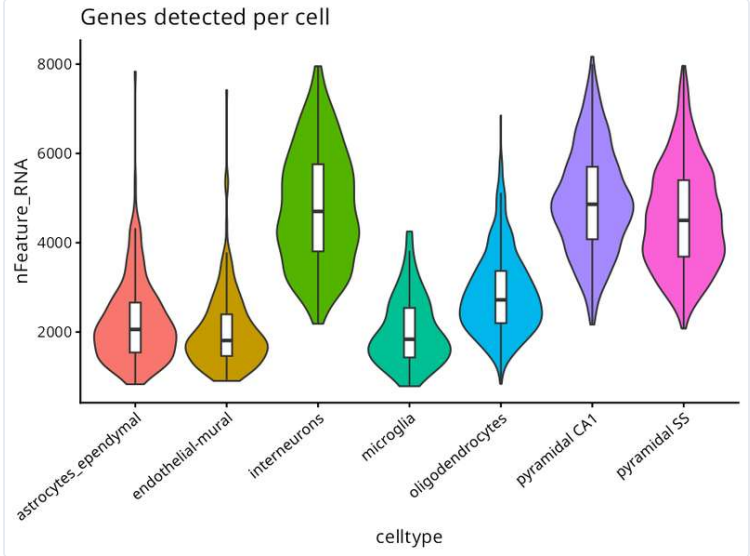
v1 volcano · DE



v2 umap · EMBEDDINGS



v3 composition_bar · COMPOSITION



v4 violin_box · COMPOSITION

GSEA (Broad) via GSEApy

DESKTOP | ENRICHMENT | ENRICHMENT & PATHWAY ACTIVITY

#5
OF 100 · PAPERS NAMING IT

2,056	11.57%	3,310	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Classic running enrichment-score plot with hit ticks and ranked metric; NES bar/dot summaries

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Broad GSEA "classic" enrichment figure, recreated with GSEApy prerank (same algorithm/plot layout as the Broad desktop app)

KEY PLOTTING CALL

RECREATIONS/GSEA_BROAD/GSEA_STYLE_RECREATE.PY — EXCERPT · ENDS LINE 8

```
res=pd.read_csv("../_data/airway_deseq2_results.csv",index_col=0).dropna(subset=["stat","symbol"])
rnk=res.groupby("symbol").stat.mean().sort_values(ascending=False)
pre=gp.prerank(rnk=rnk,gene_sets="MSigDB_Hallmark_2020",permutation_num=1000,seed=1,outdir=None,min_size=15,max_size=500,threads=4)

r=pre.res2d.sort_values("NOM p-val"); terms=r.Term.tolist()
gp.gseaplot(rank_metric=pre.ranking,term=terms[0],**pre.results[terms[0]],ofname="v1_classic_enrichment.png")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 1 — FIGURE CAPTION

Macrophage-mediated myelin recycling fuels brain cancer malignancy. *Cell, 2024*

"...ubtypes (outer circle) per model. Pseudo-location2 assignment of glioblastoma cells (inner circle) displayed as a percentage of cellular subtype.(E) UMAP representation of MG and MDM sub-clusters.(F) Gene set ..."

DOI ↗ Figure 1 on PMC ↗

Extended Data Fig. 8 — FIGURE CAPTION

Molecular and cellular dynamics of the developing human neocortex. *Nature, 2025*

"...sion levels of cell-type-specific marker genes and SSTR2. c, Scatter plots illustrating the Pearson correlation of log fold changes in individual genes between Ostreotide and L-054,264 treatments. d, Gene set ..."

DOI ↗ Extended Data Fig. 8 on PMC ↗

Fig. 1 — TEXT

Specialized astrocytes mediate glutamatergic gliotransmission in the CNS. *Nature, 2023*

"...Using gene set enrichment analysis (GSEA) databases (Methods), we found significant Gene Ontology (GO) enrichment for each cluster (Extended Data Fig."

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

- Broad desktop app is GUI; plot recreated with GSEApy prerank (same algorithm & layout)

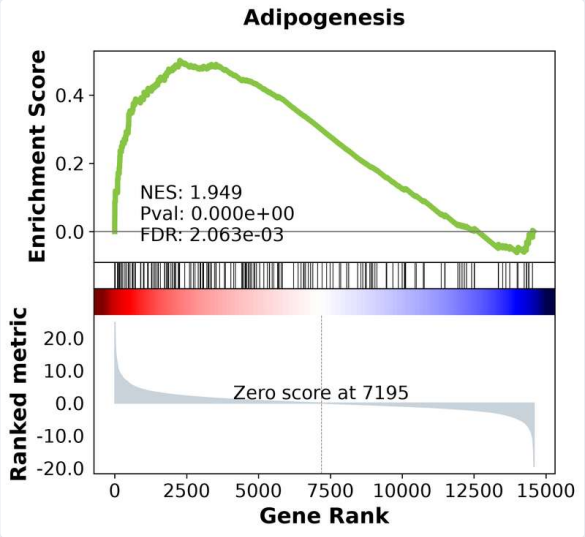
Reproduce: `python3 recreations/GSEA_Broad/gsea_style_recreate.py`

√ re-run end to end · 4 s

GSEA (Broad) via GSEApy · 3 variations

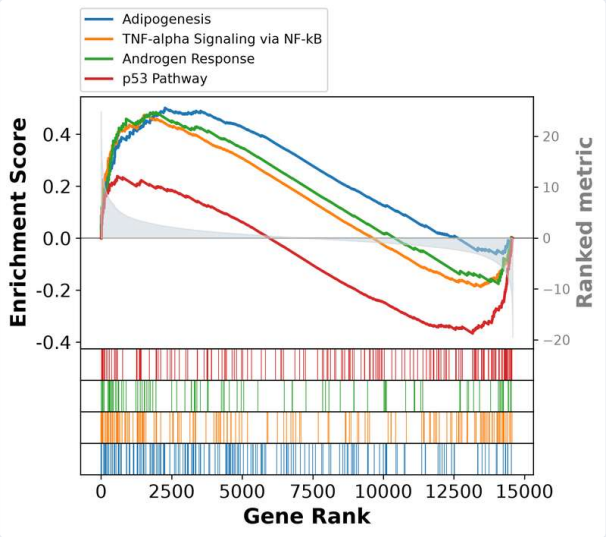
NOVAFLOW RECREATION

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



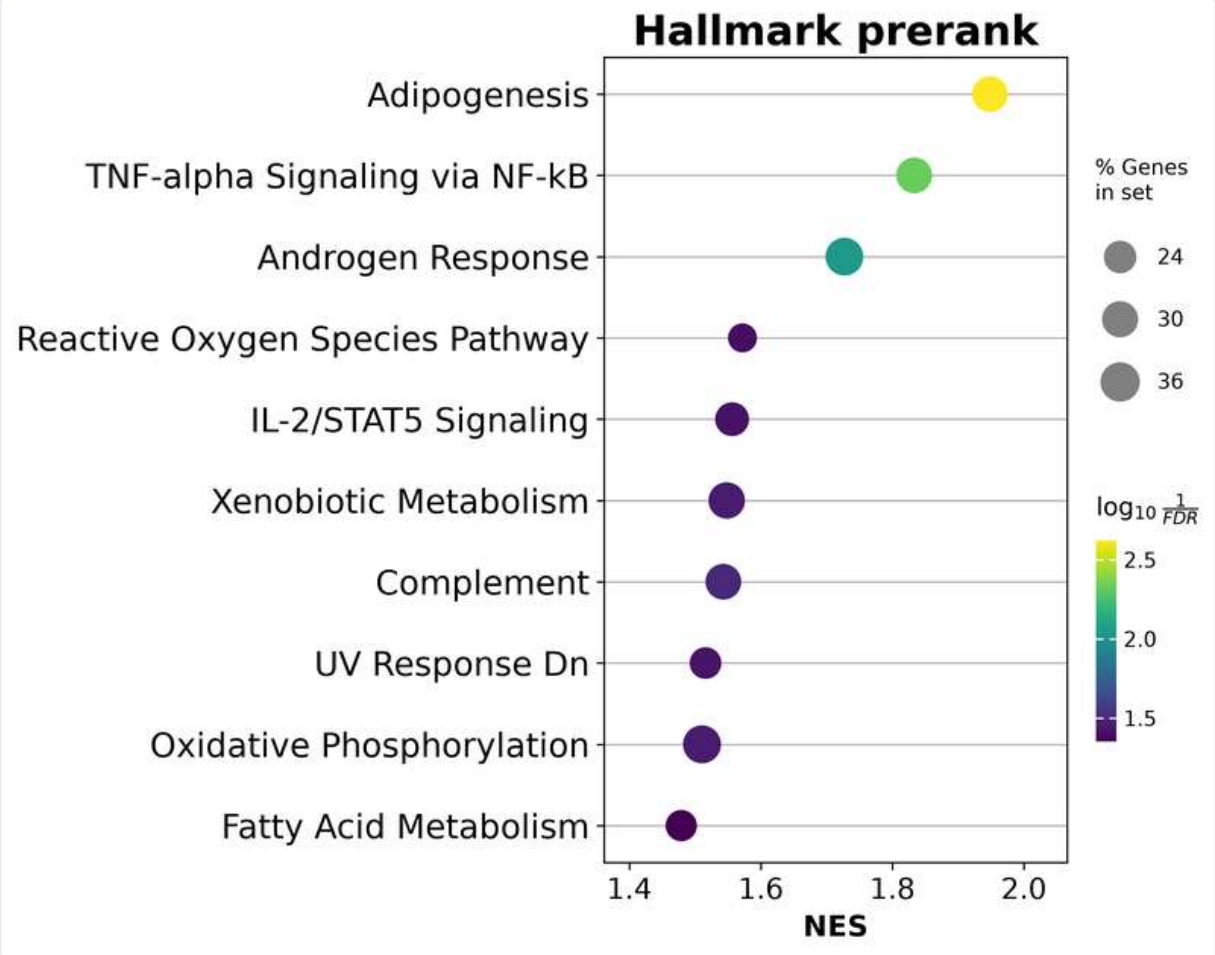
v1 classic_enrichment

ENRICHMENT



v2 multi_term

ENRICHMENT



v3 dotplot

ENRICHMENT

limma

#6

OF 100 · PAPERS NAMING IT

R | BULK QC/DE | DE, QC & SET OVERLAPS

2,047	11.52%	2,116	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

voom mean-variance, MDS, MD/MA, volcanoplot, Venn of contrasts

KEY PLOTTING CALL

RECREATIONS/LIMMA/LIMMA_RECREATE.R — EXCERPT · ENDS LINE 11

```
png("v1_voom_meanvar.png", 1000, 800, res = 150); v <- voom(y, design, plot = TRUE); dev.off()
fit <- eBayes(lmFit(v, design)); cf <- ncol(design)
# V2: MDS plot
png("v2_plotMDS.png", 1000, 850, res = 150); plotMDS(v, labels = paste(dds$dex, dds$cell), col = ifelse(dds$dex == "trt", "#984
EA3", "#4DAF4A"), main = "limma plotMDS"); dev.off()
# V3: MD plot with DE status
png("v3_plotMD.png", 1000, 800, res = 150); plotMD(fit, coef = cf, status = decideTests(fit)[,cf], main = "plotMD (dex)",
hl.col = c("#D7301F","#2B8CBE")); abline(h=0,col="grey"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 — FIGURE CAPTION

CGRP sensory neurons promote tissue healing via neutrophils and macrophages. *Nature, 2024*

"...SP-1 in neutrophils and macrophages to mediate its activity.a,b, RNA-seq analysis of CGRP-treated neutrophils and macrophages (1 nM or saline for 4 h). Differential gene expression was performed with ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 1 — FIGURE CAPTION

A TREM2-activating antibody with a blood-brain barrier transport vehicle enhances microglial metabolism in Alzheimer's disease models. *Nat Neurosci, 2023*

"...expression was generated by summing every counts per gene. Each mouse was treated as a biological replicate, and differential expression (DE) was performed for each cluster versus 'homeostatic' using limma. g ..."

DOI ↗ Fig. 1 on PMC ↗

Figure 3 — TEXT

Single-cell RNA sequencing and spatial transcriptomics reveal cancer-associated fibroblasts in glioblastoma with protumoral effects. *J Clin Invest, 2023*

"...For differential gene expression analysis, raw read counts were processed in iDEP using the limma-voom function."

DOI ↗ Figure 3 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

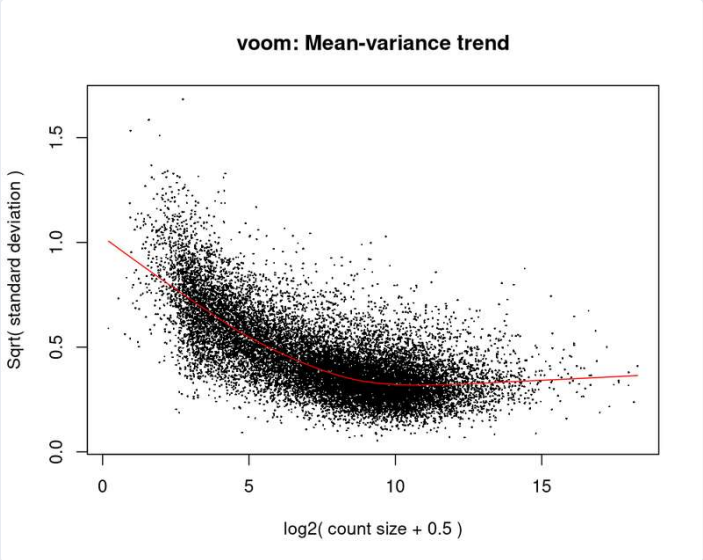
Reproduce: `Rscript recreations/limma/limma_recreate.R`

✓ re-run end to end · 9 s

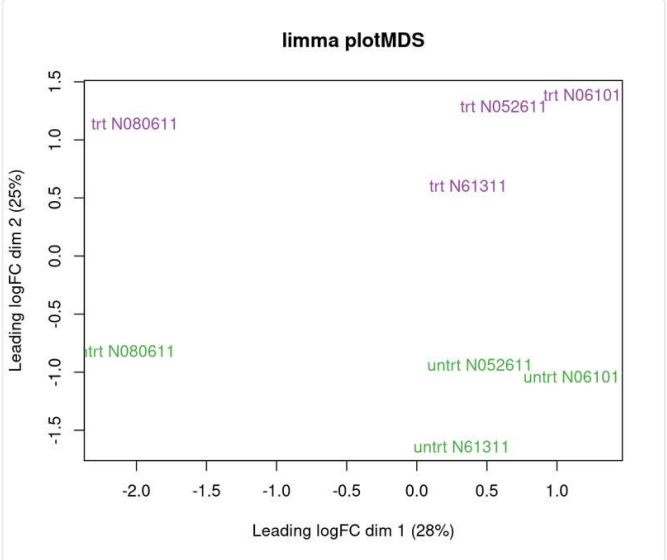
limma · 4 variations

NOVAFLOW RECREATION

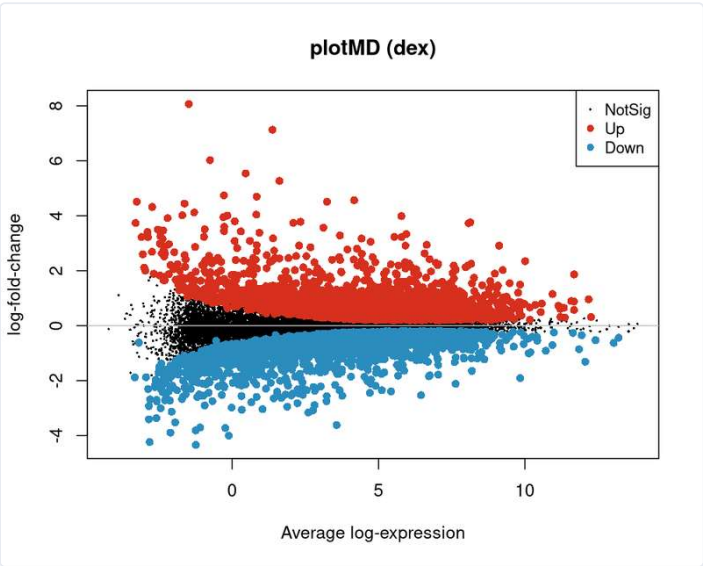
Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



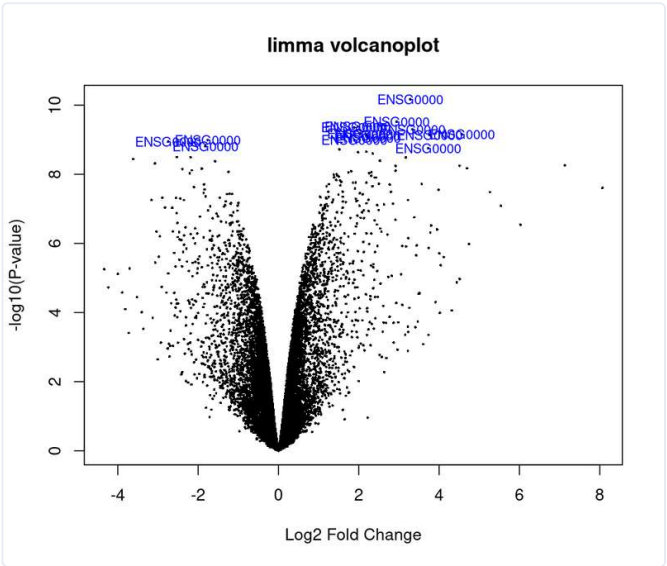
v1 voom_meanvar DE



v2 plotMDS DE



v3 plotMD DE



v4 volcanoplot DE

edgeR

#7

OF 100 · PAPERS NAMING IT

R | BULK QC/DE | DE, QC & SET OVERLAPS

1,892	10.65%	1,949	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

BCV, MDS, smear/MD plots, QL dispersion

KEY PLOTTING CALL

RECREATIONS/EDGER/EDGER_RECREATE.R — EXCERPT · ENDS LINE 6

```
dds <- readRDS("../_data/airway_deseq2.rds")$dds; grp <- dds$dex
y <- DGEList(counts(dds), group = grp); y <- y[filterByExpr(y), , keep.lib.sizes = FALSE]; y <- normLibSizes(y)
design <- model.matrix(~ dds$cell + grp); y <- estimateDisp(y, design)
png("v1_plotBCV.png", 1000, 800, res = 150); plotBCV(y, main = "edgeR plotBCV"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 — FIGURE CAPTION

Cell type specific transcriptomic differences in depression show similar patterns between males and females but implicate distinct cell types and genes. *Nat Commun, 2023*

"...ers with highest number of DEGs in the female cluster level analysis—f microglia, g inhibitory neuronal clusters. For f, g, the plotted values are pseudobulk CPMs (counts per million) calculated with edgeR and ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 1 — TEXT

Microglia ferroptosis is regulated by SEC24B and contributes to neurodegeneration. *Nat Neurosci, 2023*

"...The PCA plots were generated using edgeR and Glimma."

DOI ↗ Fig. 1 on PMC ↗

Figure 1 — TEXT

Serotonin reduction in post-acute sequelae of viral infection. *Cell, 2023*

"...Briefly, transcript quantification data were summarized to genes using the tximport package and normalized using the trimmed mean of M values (TMM) method in edgeR."

DOI ↗ Figure 1 on PMC ↗

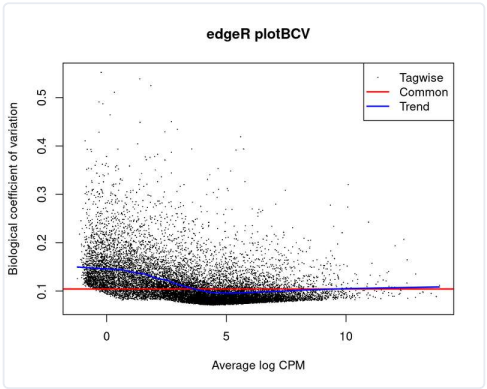
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/edgeR/edgeR_recreate.R`

✓ re-run end to end · 13 s

LEAD FIGURE

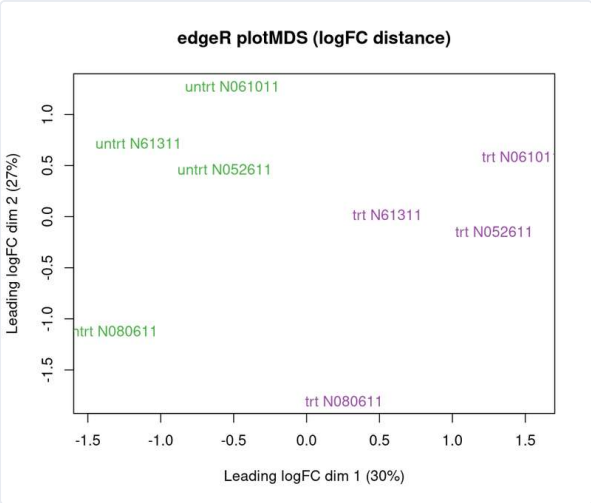


v1 plotBCV

DE

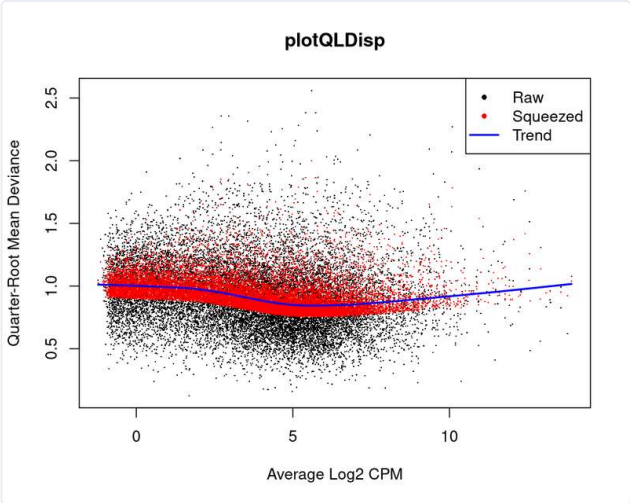
edgeR · 4 variations · v1 on facing page

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



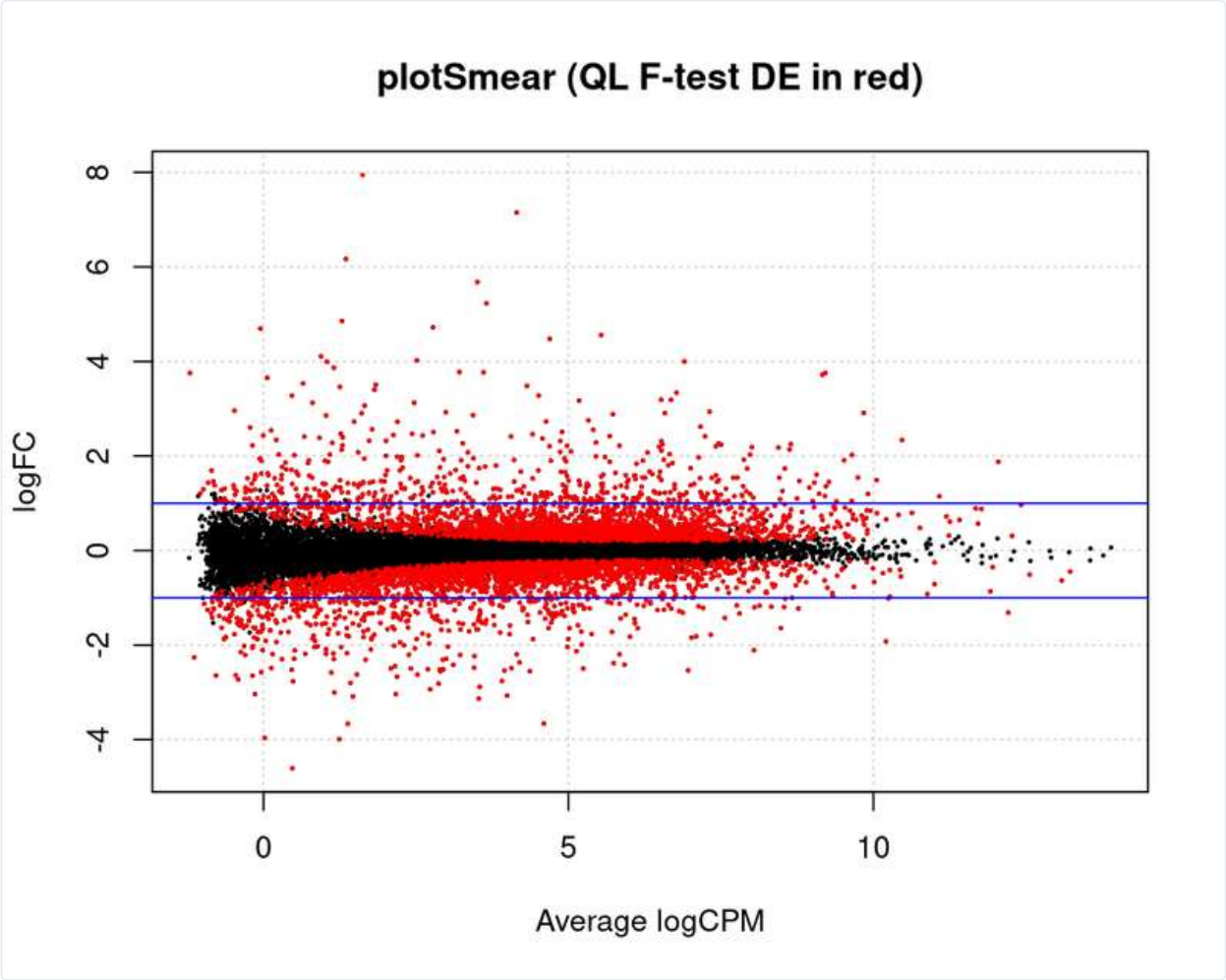
v2 plotMDS

DE



v3 plotQLDisp

DE



v4 plotSmear

DE

pheatmap

#9

OF 100 · PAPERS NAMING IT

R · HEATMAP · HEATMAPS & DOT MATRICES

1,119	6.3%	1,138	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Clustered expression heatmaps with annotation bars; sample-distance heatmaps

KEY PLOTTING CALL

RECREATIONS/PHEATMAP/PHEATMAP_RECREATE.R — EXCERPT · ENDS LINE 7

```
top <- head(order(-matrixStats::rowVars(vsd)), 50); m <- vsd[top,]; rownames(m) <- ifelse(is.na(res[rownames(m),"symbol"]), row
names(m), res[rownames(m),"symbol"])
pheatmap(m, scale = "row", annotation_col = ann, show_colnames = FALSE, fontsize_row = 6, main = "Top 50 variable genes (row z-
score)", filename = "v1_topvar_zscore.png", width = 6, height = 8)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 TEXT

Microglia ferroptosis is regulated by SEC24B and contributes to neurodegeneration. *Nat Neurosci, 2023*

"...The top 100 upregulated genes were plotted using the Pheatmap package version 1.0.12 in R using unbiased hierarchical clustering."

DOI ↗ Fig. 1 on PMC ↗

Fig. 3 TEXT

Mitophagy curtails cytosolic mtDNA-dependent activation of cGAS/STING inflammation during aging. *Nat Commun, 2024*

"...Heatmaps were generated using the tidyverse, ggplot2, viridis, RColorBrewer and pheatmap packages in R."

DOI ↗ Fig. 3 on PMC ↗

Figure 1 TEXT

APOE modulates microglial immunometabolism in response to age, amyloid pathology, and inflammatory challenge. *Cell Rep, 2023*

"...Finally, the average AUCell scores of each KEGG metabolic pathway were plotted as heatmaps using pheatmap R software (v.1.0.12) sorted by either cell subtypes and/or experimental groups."

DOI ↗ Figure 1 on PMC ↗

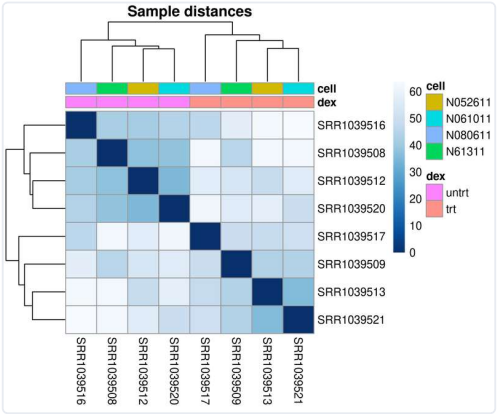
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/pheatmap/pheatmap_recreate.R`

✓ re-run end to end · 16 s

LEAD FIGURE



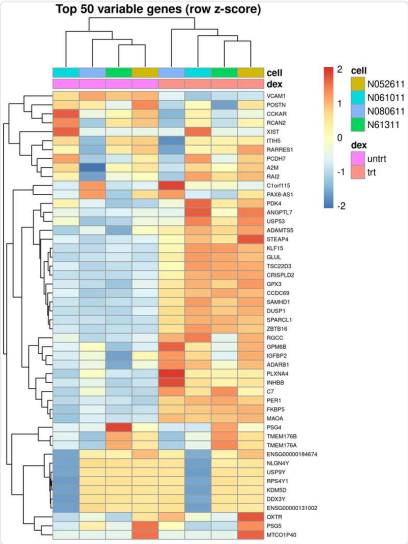
v2 sample_distance

HEATMAPS

pheatmap · 4 variations · v2 on facing page

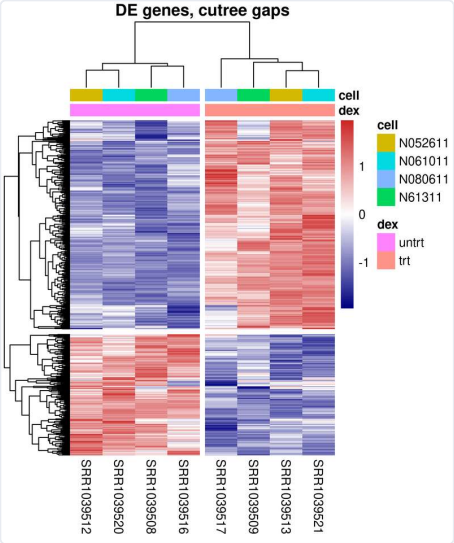
NOVAFLOW RECREATION

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*); Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



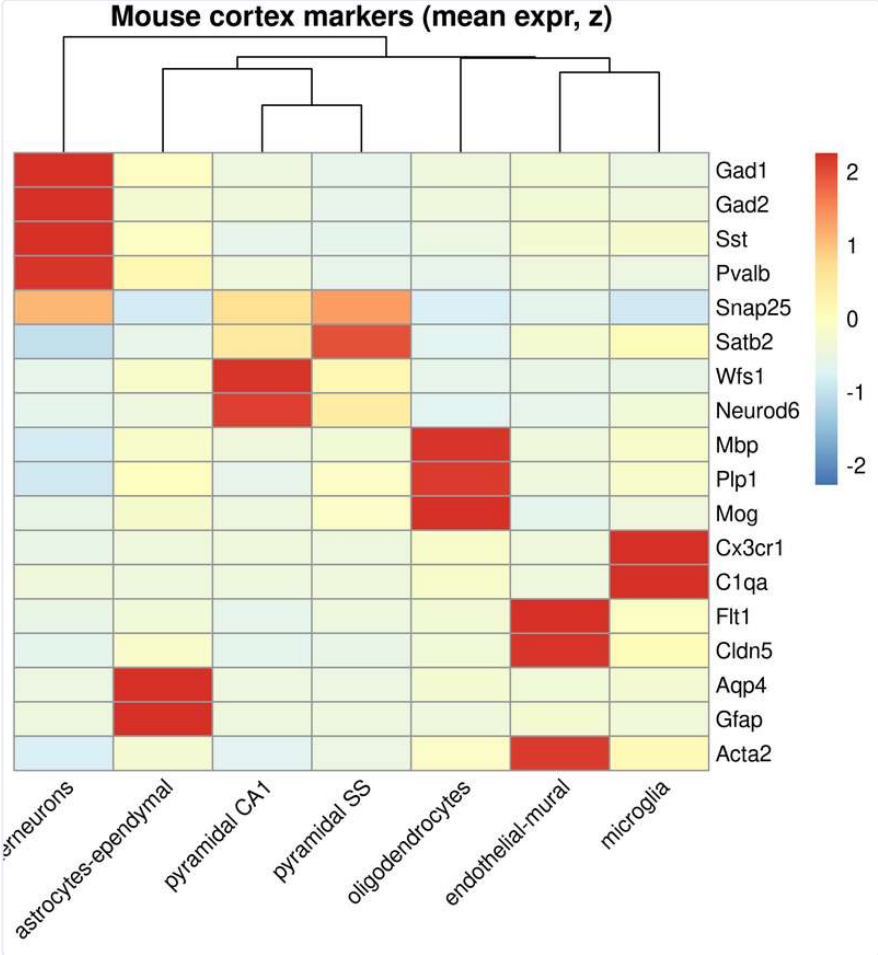
v1 topvar_zscore

HEATMAPS



v3 cutree

HEATMAPS



v4 sc_markers

HEATMAPS

WGCNA

#10

OF 100 · PAPERS NAMING IT

R · NETWORK/MODULES · NETWORKS, GRAPHS & TREES · HEATMAPS & DOT MATRICES

884	4.98%	1,025	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Soft-threshold plots, gene dendrogram + module colours, module-trait heatmap, eigengene network

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

WGCNA on pseudo-bulk samples built from Zeisel cortex/hippocampus cells (7 cell types x 10 random pseudo-replicates)

KEY PLOTTING CALL

RECREATIONS/WGCNA/WGCNA_RECREATE.R — EXCERPT · ENDS LINE 11

```
sft <- pickSoftThreshold(datExpr, powerVector = c(1:10, seq(12,20,2)), networkType = "signed", verbose = 0)
png("v1_soft_threshold.png", 1400, 650, res = 150); par(mfrow = c(1,2)); fi <- sft$fitIndices
plot(fi[,1], -sign(fi[,3])*fi[,2], type = "n", xlab = "Soft threshold (power)", ylab = "Scale-free fit, signed R^2", main = "Sc
ale independence"); text(fi[,1], -sign(fi[,3])*fi[,2], fi[,1], col = "red"); abline(h = .8, col = "red")
plot(fi[,1], fi[,5], type = "n", xlab = "Soft threshold (power)", ylab = "Mean connectivity", main = "Mean connectivity");
text(fi[,1], fi[,5], fi[,1], col = "red"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 3 — FIGURE CAPTION

Distinct molecular profiles of skull bone marrow in health and neurological disorders. *Cell, 2023*

"...(B) six bones, dura, and brain and (C) six bones in naive, sham, and MCAo conditions.(D) Dendrogram demonstrates the relation among bones and conditions. (E–J) Protein expression modules identified by WGCNA ..."

DOI ↗ Figure 3 on PMC ↗

Fig. 6 — FIGURE CAPTION

Cell type specific transcriptomic differences in depression show similar patterns between males and females but implicate distinct cell types and genes. *Nat Commun, 2023*

"...WGCNA results for microglia and PV interneurons in females.a Heatmap showing the correlation and associated p-value, in parentheses, of Mic1 WGCNA module eigengenes with case-control status and covariates ..."

DOI ↗ Fig. 6 on PMC ↗

Fig. 6 — FIGURE CAPTION

The basolateral amygdala-anterior cingulate pathway contributes to depression-like behaviors and comorbidity with chronic pain behaviors in male mice. *Nat Commun, 2023*

"...ng the level of significance of overlaps between mice and men gene modules (two-sided Fisher Exact Test). The highest overlap (P = 8.36E-49) was obtained for the man/yellow and mouse/brown modules. b WGCNA ..."

DOI ↗ Fig. 6 on PMC ↗

LIMITATIONS & NOTES

- Pseudo-bulk replicates are random partitions of cells within each cell type, so module–trait statistics are illustrative, not biological replicates.

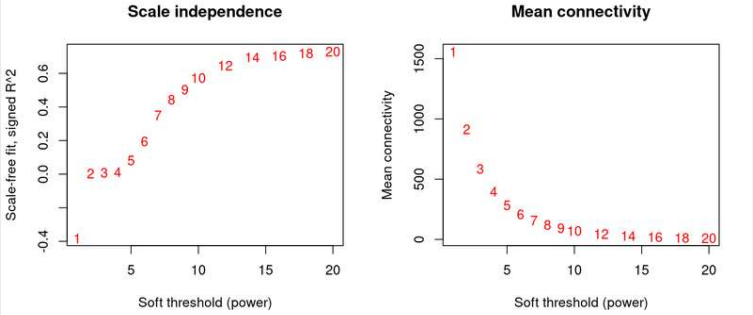
Reproduce: `Rscript recreations/WGCNA/WGCNA_recreate.R`

✓ re-run end to end · 25 s

WGCNA · 4 variations

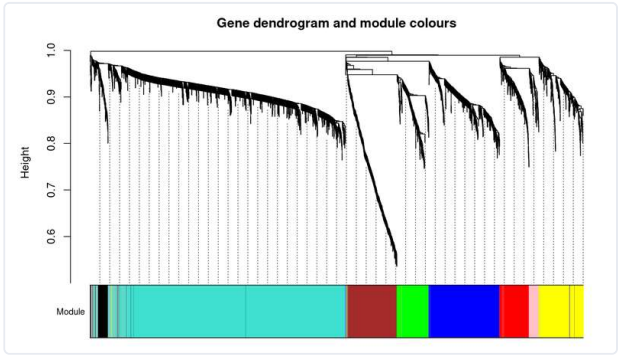
NOVAFLOW RECREATION

Data: Zeisel et al. 2015 scRNA-seq, aggregated into pseudo-bulk samples (7 cell types × 10 random cell partitions). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



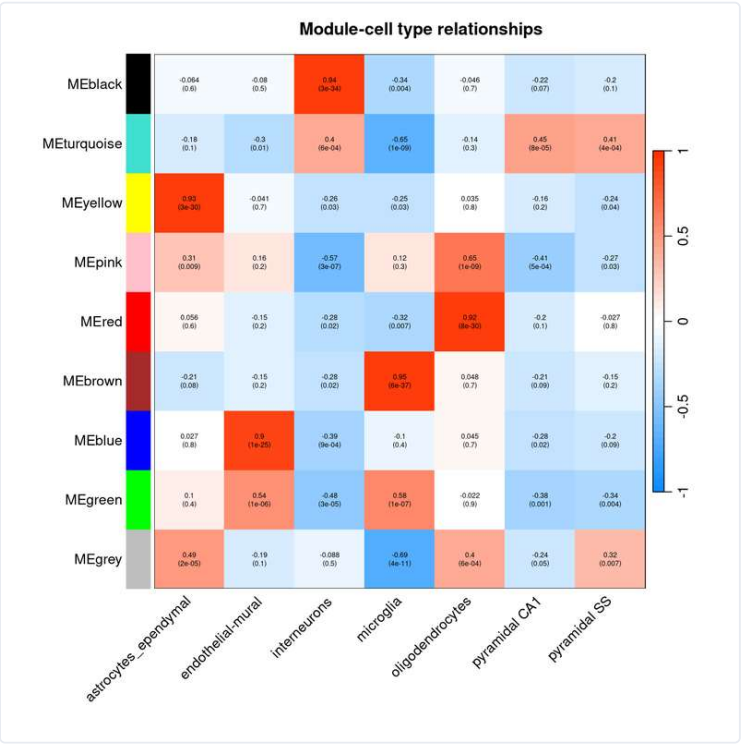
v1 soft_threshold

NETWORKS



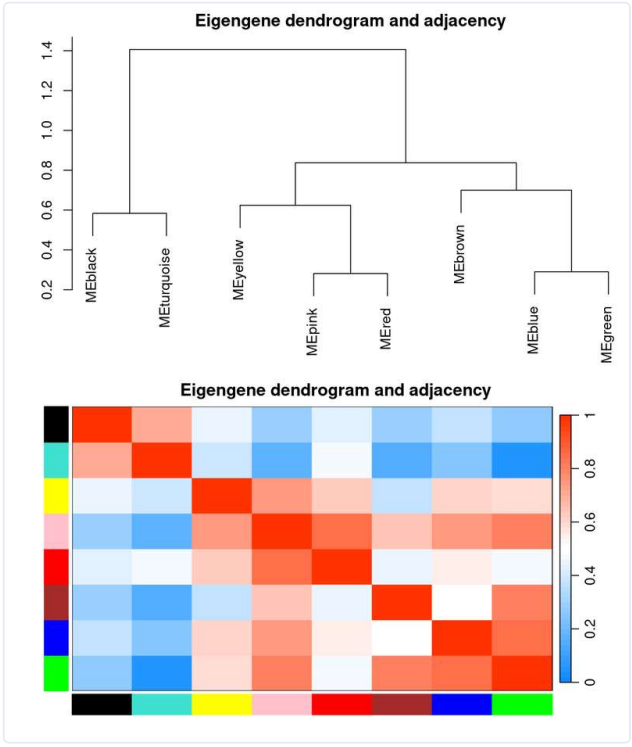
v2 dendro_modules

NETWORKS



v3 module_trait

HEATMAPS



v4 eigengene_network

NETWORKS

CellChat

R · CELL-CELL COMMUNICATION · CELL–CELL COMMUNICATION

#11

OF 100 · PAPERS NAMING IT

839	4.72%	976	analysis+plotting	5
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Circle/chord communication networks, bubble plots of L-R pairs, signalling-role scatter/heatmaps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

CellChat on Zeisel mouse cortex/hippocampus cell types

KEY PLOTTING CALL

RECREATIONS/CELLCHAT/CELLCHAT_RECREATE.R — EXCERPT · ENDS LINE 10

```
cc <- identifyOverExpressedGenes(cc); cc <- identifyOverExpressedInteractions(cc); cc <- computeCommunProb(cc, type = "triMean")
; cc <- filterCommunication(cc, min.cells = 10)
cc <- computeCommunProbPathway(cc); cc <- aggregateNet(cc); cc <- netAnalysis_computeCentrality(cc, slot.name = "netP")
gs <- as.numeric(table(cc@idents))
png("v1_circle_counts.png", 1200, 1200, res = 160); netVisual_circle(cc@net$count, vertex.weight = gs, weight.scale = TRUE, lab
el.edge = FALSE, title.name = "Number of interactions"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 8 — FIGURE CAPTION

Transcriptional and epigenetic decoding of the microglial aging process. *Nat Aging, 2023*

"...across all 16 cell types (18 populations). d, Significant ligand–receptor interactions predicted by CellPhoneDB. One-sided permutation test. e, Significant molecule–molecule interactions predicted by CellChat ..."

DOI ↗ Fig. 8 on PMC ↗

Fig. 6 — FIGURE CAPTION

A brain cell atlas integrating single-cell transcriptomes across human brain regions. *Nat Med, 2024*

"...gnificant ligand–receptor pairs in the NRG, CADM, NEGR and laminin pathways from all senders to hippocampal microglia (PCDH9high). P values are computed from a one-sided permutation test according to CellChat ..."

DOI ↗ Fig. 6 on PMC ↗

Extended Data Fig. 5: — FIGURE CAPTION

Microglial mechanisms drive amyloid-β clearance in immunized patients with Alzheimer's disease. *Nat Med, 2025*

"...A2M, APOE, FAM107A, LIPA, SPP1, and TREM2 in Aβ-associated Mg2-enriched and Mg4-enriched ST spots compared to the nAD control group for AN1792 and the lecanemab case. j, Chord plots showing inferred CellChat ..."

DOI ↗ Article ↗ figure page not on PMC

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

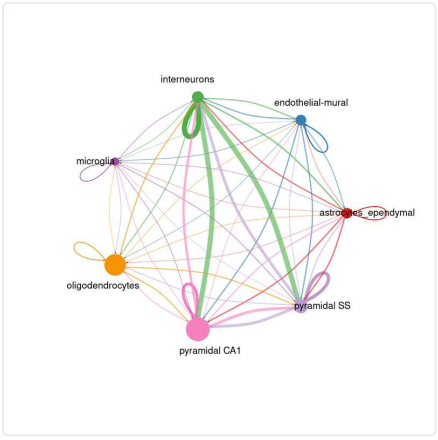
Reproduce: `Rscript recreations/CellChat/CellChat_recreate.R`

✓ re-run end to end · 145 s

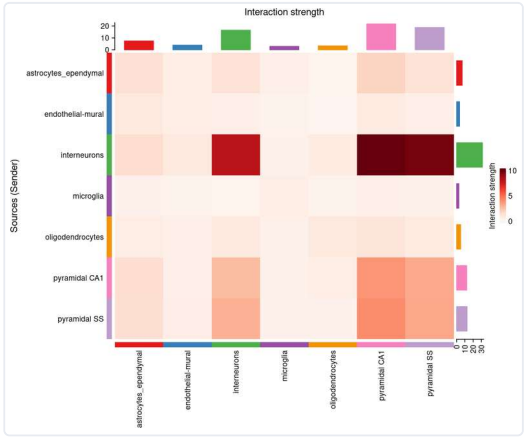
CellChat · 5 variations

NOVAFLOW RECREATION

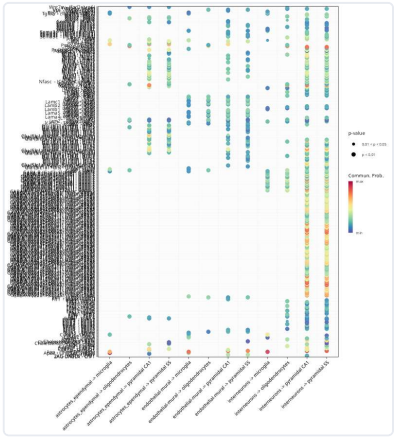
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



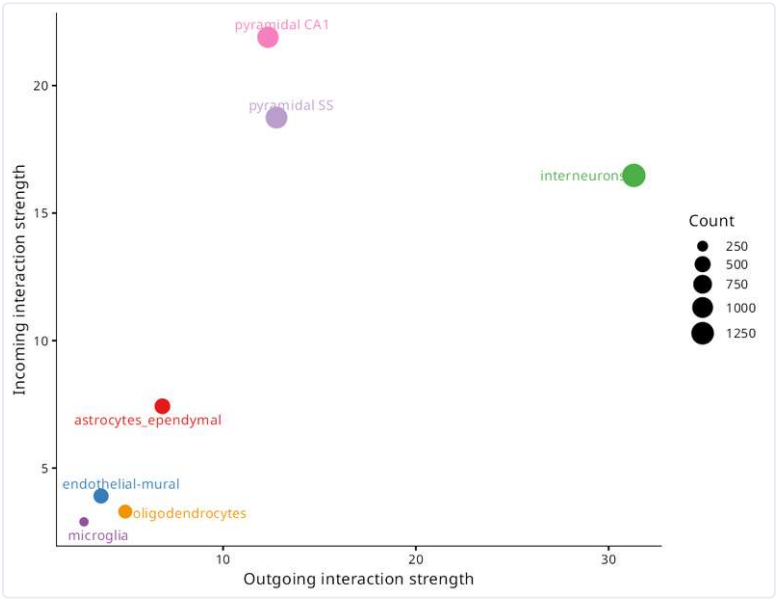
v1 circle_counts · CELL–CELL



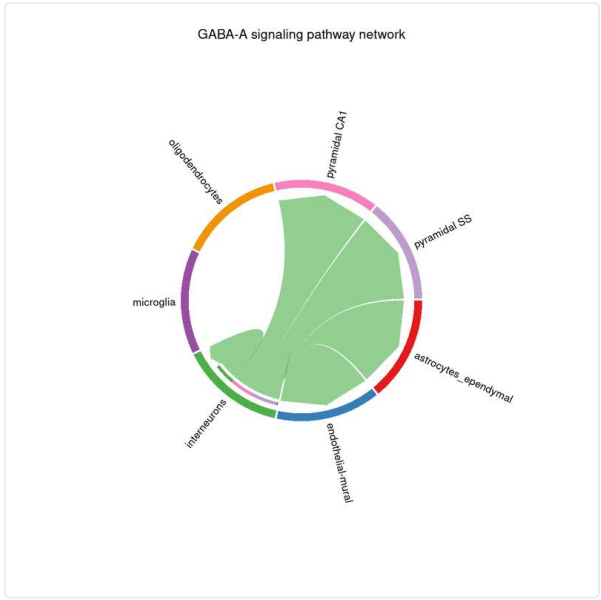
v2 heatmap · CELL–CELL



v3 bubble · CELL–CELL



v4 signalingRole_scatter · CELL–CELL



v5 chord_pathway · CELL–CELL

STRING

#12

OF 100 · PAPERS NAMING IT

WEB | PPI NETWORK | NETWORKS, GRAPHS & TREES

838	4.72%	1,142	analysis+plotting	2
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

PPI network images (web/API), often re-styled in Cytoscape

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

STRING: (V1) the canonical STRING network image via the official API; (V2) the same PPI edges redrawn with networkx, coloured by log2FC

KEY PLOTTING CALL

RECREATIONS/STRING/STRING_RECREATE.PY — EXCERPT · ENDS LINE 7

```
res=pd.read_csv("../_data/airway_deseq2_results.csv",index_col=0).dropna(subset=["padj","symbol"])
top=res[(res.padj<1e-6)&(res.log2FoldChange.abs()>1.5)].nsmallest(60,"padj")
ids="%0d".join(top.symbol); API="https://string-db.org/api"
r=requests.post(f"{API}/image/network",data=dict(identifiers=ids,species=9606,network_flavor="confidence",required_score=400,ca
ller_identity="novaflow_fig_catalog"),timeout=120); r.raise_for_status();
open("v1_string_api_network.png","wb").write(r.content)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 5 — FIGURE CAPTION

Circular RNA encoded MET variant promotes glioblastoma tumorigenesis. *Nat Commun, 2023*

"...tated by an anti-MET antibody using mass spectrometry analysis. g Reactome pathway analysis of candidate MET404-binding partners. h Protein interaction network with MET as the hub generated using the STRING ..."

DOI ↗ Fig. 5 on PMC ↗

Fig. 2 — FIGURE CAPTION

Dimethyl itaconate ameliorates cognitive impairment induced by a high-fat diet via the gut-brain axis in mice. *Microbiome, 2023*

"...supplementation. Columns with different colors represent different classification in level 2. The dotted line in the figure represents P = 0.05. h PPI network analysis of pro-cognitive genes based on STRING ..."

DOI ↗ Fig. 2 on PMC ↗

Extended Data Fig. 11 — FIGURE CAPTION

Astrocyte-neuron subproteomes and obsessive-compulsive disorder mechanisms. *Nature, 2023*

"...entified in astrocyte plasma membrane with Astro Lck-BioID2 . Node size represents the enrichment of each protein vs the GFP control (log2(BioID2/GFP)). Edges represent putative interactions from the STRING ..."

DOI ↗ Extended Data Fig. 11 on PMC ↗

LIMITATIONS & NOTES

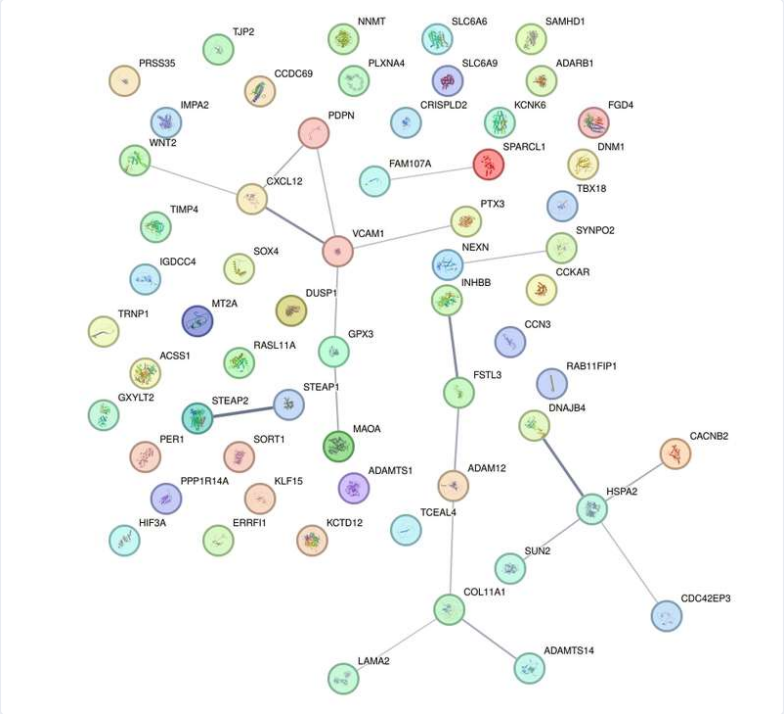
- V1 = official STRING API image; V2 = same edges redrawn in networkx
- Needs internet: string-db.org API (METHODS §4).

Reproduce: `python3 recreations/STRING/string_recreate.py`

√ re-run end to end · 3 s

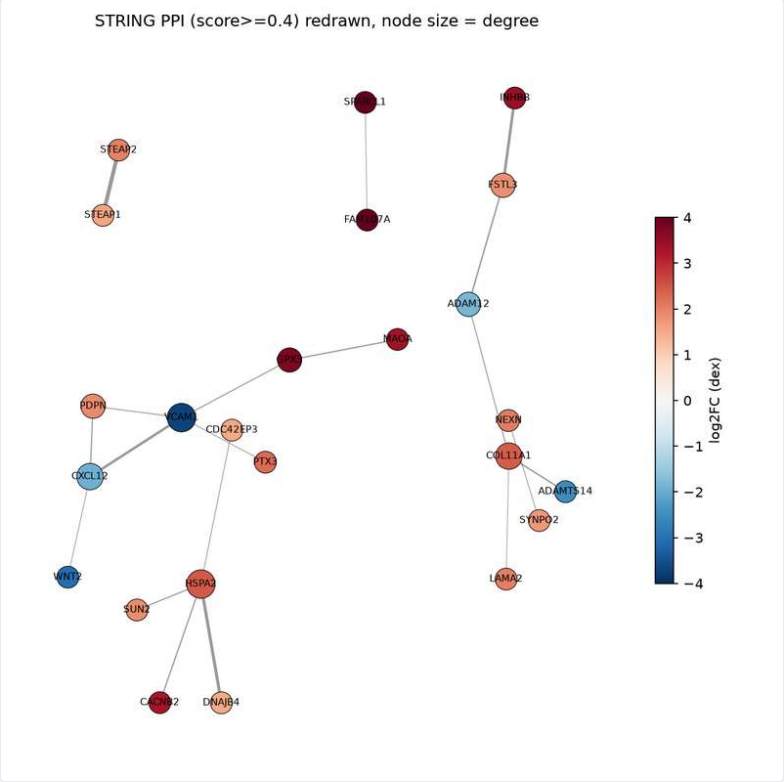
STRING · 2 variations

Data: airway dexamethasone-response genes; network from the live STRING API (string-db.org). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 string_api_network

NETWORKS



v2 networkx_fc

NETWORKS

Enrichr

#13

OF 100 · PAPERS NAMING IT

WEB | ENRICHMENT | ENRICHMENT & PATHWAY ACTIVITY

778	4.38%	912	analysis+plotting	2
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Enrichment bar charts / tables by library

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Enrichr (web API, via GSEApy) bar/dot plots for dexamethasone-up genes

KEY PLOTTING CALL

RECREATIONS/ENRICHR/ENRICHR_RECREATE.PY — EXCERPT · ENDS LINE 7

```
res=pd.read_csv("../_data/airway_deseq2_results.csv",index_col=0).dropna(subset=["padj","symbol"])
up=res[(res.padj<.01)&(res.log2FoldChange>1)].symbol.unique().tolist()
enr=gp.enrichr(gene_list=up,gene_sets=["GO_Biological_Process_2023","KEGG_2021_Human"],organism="human",outdir=None)
gp.barplot(enr.results,column="Adjusted P-value",group="Gene_set",size=10,top_term=6,figsize=(5,7),color={"GO_Biological_Proces
s_2023":"#4C72B0","KEGG_2021_Human":"#DD8452"},ofname="v1_barplot_grouped.png")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 6 — FIGURE CAPTION

GAP43-dependent mitochondria transfer from astrocytes enhances glioblastoma tumorigenicity. *Nat Cancer, 2023*

"...to-mCherry+ L1 cells; n = 3 independent co-culture experiments. Data were analyzed by paired two-tailed t-test; 2-OG, α-ketoglutarate; CDP, cytidine-5'-diphosphate. c, Phospho-array pathway analysis (Enrichr) ..."

DOI ↗ Fig. 6 on PMC ↗

Fig. 3 — TEXT

Cell-type-specific aging clocks to quantify aging and rejuvenation in neurogenic regions of the brain. *Nat Aging, 2023*

"...Statistics were exported from the Enrichr web tool and processed and visualized in R with ggplot2 (version 3.3.3) package."

DOI ↗ Fig. 3 on PMC ↗

Figure 6 — TEXT

hdWGCNA identifies co-expression networks in high-dimensional transcriptomics data. *Cell Rep Methods, 2023*

"...REAGENT or RESOURCESOURCEIDENTIFIERDeposited dataHuman Alzheimer's Disease snRNA-seq 2019Mathys et al., 201910syn18485175Human Autism spectrum disorder snRNA-seqVelmeshev et al., 20199PRJNA434002Human aging ..."

DOI ↗ Figure 6 on PMC ↗

LIMITATIONS & NOTES

- Uses the live Enrichr web API via GSEApy
- Needs internet: calls the live Enrichr web API (METHODS §4).

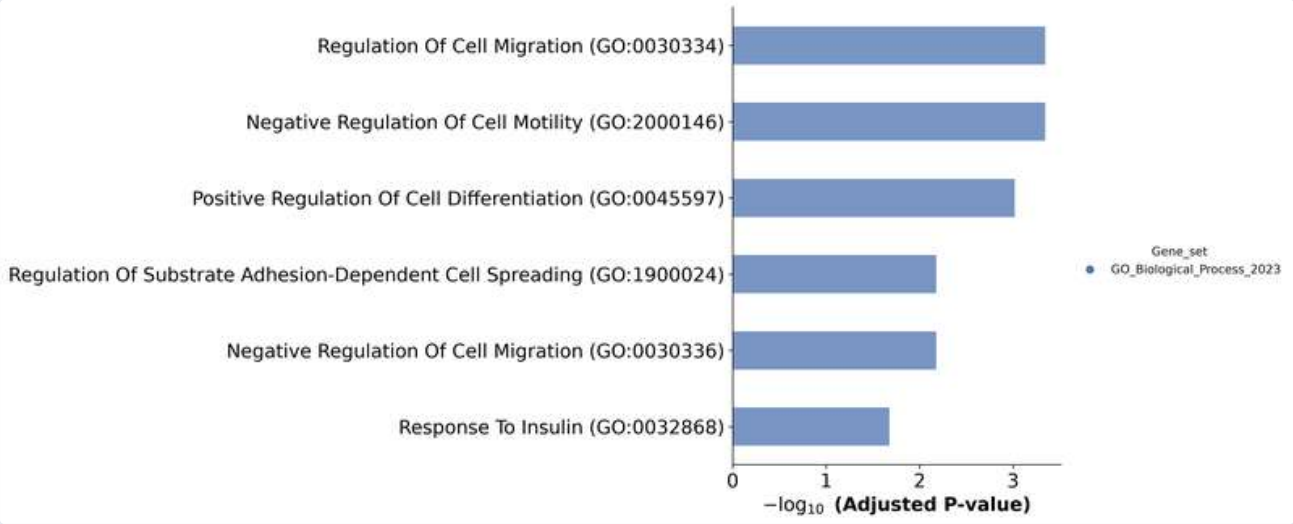
Reproduce: `python3 recreations/Enrichr/enrichr_recreate.py`

√ re-run end to end · 4 s

Enrichr · 2 variations

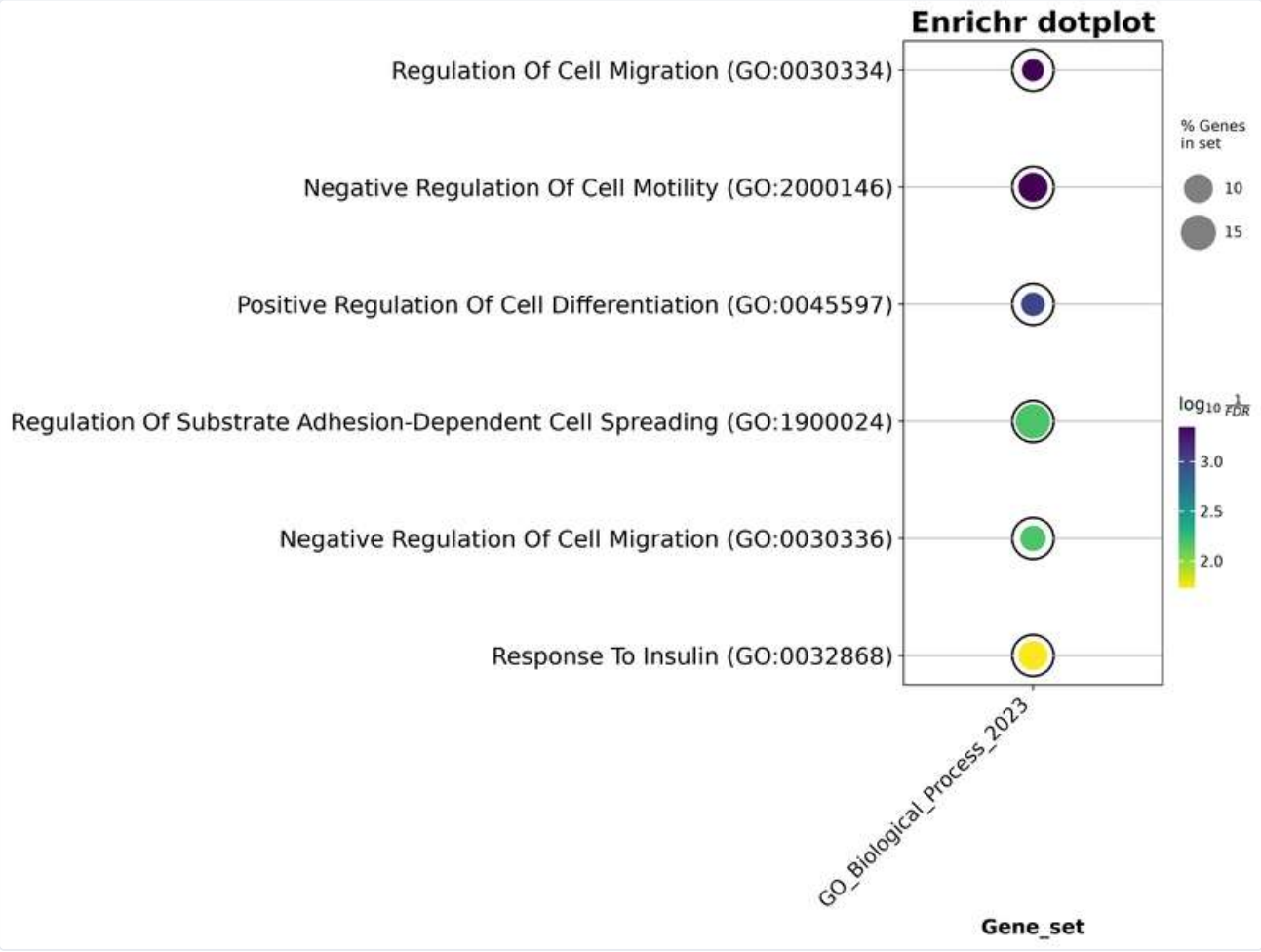
NOVAFLOW RECREATION

Data: airway dexamethasone-up genes; enrichment from the live Enrichr API. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 barplot_grouped

ENRICHMENT



v2 dotplot_by_library

ENRICHMENT

Scanpy

#14

OF 100 · PAPERS NAMING IT

PYTHON | SINGLE-CELL (EMBEDDING/DOT/VIOLIN/MATRIX) | EMBEDDINGS & FEATURE MAPS | HEATMAPS & DOT MATRICES | COMPOSITION, DISTRIBUTIONS & LAYOUTS | TRAJECTORY, PSEUDOTIME & VELOCITY

763	4.29%	789	analysis+plotting	7
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

UMAP, dotplot, matrixplot, stacked violin, rank_genes_groups, tracksplot, PAGA

Also catalogued as: **PAGA** (#75, 75 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

scanpy plotting API on Zeisel mouse cortex/hippocampus (3,005 cells)

KEY PLOTTING CALL

RECREATIONS/SCANPY/SCANPY_RECREATE.PY — EXCERPT · ENDS LINE 7

```
mk={"Neuron":["Snap25","Syt1"],"Inhib":["Gad1","Gad2","Pvalb","Sst"],"Exc":["Satb2","Neurod6","Wfs1"],"Astro":["Aqp4","Gfap"],"
Oligo":["Mbp","Plp1"],"Micro":["Cx3cr1","C1qa"],"Vasc":["Cldn5","Acta2"]}
sc.pl.umap(ad,color=["celltype","leiden","Gad1","Mbp"],ncols=2,wspace=.4,save="_v1_umap_grid.png",show=False)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 FIGURE CAPTION
Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST. *Nat Commun, 2023*
"...the Stereo-seq mouse embryo.A Tissue domain annotations of the E9.5 mouse embryo data taken from the original Stereo-seq study wherein the clusters were first identified using Leiden clustering from SCANPY and ..."
DOI ↗ Fig. 3 on PMC ↗

Fig. 1 TEXT
The molecular cytoarchitecture of the adult mouse brain. *Nature, 2023*
"...To build a graph of cell type similarity, we used Scanpy on our subsampled data to compute the connectivities over a 20 neighbour local neighbourhood using 250 principal components (the section 'Visualization ..."
DOI ↗ Fig. 1 on PMC ↗

Fig. 1 TEXT
APOE4/4 is linked to damaging lipid droplets in Alzheimer's disease microglia. *Nature, 2024*
"...We then used the batch-corrected neighbourhood graph to run Leiden clustering46 and to calculate a global UMAP embedding with default parameters in Scanpy (v.1.9.1)."
DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

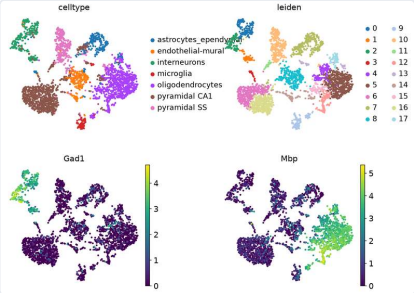
- PAGA (#75): See scanpy v7_paga.png

Reproduce: `python3 recreations/scanpy/scanpy_recreate.py` — ✓ re-run end to end · 30 s

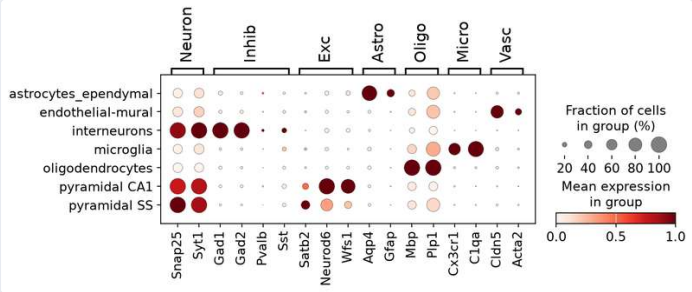
Scanpy · 7 variations

NOVAFLOW RECREATION

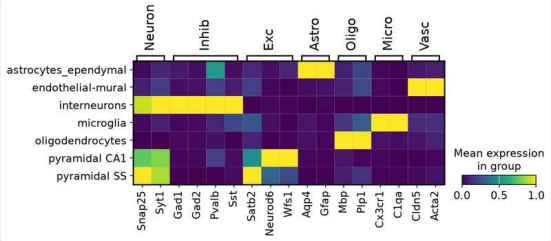
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



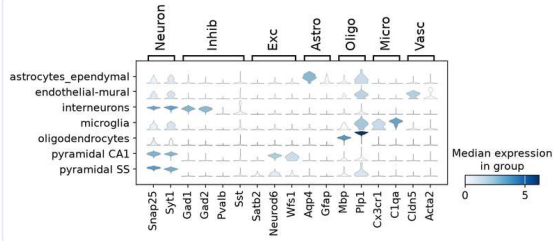
v1 umap_grid — EMBEDDINGS



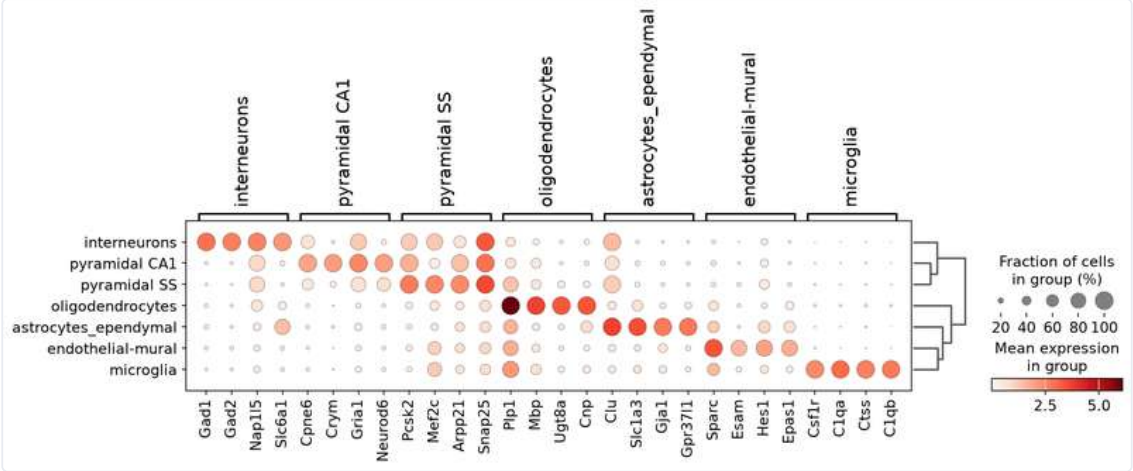
v2 dotplot — HEATMAPS



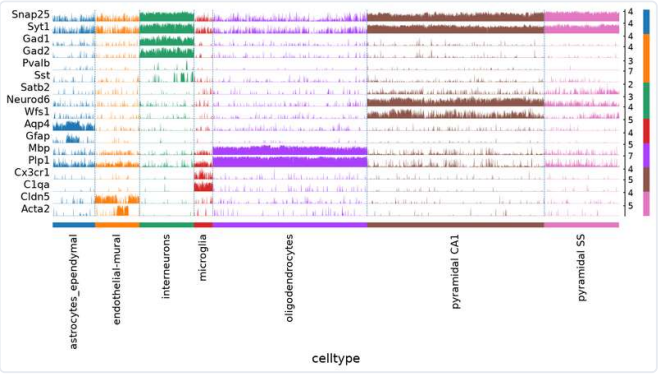
v3 matrixplot — HEATMAPS



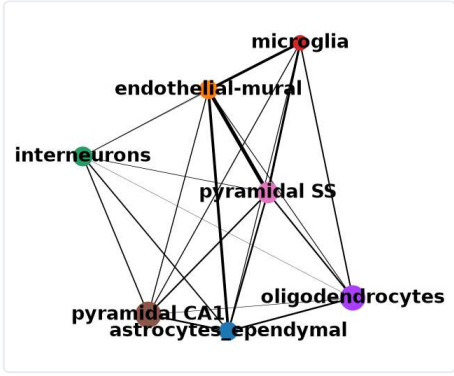
v4 stacked_violin — COMPOSITION



v5 rank_genes_dotplot — HEATMAPS



v6 tracksplot — HEATMAPS



v7 paga — TRAJECTORY

ComplexHeatmap

#17

OF 100 · PAPERS NAMING IT

R · HEATMAP · HEATMAPS & DOT MATRICES

567	3.19%	578	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Multi-annotated, split heatmaps; oncoPrint; density heatmaps; heatmap lists

KEY PLOTTING CALL

RECREATIONS/COMPLEXHEATMAP/COMPLEXHEATMAP_RECREATE.R — EXCERPT · ENDS LINE 13

```
png("v1_annotated_split.png", 1200, 1500, res = 170)
draw(Heatmap(m, name = "z", col = colorRamp2(c(-2,0,2), c("#2166AC","white","#B2182B")), top_annotation = ha, right_annotation
= ra,
  row_split = ifelse(res[sig,"log2FoldChange"] > 0, "Up in dex", "Down in dex"), column_split = dds$dex, show_row_names = FALSE,
  left_annotation = rowAnnotation(mark = anno_mark(at = match(lab, sig), labels = res[lab,"symbol"], side = "left"))) )
dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 TEXT

The molecular cytoarchitecture of the adult mouse brain. *Nature, 2023*

"...The matrix was plotted using the ComplexHeatmap package in R79."

DOI ↗ Fig. 1 on PMC ↗

Fig. 1 TEXT

Exposure of iPSC-derived human microglia to brain substrates enables the generation and manipulation of diverse transcriptional states in vitro. *Nat Immunol, 2023*

"...Heatmaps were generated using the ComplexHeatmap R package."

DOI ↗ Fig. 1 on PMC ↗

Fig. 2 TEXT

Multiomics in primary and metastatic breast tumors from the AURORA US network finds microenvironment and epigenetic drivers of metastasis. *Nat Cancer, 2023*

"...The significantly differentially expressed modules (q < 0.05) were hierarchically clustered using the ComplexHeatmap R package."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

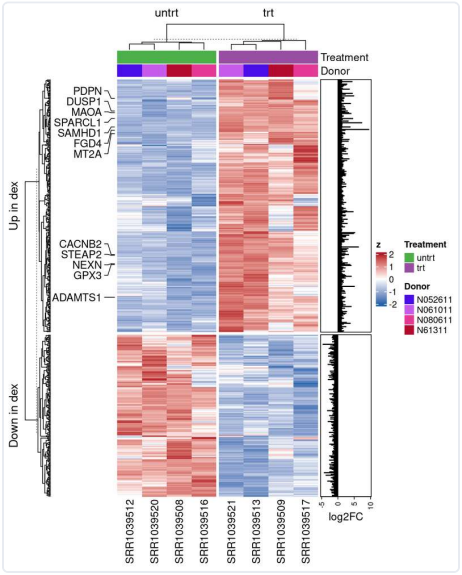
Reproduce: `Rscript recreations/ComplexHeatmap/ComplexHeatmap_recreate.R`

✓ re-run end to end · 21 s

ComplexHeatmap · 4 variations

NOVAFLOW RECREATION

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*); Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.

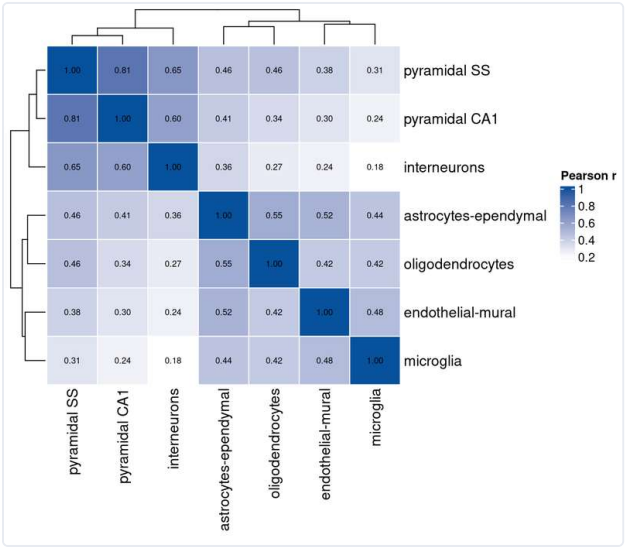


v1 annotated_split

HEATMAPS

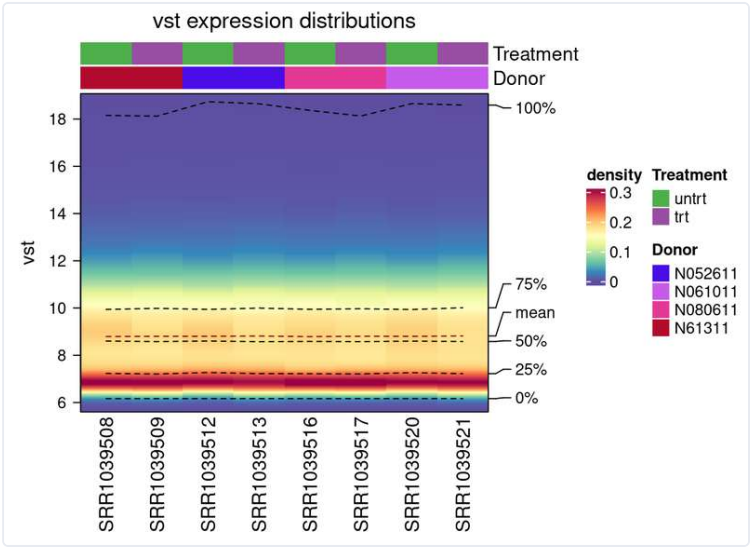
v2 singlecell_markers

HEATMAPS



v3 correlation_cellfun

HEATMAPS



v4 densityHeatmap

HEATMAPS

g:Profiler

#18

OF 100 · PAPERS NAMING IT

WEB/R/PYTHON | ENRICHMENT | ENRICHMENT & PATHWAY ACTIVITY

563	3.17%	627	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Manhattan-style enrichment plot (gostplot) + term tables

KEY PLOTTING CALL

RECREATIONS/GPROFILER/GPROFILER2_RECREATE.R — EXCERPT · ENDS LINE 9

```
set_base_url("https://biit.cs.ut.ee/gprofiler")  # use the HTTPS endpoint
res <- as.data.frame(readRDS("../_data/airway_deseq2.rds")$res); res <- res[!is.na(res$padj) & !is.na(res$symbol),]
up <- res$symbol[res$padj < .01 & res$log2FoldChange > 1]; dn <- res$symbol[res$padj < .01 & res$log2FoldChange < -1]
g <- gost(list(up = up, down = dn), organism = "hsapiens", multi_query = FALSE, sources = c("GO:BP","KEGG","REAC","TF"))
p <- gostplot(g, capped = TRUE, interactive = FALSE); ggsave("v1_manhattan.png", p, width = 9, height = 6, dpi = 130, bg = "whi
te")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Extended Data Fig. 8 — FIGURE CAPTION

Myelin dysfunction drives amyloid-β deposition in models of Alzheimer's disease. *Nature, 2023*

"...notype data. Error bars represent SD. (e) Heatmap of normalised mean expression pattern in the 10 DEG clusters across different genotypes. Each cluster of genes underwent GO enrichment analysis using ..."

DOI ↗ Extended Data Fig. 8 on PMC ↗

Extended Data Fig. 6 — FIGURE CAPTION

Single-cell multiregion dissection of Alzheimer's disease. *Nature, 2024*

"..., subtype, and covariate enrichments performed at cell level by stratifying cells with z-score > 1 and testing for regional or subtype enrichment (see Methods). Functional enrichments performed using ..."

DOI ↗ Extended Data Fig. 6 on PMC ↗

Extended Data Fig. 5 — TEXT

A NPAS4-NuA4 complex couples synaptic activity to DNA repair. *Nature, 2023*

"...Gene ontology (GO) enrichment analysis was performed using gProfiler2 in R (v.0.2.0)65, with a custom background of expression-filtered genes from neuronal cell types across all scRNA-seq datasets in this ..."

DOI ↗ Extended Data Fig. 5 on PMC ↗

LIMITATIONS & NOTES

- Calls the live g:Profiler API
- Needs internet: live g:Profiler API.gprofiler2 is pointed at its HTTPS endpoint. (METHODS §4).

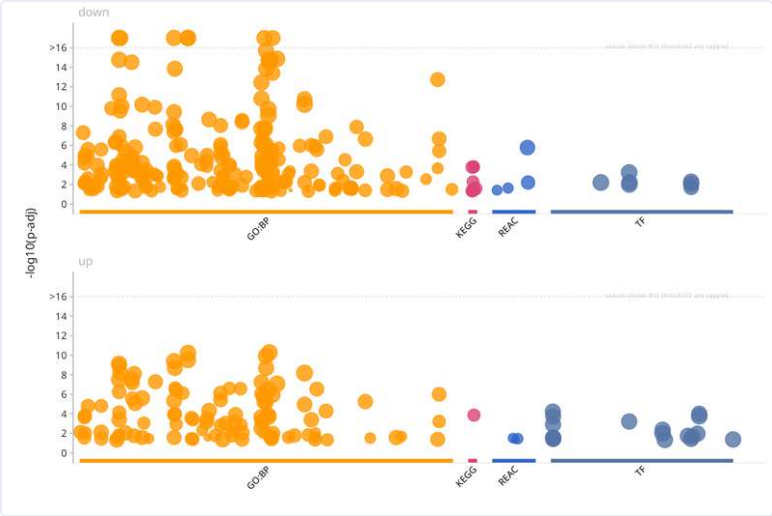
Reproduce: `Rscript recreations/gProfiler/gprofiler2_recreate.R`

√ re-run end to end · 16 s

g:Profiler · 3 variations

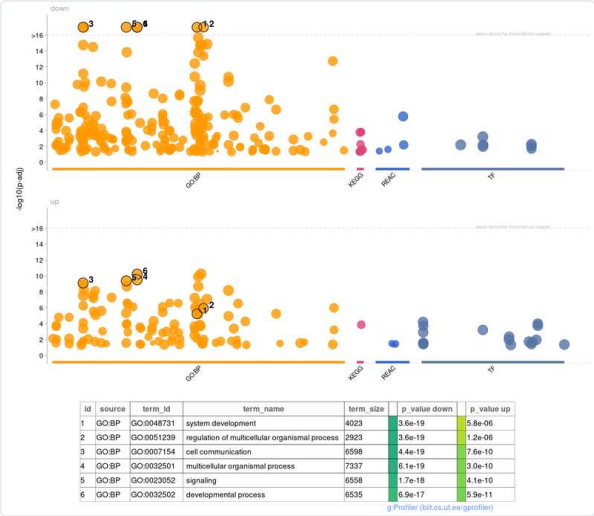
NOVAFLOW RECREATION

Data: airway DE genes; enrichment from the live g:Profiler API. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 manhattan

ENRICHMENT



v2 publish_highlight

ENRICHMENT

source	term_id	term_name	term_size	intersection_size_up	intersection_size_down	p_value_up	p_value_do
GO:BP	GO:0048731	system development	4023	122	144	5.8e-06	3.6e-19
GO:BP	GO:0051239	regulation of multicellular organismal process	2923	99	119	1.2e-06	3.6e-19
GO:BP	GO:0007154	cell communication	6598	189	195	7.6e-10	4.4e-19
GO:BP	GO:0032501	multicellular organismal process	7337	205	208	3.0e-10	6.1e-19
GO:BP	GO:0023052	signaling	6558	189	193	4.1e-10	1.7e-18
GO:BP	GO:0032502	developmental process	6535	191	189	5.9e-11	6.9e-17

g:Profiler (biit.cs.ut.ee/gprofiler)

v3 gosttable

ENRICHMENT

IGV-style tracks (Gviz)

DESKTOP/WEB · GENOME TRACKS · GENOME TRACKS & GENOMIC SIGNAL

#19
OF 100 · PAPERS NAMING IT

549	3.09%	794	viewer	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · GVIZ

Neuro-corpus papers per year (*2026 partial) · niche rank **#73**, growth **0.55×**, visual 3/5, score 0.314. Broad-corpus rate 2.229 → 1.219 per 1k papers (2023 → 2025–26).
GitHub ivanek/Gviz · created 2018-12-10 · last push 2026-08-13

BEST FOR

Genome-browser track screenshots (coverage, peaks, splicing)

Also catalogued as: **UCSC Genome Browser** (#28, 363 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

IGV / UCSC-browser-style tracks with Gviz: ideogram + axis + gene models + data tracks (hg19)

KEY PLOTTING CALL

RECREATIONS/GVIZ_IGV_STYLE/GVIZ_RECREATE.R — EXCERPT · ENDS LINE 14

```
at <- AnnotationTrack(pks, name = "peaks", fill = "#e31a1c", col = NA); set.seed(1)
cov <- DataTrack(GRanges(chr, IRanges(seq(st, en - 5000, 5000), width = 5000), score = sapply(seq(st, en - 5000, 5000),
function(x) sum(overlapsAny(pks, GRanges(chr, IRanges(x, width = 5000)))) * pmax(0, rnorm(1, 5, 1)))), name = "peak signal", ty
pe = "histogram", fill = "#41ab5d", col.histogram = NA)
ax <- GenomeAxisTrack(); bands <- data.frame(chrom = "chr1", chromStart = c(0, 2300000), chromEnd = c(2300000, 5300000), name =
c("p36.33","p36.32"), gieStain = c("gneg","gpos25"))
id <- IdeogramTrack(genome = "hg19", chromosome = chr, bands = bands)
png("v1_igv_style_tracks.png", 1600, 900, res = 160); plotTracks(list(id, ax, cov, at, gtr), from = st, to = en, background.tit
le = "#2b2b2b", sizes = c(1, 1.2, 2, .8, 4)); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 — FIGURE CAPTION

A NPAS4-NuA4 complex couples synaptic activity to DNA repair. *Nature, 2023*

"...NPAS4–NuA4-bound sites undergo recurrent DNA breaks in vivo.a, A model for the dual function of NPAS4–NuA4 in stimulating transcription and DNA repair in active neurons. b, IGV tracks displaying aggregate ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 3 — FIGURE CAPTION

Lineage-specific canonical and non-canonical activity of EZH2 in advanced prostate cancer subtypes. *Nat Commun, 2024*

"...and plots show profiles of normalized H3K27me3 and H3K4me3 peaks at ±3 kb of gene bodies or TSS of genes upregulated upon EZH2i treatment in LNCaP-abl (turquoise) or WCM154 (teal). C Screenshots from ..."

DOI ↗ Fig. 3 on PMC ↗

Extended Data Fig. 8 — FIGURE CAPTION

The complete sequence and comparative analysis of ape sex chromosomes. *Nature, 2024*

"...on JADBMG010000033.1 (positions 2 to 618,314, upper), a contig from a previous chrY assembly, and T2T chrY (positions 43,632,350 to 44,250,835, bottom), for western lowland gorilla, visualized using IGV. (e) ..."

DOI ↗ Extended Data Fig. 8 on PMC ↗

LIMITATIONS & NOTES

- Open-code look-alike: Gviz tracks (recreations/Gviz_IGV_style)
- UCSC Genome Browser (#28): Open-code look-alike: Gviz dense tracks (v3_ucsc_style_dense)
- The "peak signal" data track is illustrative (peak counts per 5-kb bin multiplied by random jitter), not a measured coverage track.

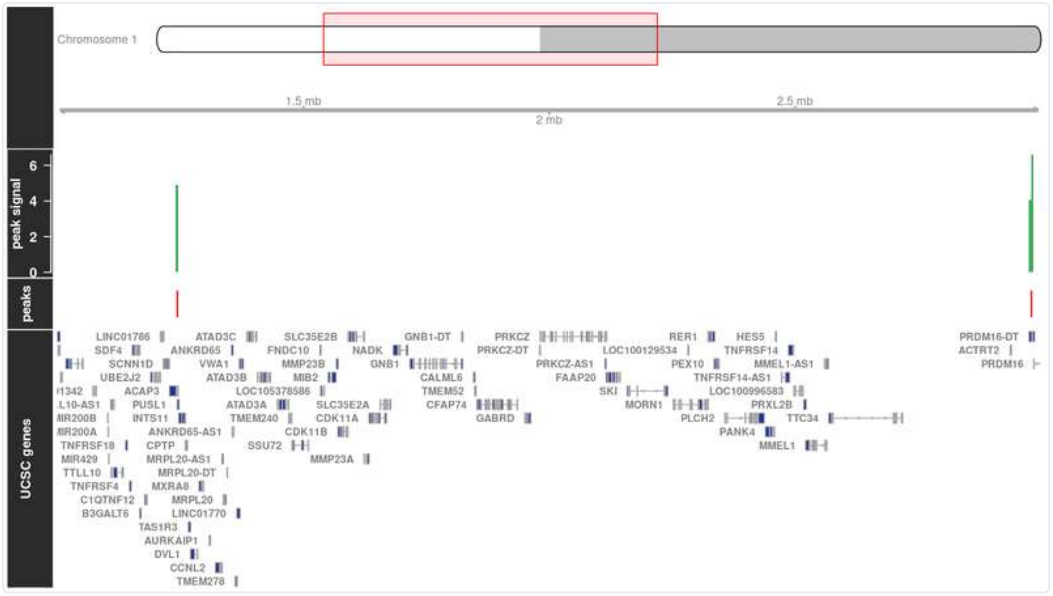
Reproduce: `Rscript recreations/Gviz_IGV_style/gviz_recreate.R`

√ re-run end to end · 94 s

IGV-style tracks (Gviz) · 3 variations

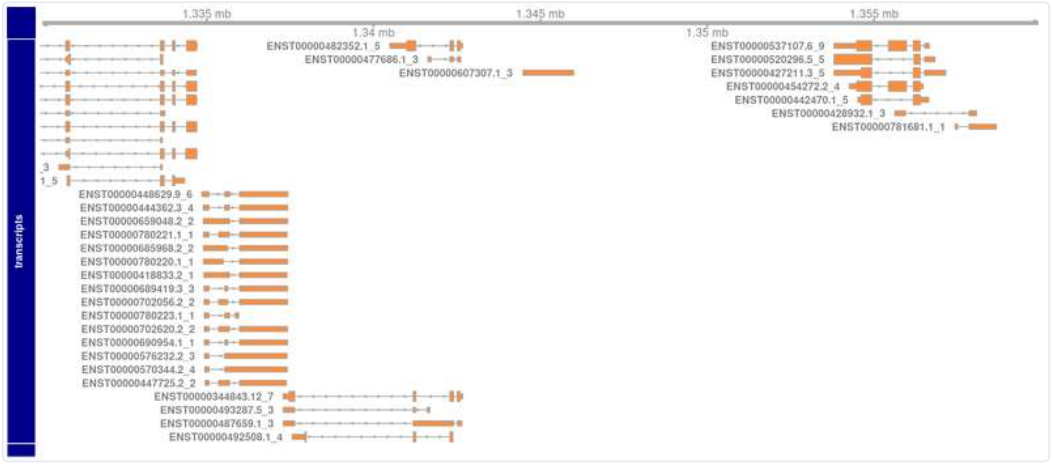
NOVAFLOW RECREATION

Data: Human hg19 gene models (UCSC knownGene) and ChIPseeker example peak file (GEO). The "peak signal" track is **illustrative**: peak counts per 5-kb bin multiplied by random jitter, not a measured coverage track.. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



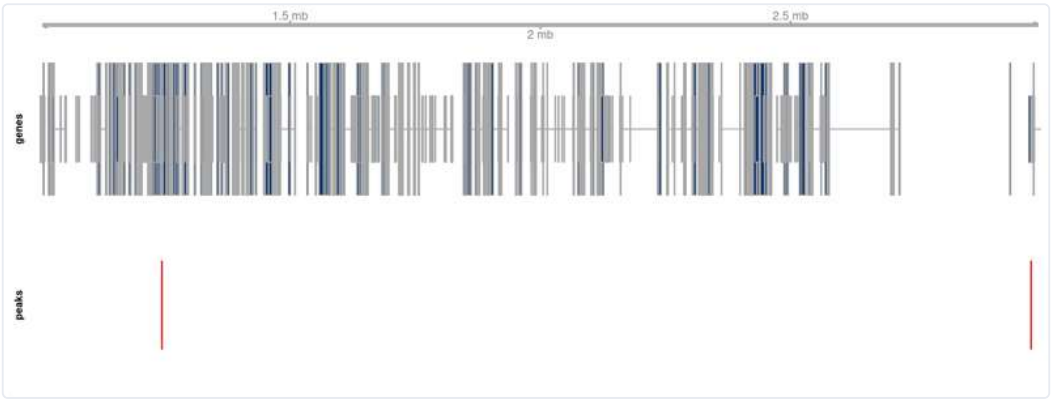
v1 igv_style_tracks

GENOME



v2 zoom_gene_model

GENOME



v3 ucsc_style_dense

GENOME

fgsea

R · ENRICHMENT · ENRICHMENT & PATHWAY ACTIVITY

#20

OF 100 · PAPERS NAMING IT

478	2.69%	498	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Enrichment curves (plotEnrichment), GSEA table plot, NES bar charts

KEY PLOTTING CALL

RECREATIONS/FGSEA/FGSEA_RECREATE.R — EXCERPT · ENDS LINE 8

```
res <- as.data.frame(readRDS("../_data/airway_deseq2.rds")$res); res <- res[!is.na(res$stat) & !is.na(res$symbol),]
rk <- sort(tapply(res$stat, res$symbol, mean), decreasing = TRUE)
h <- msigdbr(species = "Homo sapiens", collection = "H"); pw <- split(h$gene_symbol, h$gs_name)
set.seed(1); fr <- fgsea(pw, rk, minSize = 15, maxSize = 500); fr <- fr[order(pval)]
top <- fr$pathway[1]
ggsave("v1_plotEnrichment.png", plotEnrichment(pw[[top]], rk) + labs(title = top, subtitle = sprintf("NES = %.2f, padj = %.1e",
fr$NES[1], fr$padj[1])), width = 6.5, height = 4, dpi = 150, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 — FIGURE CAPTION

Exposure of iPSC-derived human microglia to brain substrates enables the generation and manipulation of diverse transcriptional states in vitro. *Nat Immunol, 2023*

"…sters defined in both the iMGL and brain biopsy datasets. Links with fewer than 200 cells have been removed for clarity. e, Heatmap illustrating the relative enrichment significance (as determined by fgsea …"

DOI ↗ Fig. 2 on PMC ↗

Fig. 4 — FIGURE CAPTION

The molecular cytoarchitecture of the adult mouse brain. *Nature, 2023*

"…ne; SRG, secondary response gene. c, Scatterplot quantifying transcription factor enrichment (P < 0.05, FDR corrected) between excitatory and inhibitory populations. Enrichment scores are computed by fgsea …"

DOI ↗ Fig. 4 on PMC ↗

Fig. 4 — TEXT

Transcriptional linkage analysis with in vivo AAV-Perturb-seq. *Nature, 2023*

"…10g), the list of expressed genes ranked by higher to lower LFC value was used as input to the R package fgsea (v.3.17)59 to run GSEA using the mouse GO:BP dataset."

DOI ↗ Fig. 4 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

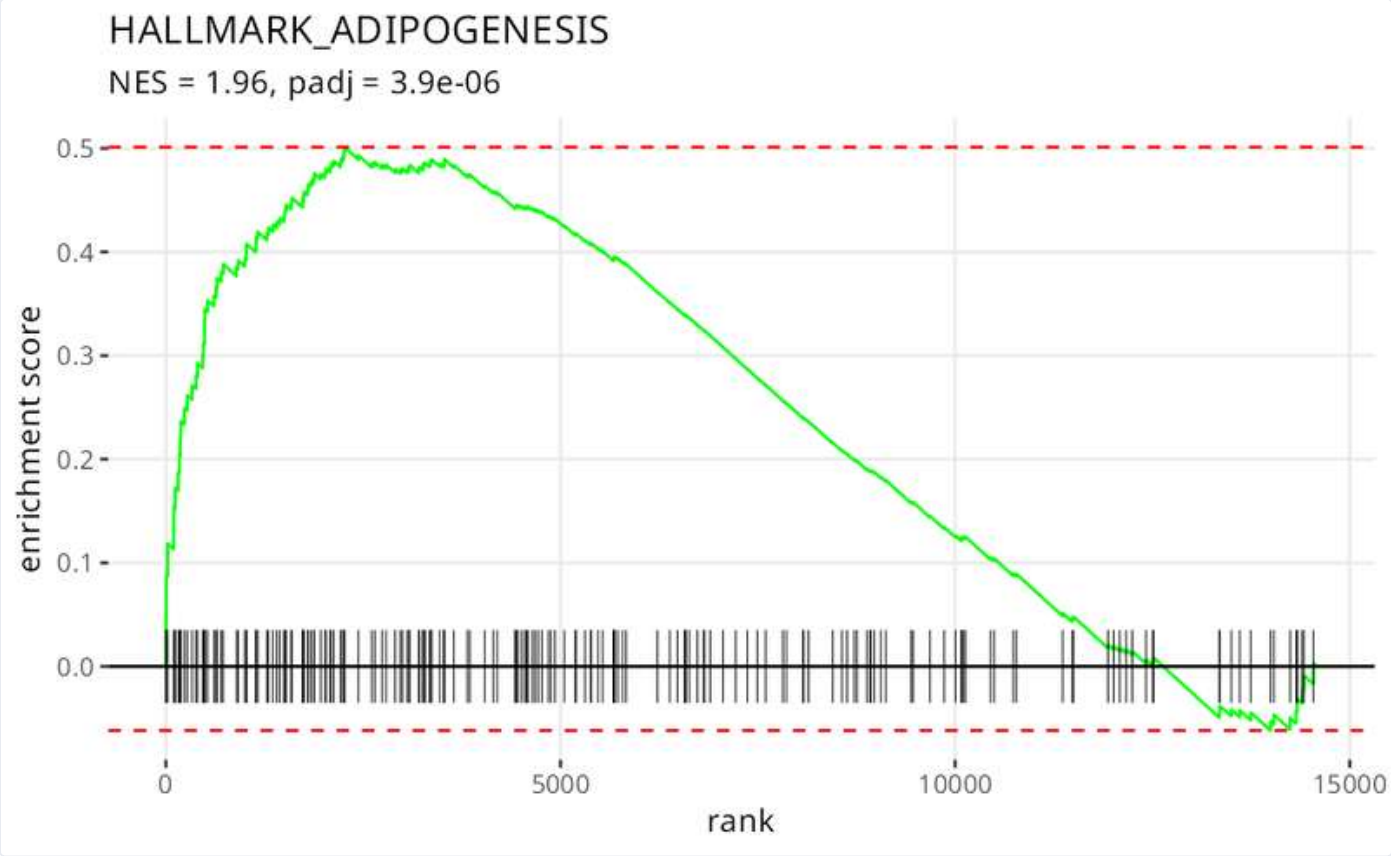
Reproduce: `Rscript recreations/fgsea/fgsea_recreate.R`

√ re-run end to end · 286 s

fgsea · 3 variations

NOVAFLOW RECREATION

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Hines et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 plotEnrichment

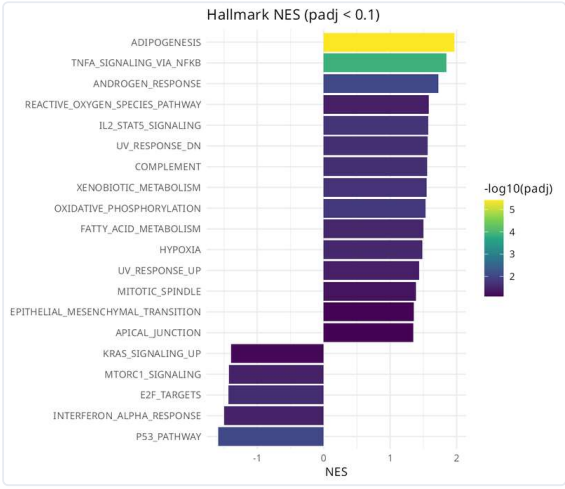
ENRICHMENT

Pathway	Gene ranks	NES	pval	padj
HALLMARK_ADIPOGENESIS		1.96	$7.8 \cdot 10^{-8}$	$3.9 \cdot 10^{-6}$
HALLMARK_TNFA_SIGNALING_VIA_NFKB		1.85	$5.5 \cdot 10^{-6}$	$1.4 \cdot 10^{-4}$
HALLMARK_ANDROGEN_RESPONSE		1.72	$7.3 \cdot 10^{-4}$	$9.1 \cdot 10^{-3}$
HALLMARK_OXIDATIVE_PHOSPHORYLATION		1.53	$1.6 \cdot 10^{-3}$	$1.6 \cdot 10^{-2}$
HALLMARK_IL2_STAT5_SIGNALING		1.57	$2.2 \cdot 10^{-3}$	$1.8 \cdot 10^{-2}$
HALLMARK_XENOBIOTIC_METABOLISM		1.55	$2.7 \cdot 10^{-3}$	$1.9 \cdot 10^{-2}$
HALLMARK_INTERFERON_GAMMA_RESPONSE		-1.33	$5.8 \cdot 10^{-2}$	$1.3 \cdot 10^{-1}$
HALLMARK_KRAS_SIGNALING_UP		-1.39	$2.3 \cdot 10^{-2}$	$6.5 \cdot 10^{-2}$
HALLMARK_INTERFERON_ALPHA_RESPONSE		-1.50	$9.3 \cdot 10^{-3}$	$3.3 \cdot 10^{-2}$
HALLMARK_MTORC1_SIGNALING		-1.42	$9.0 \cdot 10^{-3}$	$3.3 \cdot 10^{-2}$
HALLMARK_E2F_TARGETS		-1.43	$4.8 \cdot 10^{-3}$	$2.4 \cdot 10^{-2}$
HALLMARK_P53_PATHWAY		-1.58	$7.0 \cdot 10^{-4}$	$9.1 \cdot 10^{-3}$

0 5000 10000

v2 plotGseaTable

ENRICHMENT



v3 NES_barplot

ENRICHMENT

monocle3

#21

OF 100 · PAPERS NAMING IT

R | TRAJECTORY | TRAJECTORY, PSEUDOTIME & VELOCITY

430	2.42%	499	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Trajectory graph on UMAP, pseudotime colouring, genes-in-pseudotime

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

monocle3 trajectory on the oligodendrocyte lineage (Zeisel level2class Oligo1-6)

KEY PLOTTING CALL

RECREATIONS/MONOCLE3/MONOCLE3_RECREATE.R — EXCERPT · ENDS LINE 8

```
cds <- new_cell_data_set(GetAssayData(ol, layer = "counts"), cell_metadata = ol@meta.data, gene_metadata = data.frame(gene_shor
t_name = rownames(ol), row.names = rownames(ol)))
cds <- preprocess_cds(cds, num_dim = 20); cds <- reduce_dimension(cds, preprocess_method = "PCA"); cds <- cluster_cells(cds, re
solution = 1e-2); cds <- learn_graph(cds, use_partition = FALSE)
root <- colnames(cds)[cds$level2class == "Oligo1"]; cds <- order_cells(cds, root_cells = root)
ggsave("v1_trajectory_subclass.png", plot_cells(cds, color_cells_by = "level2class", label_groups_by_cluster = FALSE, label_lea
ves = TRUE, label_branch_points = TRUE, group_label_size = 3, cell_size = .8), width = 6.5, height = 5, dpi = 150, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 — FIGURE CAPTION

Xenografted human microglia display diverse transcriptomic states in response to Alzheimer's disease-related amyloid-β pathology. *Nat Neurosci, 2024*

"...croglial transcriptional trajectory in response to Aβ pathology.a, Phenotypic trajectory followed by the human microglia after exposure to Aβ in vivo, obtained by an unbiased pseudotime ordering with Monocle ..."

DOI ↗ Fig. 3 on PMC ↗

Extended Data Fig. 4 — FIGURE CAPTION

Tau activation of microglial cGAS-IFN reduces MEF2C-mediated cognitive resilience. *Nat Neurosci, 2023*

"..., disease-associated (Apoe, Itgax) and interferon (Parp14, Stat1, Trim30a, Rnf213) genes in microglia clusters. b. Microglial trajectory plots colored by cell state trajectory scores identified using Monocle 3 ..."

DOI ↗ Article ↗ figure page not on PMC

Fig. 4 — TEXT

Human microglia show unique transcriptional changes in Alzheimer's disease. *Nat Aging, 2023*

"...The data were downsampled to 1,000, 2,000, 3,000 or 5,000 nuclei per cluster and transferred to a cell dataset object, and Monocle3 'learn_graph' was run."

DOI ↗ Fig. 4 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

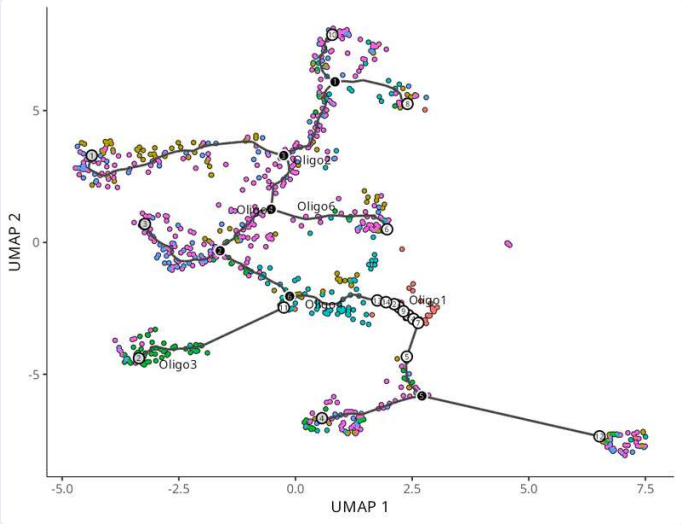
Reproduce: `Rscript recreations/monocle3/monocle3_recreate.R`

✓ re-run end to end · 31 s

monocle3 · 4 variations

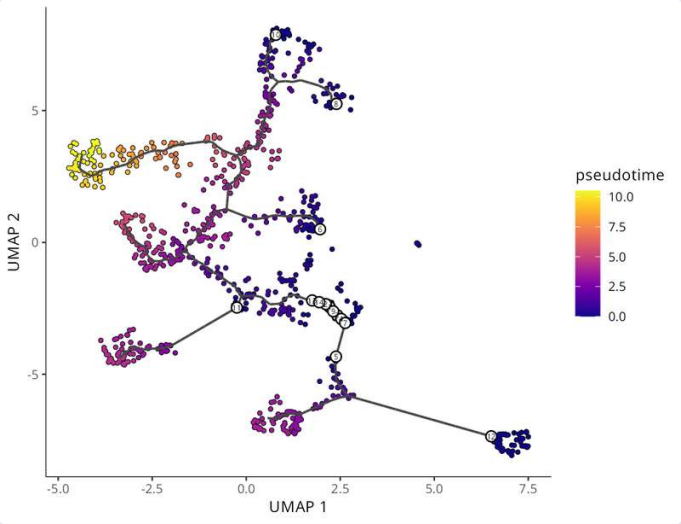
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



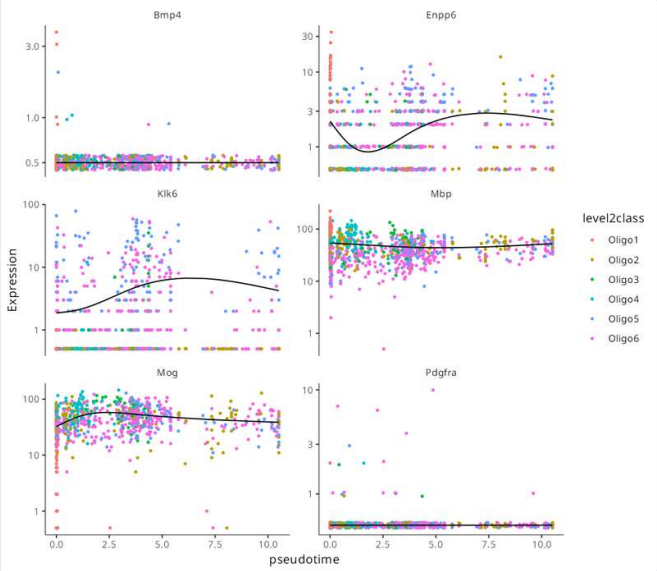
v1 trajectory_subclass

TRAJECTORY



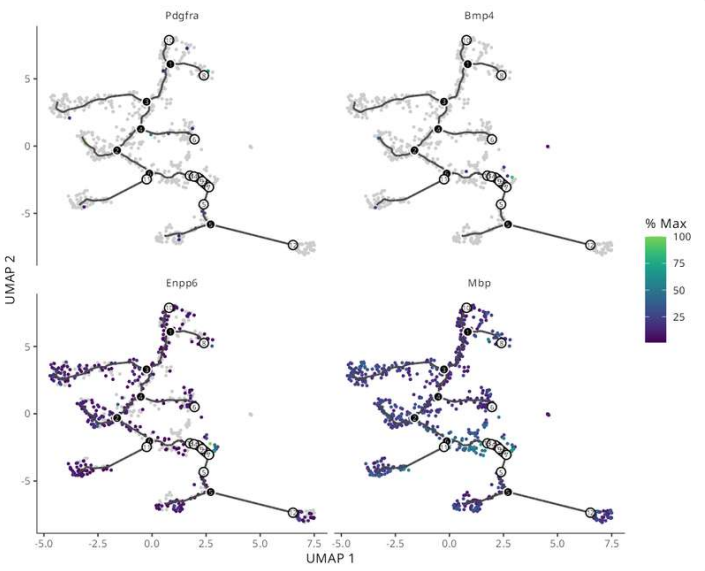
v2 pseudotime

TRAJECTORY



v3 genes_in_pseudotime

TRAJECTORY



v4 genes_on_trajectory

TRAJECTORY

deepTools

#22

OF 100 · PAPERS NAMING IT

PYTHON/CLI | HEATMAP (GENOMIC) | GENOME TRACKS & GENOMIC SIGNAL

416	2.34%	427	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

TSS/peak-centred signal heatmaps and metaprofiles (computeMatrix/plotHeatmap/plotProfile)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

deepTools computeMatrix → plotHeatmap / plotProfile. Uses SIMULATED bigWig signal (sharp TSS peaks ~H3K4me3, broad ~H3K27ac, flat input) on a toy 5-Mb chr

KEY PLOTTING CALL

RECREATIONS/DEEPTOOLS/DEEPTOOLS_RECREATE.SH — EXCERPT · ENDS LINE 6

```
python3 make_sim_bigwigs.py
computeMatrix reference-point -S sim_H3K4me3.bw sim_H3K27ac.bw sim_input.bw -R genes_active.bed genes_silent.bed --
referencePoint TSS -a 3000 -b 3000 -bs 50 -o mat.gz -p 4 --skipZeros
plotHeatmap -m mat.gz -out v1_plotHeatmap.png --colorMap Reds Blues Greys --heatmapHeight 10 --dpi 130 --plotTitle "deepTools
plotHeatmap (simulated signal)"
plotProfile -m mat.gz -out v2_plotProfile_perGroup.png --perGroup --dpi 130 --plotTitle "plotProfile --perGroup"
plotHeatmap -m mat.gz -out v3_kmeans3.png --kmeans 3 --colorMap viridis --dpi 130 --whatToShow "plot, heatmap and colorbar"
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure S1 TEXT
Macrophage-mediated myelin recycling fuels brain cancer malignancy. *Cell, 2024*
"...The coverage files were generated with the 'normalize to 1× genome coverage' methods in deepTools (version 3.5.1).89"
DOI ↗ Figure S1 on PMC ↗

Fig. 1 TEXT
Transcriptional and epigenetic decoding of the microglial aging process. *Nat Aging, 2023*
"...Heat maps of coverage at TSSs were generated using computeMatrix with the parameters computeMatrix reference-point–referencePoint TSS -b 1000 -a 1000 -R and plotHeatmap functions in deepTools 3.5.1."
DOI ↗ Fig. 1 on PMC ↗

Fig. 1 TEXT
Programs, origins and immunomodulatory functions of myeloid cells in glioma. *Nature, 2025*
"...Finally, we plotted the heatmap using the deeptools plotHeatmap function72."
DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

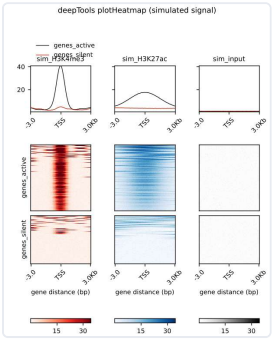
- Signal is simulated: toy bigWigs on a synthetic 5-Mb chromosome, shown only to illustrate the plot types; it carries no biological information. Use real ENCODE or GEO bigWigs for actual figures.

Reproduce: `bash recreations/deepTools/deeptools_recreate.sh`

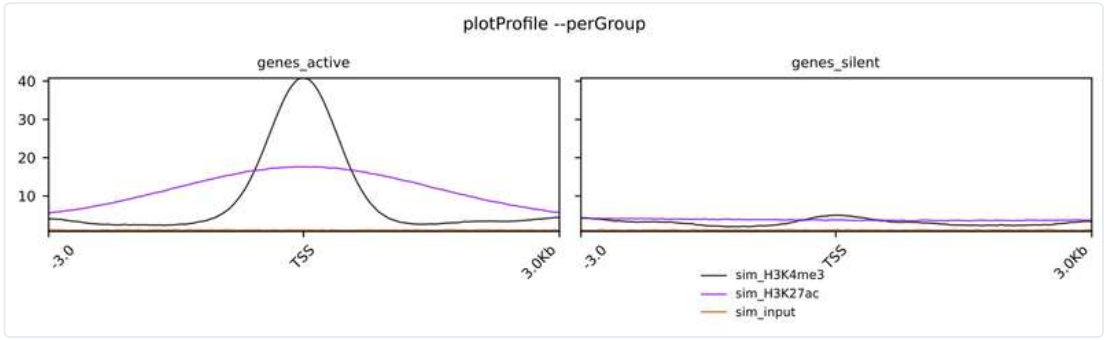
✓ re-run end to end · 225 s

deepTools · 4 variations

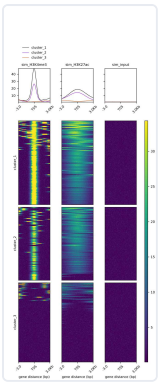
Data: Simulated bigWig signal on a toy 5-Mb chromosome (sharp TSS-like peaks, broad domains, flat input), generated by `make_sim_bigwigs.py`. Illustrates the plot types only; it is not measured data.. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



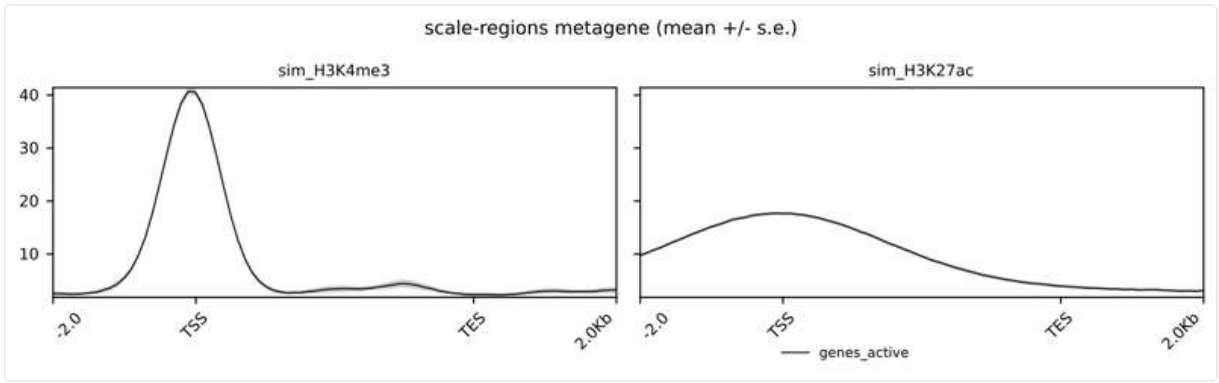
v1 `plotHeatmap` · GENOME



v2 `plotProfile_perGroup` · GENOME



v3 `kmeans3` · GENOME



v4 `scale_regions_profile` · GENOME

SingleR

#23

OF 100 · PAPERS NAMING IT

R · ANNOTATION (HEATMAP) · HEATMAPS & DOT MATRICES · COMPOSITION, DISTRIBUTIONS & LAYOUTS · EMBEDDINGS & FEATURE MAPS

416	2.34%	451	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Annotation score heatmap, delta distribution, label UMAP

KEY PLOTTING CALL

RECREATIONS/SINGLER/SINGLER_RECREATE.R — EXCERPT · ENDS LINE 5

```
so <- readRDS("../_data/zeisel_seurat.rds"); ref <- MouseRNAseqData()
pred <- SingleR(test = GetAssayData(so, layer = "data"), ref = ref, labels = ref$label.main)
png("v1_plotScoreHeatmap.png", 1400, 900, res = 140); print(plotScoreHeatmap(pred, annotation_col = data.frame(Author = so$cell
type, row.names = colnames(so)))); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 5 — FIGURE CAPTION

Molecular and cellular evolution of the amygdala across species analyzed by single-nucleus transcriptome profiling. *Cell Discov, 2023*

"...sted P-value was calculated by the Wilcoxon test, two-side comparison (adjusted P-value < 0.05). g UMAP embedding of chicken datasets, with nuclei colored by predicted cell group ID from Seurat and SingleR ..."*

DOI ↗ Fig. 5 on PMC ↗

Fig. 1 — TEXT

Nociceptor-immune interactomes reveal insult-specific immune signatures of pain. *Nat Immunol, 2024*

"...Because a full dataset of immune cells in inflamed skin has not been generated before, we complemented the singleR predictions with specific cluster annotations based on current knowledge of skin immune ..."

DOI ↗ Fig. 1 on PMC ↗

Fig. 1 — TEXT

Spatial transcriptomics reveals human cortical layer and area specification. *Nature, 2025*

"...Reference-based cell-type annotation was conducted using SingleR (v.1.8.1)68 to transfer cell-type labels from the reference data to the testing data with the following parameters: "genes": "de","

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

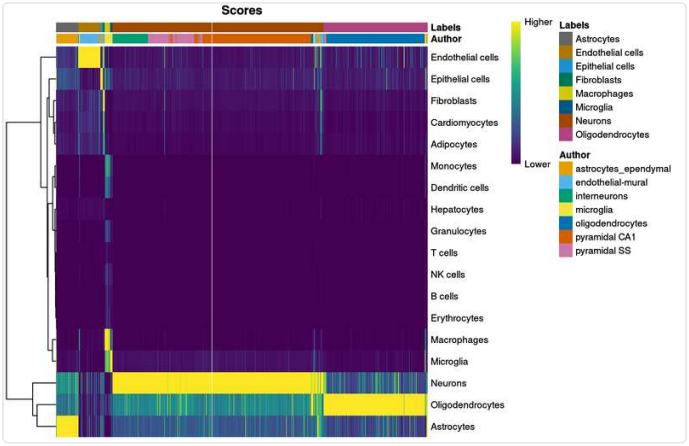
Reproduce: `Rscript recreations/SingleR/SingleR_recreate.R`

√ re-run end to end · 26 s

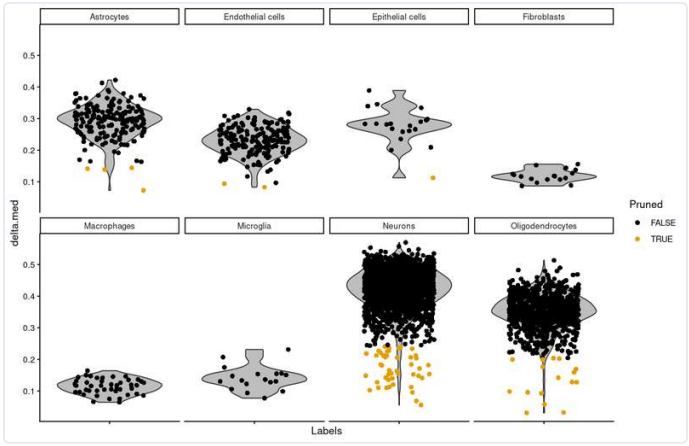
SingleR · 4 variations

NOVAFLOW RECREATION

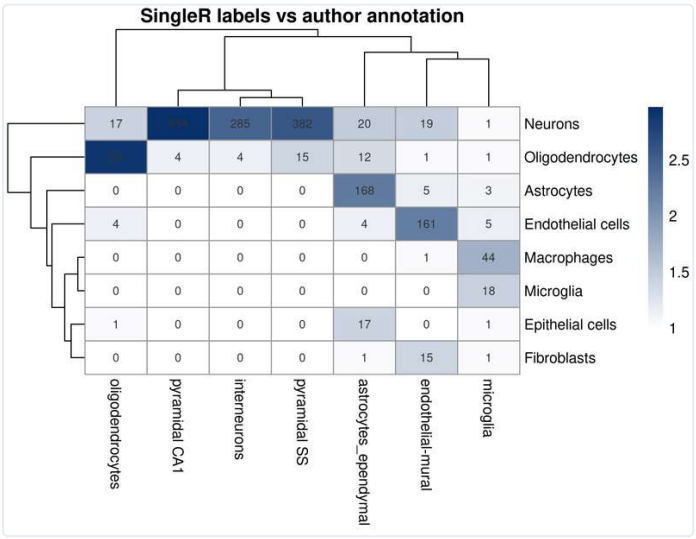
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



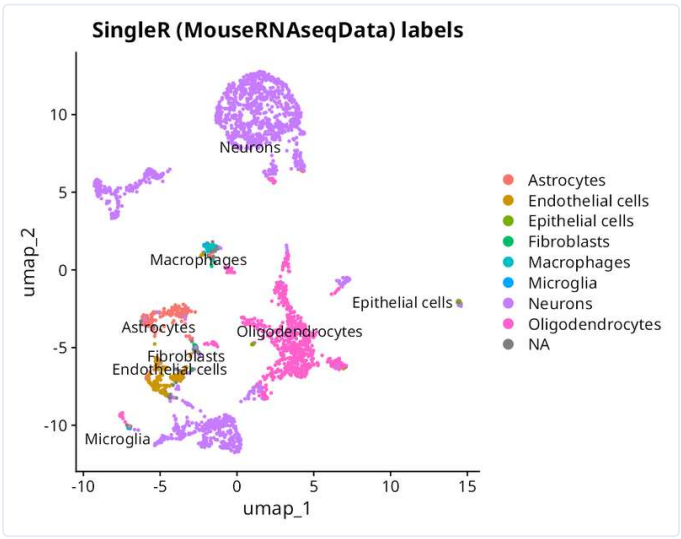
v1 plotScoreHeatmap · HEATMAPS



v2 plotDeltaDistribution · COMPOSITION



v3 confusion · HEATMAPS



v4 umap_labels · EMBEDDINGS

AUCell

#24

OF 100 · PAPERS NAMING IT

R | GENE SET SCORES | COMPOSITION, DISTRIBUTIONS & LAYOUTS | EMBEDDINGS & FEATURE MAPS | HEATMAPS & DOT MATRICES

387	2.18%	432	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Gene-set activity on UMAP, AUC histograms/thresholds

KEY PLOTTING CALL

RECREATIONS/AUCELL/AUCELL_RECREATE.R — EXCERPT · ENDS LINE 8

```
rk <- AUCell_buildRankings(m, plotStats = FALSE); auc <- AUCell_calcAUC(sets, rk)
png("v1_exploreThresholds.png", 1500, 950, res = 140); par(mfrow = c(2,3)); th <- AUCell_exploreThresholds(auc, plotHist =
TRUE, assign = TRUE); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 2 — FIGURE CAPTION

APOE modulates microglial immunometabolism in response to age, amyloid pathology, and inflammatory challenge. *Cell Rep, 2023*

"...axis) and aged versus young microglia (x axis). Genes labeled in black are common to both DAM/MgND phenotypes. Inset: ridge plot showing DAM/MgND score for each individual microglia as calculated by AUCell.(B ..."

DOI ↗ Figure 2 on PMC ↗

Extended Data Fig. 1 — FIGURE CAPTION

SCENIC+: single-cell multiomic inference of enhancers and gene regulatory networks. *Nat Methods, 2023*

"...d. Cell-topic enrichment heat map for each of the models. General topics are shown in black; mesenchymal, in red; melanocytic, in blue; cell line specific in green; and low contributing in gray. e. AUCell ..."

DOI ↗ Extended Data Fig. 1 on PMC ↗

Figure 6 — TEXT

Polygenic regression uncovers trait-relevant cellular contexts through pathway activation transformation of single-cell RNA sequencing data. *Cell Genom, 2023*

"...Additionally, when applying genes prioritized by scPagwas to two other cell-scoring methods, e.g., VISION35 and AUCell,33 we consistently found that these cell-scoring methods yielded a high performance in ..."

DOI ↗ Figure 6 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

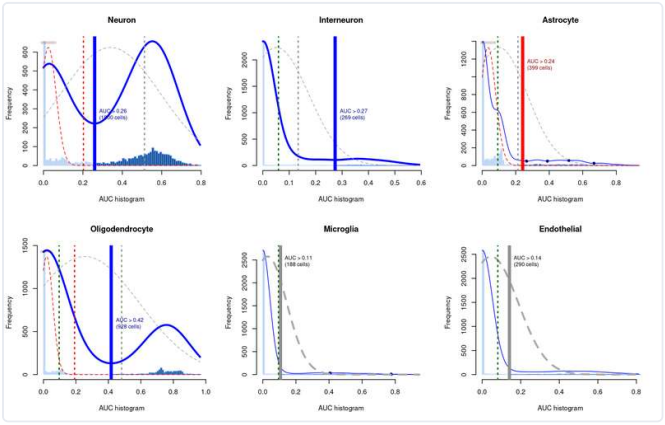
Reproduce: `Rscript recreations/AUCell/AUCell_recreate.R`

√ re-run end to end · 25 s

AUCell · 3 variations

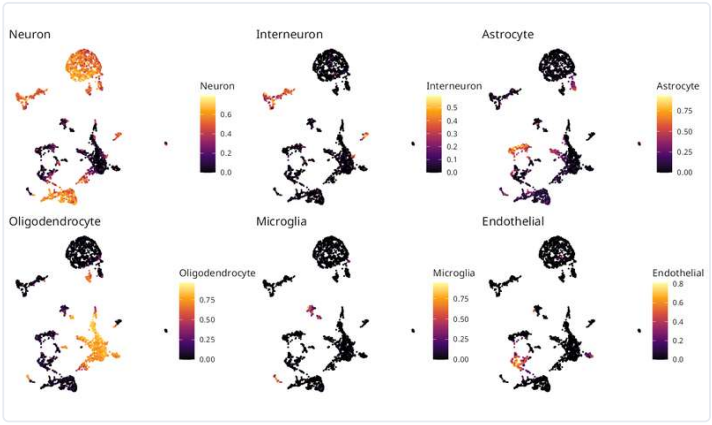
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



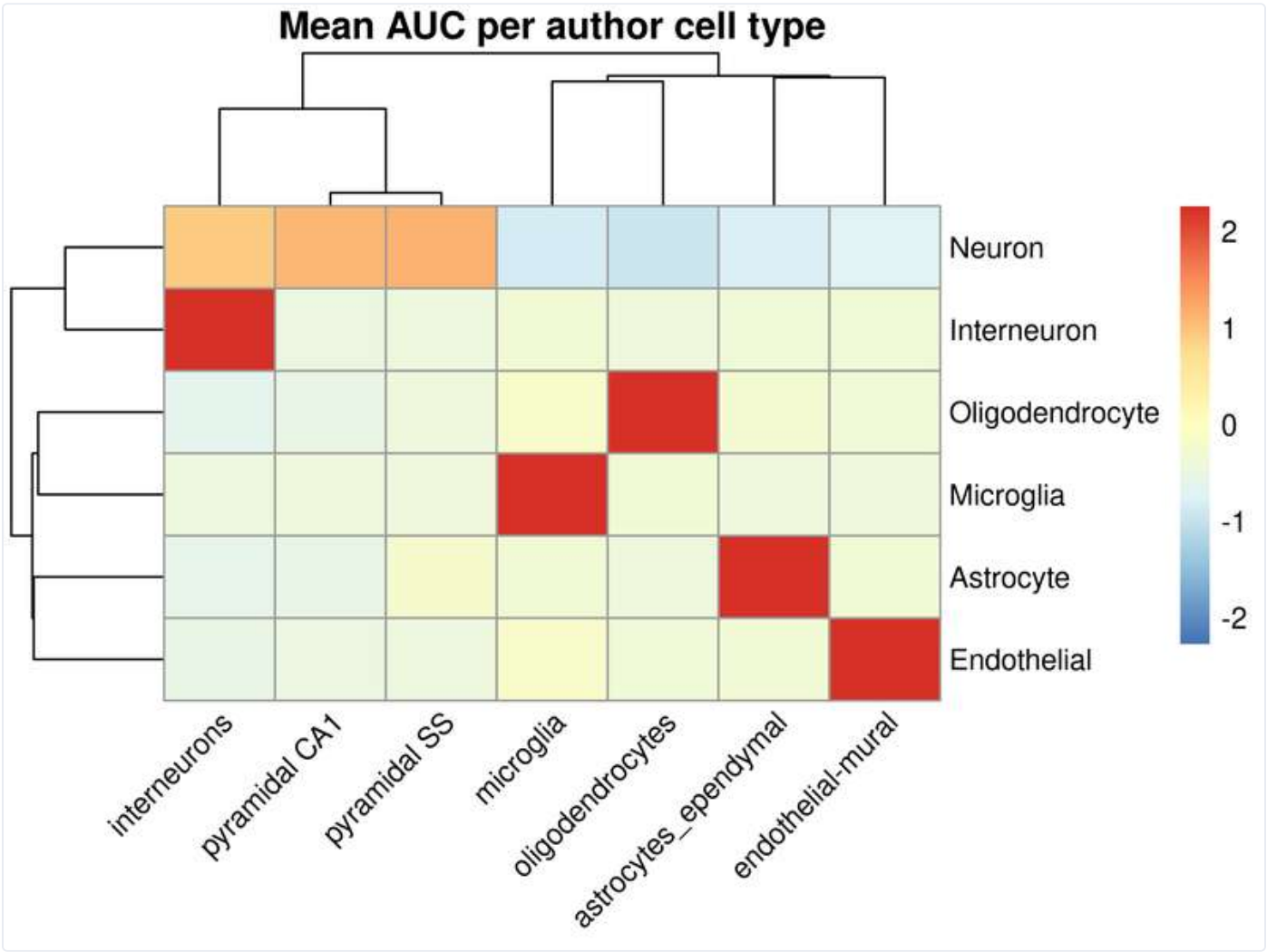
v1 exploreThresholds

COMPOSITION



v2 umap_auc

EMBEDDINGS



v3 auc_heatmap

HEATMAPS

EnhancedVolcano

#27

OF 100 · PAPERS NAMING IT

R | VOLCANO | DE, QC & SET OVERLAPS

369	2.08%	373	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Publication-ready volcano plots with labels & connectors

KEY PLOTTING CALL

RECREATIONS/ENHANCEDVOLCANO/ENHANCEDVOLCANO_RECREATE.R — EXCERPT · ENDS LINE 6

```
g <- EnhancedVolcano(res, lab = res$symbol, x = "log2FoldChange", y = "pvalue", title = "Dex vs untreated", subtitle = "Enhance
dVolcano default")
ggsave("v1_default.png", g, width = 7, height = 7, dpi = 130)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 5 TEXT
Multiomic spatial landscape of innate immune cells at human central nervous system borders. *Nat Med, 2024*
"...To this end we ran differential expression analysis of ligand–receptor pairs using the FindMarkers R function and visualized to top differentially expressed pairs using volcano plots with the EnhancedVolcano R ..."
DOI ↗ Fig. 5 on PMC ↗

Fig. 4 TEXT
Transcriptional and epigenetic decoding of the microglial aging process. *Nat Aging, 2023*
"...Venn diagrams, heat maps and volcano plots were visualized by using VennDiagram 1.6.20, pheatmap 1.0.12 and EnhancedVolcano 1.4.0, respectively."
DOI ↗ Fig. 4 on PMC ↗

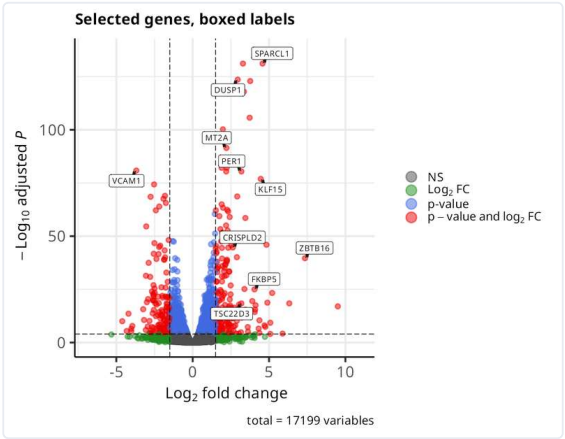
Fig. 2 TEXT
Platelet-derived exerkine CXCL4/platelet factor 4 rejuvenates hippocampal neurogenesis and restores cognitive function in aged mice. *Nat Commun, 2023*
"...EnhancedVolcano v1.16.068 produced volcano plots, with shrunken log fold change values calculated using "ashr" as an estimator."
DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/EnhancedVolcano/EnhancedVolcano_recreate.R` — √ re-run end to end · 10 s

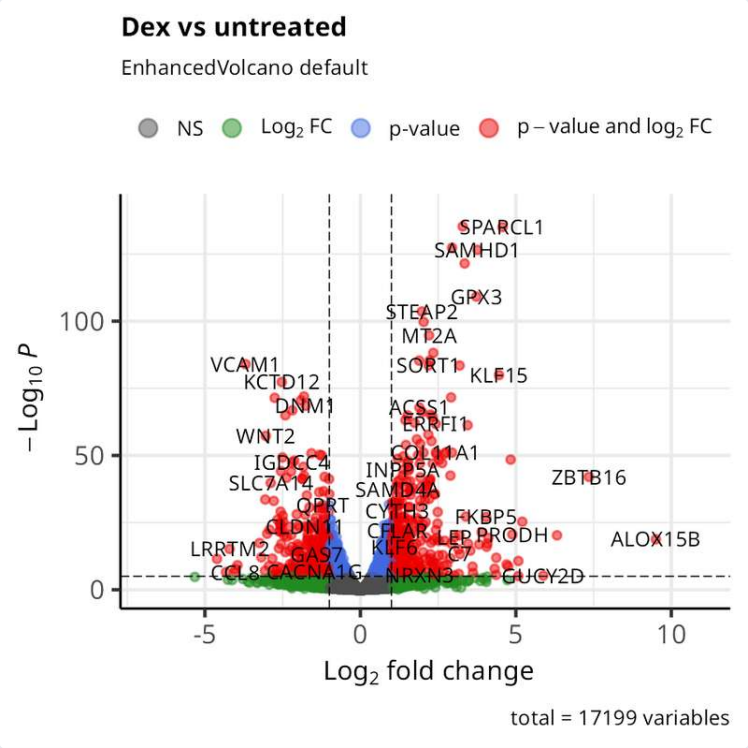
LEAD FIGURE



v2 selected_boxed — DE

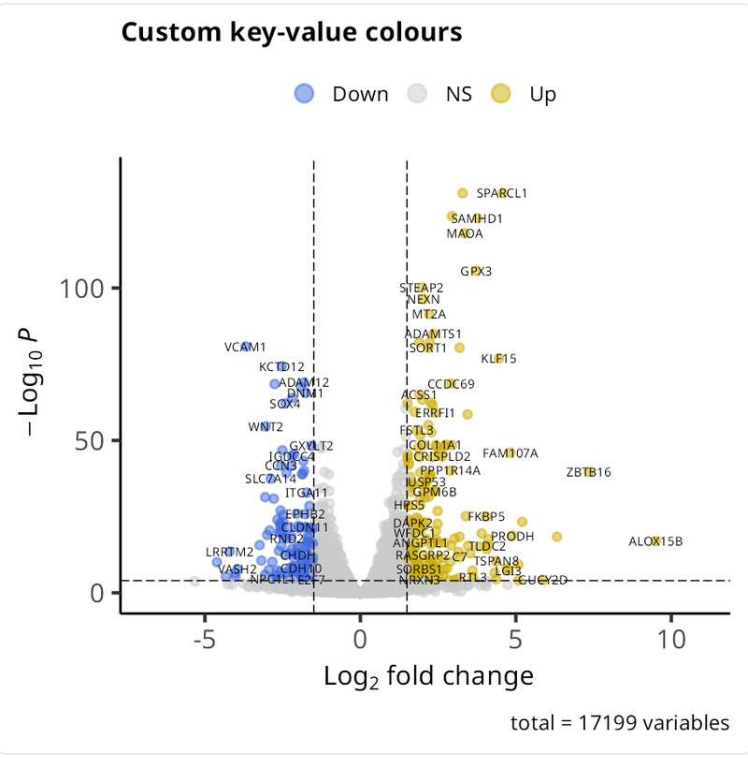
EnhancedVolcano · 3 variations · v2 on facing page

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 default

DE



v3 custom_colours

DE

survminer

#29

OF 100 · PAPERS NAMING IT

R | SURVIVAL | COMPOSITION, DISTRIBUTIONS & LAYOUTS

351	1.98%	342	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Kaplan-Meier curves + risk tables, forest plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

survminer: Kaplan-Meier + risk table; glioma survival data from coin::glioma if available, else survival::lung

KEY PLOTTING CALL

RECREATIONS/SURVMINER/SURVMINER_RECREATE.R — EXCERPT · ENDS LINE 7

```
if (requireNamespace("coin", quietly = TRUE)) { data("glioma", package = "coin"); d <- glioma; d$status <- as.numeric(d$event);
f <- Surv(time, status) ~ group; ttl <- "coin::glioma (Grana 2002): radioimmunotherapy vs control" } else { d <- lung; f <-
Surv(time, status) ~ sex; ttl <- "survival::lung" }
fit <- surv_fit(f, data = d)
p <- ggsurvplot(fit, data = d, pval = TRUE, conf.int = TRUE, risk.table = TRUE, palette = c("#2E9FDF","#E7B800"), title = ttl, g
gtheme = theme_classic())
png("v1_km_risktable.png", 1100, 1000, res = 150); print(p); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 7 TEXT

Macrophage-mediated myelin recycling fuels brain cancer malignancy. *Cell, 2024*

"...For the calculation of the survival curves in Figures 7A and 7F, the Kaplan-Meier estimator was obtained with the survfit function from the package survival.106 The statistical test applied was the log-rank ..."

DOI ↗ Figure 7 on PMC ↗

Fig. 3 TEXT

Spatial transcriptomics reveals niche-specific enrichment and vulnerabilities of radial glial stem-like cells in malignant gliomas. *Nat Commun, 2023*

"...And the survival curve was plotted using survminer (v0.4.9) R package."

DOI ↗ Fig. 3 on PMC ↗

Fig. 2 TEXT

Deciphering the longitudinal trajectories of glioblastoma ecosystems by integrative single-cell genomics. *Nat Genet, 2025*

"...Kaplan–Meier estimates were computed using the survfit function and plotted using the ggsurvplot function (survminer package)."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

- Uses coin::glioma (Grana et al. 2002, glioma radioimmunotherapy) survival data

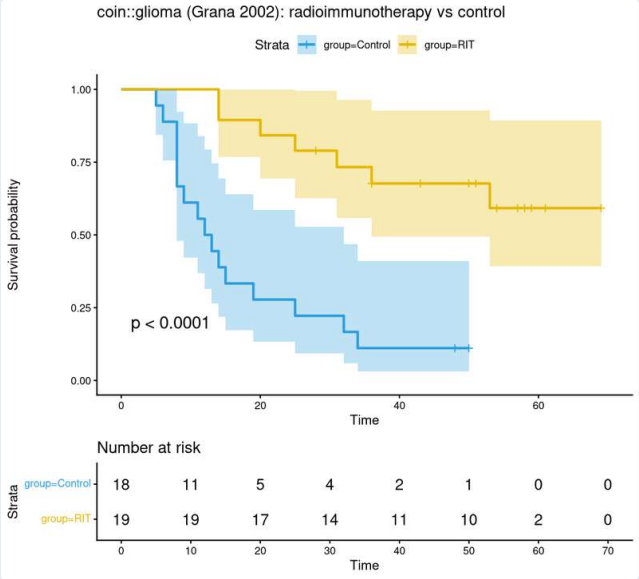
Reproduce: `Rscript recreations/survminer/survminer_recreate.R`

√ re-run end to end · 4 s

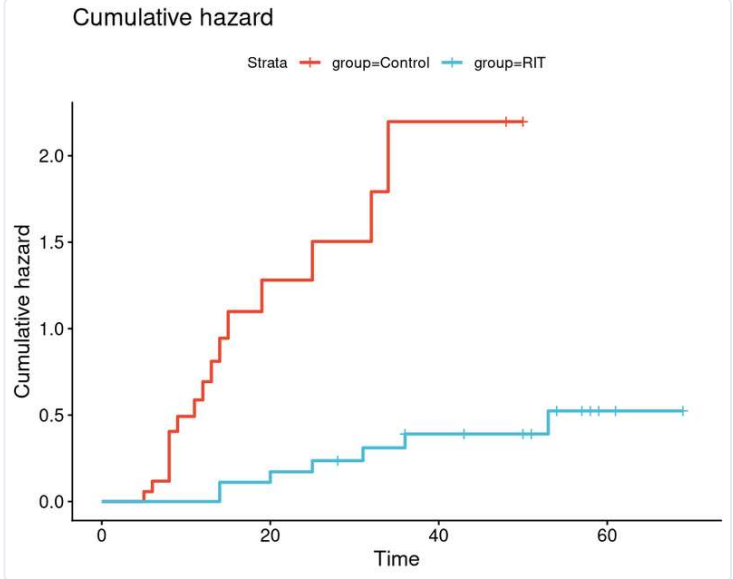
survminer · 3 variations

NOVAFLOW RECREATION

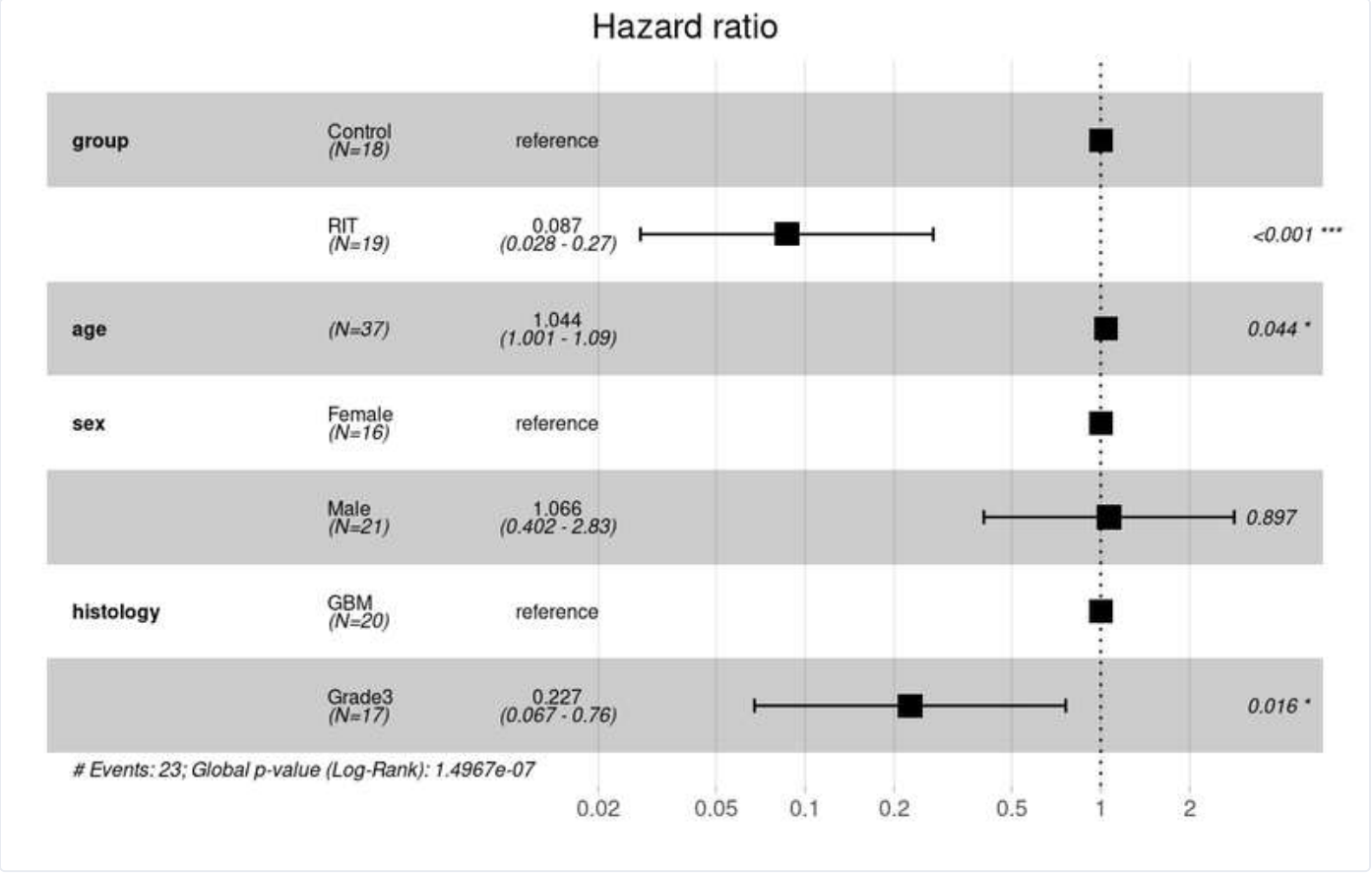
Data: *coin::glioma* survival data (Grana et al. 2002, glioma radioimmunotherapy trial). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 `km_risktable` COMPOSITION



v2 `cumhaz` COMPOSITION



v3 `forest` COMPOSITION

Space Ranger / Loupe-style

#31

OF 100 · PAPERS NAMING IT

DESKTOP | SPATIAL | SPATIAL & TISSUE MAPS

300	1.69%	307	viewer	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Visium spot/tissue overlays (Loupe)

Also catalogued as: **Loupe Browser** (#43, 166 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Space Ranger / Loupe Browser look-alike from the same Visium mouse-brain run: tissue + spot overlay, gene with opacity, side-by-side UMAP

KEY PLOTTING CALL

RECREATIONS/SPACERANGER_LOUPE_STYLE/LOUPE_STYLE_RECREATE.PY — EXCERPT · ENDS LINE 15

```
fig,ax=plt.subplots(1,2,figsize=(13,6.2),facecolor="#f4f4f4")
ax[0].imshow(img); spots(ax[0],col,.85); ax[0].set_title("Clusters (graph-based)",loc="left",fontsize=11); ax[0].axis("off")
u=v.obsm["X_umap"]; ax[1].scatter(*u.T,c=col,s=8); ax[1].set_title("UMAP",loc="left"); ax[1].axis("off")
for i,c in enumerate(cl.cat.categories): ax[1].scatter([],[],c=LOUPE[i%15],label=c)
ax[1].legend(fontsize=7,frameon=False,loc="center left",bbox_to_anchor=(1,.5)); fig.suptitle("Loupe-style linked tissue / UMAP
view",x=.02,ha="left",weight="bold"); fig.tight_layout(); fig.savefig("v1_tissue_clusters_umap.png",dpi=130)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 TEXT — **Spatial transcriptomics reveals niche-specific enrichment and vulnerabilities of radial glial stem-like cells in malignant gliomas.** *Nat Commun, 2023*

"...We used REDItools60 (v2) with default parameters to identify H3K27M mutation in DMG samples from BAM files generated by the SpaceRanger pipeline."

DOI ↗ Fig. 1 on PMC ↗

Fig. 1 TEXT — **Spatially resolved multiomics of human cardiac niches.** *Nature, 2023*

"...Visium: GRCh38-3.0.0) provided by 10x Genomics using CellRanger ARC (v.2.0.0) or SpaceRanger (v.1.1.0) with default parameters."

DOI ↗ Fig. 1 on PMC ↗

Fig. 1 TEXT — **Robust mapping of spatiotemporal trajectories and cell-cell interactions in healthy and diseased tissues.** *Nat Commun, 2023*

"...Raw BCL files were processed by bcl2fastq V2.7.0, and the fastq reads mapped to the mouse reference genome GRCm38 by SpaceRanger V1.0.0."

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

- Look-alike with matplotlib on the squidpy Visium mouse-brain run
- Loupe Browser (#43): Look-alike (see Space Ranger/Loupe)

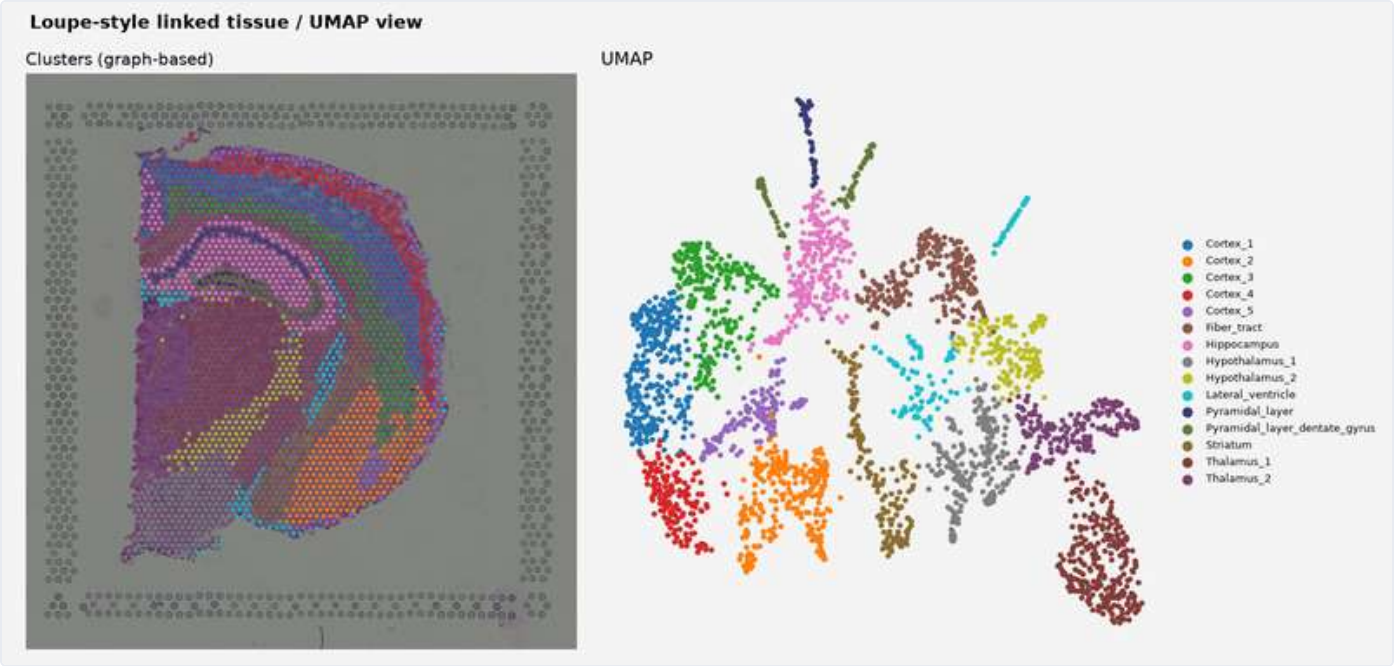
Reproduce: `python3 recreations/SpaceRanger_Loupe_style/loupe_style_recreate.py`

✓ re-run end to end · 5 s

Space Ranger / Loupe-sty... · 3 variations

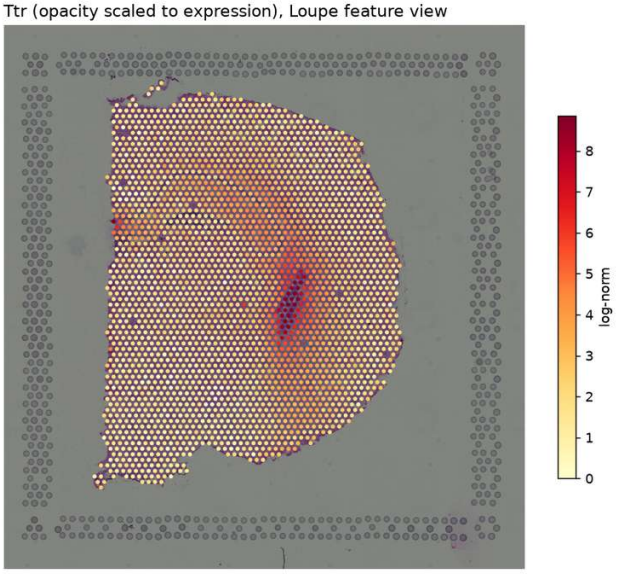
NOVAFLOW RECREATION

Data: 10x Genomics Visium mouse coronal brain section (squidpy example dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



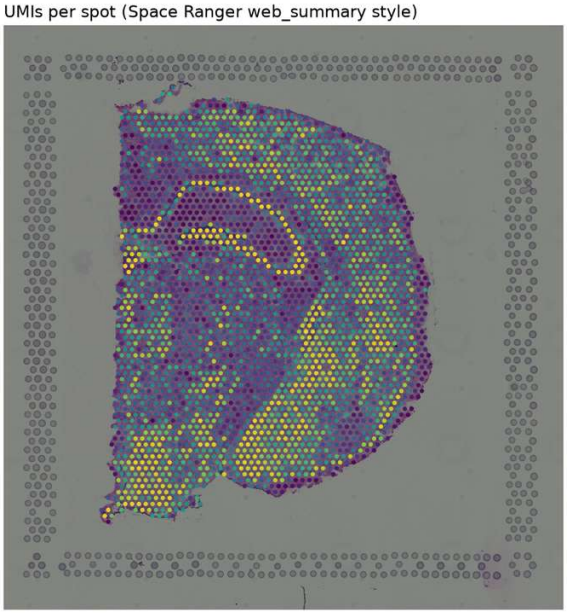
v1 tissue_clusters_umap

SPATIAL



v2 feature_opacity

SPATIAL



v3 umi_per_spot

SPATIAL

ggpubr

#33

OF 100 · PAPERS NAMING IT

R | GENERAL/STATISTICS | COMPOSITION, DISTRIBUTIONS & LAYOUTS

277	1.56%	283	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Box/violin/bar plots with p-values, correlation scatter

KEY PLOTTING CALL

RECREATIONS/GGPUBR/GGPUBR_RECREATE.R — EXCERPT · ENDS LINE 6

```
so <- readRDS("../_data/zeisel_seurat.rds"); df <- data.frame(celltype = so$celltype, Gad1 = FetchData(so,"Gad1")[,1], Snap25 =
FetchData(so,"Snap25")[,1], Mbp = FetchData(so,"Mbp")[,1], nGene = so$nFeature_RNA, tissue = so$tissue)
sub <- droplevels(df[df$celltype %in% c("interneurons","pyramidal SS","pyramidal CA1"),])
p <- ggboxplot(sub, "celltype", "Gad1", color = "celltype", add = "jitter", add.params = list(size = .3, alpha = .3), palette =
"jco") + stat_compare_means(comparisons = list(c("interneurons","pyramidal SS"), c("interneurons","pyramidal CA1"),
c("pyramidal SS","pyramidal CA1")), label = "p.signif") + stat_compare_means(label.y = 7.5)
ggsave("v1_boxplot_pvalues.png", p, width = 5.5, height = 5, dpi = 150, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure S1 TEXT
Macrophage-mediated myelin recycling fuels brain cancer malignancy. *Cell, 2024*
"...Plots were generated using the package ggplot (v.3.3.6) or ggpubr (v.0.4.0), except where mentioned."
DOI ↗ Figure S1 on PMC ↗

Fig. 3 TEXT
A concerted neuron-astrocyte program declines in ageing and schizophrenia. *Nature, 2024*
"...R (v.4.1.3): cluster (v.2.1.2)138, ComplexHeatmap (v.2.10.0)139,140, data.table (v.1.14.8)141, DescTools (v.0.99.48)142, dplyr (v.1.1.2)143, gdata (v.2.19.0)144, ggforce (v.0.4.1)145, ggplot2 (v.3.4.2)146 ..."
DOI ↗ Fig. 3 on PMC ↗

Figure 4 TEXT
Transcriptomic atlas and interaction networks of brain cells in mouse CNS demyelination and remyelination. *Cell Rep, 2023*
"...Statistics on cell type compositional analysis were calculated using scCODA package (v.0.1.8) in Python (v3.9.2).101 p values in snRNA-seq analysis were adjusted based on Bonferroni correction.106 p values in ..."
DOI ↗ Figure 4 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

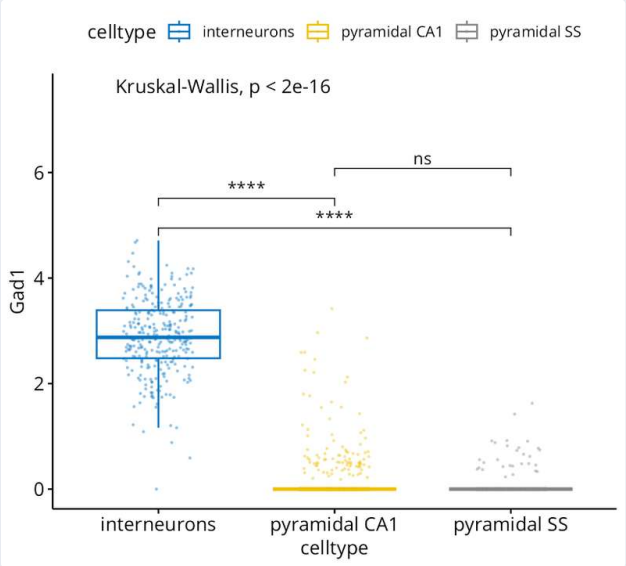
Reproduce: `Rscript recreations/ggpubr/ggpubr_recreate.R`

√ re-run end to end · 8 s

ggpubr · 4 variations

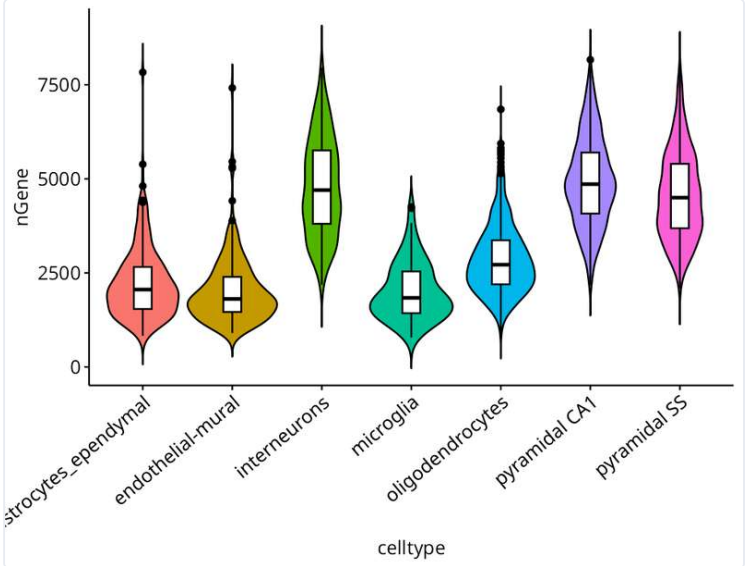
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



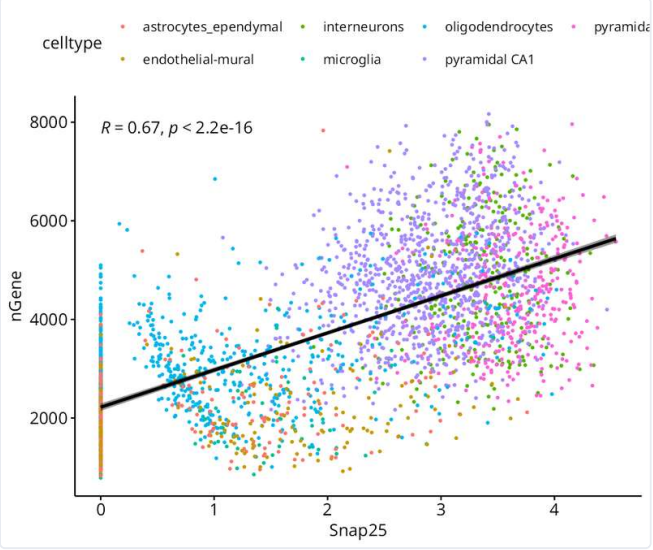
v1 boxplot_pvalues

COMPOSITION



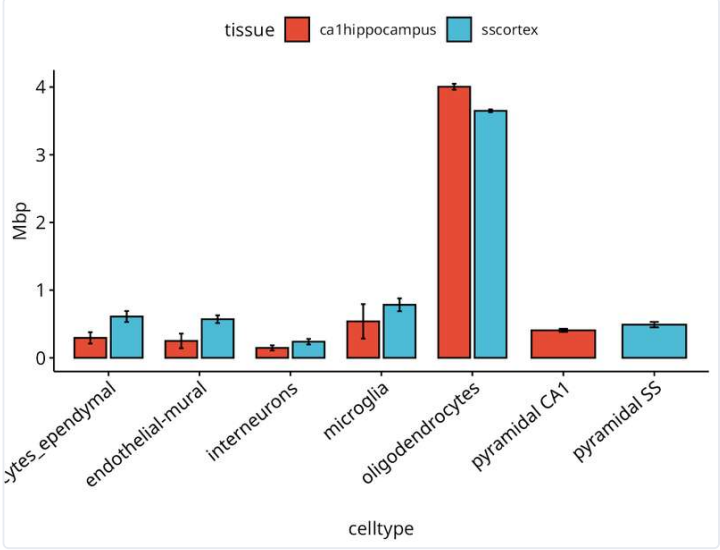
v2 violin

COMPOSITION



v3 scatter_cor

COMPOSITION



v4 barplot_mean_se

COMPOSITION

igraph

#36

OF 100 · PAPERS NAMING IT

R/PYTHON | NETWORK | NETWORKS, GRAPHS & TREES

241	1.36%	240	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Network graphs (co-expression, PPI, communities)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

igraph: gene co-expression network of variable genes in Zeisel cortex data

KEY PLOTTING CALL

RECREATIONS/IGRAPH/IGRAPH_RECREATE.R — EXCERPT · ENDS LINE 8

```
so <- readRDS("../_data/zeisel_seurat.rds"); g0 <- head(VariableFeatures(so), 150)
e <- as.matrix(GetAssayData(so, layer = "data")[g0,]); cm <- cor(t(e)); diag(cm) <- 0
adj <- (cm > .45) * cm; g <- graph_from_adjacency_matrix(adj, mode = "undirected", weighted = TRUE); g <- delete_vertices(g, de
gree(g) == 0)
set.seed(2); com <- cluster_louvain(g); pal <- hcl.colors(max(membership(com)), "Dark 3")
png("v1_fr_communities.png", 1300, 1300, res = 160)
plot(g, layout = layout_with_fr(g), vertex.color = pal[membership(com)], vertex.size = 3 + 1.2*sqrt(degree(g)),
vertex.label.cex = .5, vertex.label.color = "black", edge.width = E(g)$weight*2, edge.color = "grey80", main = "Co-expression
network (r > 0.45), Louvain communities"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 TEXT — **Inferring and perturbing cell fate regulomes in human brain organoids.** *Nature, 2023*

"...We then performed 10,000 random walks with 200 steps from each tip along edges backwards in pseudotime using the igraph R package (v.1.2.6) (https://igraph.org/)."

DOI ↗ Fig. 2 on PMC ↗

Fig. 3 TEXT — **Multimodal spatiotemporal phenotyping of human retinal organoid development.** *Nat Biotechnol, 2023*

"...The cluster graph was then visualized using the ggraph package in R with the tree layout implemented in the igraph package (the layout_as_tree function), with the edge weights represented by the edge ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 2 TEXT — **A prognostic neural epigenetic signature in high-grade glioma.** *Nat Med, 2024*

"...The adjacency matrix was then transformed into an undirected graph using the graph_from_adjacency_matrix function from the igraph package."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

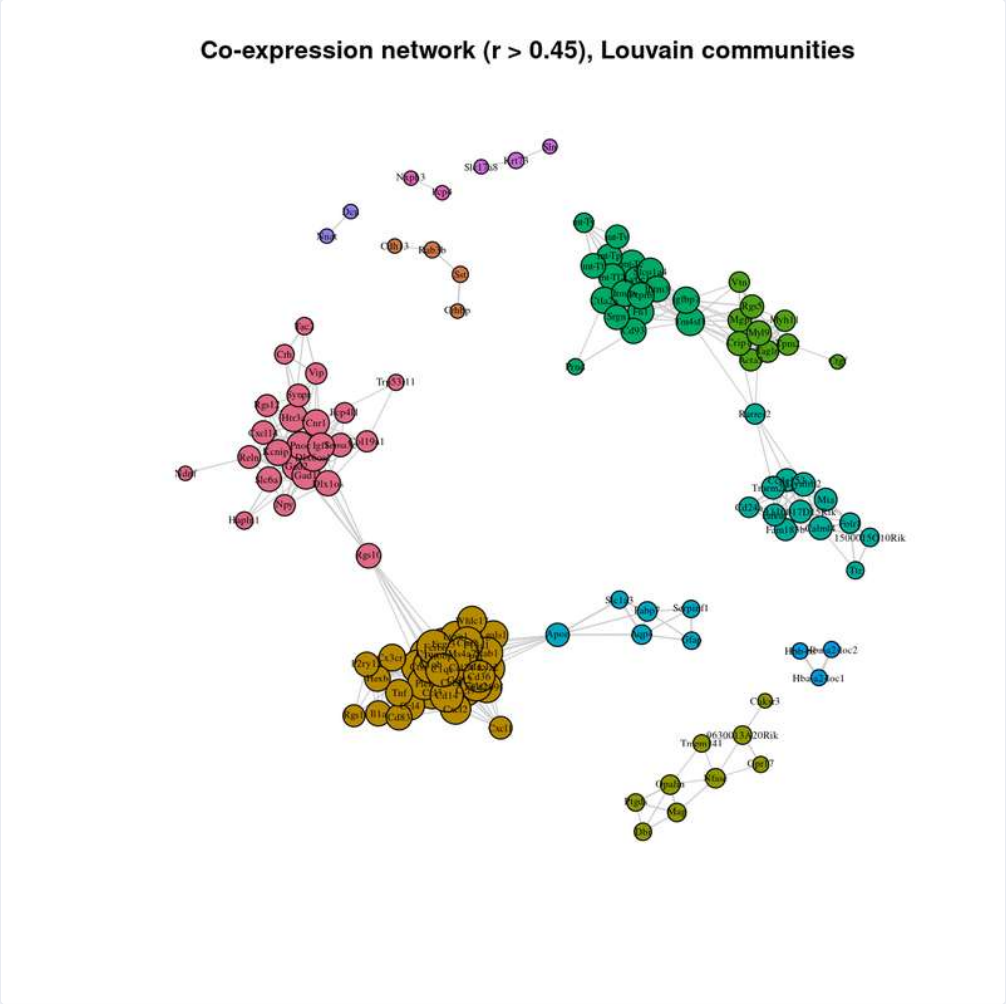
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/igraph/igraph_recreate.R`

✓ re-run end to end · 8 s

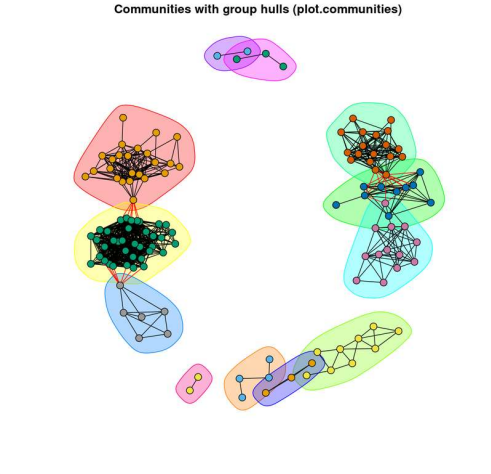
igraph · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



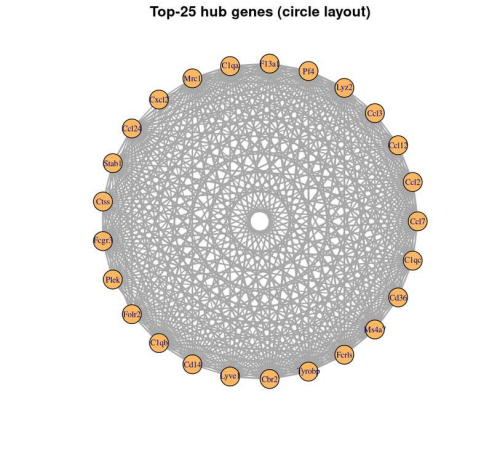
v1 fr_communities

NETWORKS



v2 mark_groups

NETWORKS



v3 circle_hubs

NETWORKS

Signac

#37

OF 100 · PAPERS NAMING IT

R | SCATAC (COVERAGE/EMBEDDING) | GENOME TRACKS & GENOMIC SIGNAL

227	1.28%	239	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · SIGNAC

Neuro-corpus papers per year (*2026 partial) · niche rank **#54**, growth **1.43×**, visual 4/5, score 0.544. Broad-corpus rate 4.876 → 6.96 per 1k papers (2023 → 2025–26).
GitHub stuart-lab/signac · created 2019-05-09 · last push 2026-09-11

BEST FOR

Coverage plots with peaks/links, motif plots, ATAC UMAP

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Signac scATAC-seq coverage/track plots. NOTE: uses Signac's bundled 'atac_small' (human PBMC, 100 cells) and its fragments file - no offline neuro scATAC small enough was available (10x E18 mouse brain fragments are >1 GB).

KEY PLOTTING CALL

RECREATIONS/SIGNAC/SIGNAC_RECREATE.R — EXCERPT · ENDS LINE 11

```
data(atac_small); fpath <- system.file("extdata", "fragments.tsv.gz", package = "Signac")
fragments <- CreateFragmentObject(fpath, cells = colnames(atac_small), validate.fragments = FALSE); Fragments(atac_small) <- NU
LL; Fragments(atac_small) <- fragments
atac_small$cluster <- factor(paste0("C", as.integer(atac_small$seurat_clusters)))
Idents(atac_small) <- "cluster"
reg <- "chr1-713500-860000"
p1 <- CoveragePlot(atac_small, region = reg, annotation = TRUE, peaks = TRUE, links = FALSE, extend.upstream = 0, extend.downst
ream = 0)
ggsave("v1_coverage_plot.png", p1 & theme(text = element_text(size = 9)), width = 9, height = 6, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 — FIGURE CAPTION

SCENIC+: single-cell multiomic inference of enhancers and gene regulatory networks. *Nat Methods, 2023*

"...r parameter optimization. c, Bar-plots showing the area under the recovery curve (AUC; enhancer recovery) on the top 10% of the ranking based on STARR-seq signal, for the top 5,000 DARs identified by Signac ..."

DOI ↗ Fig. 1 on PMC ↗

Fig. 4 — TEXT

Slide-tags enables single-nucleus barcoding for multimodal spatial genomics. *Nature, 2024*

"...The ATAC-filtered feature-barcode matrix generated by Cell Ranger was read into R (v.4.1.1) using Signac (v.1.9.0)77, and added as its own assay slot in the Seurat object containing RNA expression counts."

DOI ↗ Fig. 4 on PMC ↗

Fig. 3 — TEXT

Multimodal chromatin profiling using nanobody-based single-cell CUT&Tag. *Nat Biotechnol, 2023*

"...All dimensionality reduction and clustering was performed using package Seurat v4.0.5 and Signac v1.4.0 under R v4.1.1."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

- Non-neuro fallback: Signac's bundled atac_small (human PBMC, 100 cells) and its fragments file; tracks look thin. 10x E18 mouse-brain fragments are >1 GB, too large to use offline.

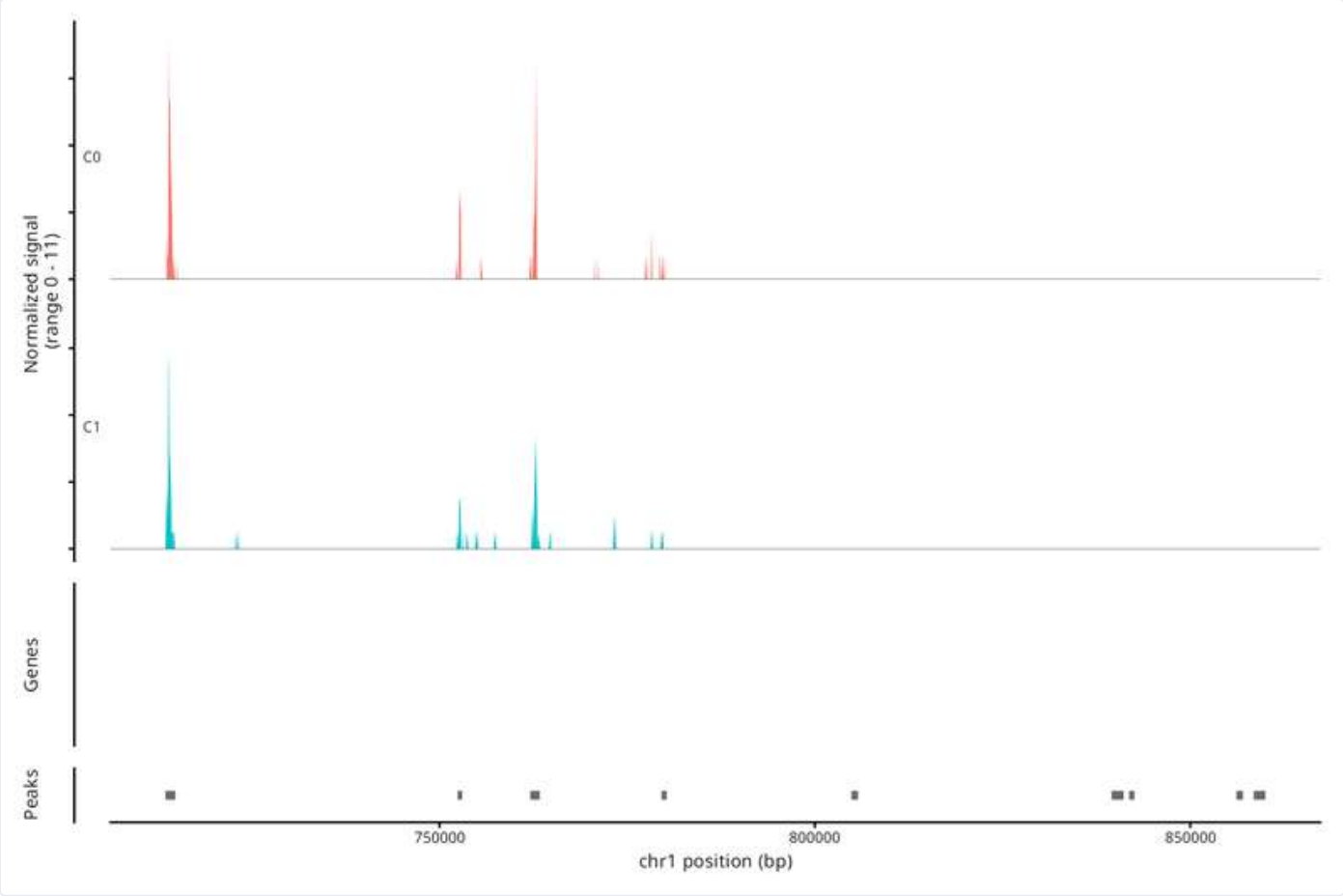
Reproduce: `Rscript recreations/Signac/Signac_recreate.R`

✓ re-run end to end · 11 s

Signac · 3 variations

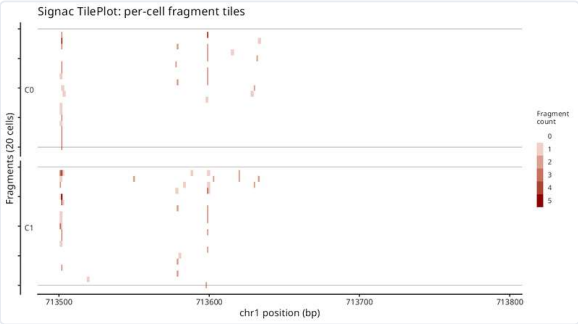
NOVAFLOW RECREATION

Data: Signac's bundled *atac_small* example (human PBMC scATAC-seq, 100 cells); not neuro data. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



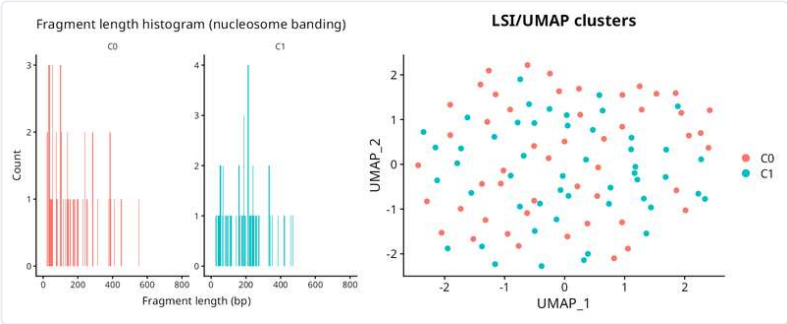
v1 coverage_plot

GENOME



v2 tile_plot

GENOME



v3 fraghist_umap

GENOME

matplotlib

#38

OF 100 · PAPERS NAMING IT

PYTHON | GENERAL | DE, QC & SET OVERLAPS | EMBEDDINGS & FEATURE MAPS | COMPOSITION, DISTRIBUTIONS & LAYOUTS

227	1.28%	238	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Base for all Python plots: scatter, volcano, multi-panel GridSpec

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

pure matplotlib versions of the most common omics panels

KEY PLOTTING CALL

RECREATIONS/MATPLOTLIB/MATPLOTLIB_RECREATE.PY — EXCERPT · ENDS LINE 11

```
fig,ax=plt.subplots(figsize=(5.5,5)); y=-np.log10(res.padj.clip(lower=1e-300))
ax.scatter(res.log2FoldChange[~(up|dn)],y[~(up|dn)],s=3,c="lightgrey",rasterized=True);
ax.scatter(res.log2FoldChange[up],y[up],s=4,c="#d7301f"); ax.scatter(res.log2FoldChange[dn],y[dn],s=4,c="#2b8cbe")
for g,r in res[up|dn].nsmallest(10,"padj").iterrows(): ax.annotate(r.symbol,(r.log2FoldChange,-np.log10(r.padj)),fontsize=7)
ax.axvline(-1,ls="--",c="k",lw=.6); ax.axvline(1,ls="--",c="k",lw=.6); ax.set(xlabel="log2 fold change",ylabel="-log10 padj",ti
tle="Volcano (matplotlib)"); [ax.spines[s].set_visible(False) for s in ("top","right")]
fig.tight_layout(); fig.savefig("v1_volcano.png",dpi=150)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 TEXT — **Specialized astrocytes mediate glutamatergic gliotransmission in the CNS.** *Nature, 2023*

"...The turbo colormap from matplotlib was used to assign amplitude values."

DOI ↗ Fig. 2 on PMC ↗

Fig. 3 TEXT — **SCENIC+: single-cell multiomic inference of enhancers and gene regulatory networks.** *Nat Methods, 2023*

"...The delta embedding is used to draw arrows on the original embedding, using the streamplot or quiverplot function from the matplotlib (v.3.5.2) Python package."

DOI ↗ Fig. 3 on PMC ↗

Fig. 3 TEXT — **A concerted neuron-astrocyte program declines in ageing and schizophrenia.** *Nature, 2024*

"...Python (v.3.8.3): matplotlib (v.3.5.2)136 and seaborn (v.0.10.1)137."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

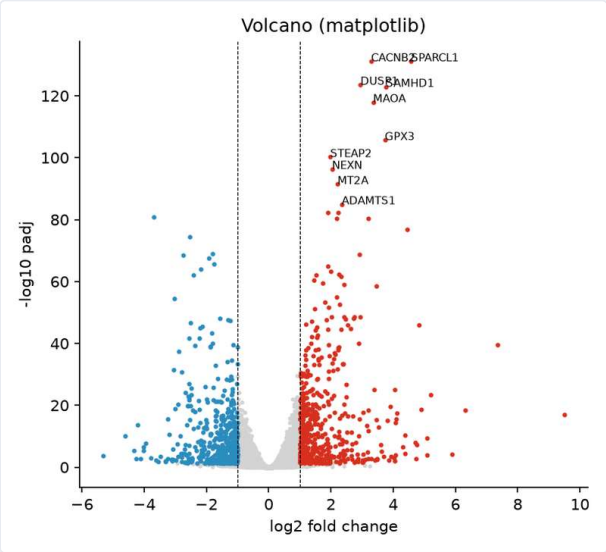
Reproduce: `python3 recreations/matplotlib/matplotlib_recreate.py`

√ re-run end to end · 2 s

matplotlib · 4 variations

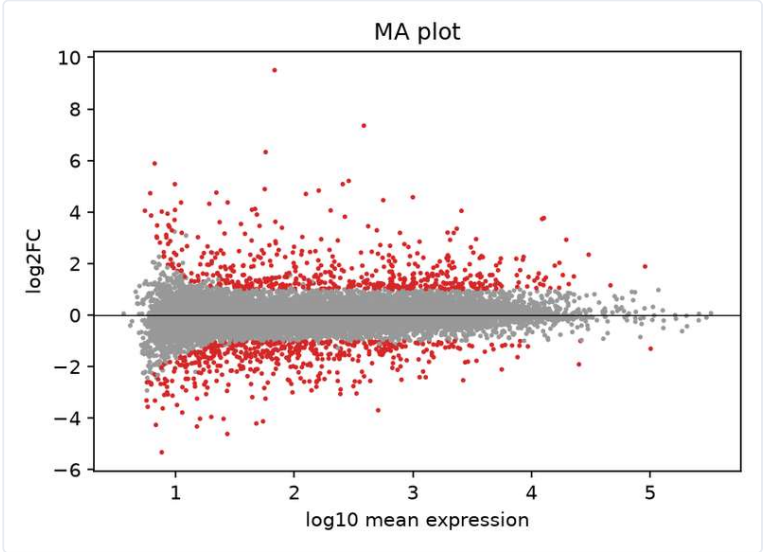
NOVAFLOW RECREATION

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*); Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



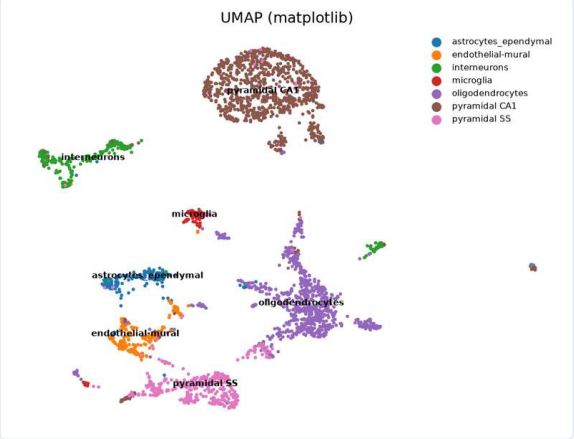
v1 volcano

DE



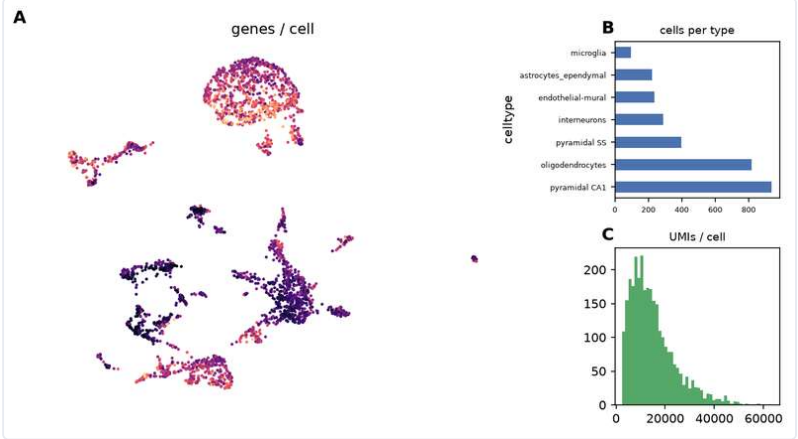
v2 MA

DE



v3 umap

EMBEDDINGS



v4 multipanel_gridspec

COMPOSITION

inferCNV

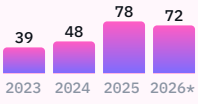
#39

OF 100 · PAPERS NAMING IT

R · CNV HEATMAP · HEATMAPS & DOT MATRICES

207	1.17%	236	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS



NICHE & COOL · INFERCNV/COPYKAT

Neuro-corpus papers per year (*2026 partial) · niche rank **#40**, growth **1.91×**, visual 4/5, score 0.622. Broad-corpus rate 10.97 → 20.93 per 1k papers (2023 → 2025–26).
GitHub broadinstitute/infercnv · created 2016-03-14 · last push 2025-11-14

BEST FOR

Inferred CNV heatmaps by chromosome

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

inferCNV on the package's downsampled oligodendroglioma scRNA-seq example (Tirosh et al. 2016, Nature) - a real brain-tumour dataset

KEY PLOTTING CALL

RECREATIONS/INFERCNV/INFERCNV_RECREATE.R — EXCERPT · ENDS LINE 6

```
o <- infercnv::run(mk(), cutoff = 1, out_dir = "out_default", cluster_by_groups = TRUE, denoise = TRUE, HMM = FALSE, no_prelim_
plot = TRUE, num_threads = 2, png_res = 150, write_expr_matrix = FALSE)
file.copy("out_default/infercnv.png", "v1_cnv_heatmap_by_group.png", overwrite = TRUE)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 — FIGURE CAPTION

A variational algorithm to detect the clonal copy number substructure of tumors from scRNA-seq data. *Nat Commun, 2023*

"...Benchmark of inferred copy number profile.a, b Copy number profile inferred with SCEVAN (segment mean (LogRatio) and CNV status), inferCNV, CopyKAT, the corresponding ground truth from low-depth WGS of sample ..."

DOI ↗ Fig. 3 on PMC ↗

Extended Data Fig. 4 — FIGURE CAPTION

Proteomic signatures improve risk prediction for common and rare diseases. *Nat Med, 2024*

"...ity and distribution is illustrated by color. c, BM UMAP from (A) highlighting the plasma cell state healthy (green), malignant (red) and unclear (yellow) based on copy number aberrations detected by ..."

DOI ↗ Extended Data Fig. 4 on PMC ↗

Fig. 1 — TEXT

Inferring histology-associated gene expression gradients in spatial transcriptomic studies. *Nat Commun, 2024*

"...The output of infercnv::plot_cnv() is stored under a specified directory from which the results are read back into the R session and stored together with metadata regarding the analysis in slot @cnv of the ..."

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

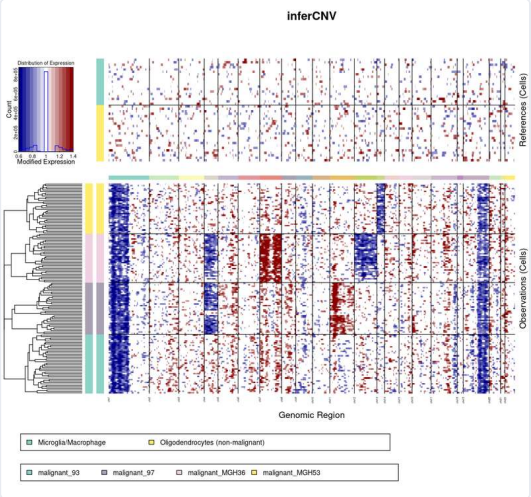
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/inferCNV/inferCNV_recreate.R`

√ re-run end to end · 23 s

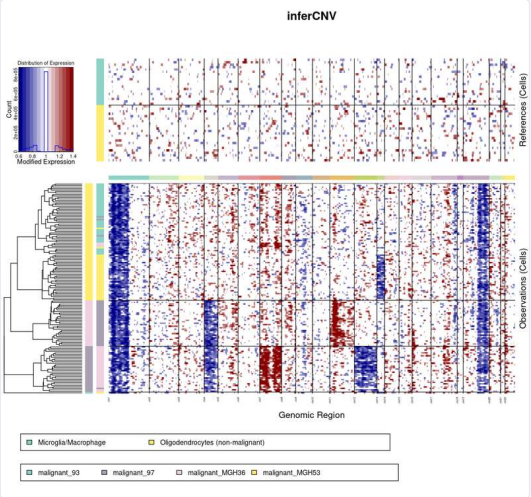
inferCNV · 3 variations

Data: inferCNV's downsampled oligodendroglioma scRNA-seq example (Tirosh et al. 2016). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



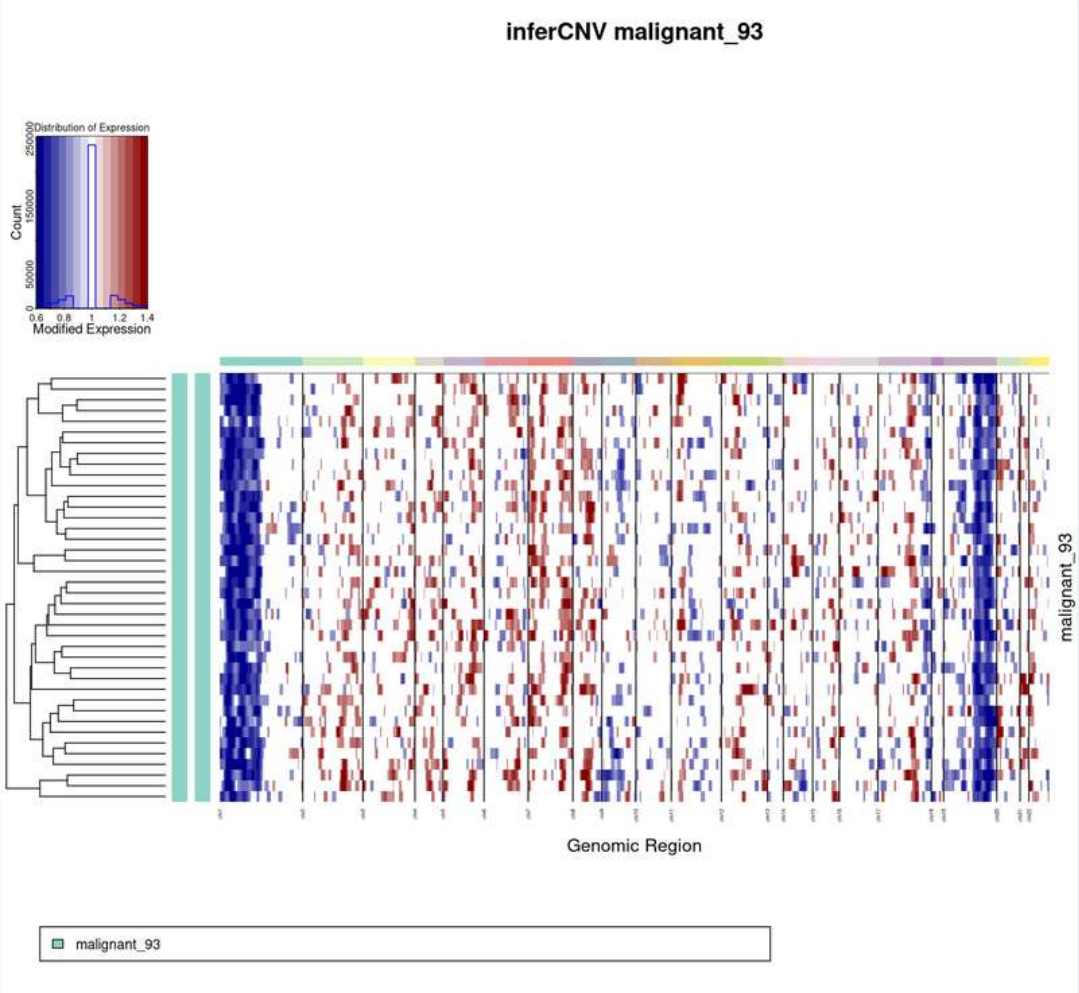
v1 cnv_heatmap_by_group

HEATMAPS



v2 cnv_heatmap_subclusters

HEATMAPS



v3 cnv_per_group

HEATMAPS

seaborn

#40

OF 100 · PAPERS NAMING IT

PYTHON | GENERAL/STATISTICS | HEATMAPS & DOT MATRICES | COMPOSITION, DISTRIBUTIONS & LAYOUTS

192	1.08%	193	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

clustermap, violin/box/strip, kde, faceted catplots

KEY PLOTTING CALL

RECREATIONS/SEABORN/SEABORN_RECREATE.PY — EXCERPT · ENDS LINE 6

```
ad=load(); genes=["Snap25","Gad1","Pvalb","Sst","Satb2","Neurod6","Aqp4","Gfap","Mbp","Plp1","Cx3cr1","C1qa","Cldn5","Acta2"]
df=pd.DataFrame(ad[:,genes].X.toarray(),columns=genes,index=ad.obs_names); df["celltype"]=ad.obs.celltype.values
av=df.groupby("celltype",observed=True).mean()
g=sns.clustermap(av.T,z_score=0,cmap="vlag",figsize=(6,7),row_cluster=True); g.fig.suptitle("seaborn clustermap (z-scored
means)",y=1.02); g.savefig("v1_clustermap.png",dpi=150)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 5 TEXT
Functional synapses between neurons and small cell lung cancer. *Nature, 2025*
"...The packages matplotlib v3.4.2 and seaborn 0.11.0 were used for visualization."
DOI ↗ Fig. 5 on PMC ↗

Fig. 3 TEXT
A concerted neuron-astrocyte program declines in ageing and schizophrenia. *Nature, 2024*
"...Python (v.3.8.3): matplotlib (v.3.5.2)136 and seaborn (v.0.10.1)137."
DOI ↗ Fig. 3 on PMC ↗

Fig. 2 TEXT
An organotypic atlas of human vascular cells. *Nat Med, 2024*
"...Python (version 3), AnnData (versions 0.8.0 and 0.9.1) Pandas (versions 1.5.2 and 1.5.3), NumPy (versions 1.21.6 and 1.23.5), SciPy (versions 1.8.0 and 1.9.3), Matplotlib (version 3.5.2), Seaborn (version ..."
DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

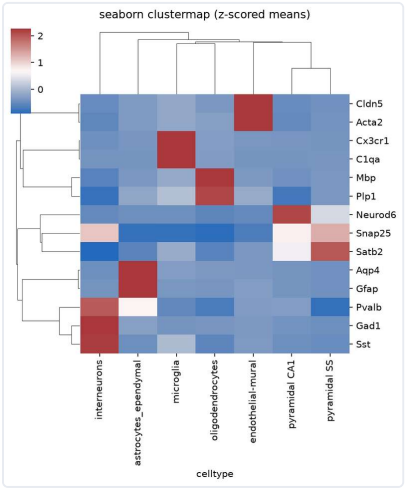
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/seaborn/seaborn_recreate.py`

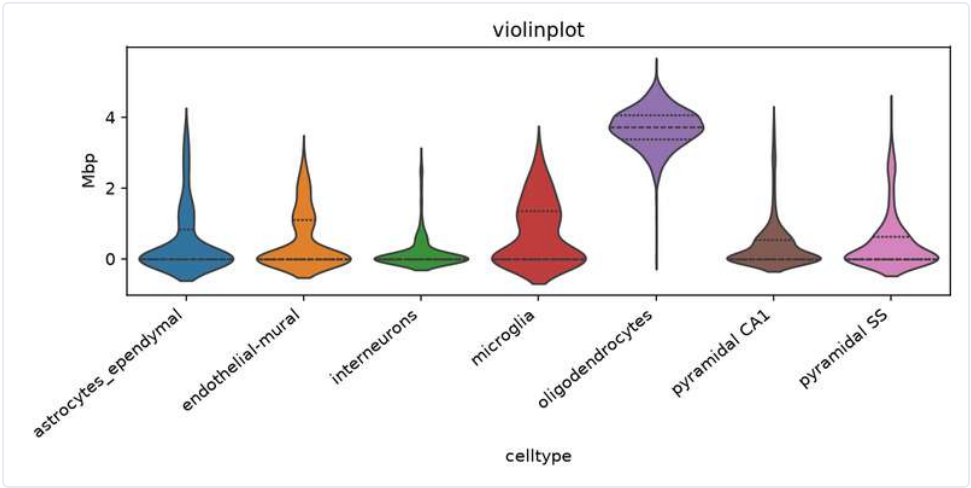
√ re-run end to end · 8 s

seaborn · 4 variations

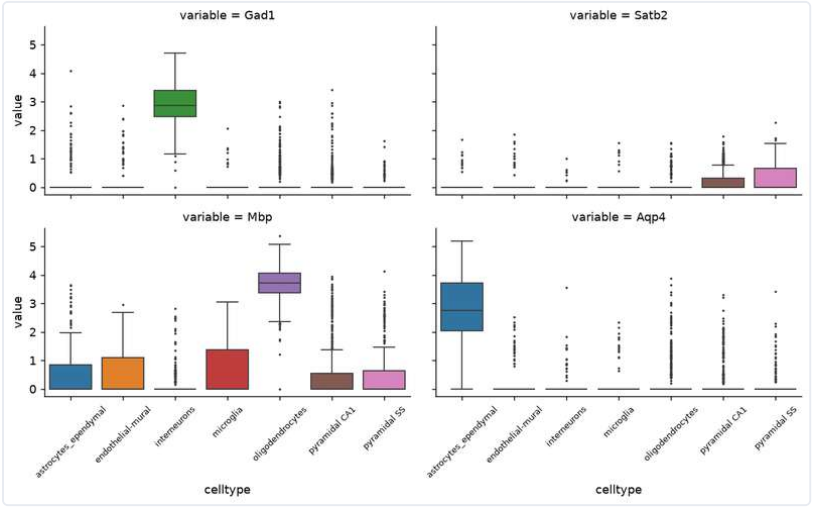
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



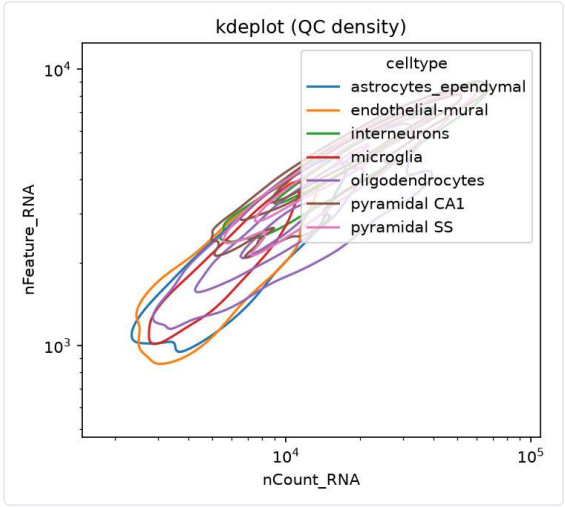
v1 clustermap — HEATMAPS



v2 violin — COMPOSITION



v3 catplot_facets — COMPOSITION



v4 kde_qc — COMPOSITION

CellPhoneDB

#41

OF 100 · PAPERS NAMING IT

PYTHON | CELL-CELL COMMUNICATION | CELL–CELL COMMUNICATION

182	1.02%	231	analysis+plotting	2
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · CELLPHONEDB/KTPLOTS

Neuro-corpus papers per year (*2026 partial) · niche rank **#62**, growth **1.1×**, visual 4/5, score 0.476. Broad-corpus rate 12.746 → 14.021 per 1k papers (2023 → 2025–26).
GitHub zktuong/ktplots · created 2019-10-11 · last push 2025-05-17

BEST FOR

L-R dot plots / heatmaps of interaction counts

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

CellPhoneDB-method results (via LIANA+) rendered with ktplotspy (the ktplots Python port): CPDB dot plot + significance heatmap

KEY PLOTTING CALL

RECREATIONS/CELLPHONEDB_KTPLOTS/KTPLOTS_RECREATE.PY — EXCERPT · ENDS LINE 14

```
means,pvals=cp(mw),cp(pw)
ad=load(); ad.obs["celltype"]=ad.obs.celltype.astype(str).astype("category")
g=kpy.plot_cpdb_heatmap(pvals=pvals,figsize=(6,6),title="# significant interactions (CellPhoneDB method)",symmetrical=False); p
lt.savefig("v1_cpdb_heatmap.png",dpi=140,bbox_inches="tight"); plt.close("all")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 6 FIGURE CAPTION

The Single-Cell Landscape of Intratumoral Heterogeneity and The Immunosuppressive Microenvironment in Liver and Brain Metastases of Breast Cancer. (Adv Sci (Weinh), 2023). ,

"...th factor genes (VEGFA, VEGFB, HGF, and IGF1) in stromal cell subtypes. m) Dot plot showing the ligand–receptor pairs of cell growth factors between cancer cells and each CAF cluster predicted by the ..."

DOI ↗ Figure 6 on PMC ↗

Fig. 1 FIGURE CAPTION

Spatially resolved multiomics of human cardiac niches. *Nature, 2023*

"...ed by cell2location. Dashed lines highlight SAN, AVN, AVB and Purkinje cells defined by histology (Extended Data Fig. 2g). Illustrations in a were created using BioRender (https://biorender.com). The ..."

DOI ↗ Fig. 1 on PMC ↗

Fig. 3 FIGURE CAPTION

Inferring neuron-neuron communications from single-cell transcriptomics through NeuronChat. *Nat Commun, 2023*

"...Comparison of NeuronChat, CellPhoneDB, and CellChat in predicting VISp and ALM projection networks.a Typical ROC curves (left panel) and PR curves for the three methods. b The boxplots of AUROC (left panel) ..."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

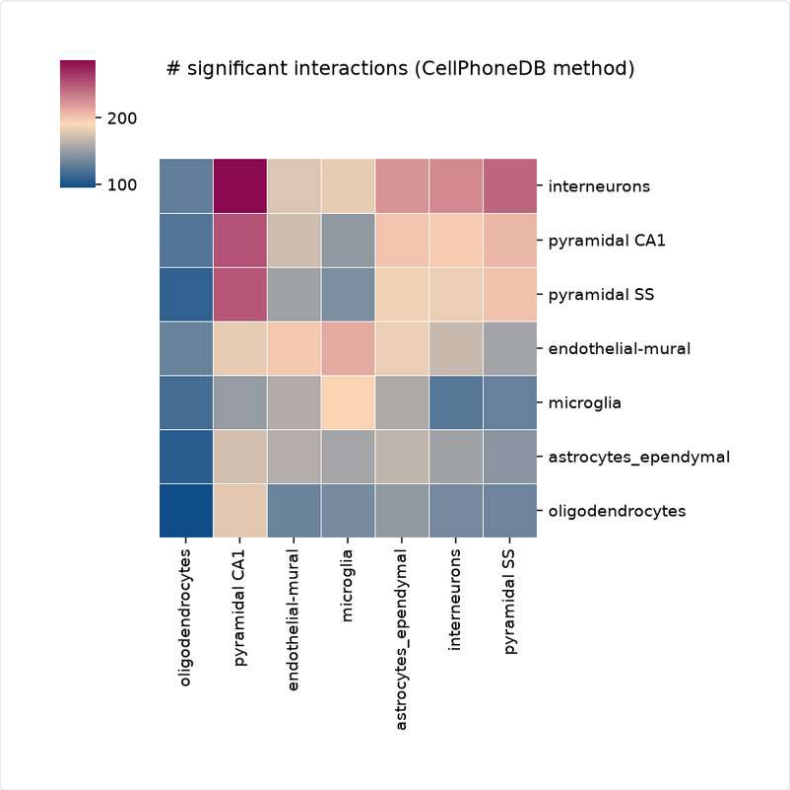
- CellPhoneDB method via LIANA+, plotted with ktplotspy
- Reads LIANA+ result tables, so run recreations/LIANA_plus first (METHODS §9).

Reproduce: `python3 recreations/CellPhoneDB_ktplots/ktplots_recreate.py`

✓ re-run end to end · 10 s

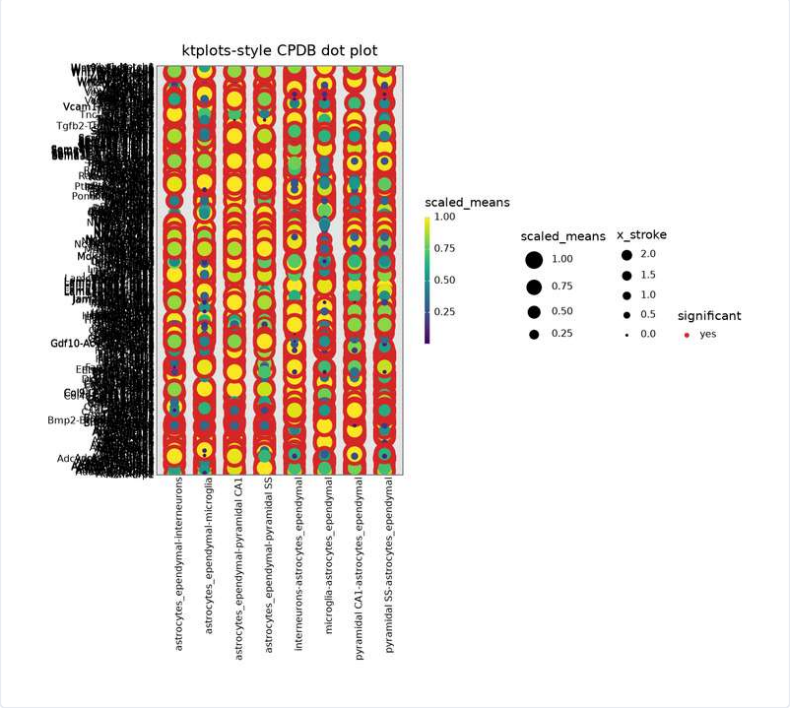
CellPhoneDB · 2 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 cpdb_heatmap

CELL-CELL



v2 cpdb_dotplot

CELL-CELL

slingshot

#42

OF 100 · PAPERS NAMING IT

R | TRAJECTORY | TRAJECTORY, PSEUDOTIME & VELOCITY

173	0.97%	225	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Lineage curves over embeddings, pseudotime

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

slingshot lineages on La Manno 2018 dentate gyrus (granule + glial branches) on UMAP: principal curves over clusters, pseudotime colouring per lineage, and the cluster MST.

KEY PLOTTING CALL

RECREATIONS/SLINGSHOT/SLINGSHOT_RECREATE.R — EXCERPT · ENDS LINE 12

```
png("v1_curves_on_umap.png", 1300, 1100, res = 150); plot(um, col = pal[cl], pch = 16, cex = .5, asp = 1, xlab = "UMAP1", ylab
= "UMAP2", main = "slingshot principal curves (start: radial glia-like)")
lines(SlingshotDataSet(sce), lwd = 3, col = "black"); legend("topright", legend = levels(cl), col = pal, pch = 16, cex = .7,
bty = "n"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 FIGURE CAPTION

A topographic atlas defines developmental origins of cell heterogeneity in the human embryonic lung. *Nat Cell Biol, 2023*

"...re is shown by H&E staining. Scale bar, 400 μm. arw, airway; tr, trachea; prox, proximal; pcw, post conception week; br-v bundle, bronchovascular bundle. c, Pseudotime analysis of the ASM cells, with Slingshot ..."

DOI ↗ Fig. 2 on PMC ↗

Fig. 1 FIGURE CAPTION

Single-cell RNA-sequencing identifies disease-associated oligodendrocytes in male APP NL-G-F and 5XFAD mice. *Nat Commun, 2023*

"...tion of oligodendrocytes from 1-, 3, 6-month-old WT, and AppNL-G-F mice. e Two lineages of oligodendrocytes with a cluster-based minimum spanning tree with line. f Continuous trajectory projection by ..."

DOI ↗ Fig. 1 on PMC ↗

Extended Data Fig. 5 FIGURE CAPTION

Defining blood-induced microglia functions in neurodegeneration through multiomic profiling. *Nat Immunol, 2023*

"...tory analysis of scRNA-seq BMDM profiles.a, UMAP plots of BMDM scRNA-seq profiles colored by treatment (left) or cluster (right) as shown in Fig. 3a, overlaid with pseudotime trajectory analysis with ..."

DOI ↗ Extended Data Fig. 5 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

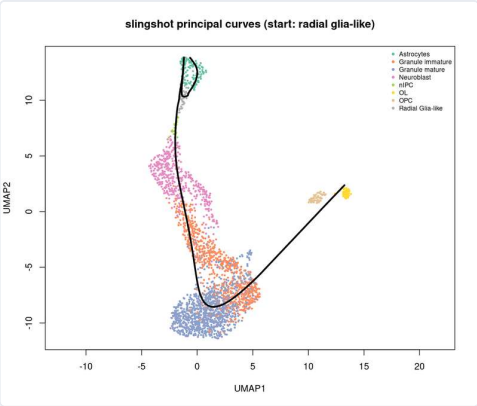
Reproduce: `Rscript recreations/slingshot/slingshot_recreate.R`

√ re-run end to end · 8 s

slingshot · 3 variations

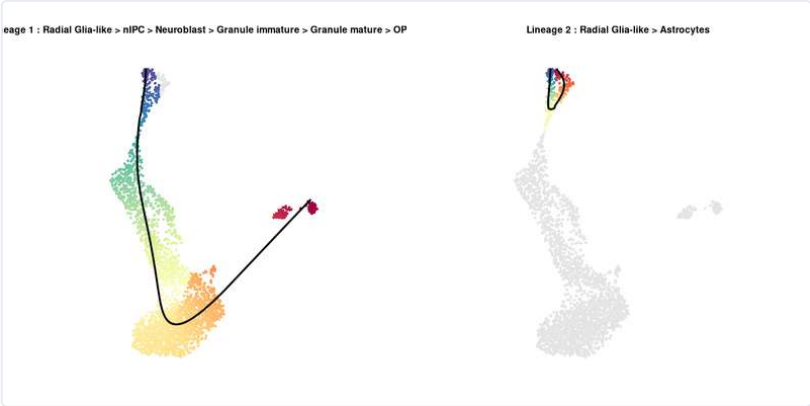
NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



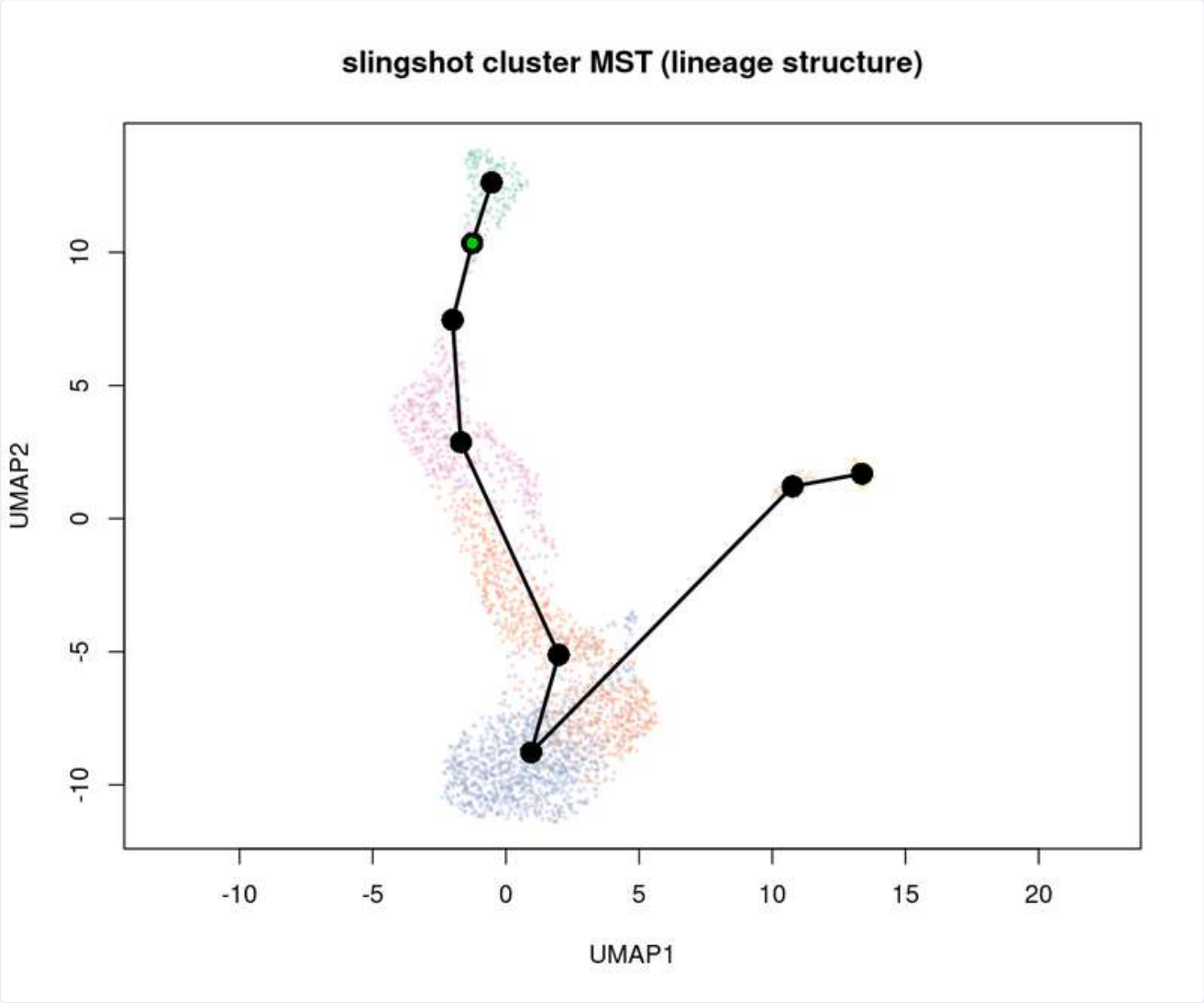
v1 curves_on_umap

TRAJECTORY



v2 pseudotime_per_lineage

TRAJECTORY



v3 cluster_MST

TRAJECTORY

rrvgo

#44

OF 100 · PAPERS NAMING IT

R | ENRICHMENT (GO REDUCE) | ENRICHMENT & PATHWAY ACTIVITY

163	0.92%	179	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

GO semantic similarity treemaps / scatter

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

rrvgo: reduce redundant GO BP terms from Zeisel 2015 astrocyte markers (treemap, scatter, wordcloud-style heatmap)

KEY PLOTTING CALL

RECREATIONS/RRVGO/RRVGO_RECREATE.R — EXCERPT · ENDS LINE 8

```
m <- read.csv("../_data/zeisel_markers.csv") %>% filter(cluster == "astrocytes_ependymal", p_val_adj < .01) %>% slice_max(avg_l
og2FC, n = 300)
ego <- as.data.frame(enrichGO(m$gene, OrgDb = org.Mm.eg.db, keyType = "SYMBOL", ont = "BP", pvalueCutoff = .01))
sim <- calculateSimMatrix(ego$ID, orgdb = "org.Mm.eg.db", ont = "BP", method = "Rel")
sc <- setNames(-log10(ego$p.adjust), ego$ID); red <- reduceSimMatrix(sim, sc, threshold = .7, orgdb = "org.Mm.eg.db")
png("v1_treemap.png", 1600, 1100, res = 150); treemapPlot(red, title = "rrvgo treemap: astrocyte marker GO BP (parents)"); dev.
off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 FIGURE CAPTION — **Copy number variation in tRNA isodecoder genes impairs mammalian development and balanced translation.** *Nat Commun, 2023*

"...s in brains a, and livers b, from tRNA-Phe-1-1-/- knockout mice (n = 3) compared to controls (n = 3), summarised by biological process gene ontologies (GOs) determined by PANTHER and visualised using REVIGO ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 3 TEXT — **iPSC-derived models of PACS1 syndrome reveal transcriptional and functional deficits in neuron activity.** *Nat Commun, 2024*

"...To more wholistically visualize GO results, the ontology hierarchy was extracted and reduced using the rrvgo package (v1.9.1)93 with the score of the term being -log10(pelim)."

DOI ↗ Fig. 3 on PMC ↗

Fig. 3 TEXT — **Single-nucleus RNA-sequencing of autosomal dominant Alzheimer disease and risk variant carriers.** *Nat Commun, 2023*

"...The terms were then loaded into rrvgo (v1.6.0), an R package that implements the Revigo94 tool for summarizing GO_BP terms."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

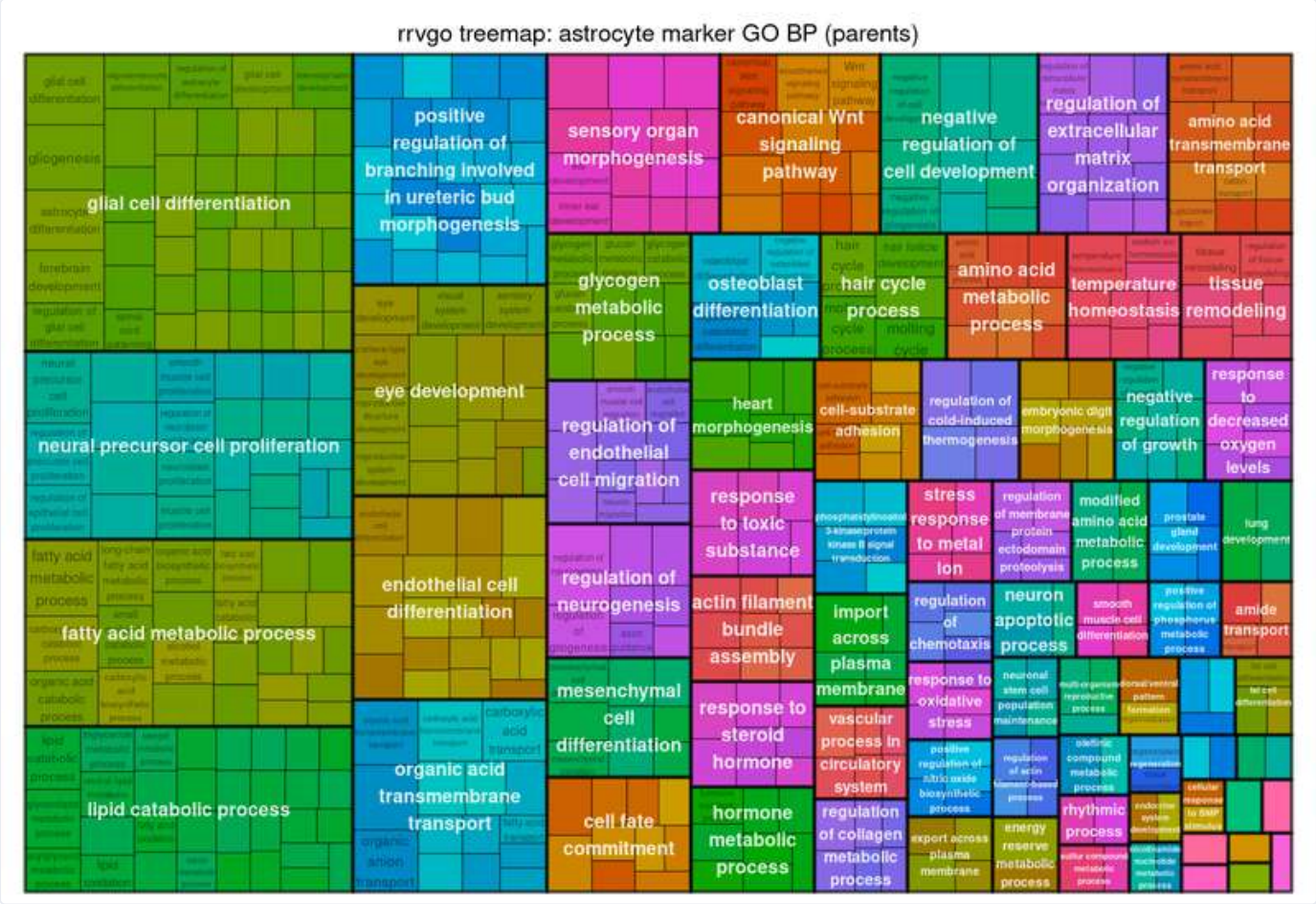
Reproduce: `Rscript recreations/rrvgo/rrvgo_recreate.R`

√ re-run end to end · 447 s

rrvgo · 3 variations

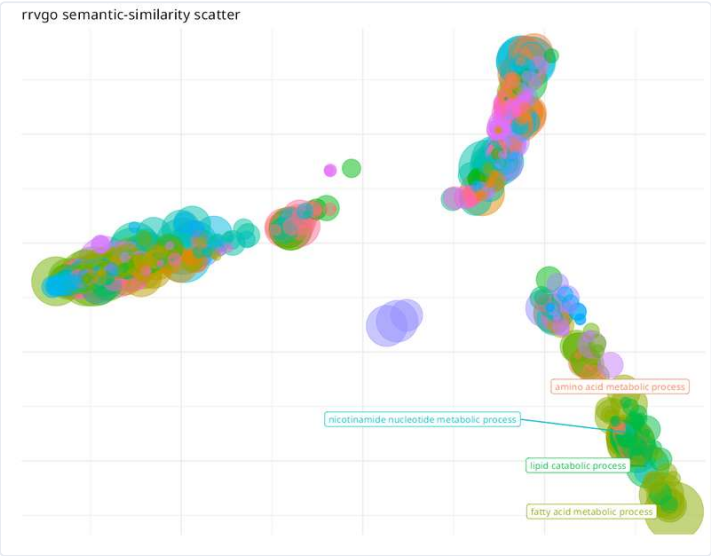
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



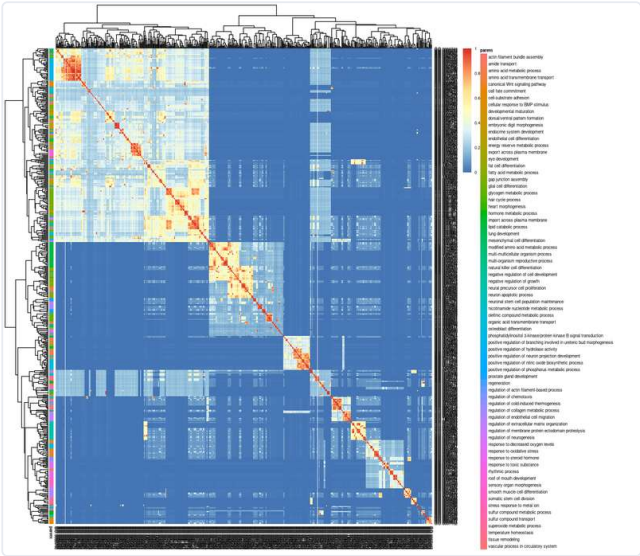
v1 treemap

ENRICHMENT



v2 scatter

ENRICHMENT



v3 heatmap

ENRICHMENT

scVelo

#45

OF 100 · PAPERS NAMING IT

PYTHON | RNA VELOCITY | TRAJECTORY, PSEUDOTIME & VELOCITY

161	0.91%	177	analysis+plotting	5
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · SCVELO STREAMLINES

Neuro-corpus papers per year (*2026 partial) · niche rank **#45**, growth **1.15×**, visual 5/5, score 0.596. Broad-corpus rate 7.244 → 8.357 per 1k papers (2023 → 2025–26).
GitHub theislab/scvelo · created 2018-08-20 · last push 2026-02-25

BEST FOR

RNA-velocity stream/arrow embeddings, phase portraits

Also catalogued as: **velocyto** (#56, 120 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

scVelo on La Manno 2018 mouse dentate gyrus: streamlines, velocyto-style grid arrows, phase portraits, latent-time heatmap

KEY PLOTTING CALL

RECREATIONS/SCVELO/SCVELO_RECREATE.PY — EXCERPT · ENDS LINE 6

```
scv.pl.velocity_embedding_stream(a,basis="umap",color="clusters",legend_loc="right margin",title="Velocity streamlines (dynamical)",save="v1_stream.png",show=False)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 7 FIGURE CAPTION
Multimodal chromatin profiling using nanobody-based single-cell CUT&Tag. *Nat Biotechnol, 2023*
"…ty visualization. The chromatin velocity was calculated by using ATAC-seq gene-by-cell matrix as input into the unspliced layer and H3K27ac gene-by-cell matrix into the spliced layer and then running scvelo …"
DOI ↗ Fig. 7 on PMC ↗

Figure 2 FIGURE CAPTION
Distinct molecular profiles of skull bone marrow in health and neurological disorders. *Cell, 2023*
"…tions, p = 0)(H) Left: in the neutrophil subpopulation, calvaria, and dura neutrophils are highlighted in region-based UMAP. Right: projected developmental trajectory of MCAo neutrophils subset using …"
DOI ↗ Figure 2 on PMC ↗

Fig. 6 FIGURE CAPTION
SCENIC+: single-cell multiomic inference of enhancers and gene regulatory networks. *Nat Methods, 2023*
"…5 cells). c–e, UMAP dimensionality reduction of 3,104 pseudocells from the fly eye based on target gene and region enrichment scores, with a schematic representation of the fly eye-antennal disc (c), scVelo …"
DOI ↗ Fig. 6 on PMC ↗

LIMITATIONS & NOTES

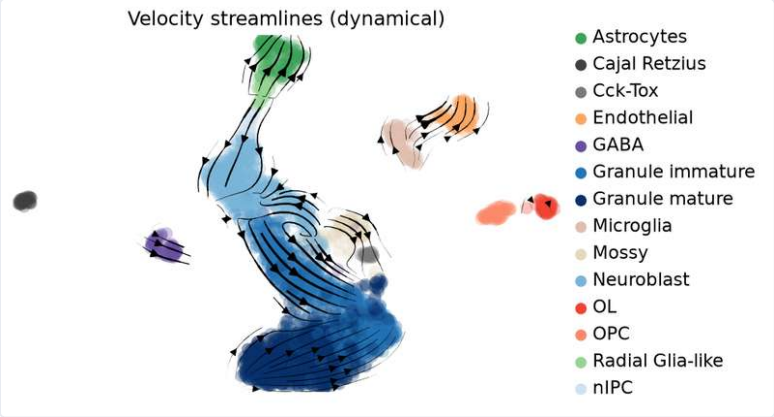
- velocyto (#56): velocyto-style grid arrows via scVelo (v2_grid_arrows)

Reproduce: `python3 recreations/scVelo/scvelo_recreate.py` — ✓ re-run end to end · 6 s

scVelo · 5 variations

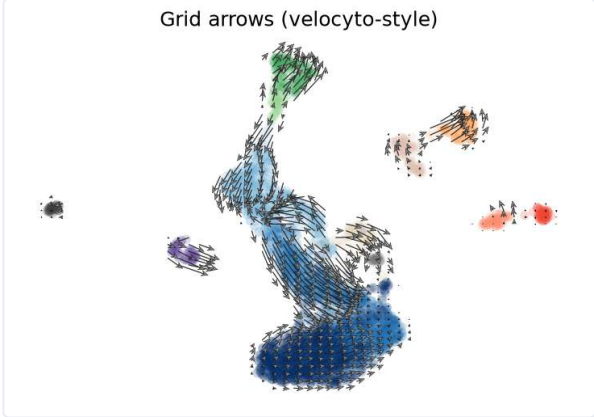
NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



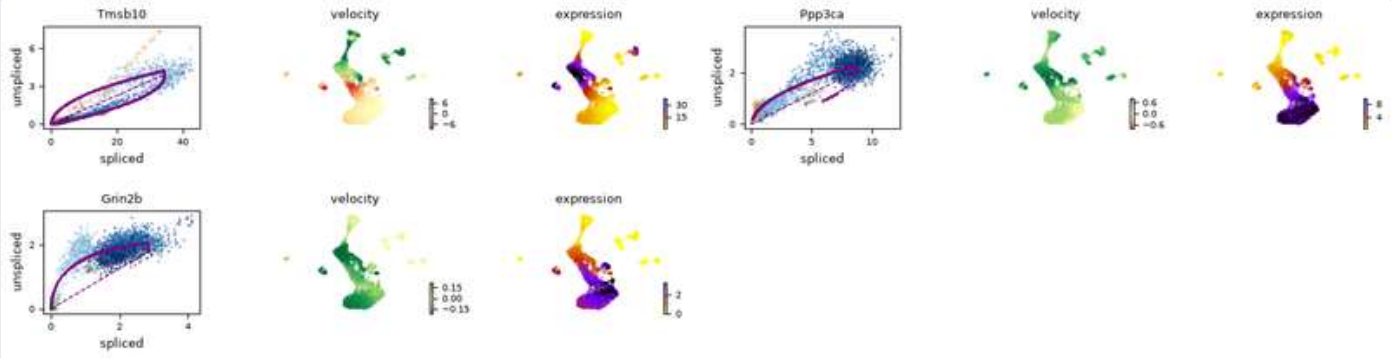
v1 stream

TRAJECTORY



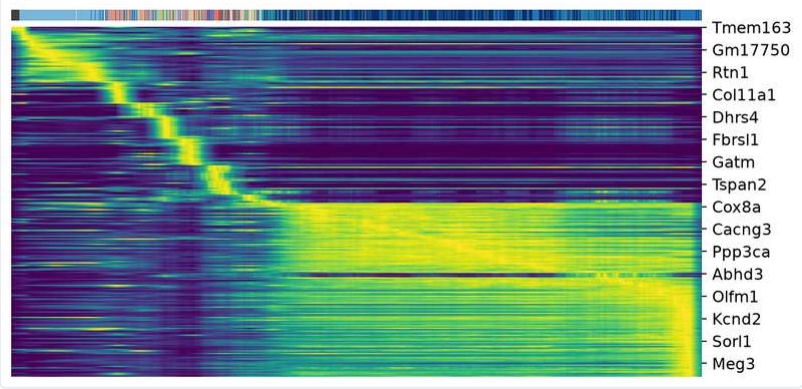
v2 grid_arrows

TRAJECTORY



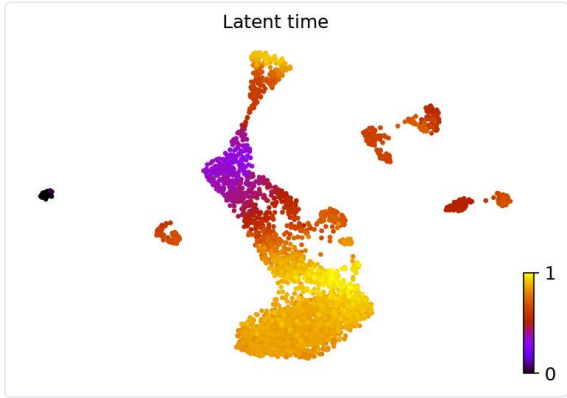
v3 phase_portraits

TRAJECTORY



v4 latent_time_heatmap

TRAJECTORY



v5 latent_time

TRAJECTORY

VennDiagram / ggvenn

#46

OF 100 · PAPERS NAMING IT

R · SET OVERLAP · DE, QC & SET OVERLAPS

157	0.88%	160	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

2-5 set Venn diagrams

Also catalogued as: **ggvenn** (#82, 61 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

VennDiagram / ggvenn / ggVennDiagram: marker overlaps between Zeisel 2015 glial cell types

KEY PLOTTING CALL

RECREATIONS/VENNDIAGRAM_GGVENN/VENN_RECREATE.R — EXCERPT · ENDS LINE 8

```
futile.logger::flog.threshold(futile.logger::ERROR, name = "VennDiagramLogger")
m <- read.csv("../_data/zeisel_markers.csv") %>% filter(p_val_adj < .01, avg_log2FC > 1); L <- split(m$gene, m$cluster)
g4 <- L[c("astrocytes_ependymal","oligodendrocytes","microglia","endothelial-mural")]
v <- venn.diagram(g4, filename = NULL, fill = c("#66c2a5","#fc8d62","#8da0cb","#e78ac3"), alpha = .5, cex = 1.1, cat.cex = 1, m
ain = "VennDiagram: glial/vascular marker overlap", category.names = c("astro","oligo","microglia","endo-mural"))
png("v1_VennDiagram_4set.png", 1200, 1200, res = 170); grid.draw(v); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 TEXT

Transcriptional and epigenetic decoding of the microglial aging process. *Nat Aging, 2023*

"...Venn diagrams, heat maps and volcano plots were visualized by using VennDiagram 1.6.20, pheatmap 1.0.12 and EnhancedVolcano 1.4.0, respectively."

DOI ↗ Fig. 1 on PMC ↗

Fig. 4 TEXT

Microglia mediate neurocognitive deficits by eliminating C1q-tagged synapses in sepsis-associated encephalopathy. *Sci Adv, 2023*

"...Venn diagrams showing the overlap of DEGs in different treatment groups were calculated and plotted with the VennDiagram package."

DOI ↗ Fig. 4 on PMC ↗

Fig. 4 TEXT

Broad rim lesions are a new pathological and imaging biomarker for rapid disease progression in multiple sclerosis. *Nat Med, 2025*

"...Any overlap of DEGs between lesion types was visualized as a Venn diagram with the R package VennDiagram v.1.7.3 (ref."

DOI ↗ Fig. 4 on PMC ↗

LIMITATIONS & NOTES

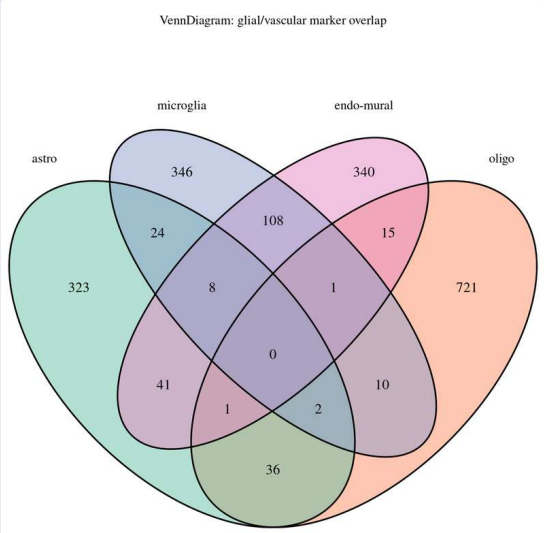
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/VennDiagram_ggvenn/Venn_recreate.R` ✓ re-run end to end · 2 s

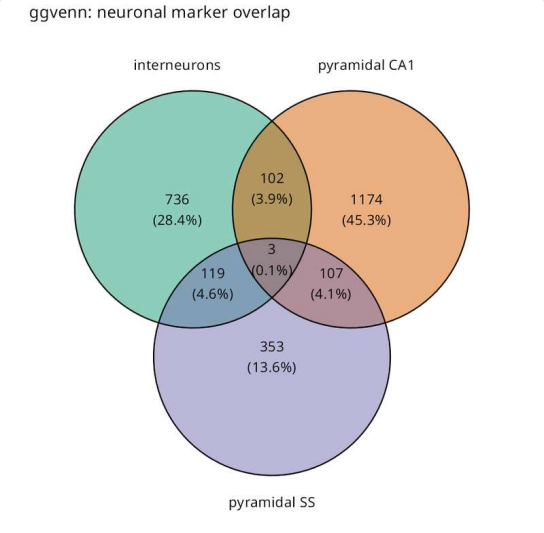
VennDiagram / ggvenn · 3 variations

NOVAFLOW RECREATION

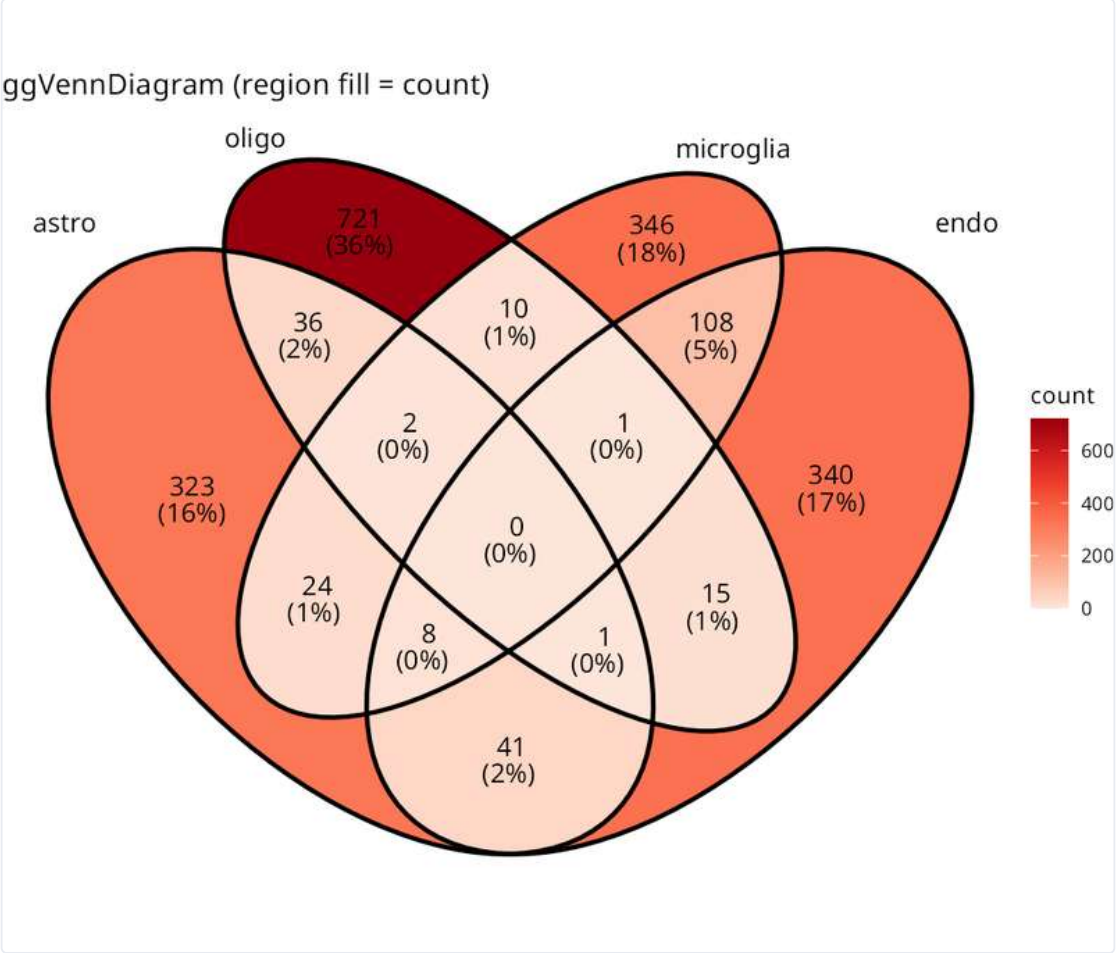
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 VennDiagram_4set · DE



v2 ggvenn_3set · DE



v3 ggVennDiagram_gradient · DE

gplots (heatmap.2)

#47

OF 100 · PAPERS NAMING IT

R HEATMAP HEATMAPS & DOT MATRICES

152	0.86%	157	dedicated plotting	2
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Classic heatmap.2 with dendrograms, Venn

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

gplots::heatmap.2 (classic: row z-score, dendrograms, colour key + histogram, side colours) on Zeisel 2015 markers

KEY PLOTTING CALL

RECREATIONS/GPLOTS_HEATMAP2/HEATMAP2_RECREATE.R — EXCERPT · ENDS LINE 8

```
png("v1_heatmap2_rowz.png", 1500, 1500, res = 150); heatmap.2(M, scale = "row", trace = "none", col = colorRampPalette(rev(brew
er.pal(11, "RdBu")))(100), ColSideColors = pal[ct], labCol = FALSE, density.info = "histogram", key.title = NA, margins = c(3,
8), cexRow = .6, main = "heatmap.2: markers x (cell type|chip) pseudobulk"); legend("topright", legend = names(pal), fill =
pal, cex = .55, bty = "n"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 TEXT

Re-convolving the compositional landscape of primary and recurrent glioblastoma reveals prognostic and targetable tissue states. *Nat Commun, 2023*

"...Heatmaps were generated using the heatmap.2 in R function from the package gplots (R Package) and scores z-scaled were indicated."

DOI ↗ Fig. 3 on PMC ↗

Figure 3 TEXT

Transcriptomic atlas and interaction networks of brain cells in mouse CNS demyelination and remyelination. *Cell Rep, 2023*

"...For heat maps of relative gene expression across cell types, average expression of each gene was Z score transformed across samples and plotted using heat.map2 function in the gplots packge (v3.1.1) in R."

DOI ↗ Figure 3 on PMC ↗

Fig. 2 TEXT

A branching model of lineage differentiation underpinning the neurogenic potential of enteric glia. *Nat Commun, 2023*

"...Data was visualised using GraphPad Prism 9.0 (GraphPad), the python SCANPY67 and scVelo packages17, and the R Seurat (v3 and 4)84, EnhancedVolcano (https://github.com/kevinblighe/EnhancedVolcano), ggplot2 ..."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

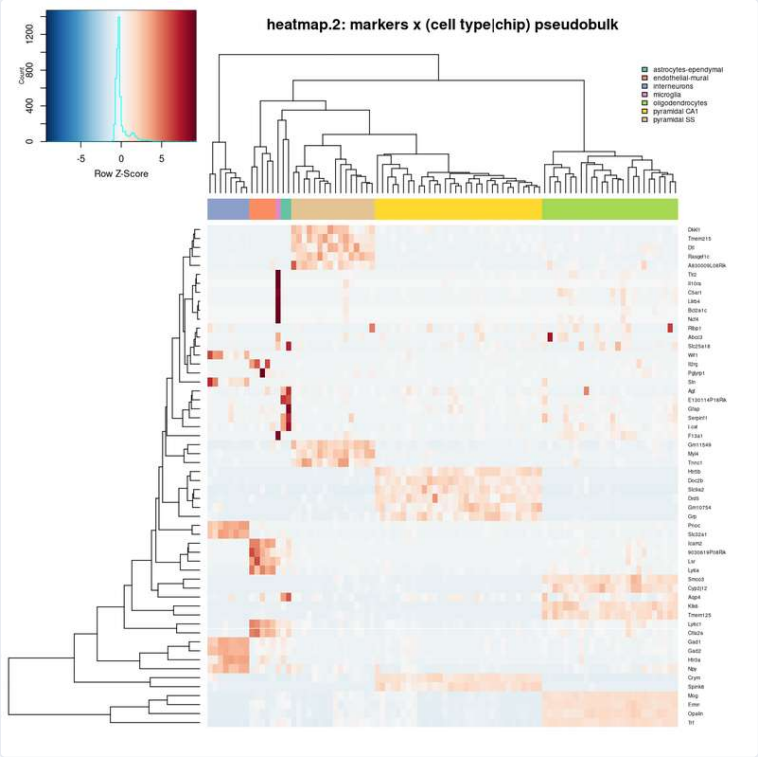
Reproduce: `Rscript recreations/gplots_heatmap2/heatmap2_recreate.R`

√ re-run end to end · 9 s

gplots (heatmap.2) · 2 variations

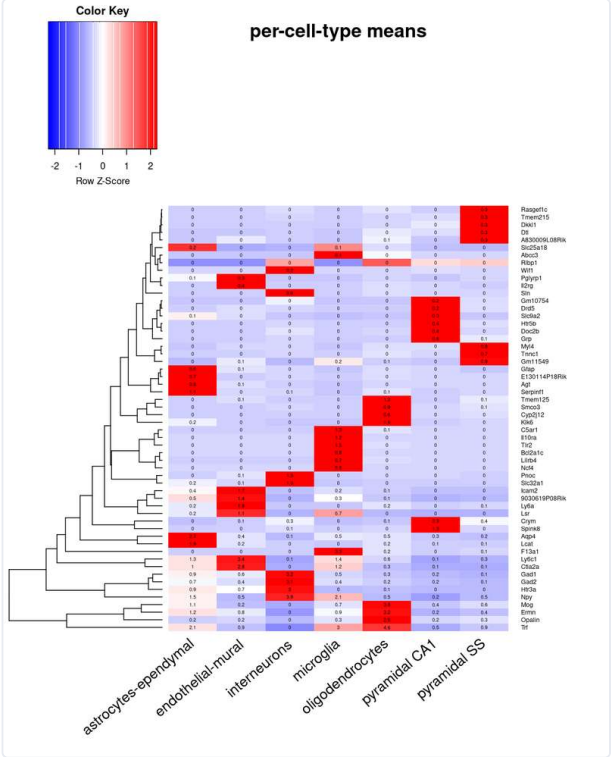
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 heatmap2_rowz

HEATMAPS



v2 heatmap2_celltype_cellnote

HEATMAPS

corrplot

#48

OF 100 · PAPERS NAMING IT

R · CORRELATION · HEATMAPS & DOT MATRICES

147	0.83%	151	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Correlation matrix plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

corrplot: correlation of pseudobulk cell-type profiles and of marker genes, Zeisel 2015

KEY PLOTTING CALL

RECREATIONS/CORRPLOT/CORRPLOT_RECREATE.R — EXCERPT · ENDS LINE 6

```
so <- readRDS("../_data/zeisel_seurat.rds"); m <- read.csv("../_data/zeisel_markers.csv") %>% filter(p_val_adj < .01) %>% group_by(cluster) %>% slice_max(avg_log2FC, n = 50)
av <- AverageExpression(so, features = unique(m$gene), group.by = "celltype", assays = "RNA")$RNA; C <- cor(log1p(as.matrix(av)), method = "spearman")
png("v1_corrplot_celltypes_ellipse.png", 1100, 1100, res = 160); corrplot(C, method = "ellipse", type = "upper", order = "hclust", addrect = 3, tl.col = "black", col = COL2("RdBu", 200), title = "Cell-type pseudobulk Spearman (marker genes)", mar = c(0,0,2,0)); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 TEXT

A proteomics analysis of 5xFAD mouse brain regions reveals the lysosome-associated protein Arl8b as a candidate biomarker for Alzheimer's disease. *Genome Med, 2023*

"...2g) using the packages correlation and corrplot in R v.4.2.0."

DOI ↗ Fig. 2 on PMC ↗

Fig. 3 TEXT

Single-cell multiregion dissection of Alzheimer's disease. *Nature, 2024*

"...The correlation of mean module scores between regions was determined using the cor.mtest function of the R package corrplot."

DOI ↗ Fig. 3 on PMC ↗

Fig. 2 TEXT

Spatial transcriptomics reveals niche-specific enrichment and vulnerabilities of radial glial stem-like cells in malignant gliomas. *Nat Commun, 2023*

"...The resultant correlation coefficient matrix was subjected to hierarchical clustering using corrplot package-based hclust method61, integrating the 48 spatially informed marker gene sets into four cluster ..."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

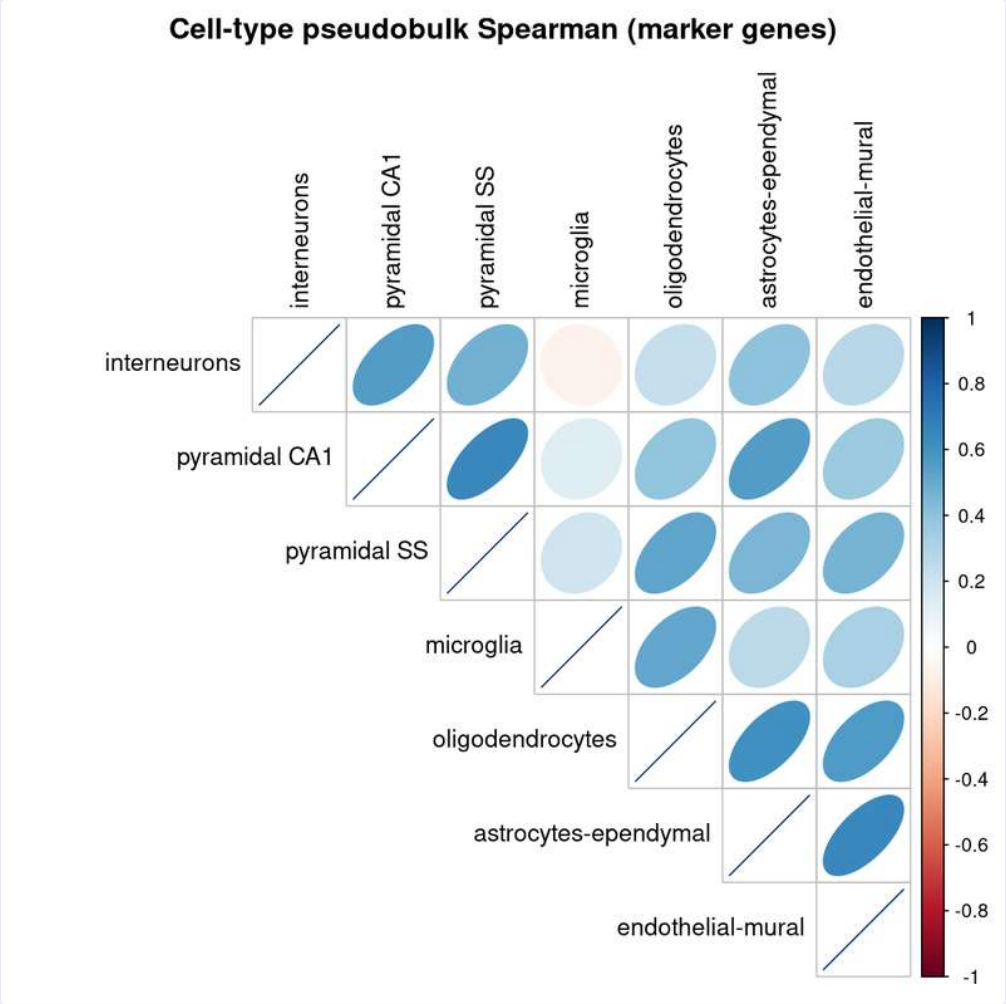
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/corrplot/corrplot_recreate.R`

✓ re-run end to end · 8 s

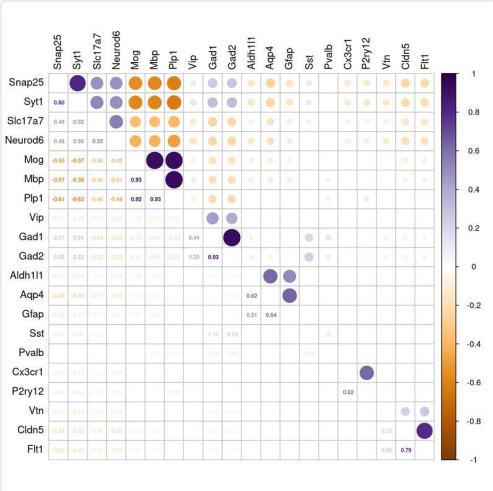
corrplot · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



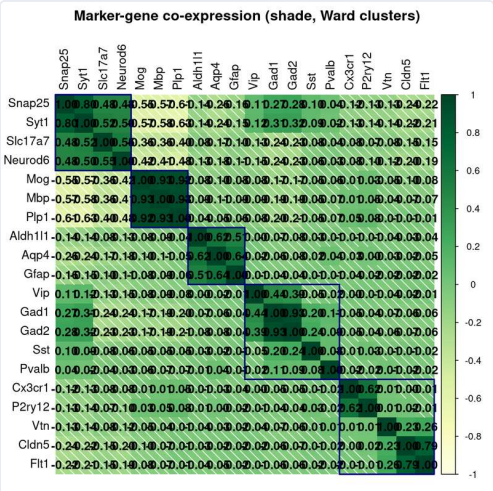
v1 corrplot_celltypes_ellipse

HEATMAPS



v2 corrplot_genes_mixed

HEATMAPS



v3 corrplot_shade_hclust

HEATMAPS

hdWGCNA

#50

OF 100 · PAPERS NAMING IT

R · NETWORK/MODULES (SC) · NETWORKS, GRAPHS & TREES · EMBEDDINGS & FEATURE MAPS · HEATMAPS & DOT MATRICES

134	0.75%	147	analysis+plotting	5
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · HDWGCNA

Neuro-corpus papers per year (*2026 partial) · niche rank **#2**, growth **6.66×**, visual 5/5, score 0.938. Broad-corpus rate 1.08 → 7.252 per 1k papers (2023 → 2025–26).
GitHub smorabit/hdWGCNA · created 2020-08-11 · last push 2026-06-16

BEST FOR

Single-cell module UMAPs, hub-gene networks, module dot plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

hdWGCNA single-cell co-expression modules on Zeisel cortex/hippocampus (metacells per cell type)

KEY PLOTTING CALL

RECREATIONS/HDWGCNA/HDWGCNA_RECREATE.R — EXCERPT · ENDS LINE 9

```
so <- readRDS("../_data/zeisel_seurat.rds")
so <- SetupForWGCNA(so, gene_select = "fraction", fraction = .25, wgcna_name = "zeisel")
so <- MetacellsByGroups(so, group.by = c("celltype","tissue"), reduction = "pca", k = 20, max_shared = 10, ident.group = "celltype", min_cells = 50)
so <- NormalizeMetacells(so); so <- SetDatExpr(so, group_name = levels(so$celltype), group.by = "celltype", assay = "RNA", layer = "data")
so <- TestSoftPowers(so, networkType = "signed")
ggsave("v1_soft_powers.png", wrap_plots(PlotSoftPowers(so), ncol = 2), width = 9, height = 6, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 1 — FIGURE CAPTION

hdWGCNA identifies co-expression networks in high-dimensional transcriptomics data. *Cell Rep Methods, 2023*

"...Overview of the hdWGCNA workflow and application in the human prefrontal cortex(A) Schematic overview of the standard hdWGCNA workflow on a scRNA-seq dataset. UMAP plot shows 36,671 cells from 11 cognitively ..."

DOI ↗ Figure 1 on PMC ↗

Fig. 3 — TEXT

Spatial and single-nucleus transcriptomic analysis of genetic and sporadic forms of Alzheimer's disease. *Nat Genet, 2024*

"...Additionally, we computed a consensus topological overlap matrix (TOM) based on the seven TOMs from the co-expression analysis in each spatial region, and we visualized this consensus TOM as a UMAP plot using ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 2 — TEXT

Cell-type brain-region specific changes in prefrontal cortex of a mouse model of alcohol dependence. *Neurobiol Dis, 2024*

"...Additionally, we used hdWGCNA (Morabito et al., 2022;"

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

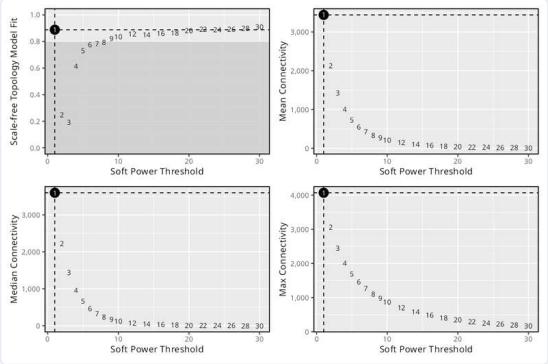
Reproduce: `Rscript recreations/hdWGCNA/hdWGCNA_recreate.R`

√ re-run end to end · 555 s

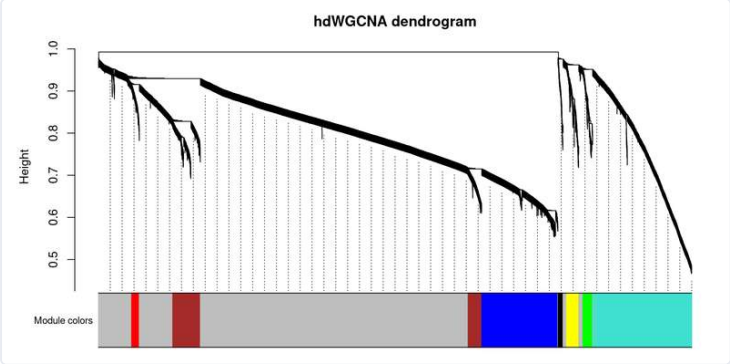
hdWGCNA · 5 variations

NOVAFLOW RECREATION

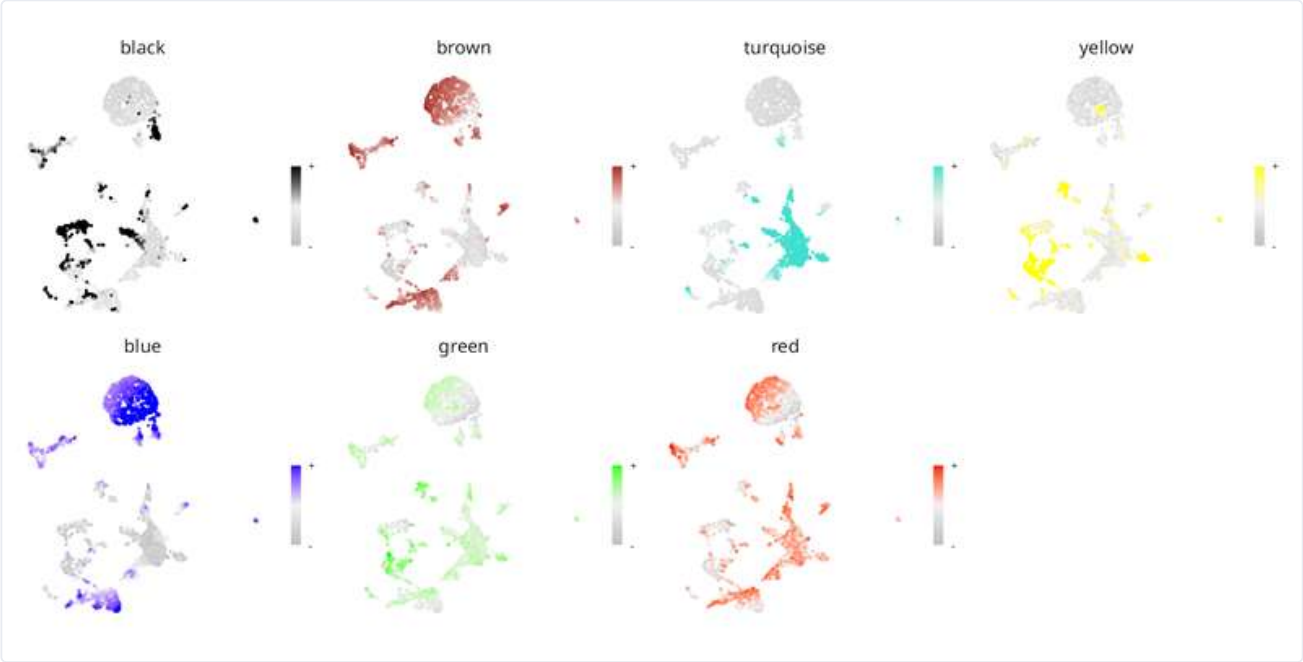
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



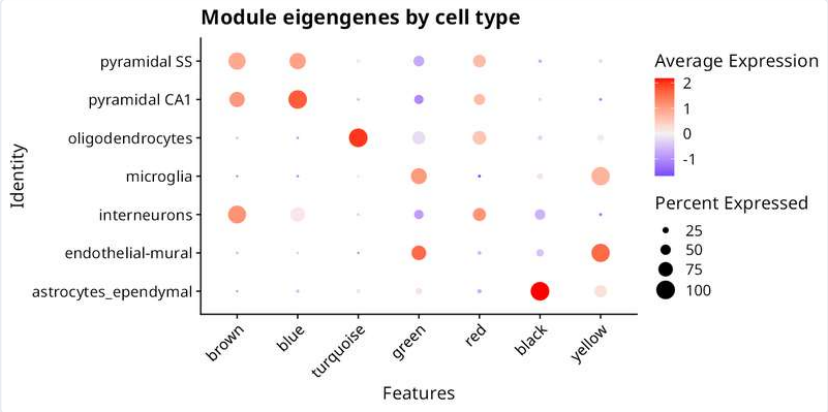
v1 soft_powers — NETWORKS



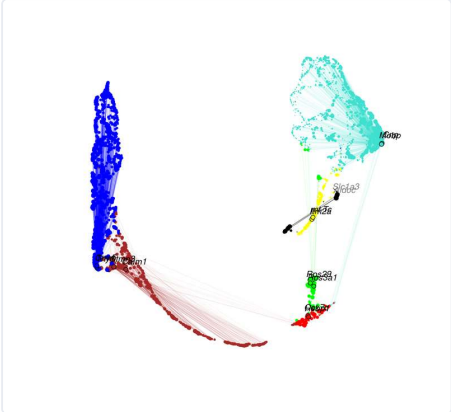
v2 dendrogram — NETWORKS



v3 ModuleFeaturePlot — EMBEDDINGS



v4 module_dotplot — HEATMAPS



v5 ModuleUMAPPlot — NETWORKS

clustree

#52

OF 100 · PAPERS NAMING IT

R | CLUSTER TREE | NETWORKS, GRAPHS & TREES | EMBEDDINGS & FEATURE MAPS

133	0.75%	144	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Cluster-resolution trees

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

clustree: Seurat Louvain resolution sweep on Zeisel 2015 (tree of cluster splits, overlaid on UMAP, marker-coloured)

KEY PLOTTING CALL

RECREATIONS/CLUSTREE/CLUSTREE_RECREATE.R — EXCERPT · ENDS LINE 5

```
so <- readRDS("../_data/zeisel_seurat.rds"); so <- FindClusters(so, resolution = c(.1,.2,.4,.6,.8,1.2), verbose = FALSE)
ggsave("v1_clustree_tree.png", clustree(so, prefix = "RNA_snn_res.") + ggtitle("clustree: resolution sweep (Zeisel 2015)"), wid
th = 9, height = 9, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Extended Data Fig. 9 — FIGURE CAPTION

Microglia regulate central nervous system myelin growth and integrity. *Nature, 2023*

"...ple = 4, orange = 5, red = 6). (k) Clustering with four different resolutions, obtained by modifying the number of nearest neighbours (k). Each resolution is represented in a separate t-SNE plot. (l) Clustree ..."

DOI ↗ Extended Data Fig. 9 on PMC ↗

Figure 4 — TEXT

Trem2 expression in microglia is required to maintain normal neuronal bioenergetics during development. *Immunity, 2024*

"...Optimal resolution was chosen by analyzing the cluster tree plot (clustree)."

DOI ↗ Figure 4 on PMC ↗

Fig. 5 — TEXT

Divergent single cell transcriptome and epigenome alterations in ALS and FTD patients with C9orf72 mutation. *Nat Commun, 2023*

"...We used SC3 stability index101 and Clustree (v0.5.0)102 to evaluate and visualize the stability of the clusters across a range of resolutions for the Leiden community detection analysis (Supplementary Fig."

DOI ↗ Fig. 5 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

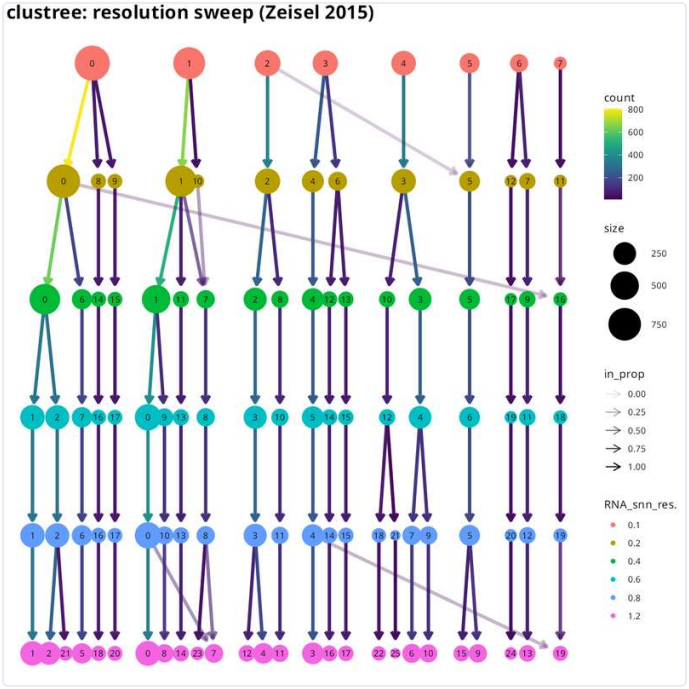
Reproduce: `Rscript recreations/clustree/clustree_recreate.R`

✓ re-run end to end · 14 s

clustree · 4 variations

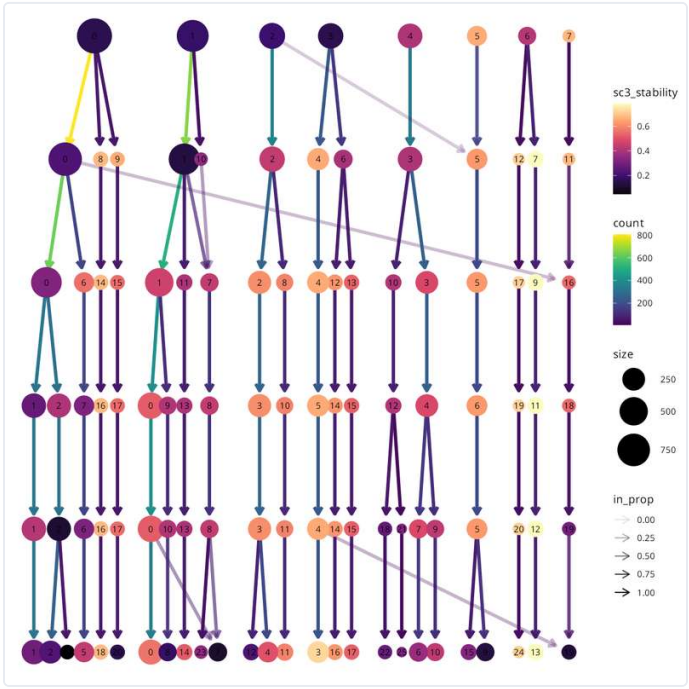
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



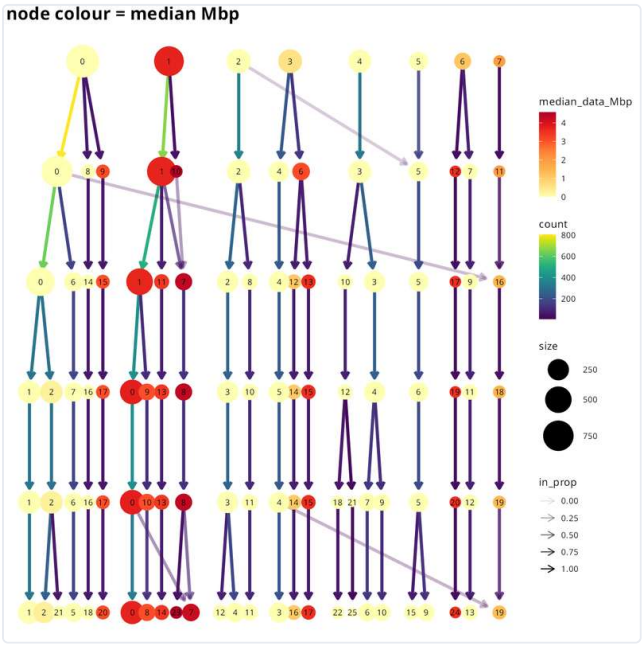
v1 clustree_tree

NETWORKS



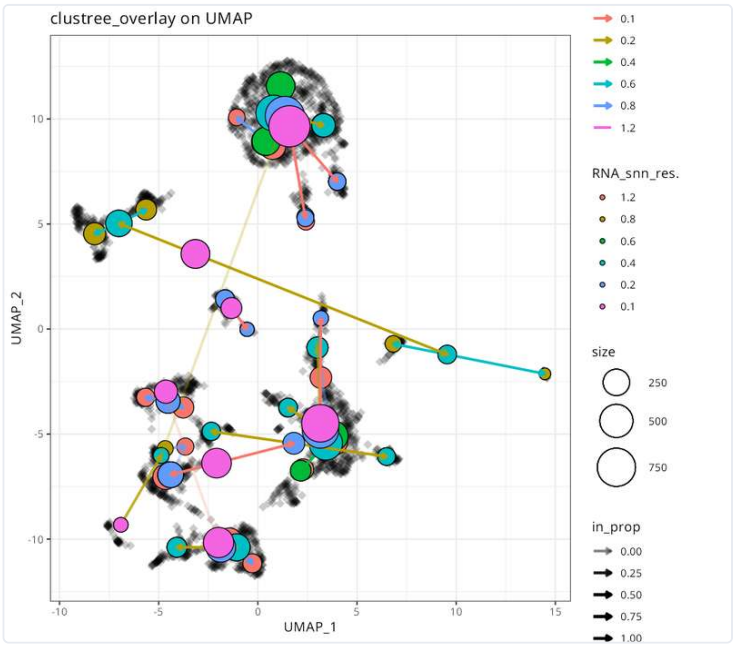
v2 clustree_sc3stability

NETWORKS



v3 clustree_gene_Mbp

NETWORKS



v4 clustree_overlay

EMBEDDINGS

circlize

#54

OF 100 · PAPERS NAMING IT

R | CIRCOS/CHORD | CELL-CELL COMMUNICATION

124	0.7%	130	dedicated plotting	2
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · CIRCLIZE (CHORD)

Neuro-corpus papers per year (*2026 partial) · niche rank **#43**, growth **1.17×**, visual 5/5, score 0.602. Broad-corpus rate 7.697 → 8.979 per 1k papers (2023 → 2025–26).
GitHub jokergoo/circlize · created 2013-04-18 · last push 2023-11-11

BEST FOR

Circos/chord diagrams; colour mapping for ComplexHeatmap

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

circlize: chord diagram of LIANA+ LR counts (Zeisel) + circos genome rings (airway DE genes, hg19)

KEY PLOTTING CALL

RECREATIONS/CIRCLIZE_CHORD/CIRCLIZE_RECREATE.R — EXCERPT · ENDS LINE 5

```
m <- as.matrix(read.csv("../LIANA_plus/lr_counts_matrix.csv", row.names = 1, check.names = FALSE)); grid.col <- setNames(hcl.co
lors(length(union(rownames(m), colnames(m))), "Dark 3"), union(rownames(m), colnames(m)))
png("v1_chord_directional.png", 1300, 1300, res = 170); circos.par(gap.after = 4); chordDiagram(m, grid.col = grid.col, directi
onal = 1, direction.type = c("diffHeight","arrows"), link.arr.type = "big.arrow", annotationTrack = c("grid","name"), transpare
ncy = .3); title("Cell-cell LR pair counts (LIANA+)"); circos.clear(); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 6 TEXT
Diagnostic utility of DNA methylation analysis in genetically unsolved pediatric epilepsies and CHD2 episignature refinement. *Nat Commun, 2024*
"...Heatmaps were plotted using the R package pheatmap (version 1.0.12) and circos plots using the R package circlize (version 0.4.15)95."
DOI ↗ Fig. 6 on PMC ↗

Fig. 7 TEXT
Synaptic connectome of the Drosophila circadian clock. *Nat Commun, 2024*
"...Data was visualized using ggplot2 (v 3.4.2, Wickham, 2016) and circlize (v 0.4.15) for R (v 4.2.2) in RStudio (2022.12.0)91."
DOI ↗ Fig. 7 on PMC ↗

Figure 4 TEXT
The meningeal transcriptional response to traumatic brain injury and aging. *Elife, 2023*
"...Data was organized and graphs were created using ggplot2, tidyverse, treemapify, circlize, Seurat and dplyr (Wickham, 2016;"
DOI ↗ Figure 4 on PMC ↗

LIMITATIONS & NOTES

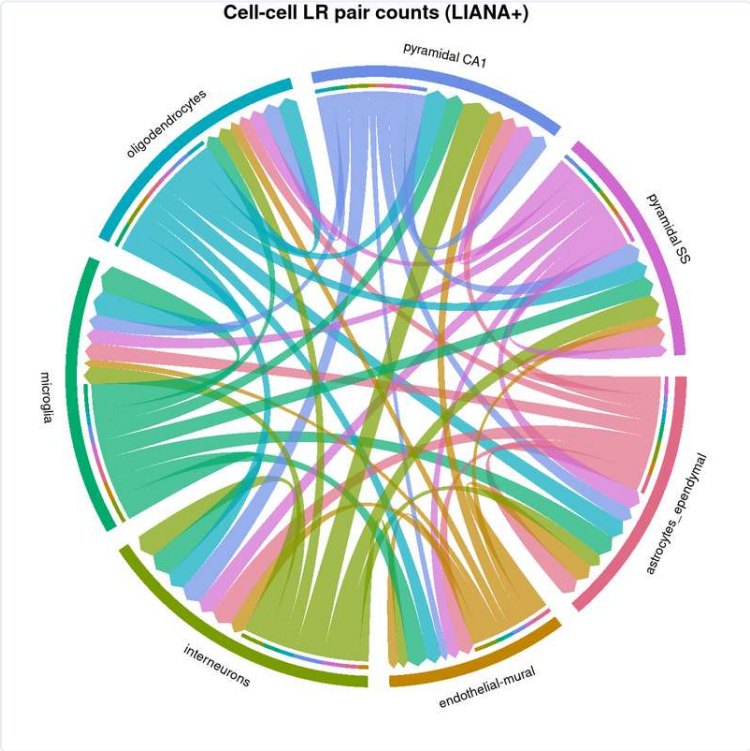
- Chord + circos genome rings
- Reads LIANA+ result tables, so run recreations/LIANA_plus first (METHODS §9).

Reproduce: `Rscript recreations/circlize_chord/circlize_recreate.R`

√ re-run end to end · 18 s

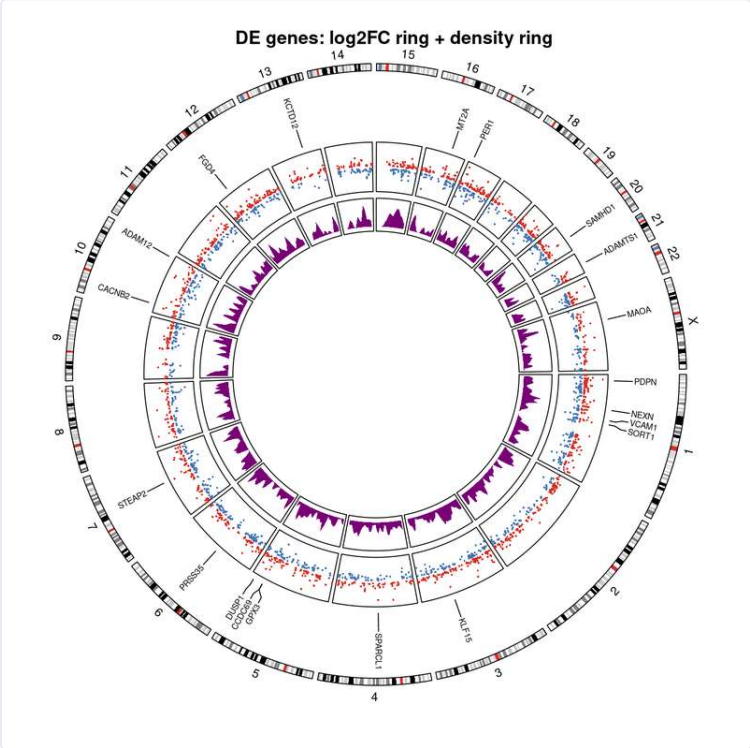
circlize · 2 variations

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*); Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 chord_directional

CELL-CELL



v2 circos_genome_rings

CELL-CELL

UpSetR

#57

OF 100 · PAPERS NAMING IT

R | SET OVERLAP | DE, QC & SET OVERLAPS

112	0.63%	130	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

UpSet set-intersection plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

UpSetR: overlap of Zeisel 2015 cell-type marker sets (FindAllMarkers, adj p<0.01, log2FC>1)

KEY PLOTTING CALL

RECREATIONS/UPSETR/UPSETR_RECREATE.R — EXCERPT · ENDS LINE 5

```
m <- read.csv("../_data/zeisel_markers.csv") %>% filter(p_val_adj < .01, avg_log2FC > 1); L <- split(m$gene, m$cluster)
png("v1_upset_frequency.png", 1600, 900, res = 150); print(upset(fromList(L), nsets = length(L), order.by = "freq", nintersects
= 30, main.bar.color = "#2c7fb8", sets.bar.color = "#de2d26", text.scale = 1.2, mb.ratio = c(.6,.4))); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 — FIGURE CAPTION

Cell-type-specific Alzheimer's disease polygenic risk scores are associated with distinct disease processes in Alzheimer's disease. *Nat Commun, 2023*

"...Cell-type-specific Alzheimer's disease polygenic risk scores (ADPRS).a A schematic of cell-type-specific PRS derivation. b An UpSetR plot of cell-type-specific gene sets used to define cell-type-specific ..."

DOI ↗ Fig. 1 on PMC ↗

Extended Data Fig. 9 — TEXT

Characteristics of blood-brain barrier heterogeneity between brain regions revealed by profiling vascular and perivascular cells. *Nat Neurosci, 2024*

"...Differentially expressed genes were compared to enriched genes from human mural cells from the lung45, gut46 kidney47 and brain40 and visualized by UpSetR (v.1.4.0)97."

DOI ↗ Extended Data Fig. 9 on PMC ↗

Fig. 3 — TEXT

Exposure of iPSC-derived human microglia to brain substrates enables the generation and manipulation of diverse transcriptional states in vitro. *Nat Immunol, 2023*

"...To calculating the number of unique differentially expressed genes, UpSet plots were created for each state using all differentially expressed genes with the UpSetR R package."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

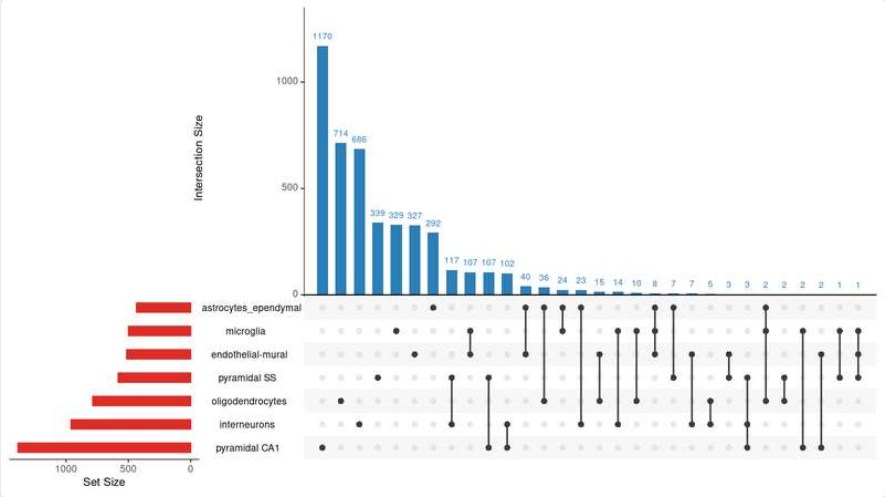
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/UpSetR/UpSetR_recreate.R`

√ re-run end to end · 6 s

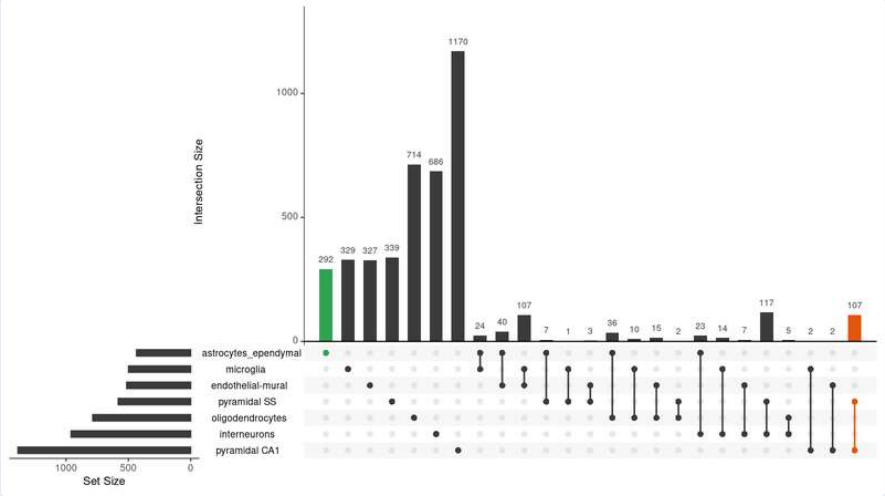
UpSetR · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



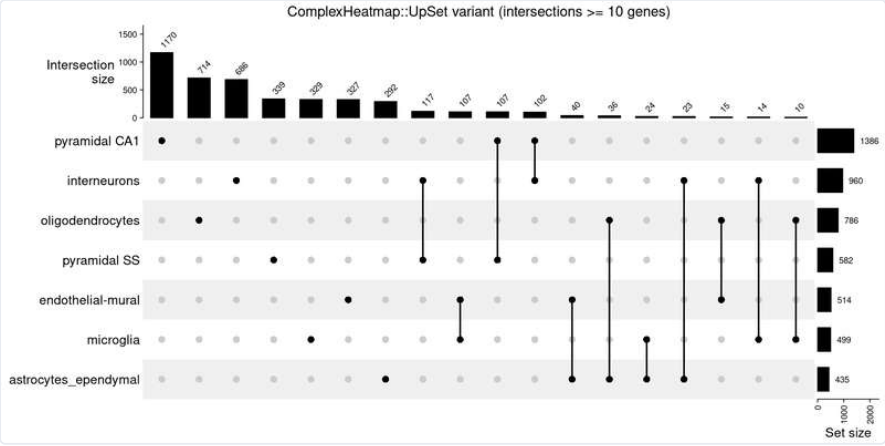
v1 upset_frequency

DE



v2 upset_degree_queries

DE



v3 complexheatmap_UpSet

DE

RCTD/spacexr

#59

OF 100 · PAPERS NAMING IT

R | SPATIAL DECONVOLUTION | SPATIAL & TISSUE MAPS

105	0.59%	130	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Spatial deconvolution pie/proportion maps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

RCTD (spacexr) deconvolution of the 10x Visium mouse coronal brain (squidpy visium_hne; integer UMIs recovered, see _data/export_visium_for_R.py) with the Zeisel 2015 cortex/hippocampus reference. full mode. Cell-type weight maps, dominant-type map, and a spatial pie-chart (scatterpie) view.

KEY PLOTTING CALL

RECREATIONS/RCTD_SPACEXR/RCTD_RECREATE.R — EXCERPT · ENDS LINE 14

```
W <- as.matrix(normalize_weights(r@results$weights)); df <- cbind(obs[rownames(W), c("x","y")], W)
ps <- lapply(colnames(W), function(k) ggplot(df, aes(x, -y, colour = .data[[k]])) + geom_point(size = .6) + scale_colour_viridis_c(option = "inferno", limits = c(0, 1)) + coord_fixed() + theme_void() + ggtitle(k) + theme(legend.position = "none", plot.title = element_text(size = 9)))
ggsave("v1_weight_maps.png", wrap_plots(ps, ncol = 4) + plot_annotation(title = "RCTD cell-type weights (Zeisel ref -> Visium mouse brain)"), width = 12, height = 6.5, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Extended Data Fig. 6 — FIGURE CAPTION

Programs, origins and immunomodulatory functions of myeloid cells in glioma. *Nature, 2025*

"...was applied to spatial transcriptomic data from Ravi et al. to define broad transcriptomic tumor niche programs; in parallel, the cell content of each spot was inferred based on scRNA-seq signatures (RCTD ..."

DOI ↗ Extended Data Fig. 6 on PMC ↗

Fig. 4 — FIGURE CAPTION

An integrated single-nucleus and spatial transcriptomics atlas reveals the molecular landscape of the human hippocampus. *Nat Neurosci, 2025*

"...NMF reveals cell type heterogeneity and biologically relevant pathways in hippocampal subfields compared to RCTD deconvolution results.For all spot plots of example capture areas from donor Br3942, spot ..."

DOI ↗ Fig. 4 on PMC ↗

Fig. 2 — FIGURE CAPTION

BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. *Nat Genet, 2024*

"...2 and 3). Top row, BANKSY clustering. Second row, nonspatial clustering (BANKSY with λ = 0). Third row, RCTD4 weights. The r values indicate the point-biserial correlation of cluster assignments with RCTD ..."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

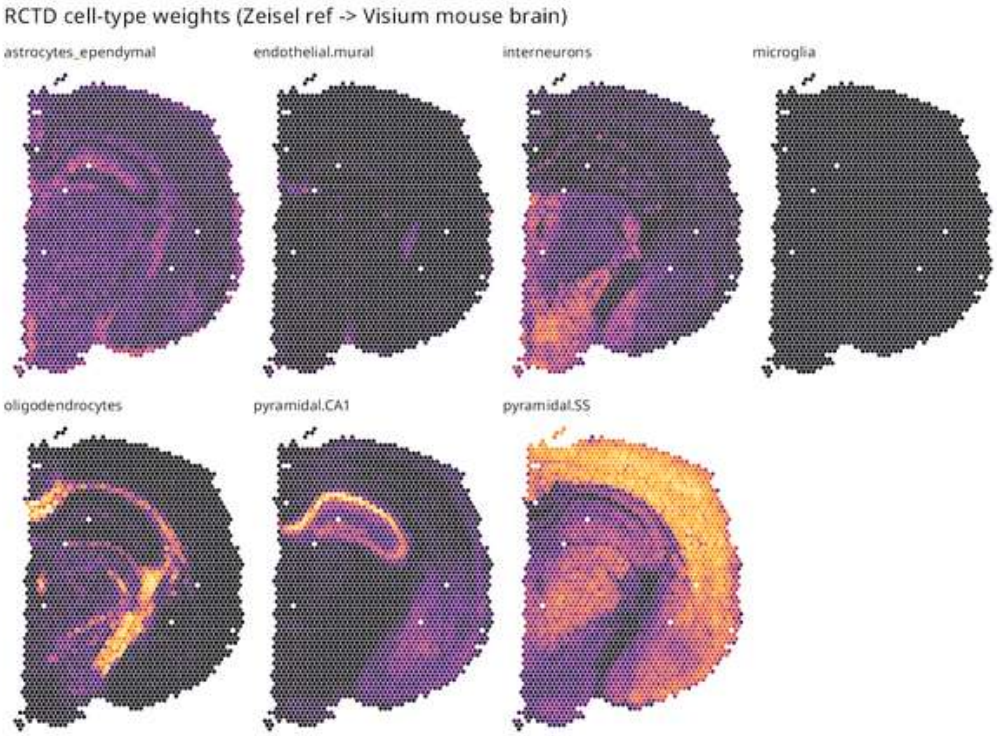
Reproduce: `Rscript recreations/RCTD_spacexr/RCTD_recreate.R`

√ re-run end to end · 19 s

RCTD/spacexr · 3 variations

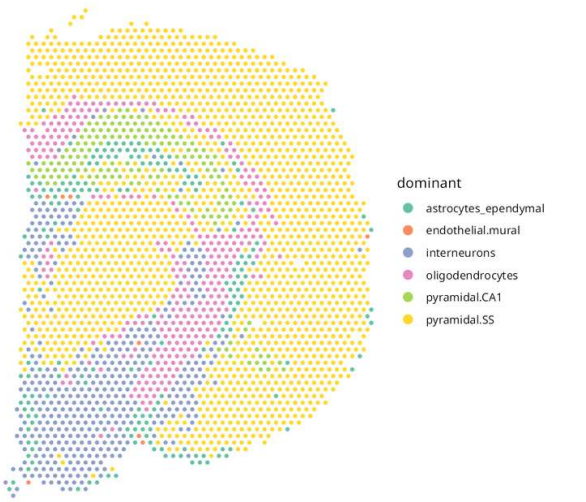
NOVAFLOW RECREATION

Data: 10x Visium mouse coronal brain section deconvolved with a Zeisel et al. 2015 reference. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



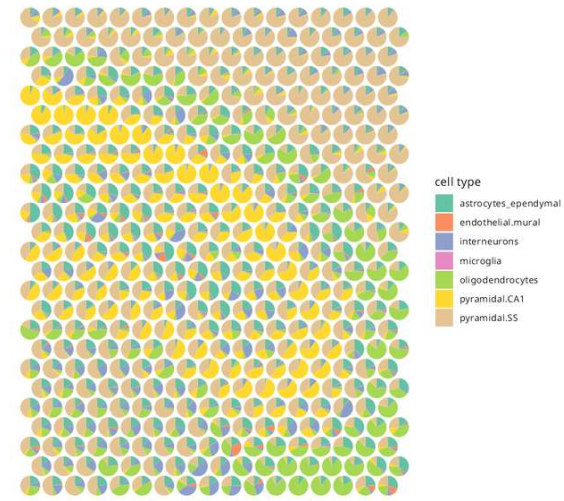
v1 weight_maps

SPATIAL



v2 dominant_type

SPATIAL



v3 scatterpie_zoom

SPATIAL

CytoTRACE 2

#60

OF 100 · PAPERS NAMING IT

R/PYTHON · DIFFERENTIATION POTENCY · TRAJECTORY, PSEUDOTIME & VELOCITY · COMPOSITION, DISTRIBUTIONS & LAYOUTS

103	0.58%	131	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · CYTOTRACE2

Neuro-corpus papers per year (*2026 partial) · niche rank **#5**, growth **324.85×**†, visual 4/5, score 0.9. Broad-corpus rate 0 → 3.239 per 1k papers (2023 → 2025–26).

GitHub digitalcytometry/cytotrace2 · created 2024-03-05 · last push 2026-07-08

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

Differentiation-potency score on UMAP / boxplots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

CytoTRACE 2 (interpretable deep model of developmental potency) on La Manno 2018 dentate gyrus: potency score on the UMAP, potency category composition per cell type, and score-vs-cell-type ranking.

KEY PLOTTING CALL

RECREATIONS/CYTOTRACE2/CYTOTRACE2_RECREATE.PY — EXCERPT · ENDS LINE 19

```
fig,axs=plt.subplots(1,2,figsize=(13,5.5))
sc.pl.umap(a,color="clusters",ax=axs[0],show=False,frameon=False,legend_loc="on data",legend_fontsize=7,title="Dentate gyrus cell types")
o=np.argsort(a.obs.CytoTRACE2_Score.values); xy=a.obsm["X_umap"]
s=axs[1].scatter(xy[o,0],xy[o,1],c=a.obs.CytoTRACE2_Score.values[o],cmap="Spectral_r",s=6,vmin=0,vmax=max(.35,np.quantile(a.obs.CytoTRACE2_Score,.995))); axs[1].set_axis_off(); axs[1].set_title("CytoTRACE 2 potency score (colour range clipped to data)"); fig.colorbar(s,ax=axs[1],shrink=.6)
fig.tight_layout(); fig.savefig("v1_potency_umap.png",dpi=140)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 4 — FIGURE CAPTION

Decoding multiple myeloma: single-cell insights into tumor heterogeneity, immune dynamics, and disease progression. *Front Immunol, 2025*

"...Proposed time-series analysis of myeloma cell subpopulations. (A, B) The UMAP and Bar plots showed the results of the stemness ability of four subtypes of myeloma cells calculated based on CytoTRACE. (C) Bar ..."

DOI ↗ Figure 4 on PMC ↗

Fig. 3 — FIGURE CAPTION

Cranioencephalic functional lymphoid units in glioblastoma. *Nat Med, 2024*

"...dition and tissue source. CBe T cell types in dark blue. g, UMAP of 3′ GEX CB data from e generated by Palantir, based on diffusion map dimensionality reduction, color coded as in e. Left, continuous CytoTRACE ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 6 — FIGURE CAPTION

Single-cell RNA sequencing unveils Lrg1's role in cerebral ischemia–reperfusion injury by modulating various cells. *J Neuroinflammation, 2023*

"...he direction of arrows indicates the direction of differentiation and development. H The dot plot illustrates the differentiation trajectories of microglial subpopulations in Sham + WT group drawn by ..."

DOI ↗ Fig. 6 on PMC ↗

LIMITATIONS & NOTES

- cytotrace2_py is installed from a GitHub subdirectory (METHODS §10).

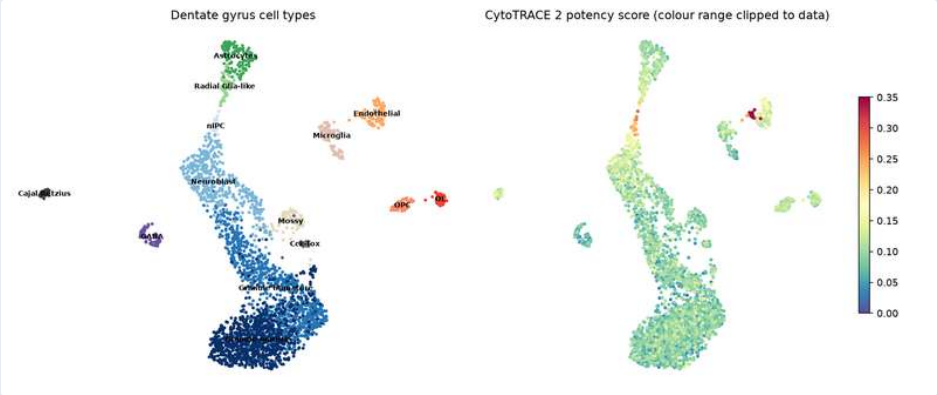
Reproduce: `python3 recreations/CytoTRACE2/cytotrace2_recreate.py`

√ re-run end to end · 102 s

CytoTRACE 2 · 3 variations

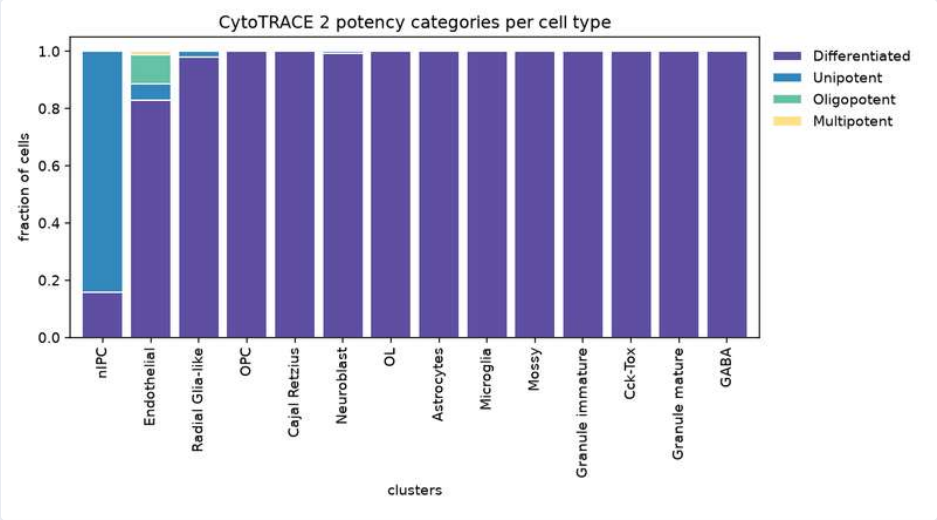
NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



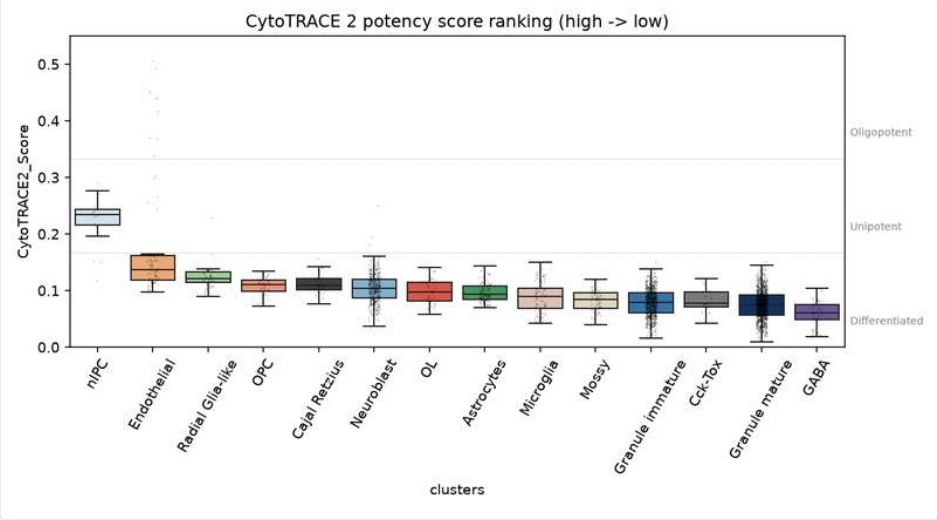
v1 potency_umap

TRAJECTORY



v2 potency_composition

COMPOSITION



v3 score_ranking

COMPOSITION

decoupleR

#61

OF 100 · PAPERS NAMING IT

R/PYTHON | TF/PATHWAY ACTIVITY | ENRICHMENT & PATHWAY ACTIVITY

102	0.57%	112	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

TF/pathway activity heatmaps, UMAPs

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

decoupleR: PROGENy pathway and CollecTRI TF activities (ULM) on Zeisel 2015 (OmniPath networks fetched at run time): pathway activity heatmap per cell type, TF activity heatmap (top variable TFs), pathway score on UMAP.

KEY PLOTTING CALL

RECREATIONS/DECOUPLER/DECOUPLER_RECREATE.R — EXCERPT · ENDS LINE 14

```
ap <- run_ulm(X, net_p, .source = "source", .target = "target", .mor = "weight", minsize = 5)
at <- run_ulm(X, net_t, .source = "source", .target = "target", .mor = "weight", minsize = 10)
agg <- function(a) { M <- a %>% pivot_wider(id_cols = condition, names_from = source, values_from = score) %>% tibble::column_t
o_rownames("condition") %>% as.matrix(); M <- M[colnames(so), ]
  t(sapply(split(seq_len(ncol(so)), so$celltype), function(i) colMeans(M[i, , drop = FALSE]))) }
P <- agg(ap); Tf <- agg(at); pal <- colorRampPalette(c("#2166ac","white","#b2182b"))(100)
png("v1_progeny_heatmap.png", 1300, 800, res = 150); pheatmap(P, scale = "column", color = pal, border_color = NA, main = "deco
upleR PROGENy pathway activity (ULM, mean per cell type)"); dev.off()
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 FIGURE CAPTION — **Single-nucleus RNA sequencing reveals glial cell type-specific responses to ischemic stroke in male rodents.** *Nat Commun, 2024*

"...The Top 25 module-defining genes, as sorted by descending Moran´s I, are depicted in boxes on the right side of the respective gene module feature plots. l Heatmap depicting the top 100 most variable ..."

DOI ↗ Fig. 2 on PMC ↗

Fig. 8 FIGURE CAPTION — **The spatial landscape of glial pathology and T cell response in Parkinson's disease substantia nigra.** *Nat Commun, 2025*

"...ed based on two-sided adjusted p-values. d Bar plot showing activation scores of select pathways in differentially expressed genes in CD44 KD astrocytes. The scores were calculated using Omnipath and decoupleR ..."

DOI ↗ Fig. 8 on PMC ↗

Fig. 2 TEXT — **Cell type mapping reveals tissue niches and interactions in subcortical multiple sclerosis lesions.** *Nat Neurosci, 2024*

"...Pseudobulk transcriptomic profiles were generated for both atlases at the cell type level using decoupler-py (v.1.5.1, commit b6b430e)73."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

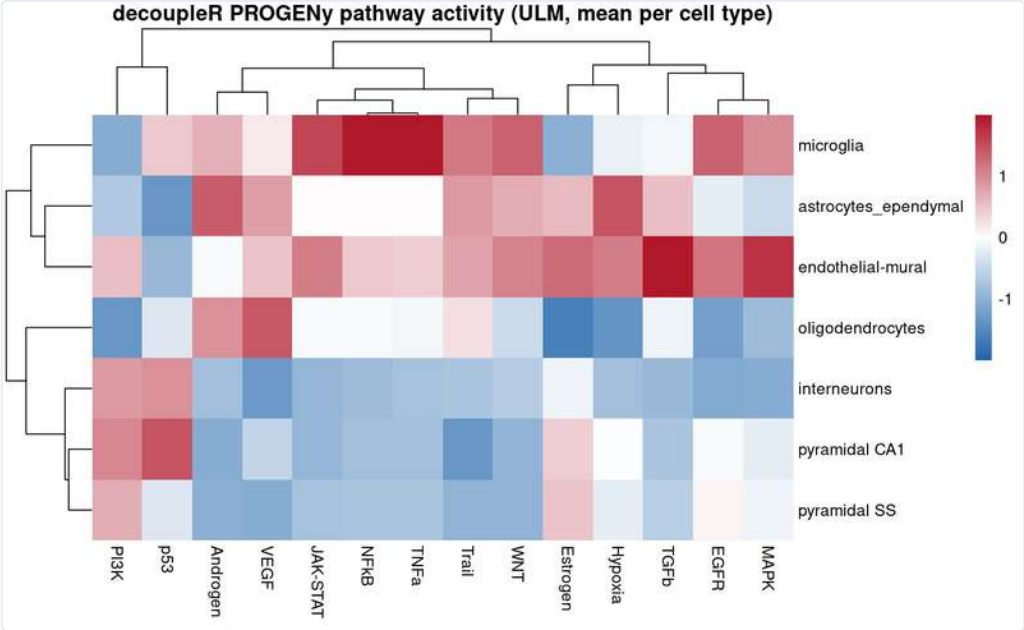
- OmnipathR's PROGENy/CollecTRI download failed, so the networks are fetched at run time with the Python decoupler 2.x (dc.op.*).

Reproduce: `Rscript recreations/decoupleR/decoupleR_recreate.R`

✓ re-run end to end · 78 s

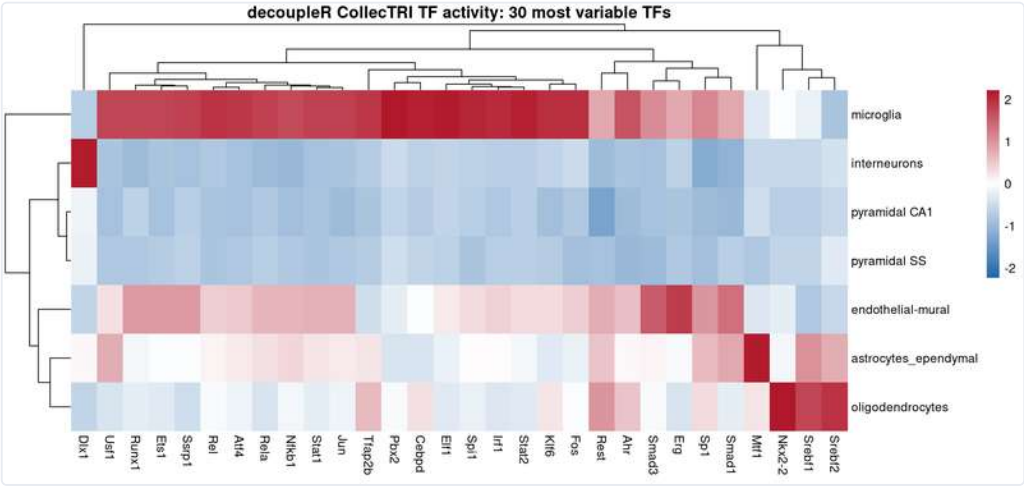
decoupleR · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



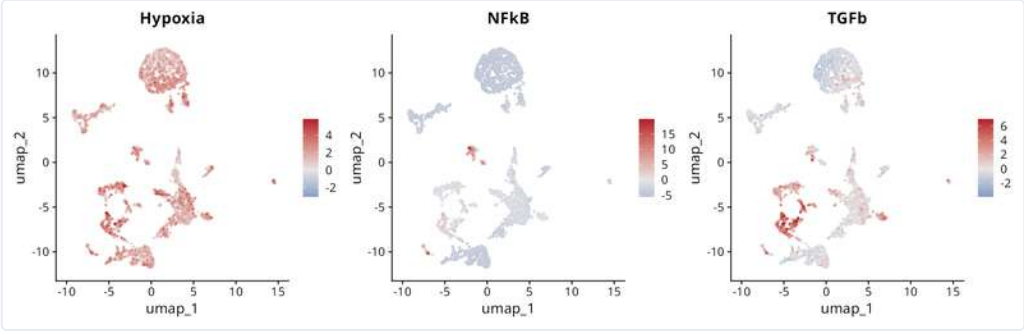
v1 progeny_heatmap

ENRICHMENT



v2 collectri_tf_heatmap

ENRICHMENT



v3 pathway_umaps

ENRICHMENT

Allen CCF atlas (BrainGlobe)

#63

OF 100 · PAPERS NAMING IT

WEB | ATLAS | BRAIN ATLAS, 3D & NEUROIMAGING

96	0.54%	169	viewer	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · BRAINGLOBE ATLAS API

Neuro-corpus papers per year (*2026 partial) · niche rank **#18**, growth **2.27×**†, visual 5/5, score 0.798. Broad-corpus rate 0 → 0.013 per 1k papers (2023 → 2025–26).

GitHub braingloBe/brainglobe-atlasapi · created 2020-04-30 · last push 2026-10-14 · also scored: Allen CCF overlays (#61, growth 1.14×)

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

Atlas reference views / cell-type taxonomies

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Allen Mouse CCFv3 (BrainGlobe atlas API, allen_mouse_25um): coronal annotation overlay, region outlines, 3D region point cloud

KEY PLOTTING CALL

RECREATIONS/BRAINGLOBE_ATLAS/BRAINGLOBE_RECREATE.PY — EXCERPT · ENDS LINE 12

```
a=BrainGlobeAtlas("allen_mouse_25um",check_latest=False); ref=a.reference; ann=a.annotation; z=ann.shape[0]//2+20
cols=np.zeros(ann.shape[1:]+(3,)); sl=ann[z]
for rid in np.unique(sl):
    if rid==0: continue
    try: cols[sl==rid]=np.array(a.structures[rid]["rgb_triplet"])/255
    except KeyError: pass
fig,ax=plt.subplots(1,2,figsize=(12,4.8)); ax[0].imshow(ref[z],cmap="gray"); ax[0].set_title("CCFv3 template, coronal slice"); a
x[1].imshow(ref[z],cmap="gray"); ax[1].imshow(cols,alpha=.6); ax[1].set_title("Allen annotation overlay (official colours)")
[x.axis("off") for x in ax]; fig.tight_layout(); fig.savefig("v1_coronal_overlay.png",dpi=140)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 4 — FIGURE CAPTION

Conservation and alteration of mammalian striatal interneurons. *Nature, 2025*

"...parent clusters 0840–0842. h, Distribution of highlighted ABC clusters in the adult mouse brain and Th, Tac2 and Lhx8 expression (in counts per million reads (CPM)) found within those populations27. ABC atlas ..."

DOI ↗ Fig. 4 on PMC ↗

Figure S7 — FIGURE CAPTION

LRRC37B is a human modifier of voltage-gated sodium channels and axon excitability in cortical neurons. *Cell, 2023*

"...m HEK293T cells transfected for their cDNA and SCN1B cDNA: all LRRC37B proteins bind to SCN1B but not LRRC37A2.(C) LRRC37B, FGF13, SCN1B, and SCN8A transcripts detection in human cortical cells (from Allen ..."

DOI ↗ Figure S7 on PMC ↗

Figure 1 — TEXT

The expression profile and tumorigenic mechanisms of CD97 (ADGRE5) in glioblastoma render it a targetable vulnerability. *Cell Rep, 2023*

"...First, we analyzed the mRNA expression levels of all 32 aGPCRs in healthy brain tissue using the Allen Brain Map database and in PDGCs collected in-house (Figure 1A)."

DOI ↗ Figure 1 on PMC ↗

LIMITATIONS & NOTES

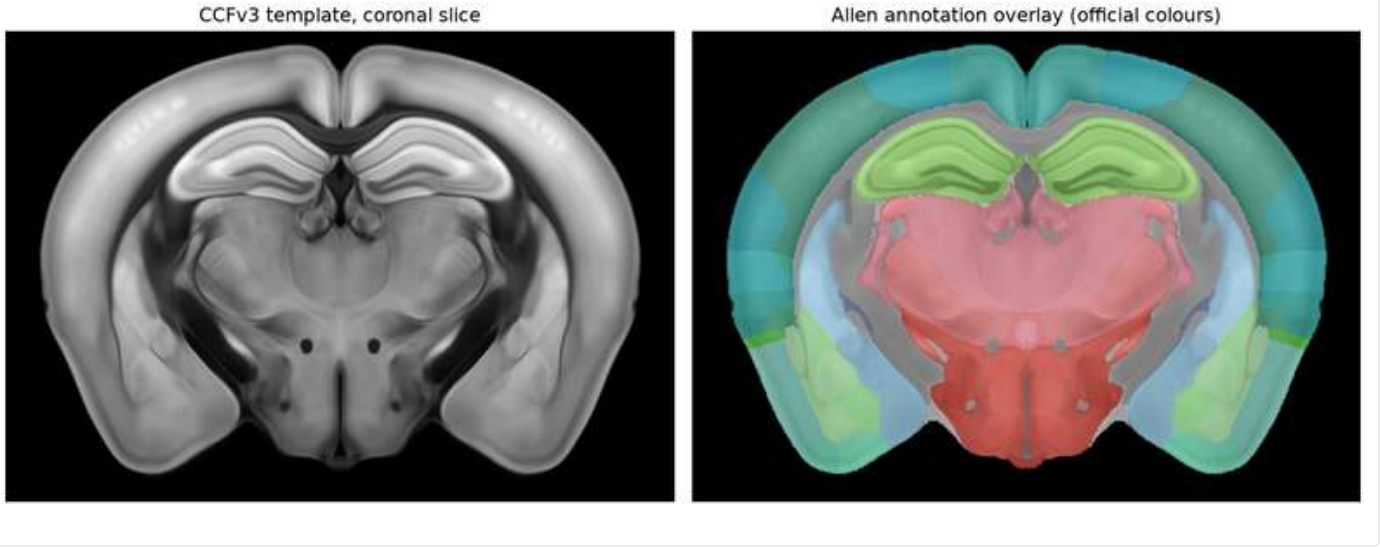
- Allen CCFv3 atlas renders via BrainGlobe atlas API

Reproduce: `python3 recreations/BrainGlobe_atlas/brainglobe_recreate.py` — √ re-run end to end · 6 s

Allen CCF atlas (BrainGl… · 3 variations

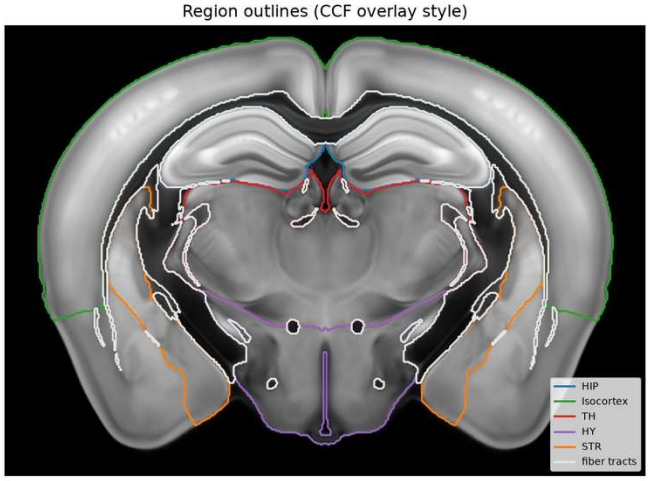
NOVAFLOW RECREATION

Data: Allen Mouse Brain Common Coordinate Framework v3, 25 µm (BrainGlobe atlas API). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



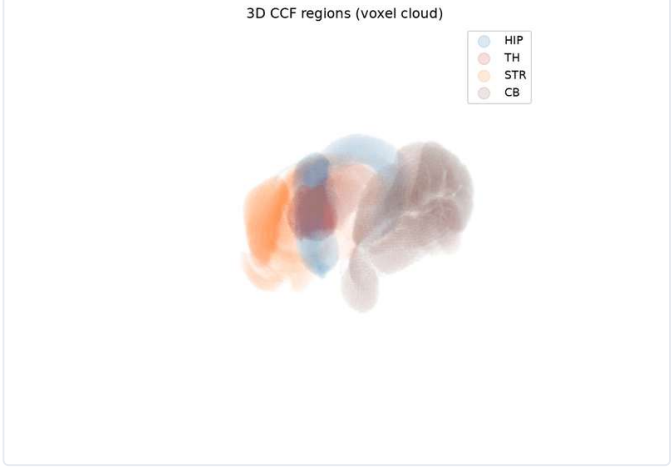
v1 coronal_overlay

BRAIN



v2 region_outlines

BRAIN



v3 ccf_3d_regions

BRAIN

NicheNet

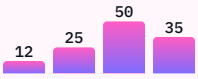
#66

OF 100 · PAPERS NAMING IT

R · LIGAND-TARGET · CELL-CELL COMMUNICATION

90	0.51%	122	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS



NICHE & COOL · NICHENET

Neuro-corpus papers per year (*2026 partial) · niche rank **#46**, growth **1.73×**, visual 4/5, score 0.592. Broad-corpus rate 4.632 → 8.026 per 1k papers (2023 → 2025–26).
GitHub saeyslab/nichenetr · created 2018-02-05 · last push 2026-09-25

BEST FOR

Ligand activity / ligand-target heatmaps, circos

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

NicheNet (nichenetr v2 priors, mouse, Zenodo 7074291): which glial/vascular ligands best explain the cortex-vs-CA1 programme in oligodendrocytes (Zeisel 2015)? Ligand activity, ligand→target heatmap, ligand→receptor heatmap, and a circlize circos of top links coloured by sender.

KEY PLOTTING CALL

RECREATIONS/NICHENET/NICHENET_RECREATE.R — EXCERPT · ENDS LINE 18

```
la <- nn$ligand_activities %>% filter(test_ligand %in% nn$top_ligands) %>% arrange(aupr_corrected)
g1 <- ggplot(la, aes(aupr_corrected, factor(test_ligand, levels = test_ligand))) + geom_segment(aes(x = 0, xend =
aupr_corrected, yend = factor(test_ligand, levels = test_ligand)), colour = "grey70") + geom_point(aes(colour =
aupr_corrected), size = 4) + scale_colour_gradient(low = "#fdd49e", high = "#d7301f") + theme_bw() + labs(x = "ligand activity
(corrected AUPR)", y = NULL, colour = "AUPR", title = "NicheNet ligand activity")
ggsave("v1_ligand_activity.png", g1, width = 5.5, height = 6, dpi = 150, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 8 — FIGURE CAPTION

A single nuclear transcriptomic characterisation of mechanisms responsible for impaired angiogenesis and blood-brain barrier function in Alzheimer's disease. *Nat Commun, 2024*

"...NicheNet intercellular communication analysis identified potential regulators of EC DEG associated with proinflammatory and anti-angiogenic gene expression in astrocytes and perivascular macrophages.A ..."

DOI ↗ Fig. 8 on PMC ↗

Figure 8 — FIGURE CAPTION

Single-cell and Spatial Transcriptomics Reveals Ferroptosis as The Most Enriched Programmed Cell Death Process in Hemorrhage Stroke-induced Oligodendrocyte-mediated White Matter Injury. *Int J Biol Sci, 2024*

"...LCN2+ microglia induce oligodendrocyte ferroptosis via the CSF1/CSF1R pathway. (A) Top-ranked ligands that were inferred to regulate oligodendrocytes by LCN2+ microglia (Mg5) according to NicheNet. Dot plots ..."

DOI ↗ Figure 8 on PMC ↗

Figure 6 — FIGURE CAPTION

Transcriptomic atlas and interaction networks of brain cells in mouse CNS demyelination and remyelination. *Cell Rep, 2023*

"...NicheNet analysis reveals cell-cell interactions elicited by demyelination(A–C) Circos plot showing links between predicted ligands from sender cells (bottom) with their associated receptors on receiver ..."

DOI ↗ Figure 6 on PMC ↗

LIMITATIONS & NOTES

- NicheNet v2 mouse priors are downloaded from Zenodo 7074291.

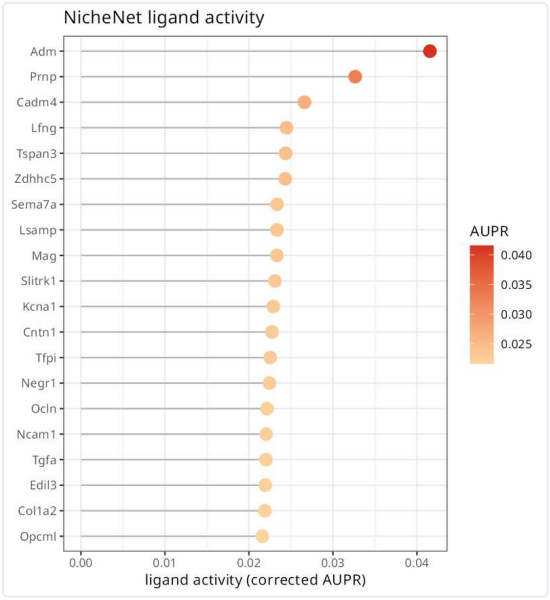
Reproduce: `Rscript recreations/NicheNet/NicheNet_recreate.R`

√ re-run end to end · 42 s

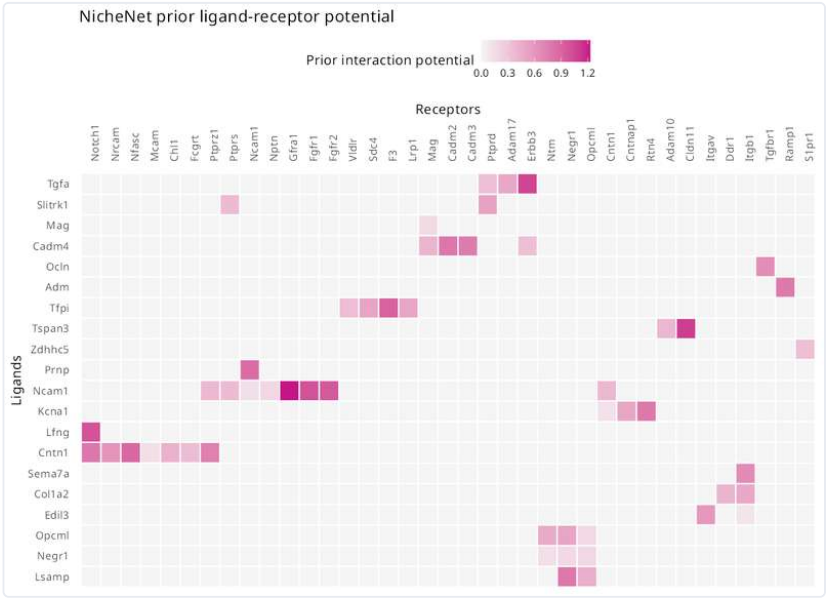
NicheNet · 4 variations

NOVAFLOW RECREATION

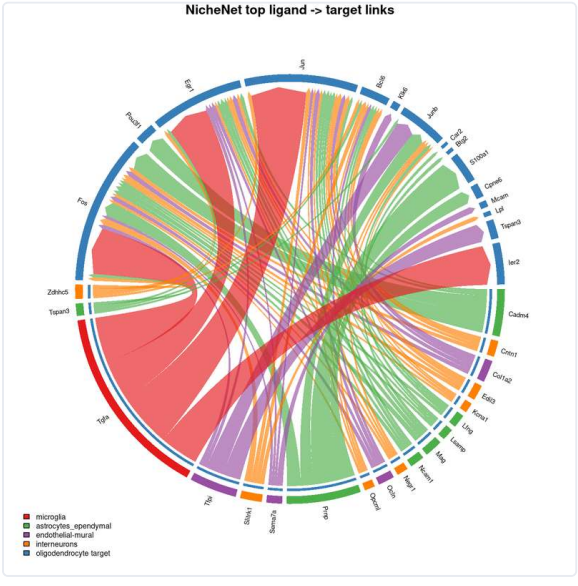
Data: Zeisel et al. 2015 scRNA-seq with NicheNet v2 mouse prior networks (Zenodo 7074291). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



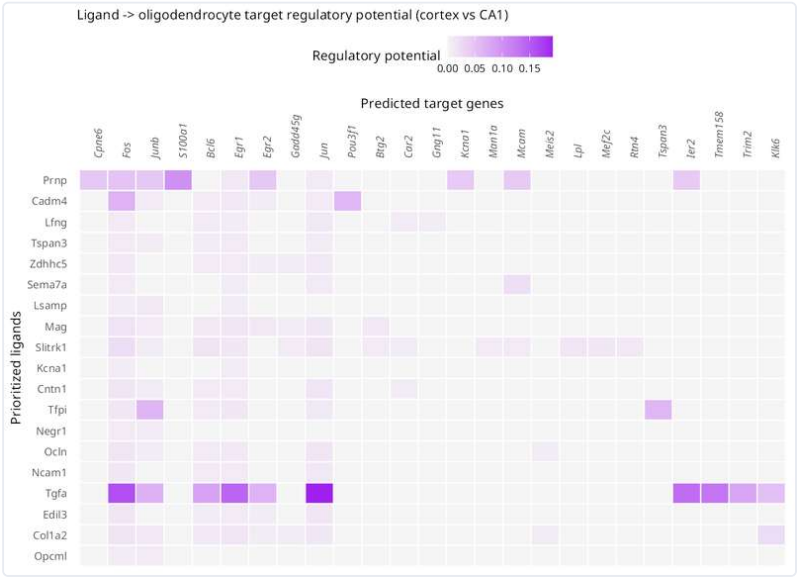
v1 ligand_activity · CELL–CELL



v2 ligand_receptor_heatmap · CELL–CELL



v3 circos · CELL–CELL



v4 ligand_target_heatmap · CELL–CELL

ReactomePA

#68

OF 100 · PAPERS NAMING IT

R · ENRICHMENT · ENRICHMENT & PATHWAY ACTIVITY

84	0.47%	87	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Reactome enrichment dot/network plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

ReactomePA: Reactome pathway enrichment of Zeisel 2015 cell-type markers (mouse; compareCluster dotplot, enrichment map and gene-concept network for interneurons)

KEY PLOTTING CALL

RECREATIONS/REACTOMEPA/REACTOMEPA_RECREATE.R — EXCERPT · ENDS LINE 9

```
m <- read.csv("../_data/zeisel_markers.csv") %>% filter(p_val_adj < .01) %>% group_by(cluster) %>% slice_max(avg_log2FC, n =
200) %>% ungroup()
ids <- bitr(unique(m$gene), "SYMBOL", "ENTREZID", OrgDb = org.Mm.eg.db); m <- inner_join(m, ids, by = c(gene = "SYMBOL"))
L <- split(m$ENTREZID, m$cluster)
cc <- compareCluster(L, fun = "enrichPathway", organism = "mouse", pvalueCutoff = .05)
ggsave("v1_compareCluster_dotplot.png", dotplot(cc, showCategory = 4, label_format = 45) +
ggtitle("ReactomePA: cell-type marker pathways") + theme(axis.text.x = element_text(angle = 30, hjust = 1)), width = 11, height
= 9, dpi = 120, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 TEXT

The Global Neurodegeneration Proteomics Consortium: biomarker and drug target discovery for common neurodegenerative diseases and aging. *Nat Med, 2025*

"...Proteins identified in the meta-analytic differential abundance analysis were filtered into gene set enrichment analysis using the R libraries 'ReactomePA' and 'clusterProfiler'."

DOI ↗ Fig. 1 on PMC ↗

Fig. 1 TEXT

Shared and disease-specific pathways in frontotemporal dementia and Alzheimer's and Parkinson's diseases. *Nat Med, 2025*

"...To investigate the biological processes associated specifically with each ND and those common to all, we performed pathway enrichment analysis using the ReactomePA (version 1.48)80 and clusterProfiler (version ..."

DOI ↗ Fig. 1 on PMC ↗

Fig. 2 TEXT

MicroRNA-335-5p suppresses voltage-gated sodium channel expression and may be a target for seizure control. *Proc Natl Acad Sci U S A, 2023*

"...Pathway enrichment analysis of the remaining 267 MTIs (Dataset S3) was performed using the ReactomePA (74) and clusterProfiler (75) packages, with an adjusted p-value (Benjamini–Hochberg) <0.05 considered ..."

DOI ↗ Fig. 2 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

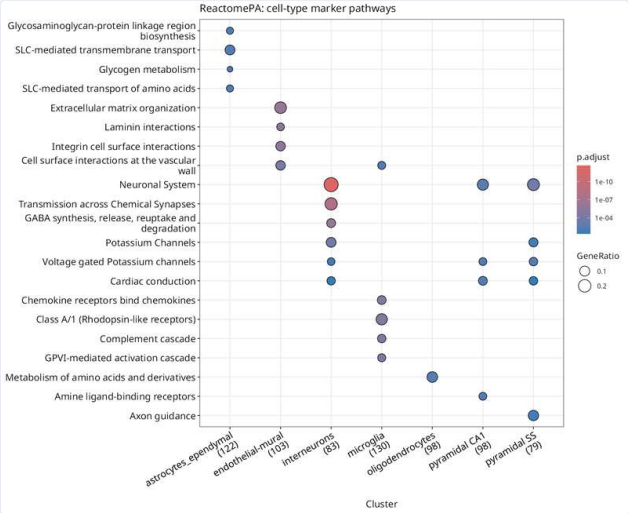
Reproduce: `Rscript recreations/ReactomePA/ReactomePA_recreate.R`

√ re-run end to end · 332 s

ReactomePA · 3 variations

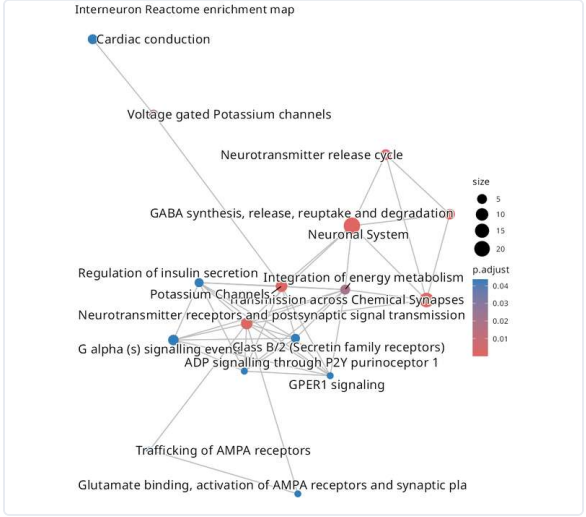
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



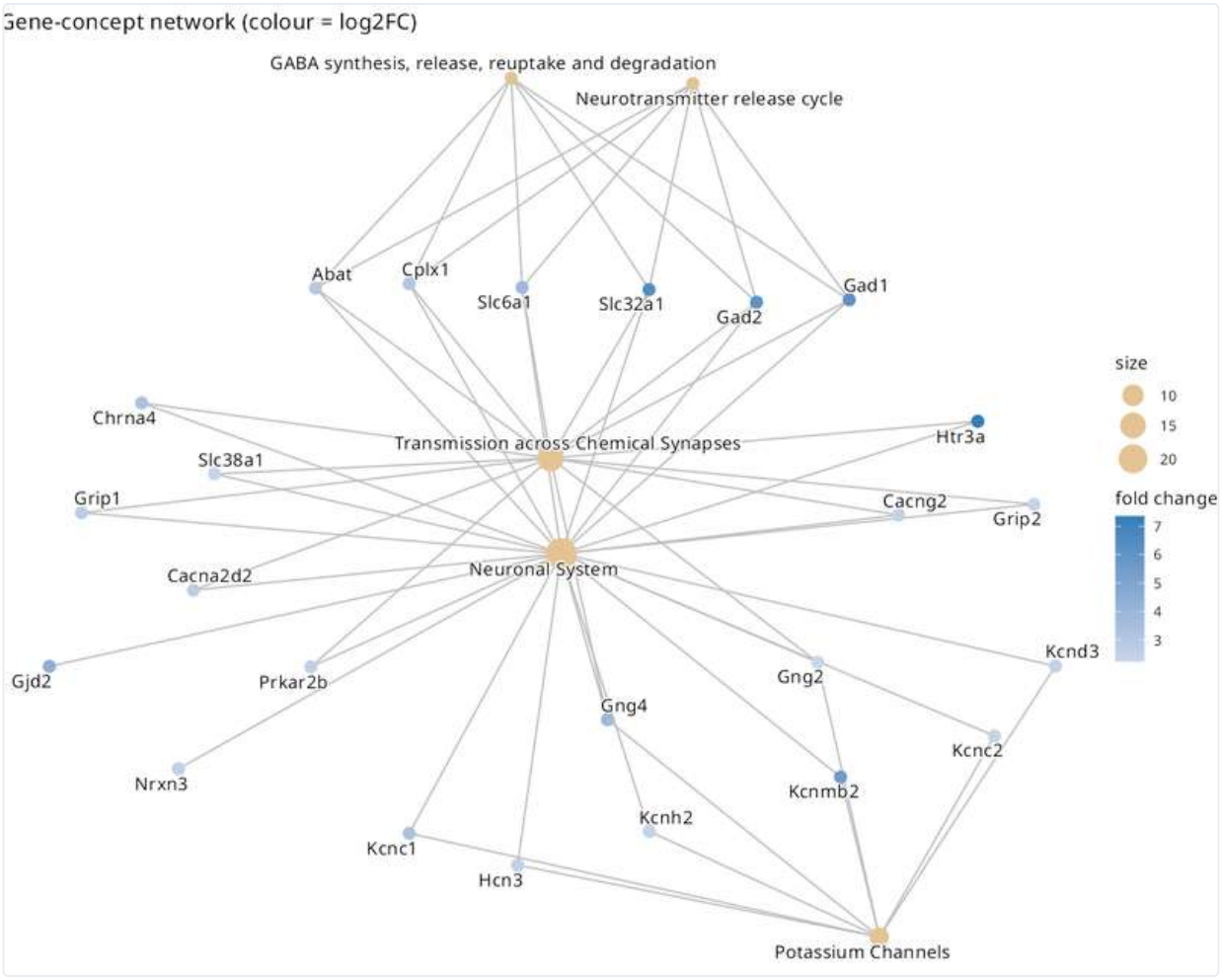
v1 compareCluster_dotplot

ENRICHMENT



v2 emapplot

ENRICHMENT



v3 cnetplot

ENRICHMENT

pathview

#70

OF 100 · PAPERS NAMING IT

R · PATHWAY MAPS · ENRICHMENT & PATHWAY ACTIVITY

82	0.46%	99	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

KEGG pathway maps coloured by fold-change

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

pathview: Zeisel 2015 interneuron-vs-pyramidal-SS log2FC painted onto KEGG GABAergic (mmu04727) and glutamatergic (mmu04724) synapse maps (KEGG XML/PNG fetched once from KEGG REST at run time).

KEY PLOTTING CALL

RECREATIONS/PATHVIEW/PATHVIEW_RECREATE.R — EXCERPT · ENDS LINE 9

```
for (pw in c("04727","04724")) pathview(gene.data = fc, pathway.id = pw, species = "mmu", out.suffix = "interneuron_vs_pyrSS", l
imit = list(gene = 3), low = list(gene = "#2166ac"), high = list(gene = "#b2182b"), kegg.native = TRUE)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 4 TEXT

Sildenafil as a Candidate Drug for Alzheimer's Disease: Real-World Patient Data Observation and Mechanistic Observations from Patient-Induced Pluripotent Stem Cell-Derived Neurons. *J Alzheimers Dis, 2024*

"...Significant KEGG pathways were depicted using Pathview tool with highlighted sildenafil regulated key genes (Supplementary Figures 2–4)."

DOI ↗ Fig. 4 on PMC ↗

Fig. 1 TEXT

Integrating lipid metabolism, pheromone production and perception by Fruitless and Hepatocyte Nuclear Factor 4. *Sci Adv, 2023*

"...The KEGG pathway maps were visualized using pathview (111)."

DOI ↗ Fig. 1 on PMC ↗

Fig. 11 TEXT

Sex differences in the transcriptome of extracellular vesicles secreted by fetal neural stem cells and effects of chronic alcohol exposure. *Biol Sex Differ, 2023*

"...Pathway analysis was conducted on differentially expressed genes using ReactomePA R package [69] which utilizes the KEGG database [70] and the Pathview R package [71] to visualize differentially regulated ..."

DOI ↗ Fig. 11 on PMC ↗

LIMITATIONS & NOTES

- KEGG XML/PNG is fetched once from KEGG REST at run time, so this needs internet.

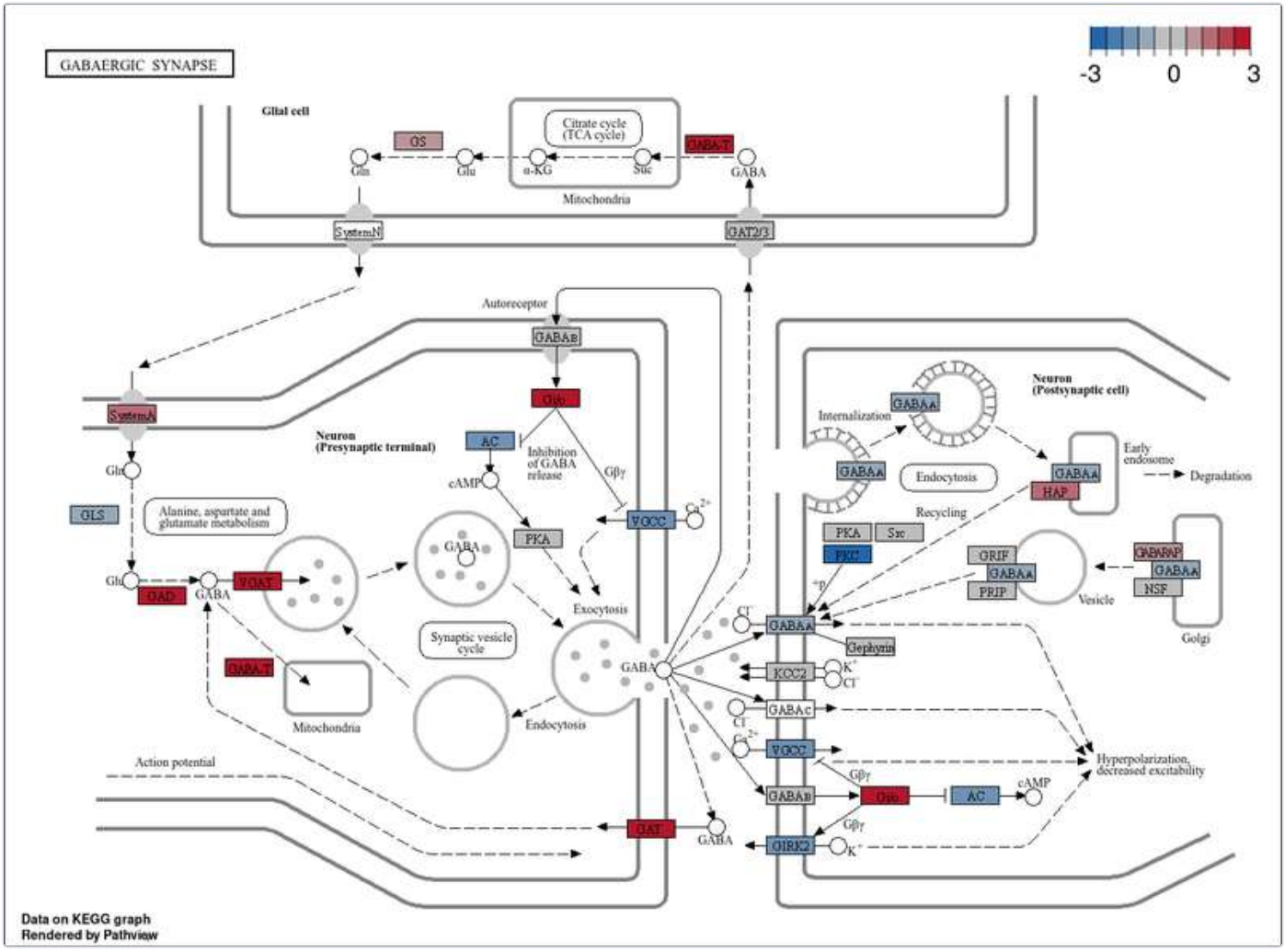
Reproduce: `Rscript recreations/pathview/pathview_recreate.R`

√ re-run end to end · 20 s

pathview · 3 variations

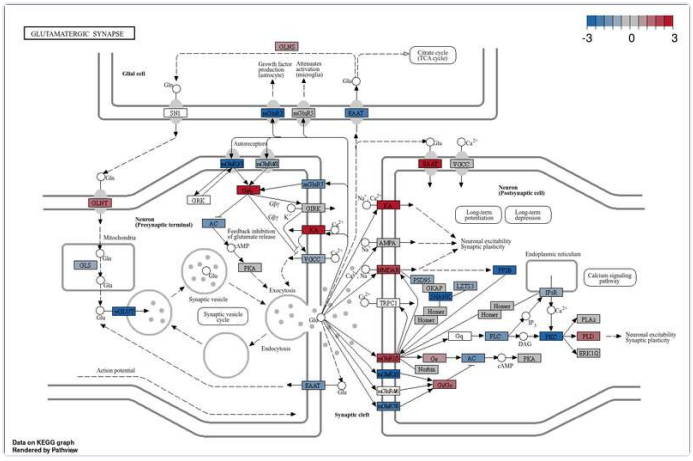
NOVAFLOW RECREATION

Data: Zeisel et al. 2015 interneuron vs pyramidal log2FC on KEGG pathway maps (KEGG REST). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



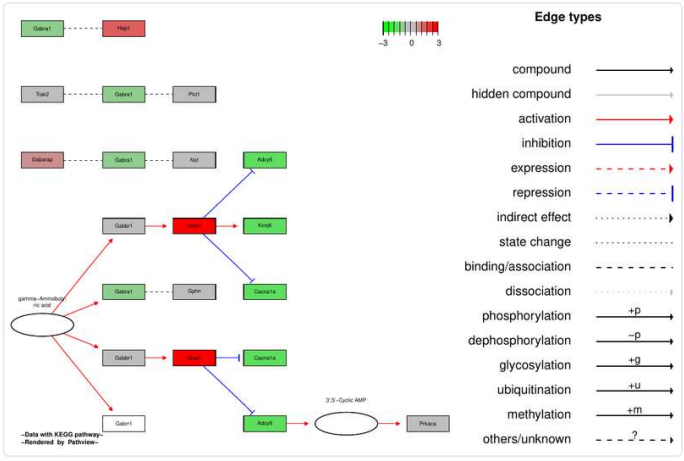
v1 GABAergic_synapse_native

ENRICHMENT



v2 glutamatergic_synapse_native

ENRICHMENT



v3 GABAergic_graphviz

ENRICHMENT

GOplot

#71

OF 100 · PAPERS NAMING IT

R · ENRICHMENT · ENRICHMENT & PATHWAY ACTIVITY

81	0.46%	85	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

GOCircle, GOChord, GOBubble

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

GOplot (GOBubble, GOCircle, GOChord, GOCluster) on oligodendrocyte markers vs all other Zeisel 2015 cell types (clusterProfiler enrichGO BP, mouse) - logFC from FindAllMarkers.

KEY PLOTTING CALL

RECREATIONS/GOPLOT/GOPLOT_RECREATE.R — EXCERPT · ENDS LINE 9

```
m <- read.csv("../_data/zeisel_markers.csv") %>% filter(cluster == "oligodendrocytes", p_val_adj < .01) %>%
slice_max(avg_log2FC, n = 300)
ego <- enrichGO(m$gene, OrgDb = org.Mm.eg.db, keyType = "SYMBOL", ont = "BP", pvalueCutoff = .01, readable = FALSE) %>%
simplify(cutoff = .6)
e <- as.data.frame(ego) %>% head(12); terms <- data.frame(category = "BP", ID = e$ID, term = e$Description, genes = gsub("/",
", ", e$geneID), adj_pval = e$p.adjust)
genes <- data.frame(ID = m$gene, logFC = m$avg_log2FC); circ <- circle_dat(terms, genes)
ggsave("v1_GOBubble.png", GOBubble(circ, labels = 3, table.legend = FALSE, display = "single") + ggtitle("GOBubble:
oligodendrocyte marker BP terms"), width = 8, height = 7, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 3 TEXT

Human Brain Microvascular Endothelial Cells Exposure to SARS-CoV-2 Leads to Inflammatory Activation through NF-κB Non-Canonical Pathway and Mitochondrial Remodeling. *Viruses, 2023*

"...Overlapping gene sets from reactome pathway terms were visualized as a chord plot using the GOplot R [44]."

DOI ↗ Figure 3 on PMC ↗

Figure 2 TEXT

A novel phenotype of B cells associated with enhanced phagocytic capability and chemotactic function after ischemic stroke. *Neural Regen Res, 2023*

"...The activation Z-score was calculated for each term by the R package GOplot to predict the activation state (Walter et al., 2015)."

DOI ↗ Figure 2 on PMC ↗

Fig. 6 TEXT

Choroid plexus defects in Down syndrome brain organoids enhance neurotropism of SARS-CoV-2. *Sci Adv, 2024*

"...Enrichment data were then combined with expression values, and z scores were calculated using the GOplot v.1.0.2 R package and plotted using ggplot2 v.3.3.2."

DOI ↗ Fig. 6 on PMC ↗

LIMITATIONS & NOTES

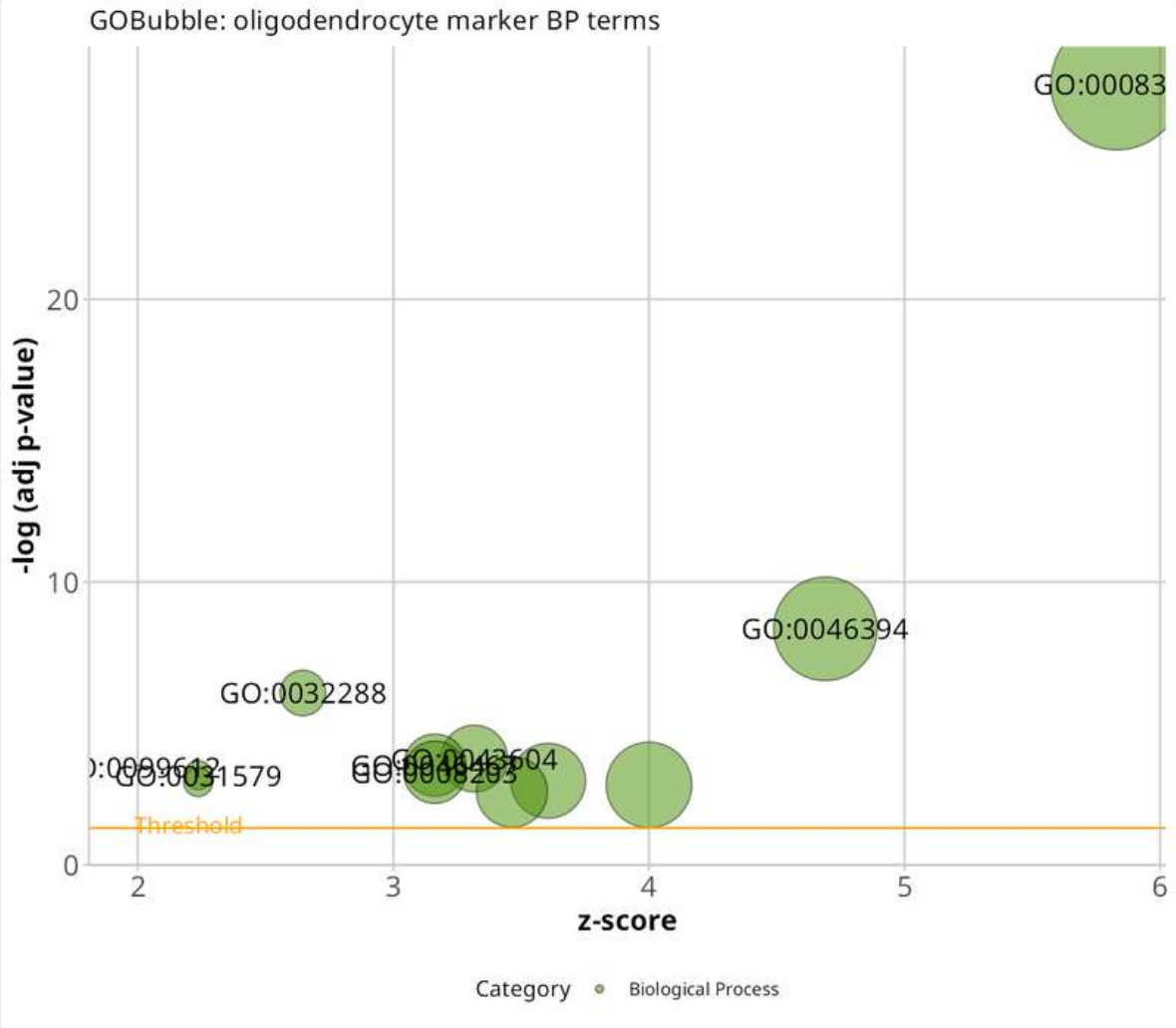
- GOCluster / GOHeat omitted, and GOChord drawn without the logFC ring: GOplot logFC gradients fail with ggplot2 ≥ 3.5.

Reproduce: `Rscript recreations/GOplot/GOplot_recreate.R`

√ re-run end to end · 50 s

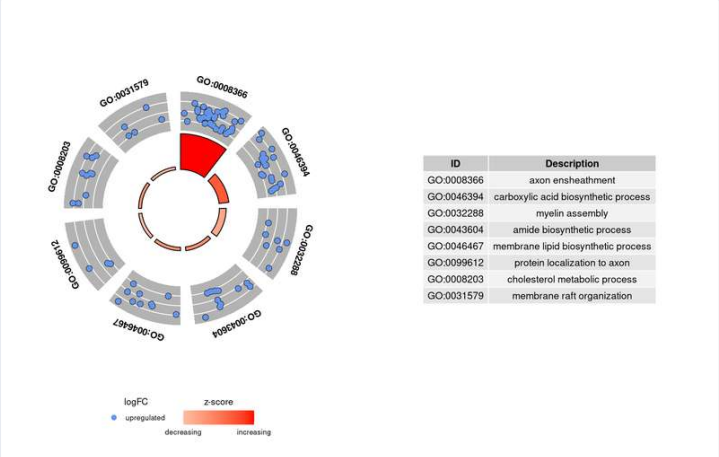
GOplot · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 GOBubble

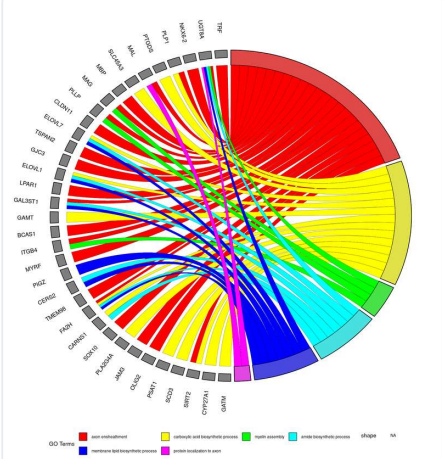
ENRICHMENT



ID	Description
GO:0008366	axon ensheathment
GO:0046394	carboxylic acid biosynthetic process
GO:0032288	myelin assembly
GO:0043604	amide biosynthetic process
GO:0046467	membrane lipid biosynthetic process
GO:0099612	protein localization to axon
GO:0008203	cholesterol metabolic process
GO:0031579	membrane raft organization

v2 GOCircle

ENRICHMENT



v3 GOChord

ENRICHMENT

scCustomize

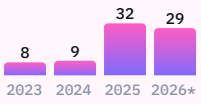
#74

OF 100 · PAPERS NAMING IT

R · SINGLE-CELL · EMBEDDINGS & FEATURE MAPS · HEATMAPS & DOT MATRICES · COMPOSITION, DISTRIBUTIONS & LAYOUTS

75	0.42%	78	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS



NICHE & COOL · SCCUSTOMIZE

Neuro-corpus papers per year (*2026 partial) · niche rank **#30**, growth **4.24×**, visual 3/5, score 0.684. Broad-corpus rate 0.627 → 2.692 per 1k papers (2023 → 2025–26).
GitHub samuel-marsh/scCustomize · created 2021-09-29 · last push 2026-06-30

BEST FOR

Enhanced Seurat plots (FeaturePlot_scCustom, Clustered_DotPlot)

KEY PLOTTING CALL

RECREATIONS/SCCUSTOMIZE/SCCUSTOMIZE_RECREATE.R — EXCERPT · ENDS LINE 4

```
so <- readRDS("../_data/zeisel_seurat.rds"); mk <-
c("Snap25","Gad1","Gad2","Pvalb","Sst","Satb2","Neurod6","Wfs1","Aqp4","Gfap","Mbp","Plp1","Cx3cr1","C1qa","Cldn5","Acta2")
ggsave("v1_DimPlot_scCustom_figureplot.png", DimPlot_scCustom(so, figure_plot = TRUE, label = TRUE), width = 7, height = 6, dpi
= 140, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 TEXT

Characterisation of premature cell senescence in Alzheimer's disease using single nuclear transcriptomics. *Acta Neuropathol, 2024*

"...Gene set or module feature plots were generated by first calculating aggregated gene set scores using AddModuleScore() function from Seurat and then plotted using FeaturePlot scCustom() from scCustomize (v."

DOI ↗ Fig. 2 on PMC ↗

Fig. 4 TEXT

cGAS-STING drives ageing-related inflammation and neurodegeneration. *Nature, 2023*

"...9a,f) and visualized using the FeaturePlot_scCustom function from the R package scCustomize (v.0.7.0), allowing for subsequent analysis of specific cell types (such as microglia, oligodendrocytes) ..."

DOI ↗ Fig. 4 on PMC ↗

Fig. 1 TEXT

Microglia ferroptosis is regulated by SEC24B and contributes to neurodegeneration. *Nat Neurosci, 2023*

"...UMAP plots and gene expression plots were generated using built-in Seurat/ggplot2 plotting functions or scCustomize version 0.7.0 unless otherwise described."

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

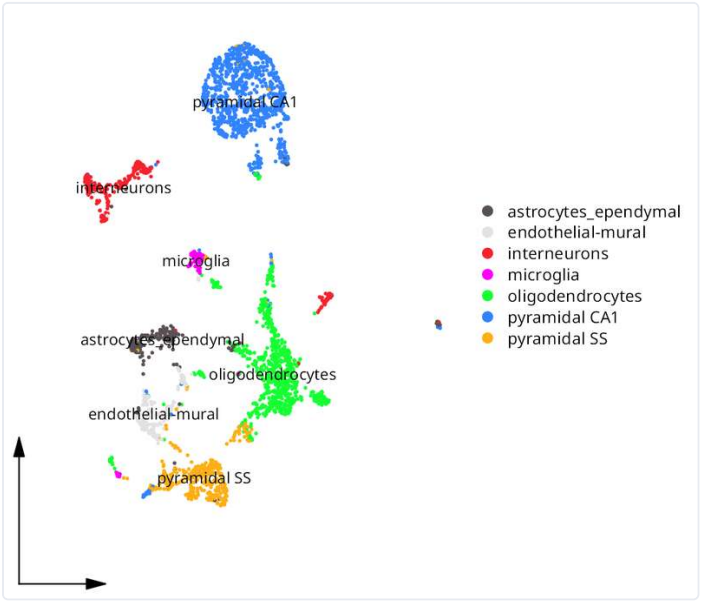
Reproduce: `Rscript recreations/scCustomize/scCustomize_recreate.R`

√ re-run end to end · 17 s

scCustomize · 4 variations

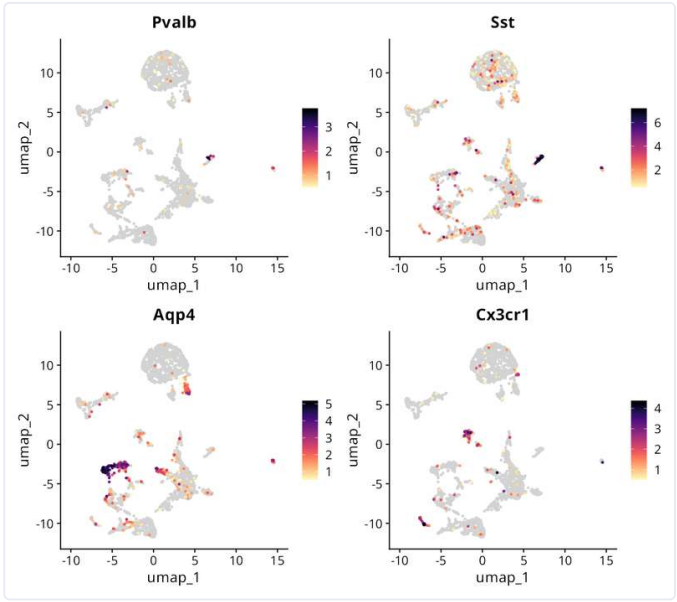
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



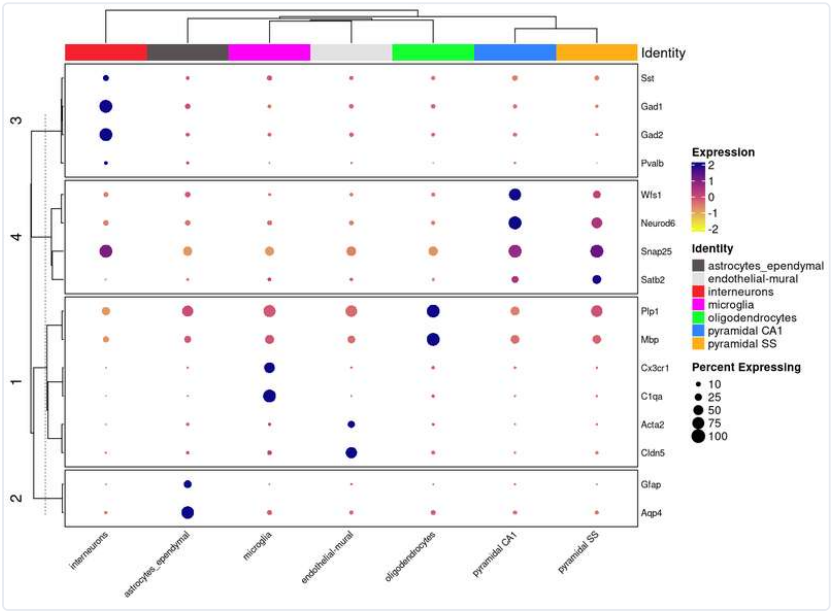
v1 DimPlot_scCustom_figureplot

EMBEDDINGS



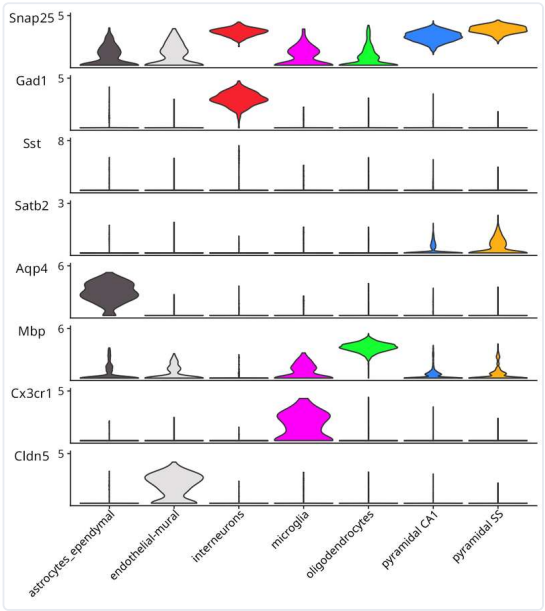
v2 FeaturePlot_scCustom

EMBEDDINGS



v3 Clustered_DotPlot

HEATMAPS



v4 Stacked_VlnPlot

COMPOSITION

patchwork

#79

OF 100 · PAPERS NAMING IT

R | LAYOUT | COMPOSITION, DISTRIBUTIONS & LAYOUTS

65	0.37%	71	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Multi-panel layout of ggplots

Also catalogued as: **cowplot** (#94, 44 papers)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

patchwork + cowplot multi-panel composites (typical Fig. 1 layout) from Zeisel 2015

KEY PLOTTING CALL

RECREATIONS/PATCHWORK_COWPLOT/COMPOSITE_RECREATE.R — EXCERPT · ENDS LINE 11

```
d <- so@meta.data %>% count(tissue, celltype) %>% group_by(tissue) %>% mutate(f = n / sum(n)) %>% ggplot(aes(tissue, f, fill = c
elltype)) + geom_col() + theme_classic() + labs(y = "fraction", x = NULL)
e <- DotPlot(so, features = c("Gad1","Slc17a7","Neurod6","Mbp","Aqp4","Cx3cr1","Cldn5"), group.by = "celltype") + RotatedAxis()
+ theme(axis.title = element_blank())
p <- (a | b) / (c | d | e) + plot_layout(heights = c(1.1, 1), widths = c(1, 1, 1.5)) + plot_annotation(tag_levels = "A", title
= "patchwork composite: Zeisel 2015 overview") & theme(plot.tag = element_text(face = "bold", size = 16))
ggsave("v1_patchwork_composite.png", p, width = 15, height = 10, dpi = 110, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 TEXT

TDP-43 nuclear loss in FTD/ALS causes widespread alternative polyadenylation changes. *Nat Neurosci, 2025*

"...All plots were prepared using tidyplots, ggplot2, ggpubr, patchwork and ggrepel in R."

DOI ↗ Fig. 3 on PMC ↗

Fig. 4 TEXT

Prospective, multicenter validation of a platform for rapid molecular profiling of central nervous system tumors. *Nat Med, 2025*

"...Plots were generated using ggplot2 (v.3.5.1), ggsankey (v.0.0.99999), ggridges (v.0.5.4), patchwork (v.1.1.3) and related R packages for visualization."

DOI ↗ Fig. 4 on PMC ↗

Fig. 3 TEXT

Repression of developmental transcription factor networks triggers aging-associated gene expression in human glial progenitor cells. *Nat Commun, 2024*

"...Graphs were further edited or generated anew using ggplot2 and aligned using patchwork."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

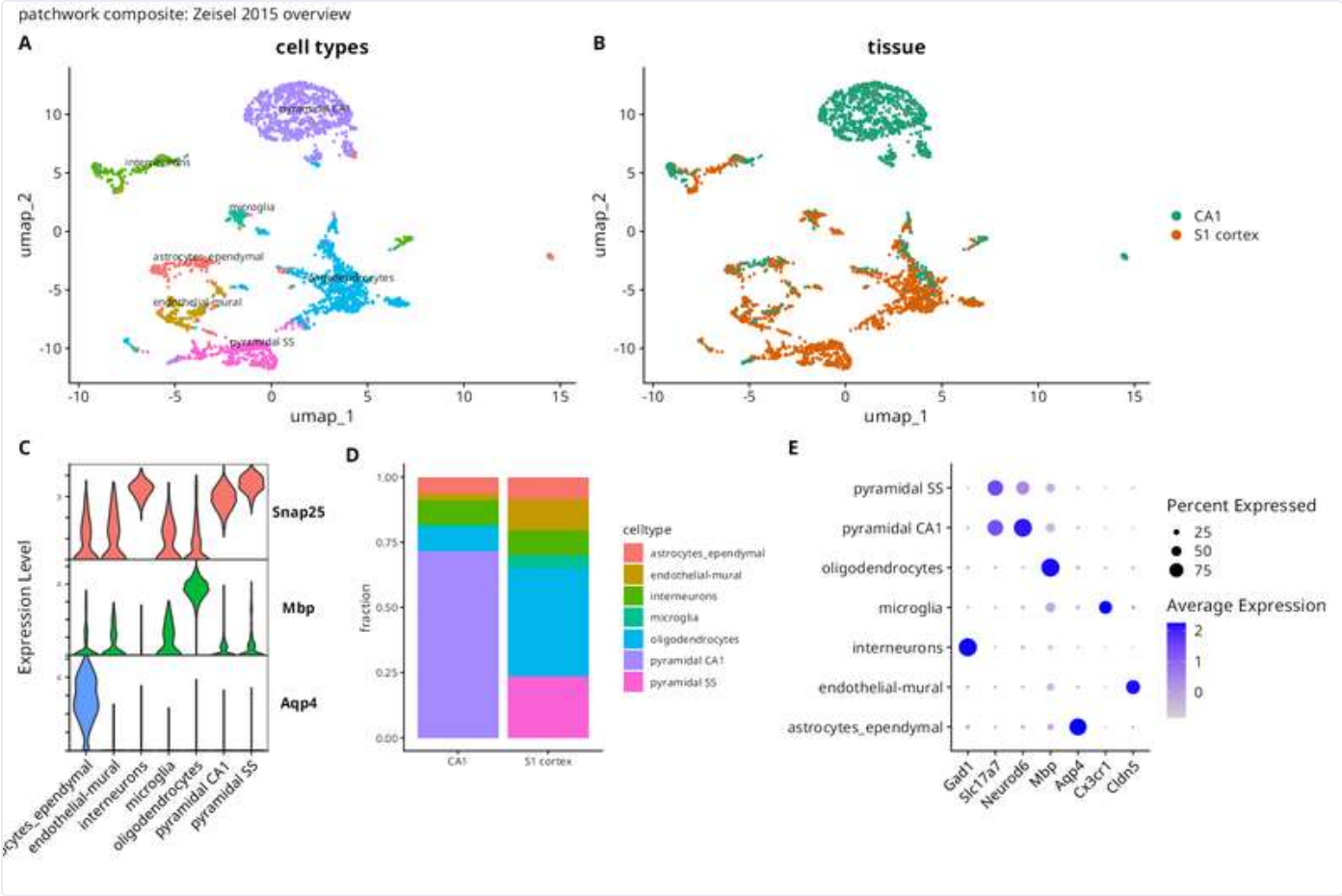
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/patchwork_cowplot/composite_recreate.R`

✓ re-run end to end · 12 s

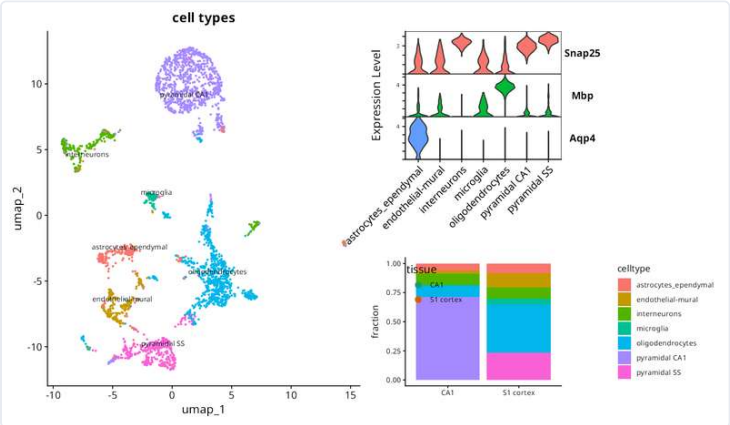
patchwork · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



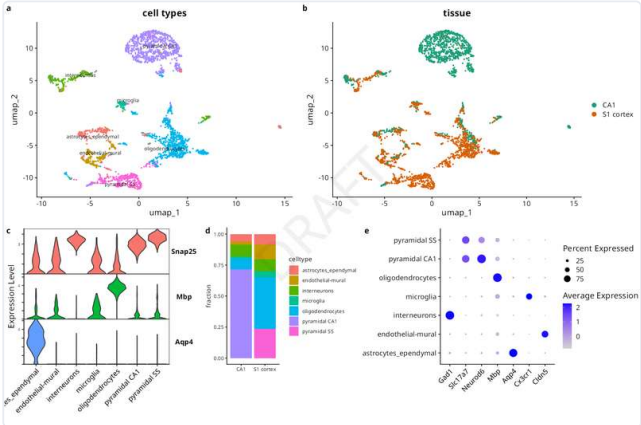
v1 patchwork_composite

COMPOSITION



v2 patchwork_design_inset

COMPOSITION



v3 cowplot_grid_watermark

COMPOSITION

Squidpy

#83

OF 100 · PAPERS NAMING IT

PYTHON | SPATIAL | SPATIAL & TISSUE MAPS

60	0.34%	74	analysis+plotting	6
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · SQUIDPY

Neuro-corpus papers per year (*2026 partial) · niche rank **#28**, growth **2.72×**, visual 4/5, score 0.736. Broad-corpus rate 1.358 → 3.708 per 1k papers (2023 → 2025–26).
GitHub scverse/squidpy · created 2020-08-21 · last push 2026-10-13

BEST FOR

Spatial scatter, neighbourhood enrichment, co-occurrence

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Squidpy: 10x Visium mouse coronal brain (H&E) + Moffitt 2018 MERFISH hypothalamus

KEY PLOTTING CALL

RECREATIONS/SQUIDPY/SQUIDPY_RECREATE.PY — EXCERPT · ENDS LINE 5

```
v=sc.read_h5ad("../_data/py/visium_hne.h5ad")
sq.pl.spatial_scatter(v,color="cluster",size=1.3,title="Visium mouse brain: annotated clusters on H&E"); plt.savefig("v1_clusters_on_HE.png",dpi=140,bbox_inches="tight"); plt.close("all")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 1 TEXT

Spatial transcriptomic clocks reveal cell proximity effects in brain ageing. *Nature, 2025*

"...To compute the distribution of nearest neighbour distances, we constructed a triangulation mesh graph connecting neighbouring cells on a given sample using squidpy.gr.spatial_neighbors with delaunay=True103."

DOI ↗ Article ↗ figure page not on PMC

Extended Data Fig. 4 TEXT

Spatial atlas of the mouse central nervous system at molecular resolution. *Nature, 2023*

"...The following packages and software were used in data analysis: UCSF ChimeraX 1.0, ImageJ 1.51, MATLAB R2019b, R 4.0.4, RStudio 1.4.1106, Jupyter Notebook 6.0.3, Anaconda 2-2-.02, h5py 3.1.0, hdbscan 0.8.36 ..."

DOI ↗ Extended Data Fig. 4 on PMC ↗

Fig. 3 TEXT

Mapping cells through time and space with moscot. *Nature, 2025*

"...To this end, we computed the neighbour graph in the aligned space using squidpy (kNN mode with at least 30 neighbours for each observation)."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

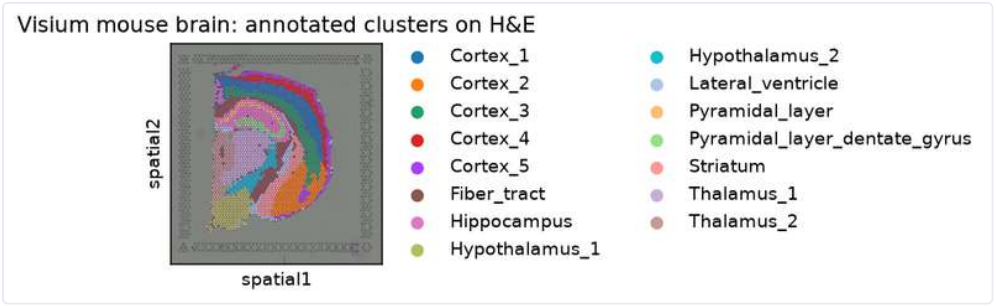
Reproduce: `python3 recreations/Squidpy/squidpy_recreate.py`

√ re-run end to end · 16 s

Squidpy · 6 variations

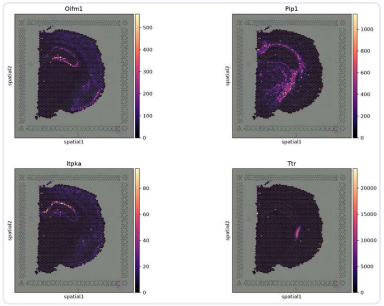
NOVAFLOW RECREATION

Data: 10x Visium mouse coronal brain section and Moffitt et al. 2018 MERFISH hypothalamus (squidpy example datasets). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



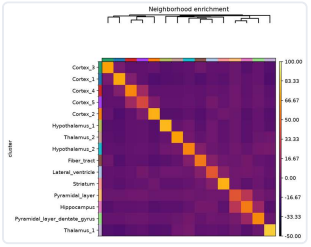
v1 clusters_on_HE

SPATIAL



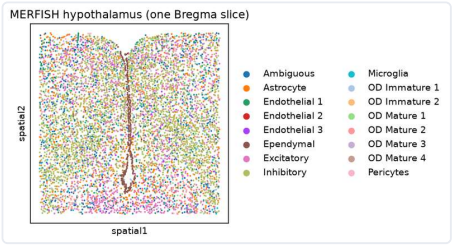
v2 genes_on_HE

SPATIAL



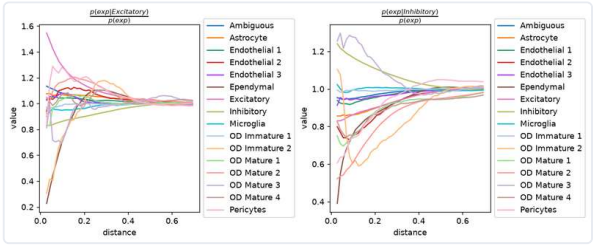
v3 nhood_enrichment

SPATIAL



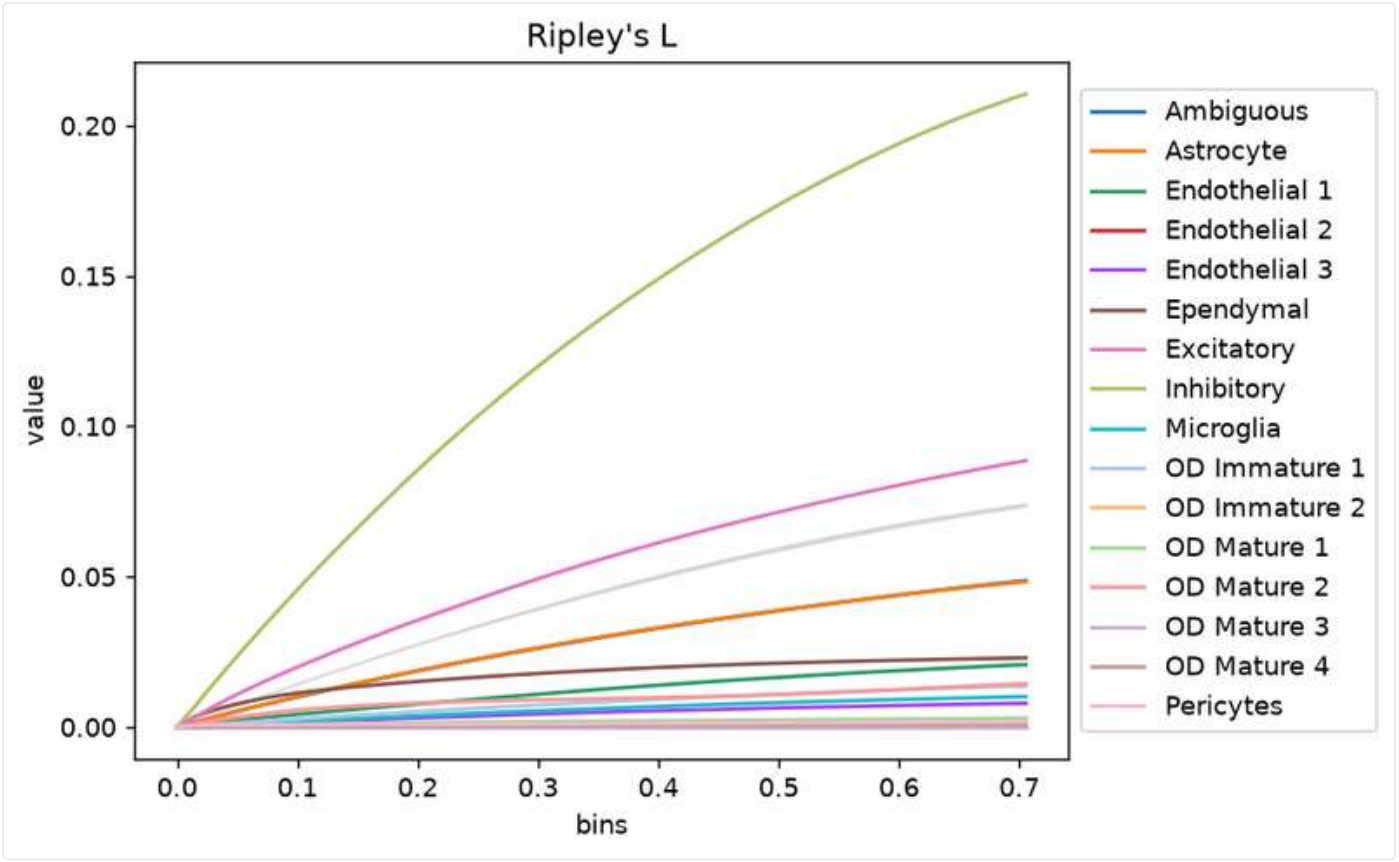
v4 merfish_cell_classes

SPATIAL



v5 co_occurrence

SPATIAL



v6 ripley_L

SPATIAL

plotly (3D)

#84

OF 100 · PAPERS NAMING IT

R/PYTHON/JS · INTERACTIVE · EMBEDDINGS & FEATURE MAPS · SPATIAL & TISSUE MAPS

56	0.32%	60	dedicated plotting	2
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · PLOTLY/DASH 3D

Neuro-corpus papers per year (*2026 partial) · niche rank **#56**, growth **1.36×**, visual 4/5, score 0.537. Broad-corpus rate 2.368 → 3.226 per 1k papers (2023 → 2025–26).
GitHub plotly/plotly.py · created 2013-11-21 · last push 2026-10-14

BEST FOR

Interactive 2D/3D plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Interactive 3D UMAP (HTML) + static PNG export (kaleido) on Zeisel; 3D MERFISH stack of hypothalamic slices

KEY PLOTTING CALL

RECREATIONS/PLOTLY_3D/PLOTLY3D_RECREATE.PY — EXCERPT · ENDS LINE 7

```
ad=load(); sc.pp.highly_variable_genes(ad,n_top_genes=2000); sc.pp.pca(ad); sc.pp.neighbors(ad); sc.tl.umap(ad,n_components=3)
u=ad.obsm["X_umap"]; f=px.scatter_3d(x=u[:,0],y=u[:,1],z=u[:,2],color=ad.obs.celltype.astype(str),opacity=.8,title="3D UMAP,
Zeisel cortex/hippocampus")
f.update_traces(marker_size=2); f.update_layout(scene=dict(xaxis_title="UMAP1",yaxis_title="UMAP2",zaxis_title="UMAP3"),legend_
itemsizing="constant",width=900,height=700)
f.write_html("v1_umap3d.html"); f.write_image("v1_umap3d.png",scale=1.5)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 6 TEXT

A brain cell atlas integrating single-cell transcriptomes across human brain regions. *Nat Med, 2024*

"...The 'Data Viewer' page and other visualization modules were built with Plotly (https://plot.ly)."

DOI ↗ Fig. 6 on PMC ↗

Fig. 2 TEXT

Mapping medically relevant RNA isoform diversity in the aged human frontal cortex with deep long-read RNA-seq. *Nat Biotechnol, 2025*

"...We used the following Python libraries: numpy (v.1.24.1), pandas (v.1.5.2), regex (v.2022.10.31), matplotlib (v.3.6.2), seaborn (v.0.12.2), matplotlib_venn (v.0.11.7), wordcloud (v.1.8.2.2), plotly (v.5.11.0) ..."

DOI ↗ Fig. 2 on PMC ↗

Fig. 1 TEXT

Multi-omic analysis of Huntington's disease reveals a compensatory astrocyte state. *Nat Commun, 2024*

"...Each t-SNE plot was created using the first three coordinates through the package plotly (version 4.10.0)."

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

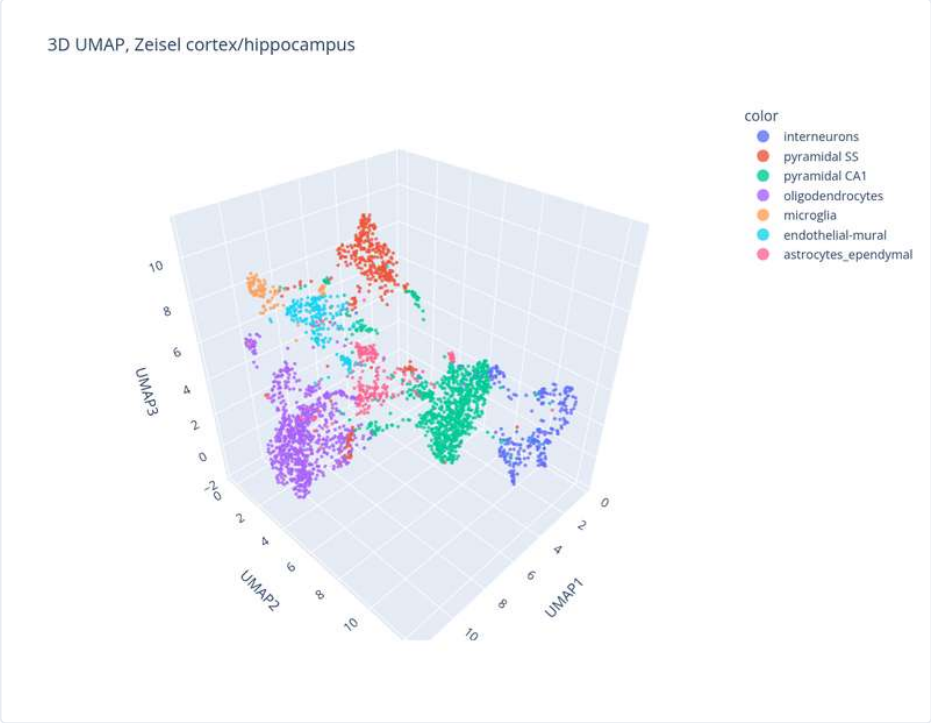
- The primary output is interactive HTML; the book shows the static PNG export (kaleido).

Reproduce: `python3 recreations/plotly_3D/plotly3d_recreate.py`

√ re-run end to end · 34 s

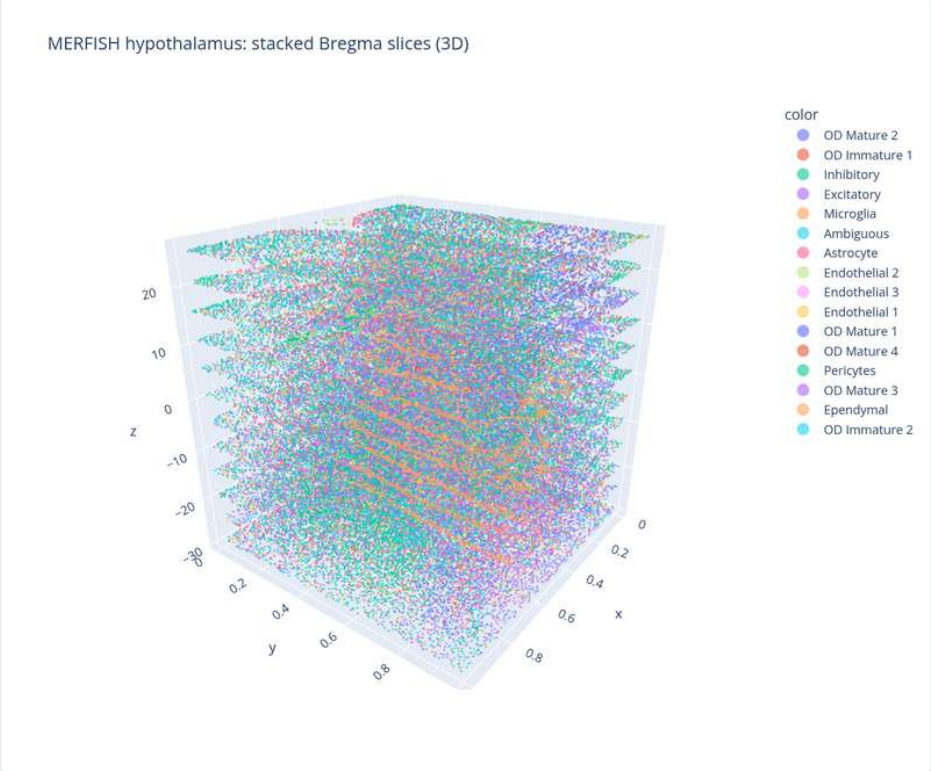
plotly (3D) · 2 variations

Data: Zeisel et al. 2015 scRNA-seq (3D UMAP) and Moffitt et al. 2018 MERFISH hypothalamus (stacked slices). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 umap3d

EMBEDDINGS



v2 merfish3d

SPATIAL

ggraph

#86

OF 100 · PAPERS NAMING IT

R · NETWORK · NETWORKS, GRAPHS & TREES

50	0.28%	49	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Network/dendrogram plots in ggplot grammar

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

ggraph / tidygraph: Zeisel 2015 marker co-expression network (force layout, communities), circular cell-type dendrogram, and circle-packing of cell-type → Louvain sub-cluster hierarchy.

KEY PLOTTING CALL

RECREATIONS/GGRAPH/GGRAPH_RECREATE.R — EXCERPT · ENDS LINE 10

```
X <- as.matrix(GetAssayData(so, layer = "data")[m$gene, ]); C <- cor(t(X)); C[lower.tri(C, diag = TRUE)] <- NA
e <- as.data.frame(as.table(C)) %>% filter(!is.na(Freq), Freq > .35); names(e) <- c("from","to","r")
g <- tbl_graph(nodes = data.frame(name = m$gene, celltype = m$cluster), edges = e, directed = FALSE) %>% activate(nodes) %>% mu
tate(deg = centrality_degree(), comm = as.factor(group_louvain()))
p1 <- ggraph(g, "fr") + geom_edge_link(aes(alpha = r), colour = "grey50") + geom_node_point(aes(colour = celltype, size = deg))
+ geom_node_text(aes(label = name), repel = TRUE, size = 2.4) + theme_graph(base_family = "sans") + ggtitle("Marker co-
expression network (r > 0.35)")
ggsave("v1_coexpression_network.png", p1, width = 10, height = 8, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 4 TEXT
Single-cell epigenomic reconstruction of developmental trajectories from pluripotency in human neural organoid systems. *Nat Neurosci, 2024*
"...Network visualization and analysis was performed with ggraph (TLP;"
DOI ↗ Fig. 4 on PMC ↗

Fig. 3 TEXT
Single-cell brain organoid screening identifies developmental defects in autism. *Nature, 2023*
"...The resulting graph was visualized with the R package ggraph (https://github.com/thomasp85/ggraph) using the circular tree layout."
DOI ↗ Fig. 3 on PMC ↗

Fig. 3 TEXT
Multimodal spatiotemporal phenotyping of human retinal organoid development. *Nat Biotechnol, 2023*
"...The cluster graph was then visualized using the ggraph package in R with the tree layout implemented in the igraph package (the layout_as_tree function), with the edge weights represented by the edge ..."
DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

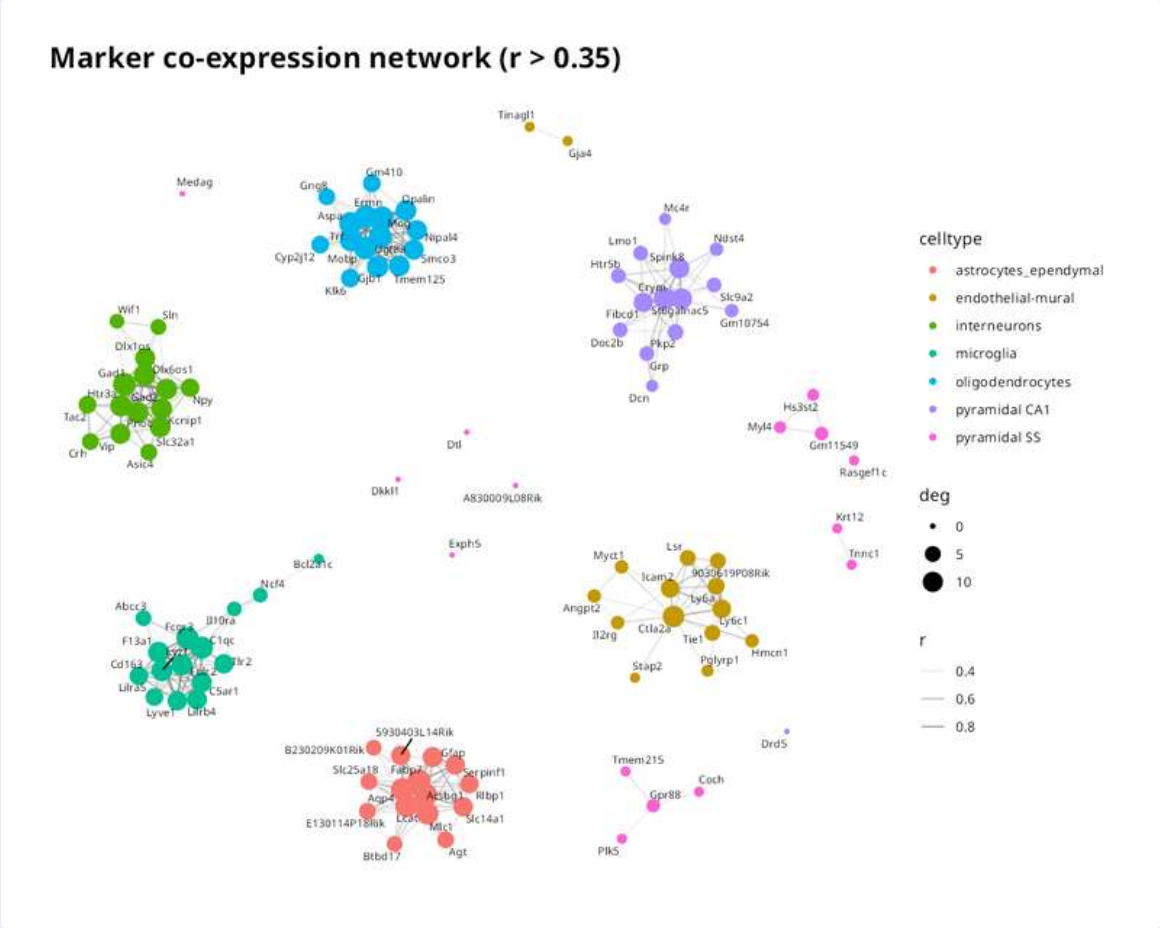
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/ggraph/ggraph_recreate.R`

✓ re-run end to end · 11 s

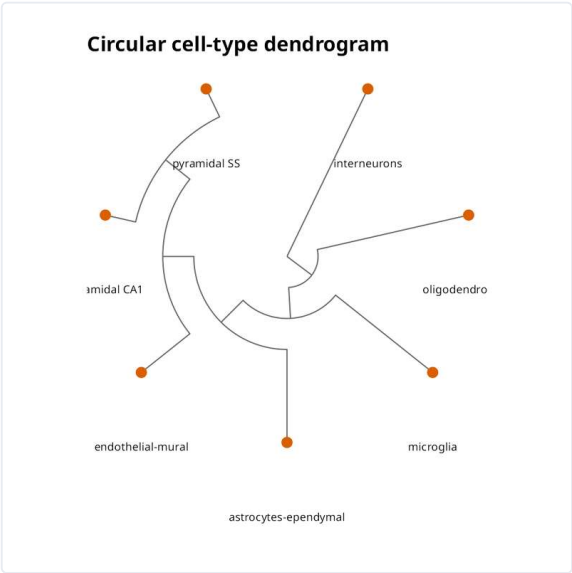
ggraph · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



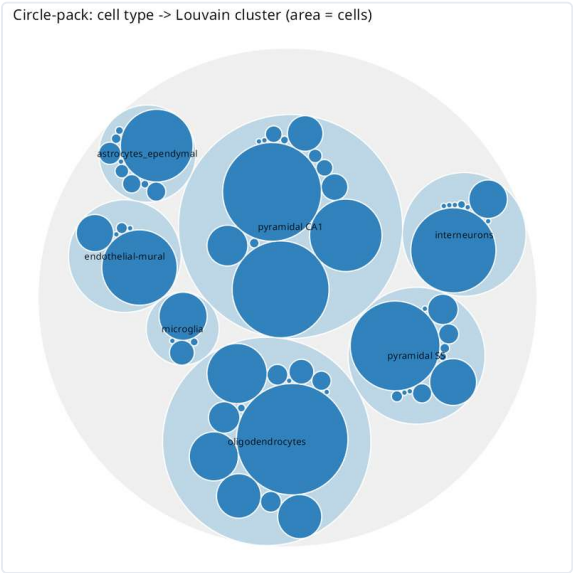
v1 coexpression_network

NETWORKS



v2 circular_dendrogram

NETWORKS



v3 circlepack

NETWORKS

Tangram

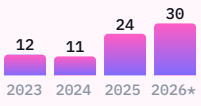
#87

OF 100 · PAPERS NAMING IT

PYTHON | SPATIAL MAPPING | SPATIAL & TISSUE MAPS

49	0.28%	77	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS



NICHE & COOL · TANGRAM

Neuro-corpus papers per year (*2026 partial) · niche rank **#44**, growth **1.83×**, visual 4/5, score 0.598. Broad-corpus rate 1.498 → 2.756 per 1k papers (2023 → 2025–26).
GitHub broadinstitute/Tangram · created 2020-08-21 · last push 2025-07-01

BEST FOR

Projected cell types on spatial maps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Tangram: map Zeisel 2015 cortex/hippocampus single cells onto the 10x Visium mouse-brain H&E section (squidpy), then project cell-type probabilities onto the tissue; plus training-gene score diagnostics.

KEY PLOTTING CALL

RECREATIONS/TANGRAM/TANGRAM_RECREATE.PY — EXCERPT · ENDS LINE 17

```
tg.pp_adatas(ad_sc,ad_sp,genes=markers)
m=tg.map_cells_to_space(ad_sc,ad_sp,mode="clusters",cluster_label="celltype",device="cpu",num_epochs=500,density_prior="rna_cou
nt_based",verbose=False)
tg.project_cell_annotations(m,ad_sp,annotation="celltype")
ct=list(ad_sp.obsm["tangram_ct_pred"].columns)
P=ad_sp.obsm["tangram_ct_pred"]; P=(P-P.min())/(P.max()-P.min())
for c in ct: ad_sp.obs[f"tg_{c}"]=P[c].values
sq.pl.spatial_scatter(ad_sp,color=[f"tg_{c}" for c in ct],ncols=4,cmap="magma",size=1.4,img_alpha=.5,figsize=(4,4),save=None)
plt.gcf().suptitle("Tangram: Zeisel cell types mapped onto Visium mouse brain",y=1.02);
plt.savefig("v1_celltype_projection.png",dpi=120,bbox_inches="tight"); plt.close("all")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Extended Data Fig. 12 — FIGURE CAPTION

Spatial transcriptomic clocks reveal cell proximity effects in brain ageing. *Nature, 2025*

"...Identification and validation of potential mediators of cell proximity effects.a, Schematic of the computational pipeline using SpaGE and Tangram algorithms for spatial gene expression imputation and using ..."

DOI ↗ Article ↗ figure page not on PMC

Fig. 3 — FIGURE CAPTION

A single-cell time-lapse of mouse prenatal development from gastrula to birth. *Nature, 2024*

"...mental stage. SMC, smooth muscle cell. e, The spatial origin of each lateral plate and intermediate mesoderm derivative was inferred based on a public dataset, Mosta46, together with our data and the Tangram ..."

DOI ↗ Fig. 3 on PMC ↗

Fig. 3 — FIGURE CAPTION

Inferring histology-associated gene expression gradients in spatial transcriptomic studies. *Nat Commun, 2024*

"...se the expression profiles of co-expressed genes identified via spatial annotation screening. k UMAP of the scRNA-seq dataset, from the same mouse model as the Visium sample. l and m Visualization of Tangram ..."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

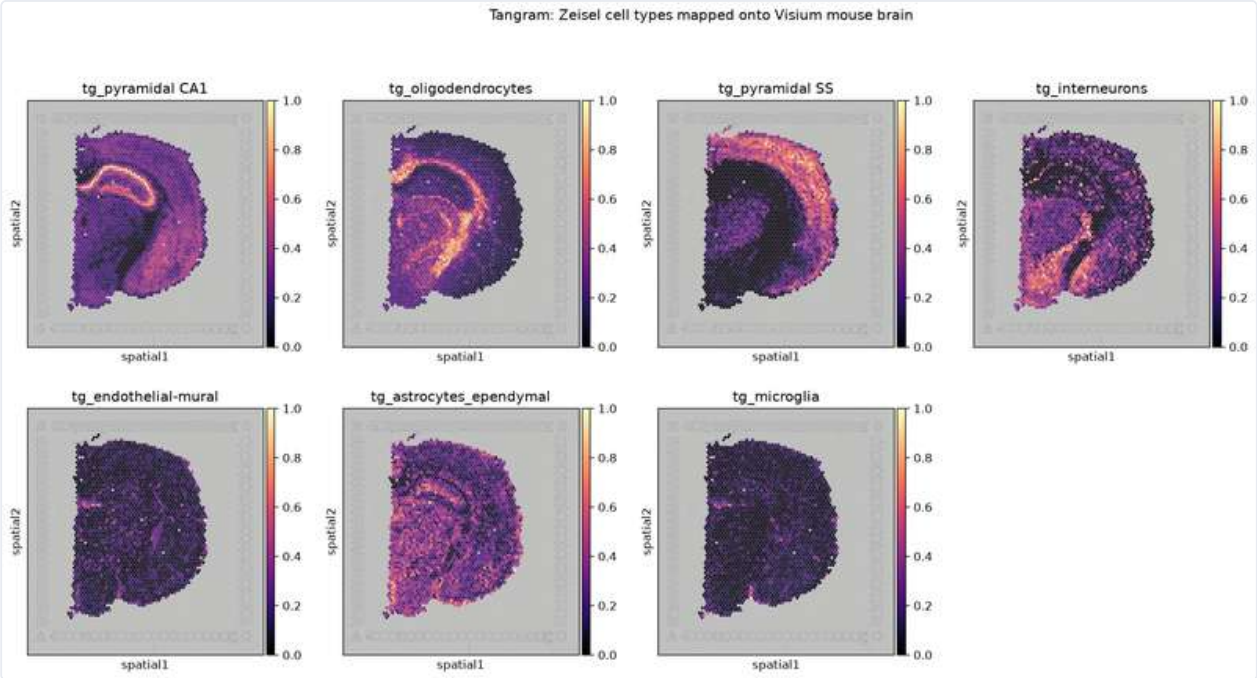
Reproduce: `python3 recreations/Tangram/tangram_recreate.py`

√ re-run end to end · 24 s

Tangram · 3 variations

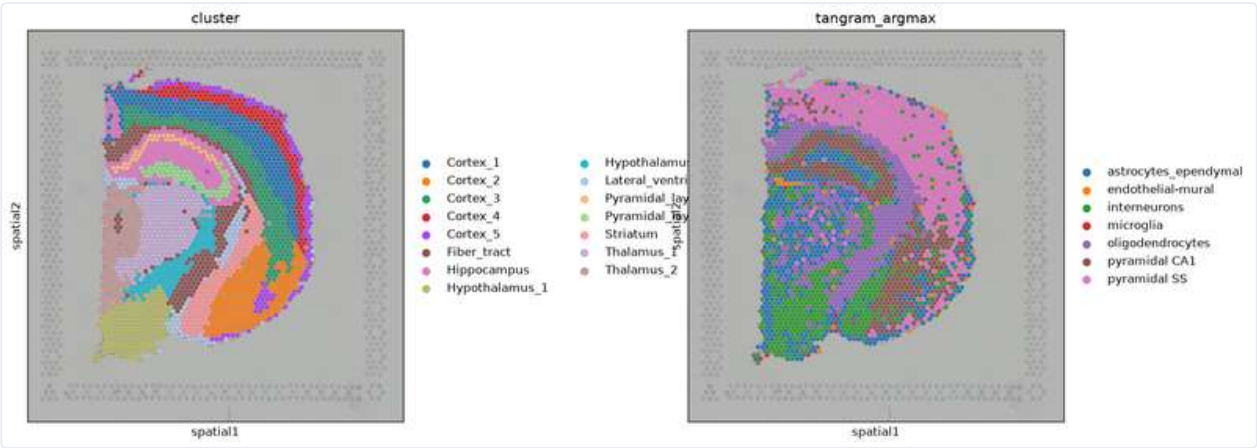
NOVAFLOW RECREATION

Data: Zeisel et al. 2015 scRNA-seq mapped onto the 10x Visium mouse coronal brain section (squidpy example). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



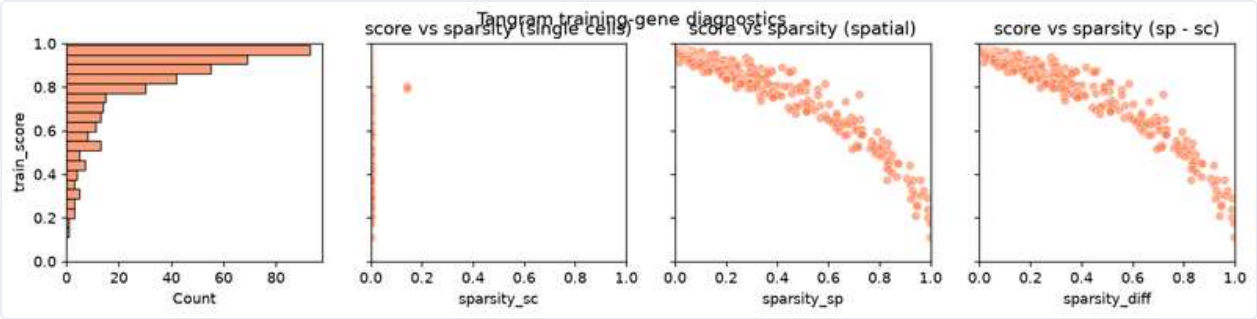
v1 celltype_projection

SPATIAL



v2 argmax_vs_clusters

SPATIAL



v3 training_scores

SPATIAL

PCAtools

#88

OF 100 · PAPERS NAMING IT

R · PCA · DE, QC & SET OVERLAPS · HEATMAPS & DOT MATRICES

48	0.27%	48	dedicated plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Biplots, scree, eigencor plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

PCAtools on Zeisel 2015 pseudobulk (C1 chip x cell type, >=10 cells): scree + Horn/elbow, biplot, eigencor, pairs, loadings

KEY PLOTTING CALL

RECREATIONS/PCATOOLS/PCATOOLS_RECREATE.R — EXCERPT · ENDS LINE 10

```
md <- unique(so@meta.data[, c("pb","chip","celltype","tissue")]); rownames(md) <- md$pb; md <- md[colnames(pb), ]; md$n_cells
<- as.numeric(table(so$pb)[rownames(md)]); md$log_lib <- log10(colSums(pb))
lc <- cpm(pb[rowSums(pb >= 10) >= 5, ], log = TRUE, prior.count = 2); p <- pca(lc, metadata = md, removeVar = .9)
horn <- parallelPCA(lc, removeVar = .9); elbow <- findElbowPoint(p$variance)
ggsave("v1_screeplot.png", screeplot(p, components = getComponents(p, 1:15), vline = c(horn$n, elbow), title = "Scree (Horn +
elbow)") + geom_label(aes(x = horn$n + 1, y = 50, label = "Horn", vjust = -1, size = 8)) + geom_label(aes(x = elbow + 1, y =
50, label = "Elbow", vjust = -1, size = 8)), width = 8, height = 5, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 4 TEXT

Metabolic switch in the aging astrocyte supported via integrative approach comprising network and transcriptome analyses. (Aging (Albany NY), 2023). ,

"...In the same manner, PCA was carried out in R by using the library PCAtools (https://github.com/kevinblighe/PCAtools)."

DOI ↗ Figure 4 on PMC ↗

Figure 1 TEXT

Brain transcriptomics reveal the activation of neuroinflammation pathways during acute Orientia tsutsugamushi infection in mice. *Front Immunol, 2023*

"...Principal component analysis and volcano plot were done in R using PCAtools (34) and EnhancedVolcano (35) packages."

DOI ↗ Figure 1 on PMC ↗

Fig. 1 TEXT

Translatome analysis of tuberous sclerosis complex 1 patient-derived neural progenitor cells reveals rapamycin-dependent and independent alterations. *Mol Autism, 2023*

"...parameters removeVar = 0.75 and scale = T) and visualized using eigencorplot, screeplot and biplot functions from the PCAtools R package (v.2.4.0;"

DOI ↗ Fig. 1 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

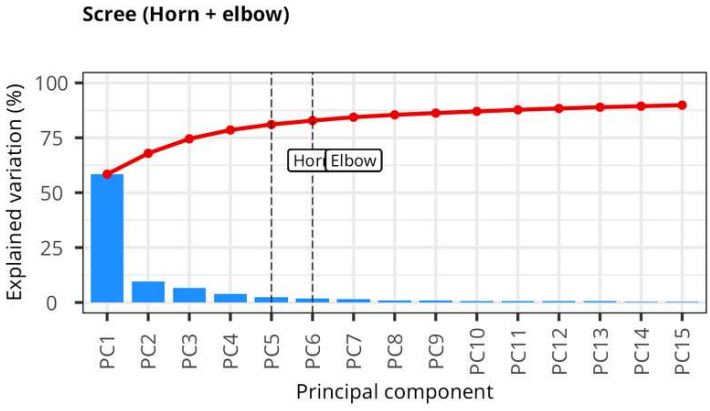
Reproduce: `Rscript recreations/PCAtools/PCAtools_recreate.R`

√ re-run end to end · 21 s

PCAtools · 4 variations

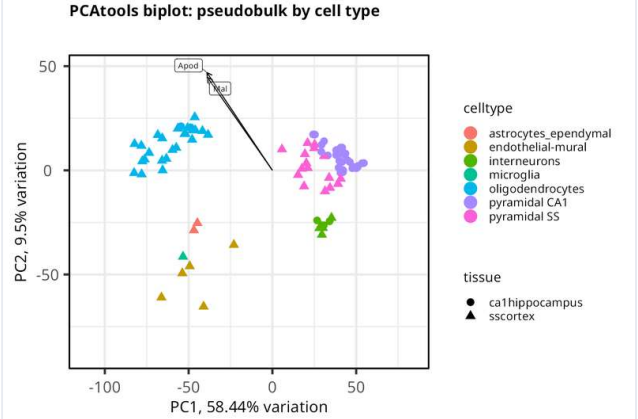
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



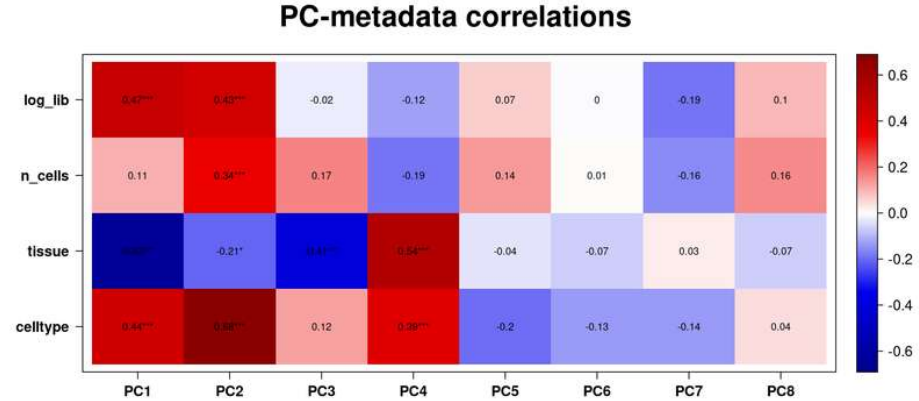
v1 screeplot

DE



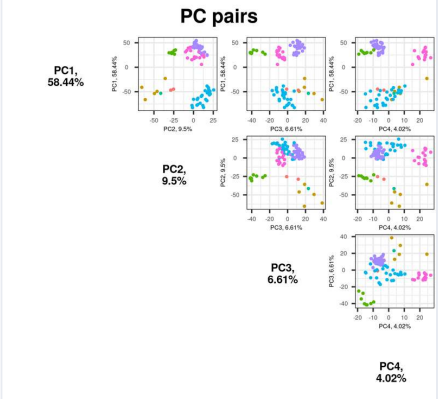
v2 biplot

DE



v3 eigencorplot

HEATMAPS



v4 pairsplot

DE

tradeSeq

#91

OF 100 · PAPERS NAMING IT

R | TRAJECTORY DE | TRAJECTORY, PSEUDOTIME & VELOCITY

46	0.26%	51	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

BEST FOR

Smoothed gene expression along lineages

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

tradeSeq GAM smoothers along slingshot lineages (La Manno 2018 dentate gyrus): smoothed gene trends, association-test heatmap of top dynamic genes, and a single-gene smoother on UMAP. Uses slingshot's fit (runs it if absent).

KEY PLOTTING CALL

RECREATIONS/TRADESEQ/TRADESEQ_RECREATE.R — EXCERPT · ENDS LINE 13

```r
show <- head(unique(c(intersect(c("Dcx","Sox11","Calb1","Eomes","Neurod1","Prox1"), genes), top)), 4)
ps <- lapply(show, function(g) plotSmoothers(fit, cnt[genes, ], gene = g, alpha = .4, border = TRUE) + ggtitle(g))
ggsave("v1_plotSmoothers.png", wrap_plots(ps, ncol = 2), width = 10, height = 7.5, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 5 — FIGURE CAPTION

A topographic atlas defines developmental origins of cell heterogeneity in the human embryonic lung. *Nat Cell Biol, 2023*

"...es module 1: 0.60, module 2: 0.69, module 3: 0.84, module 4: 0.57, module 5: 0.80, module 6: 0.73, module 7: 0.61, module 8: 0.55, module 9: 0.85) along the two trajectories, shown in a, according to tradeSeq ..."

DOI ↗ Fig. 5 on PMC ↗

Extended Data Fig. 6 — FIGURE CAPTION

Multiplexing cortical brain organoids for the longitudinal dissection of developmental traits at single-cell resolution. *Nat Methods, 2025*

"...of cells per annotated cell type (shown as reference on the right side), weighted edges among PAGA clusters encode their transcriptional similarity. b) Plot of smoothed gene expression - obtained via tradeseq ..."

DOI ↗ Extended Data Fig. 6 on PMC ↗

Fig. 6 — FIGURE CAPTION

Inferring histology-associated gene expression gradients in spatial transcriptomic studies. *Nat Commun, 2024*

"...Statistical comparison of the test statistic from SPATA2's spatial gradient screening with those of analogous methods PseudotimeDE and tradeSeq.a–c Scatterplots display the relationship between different test ..."

DOI ↗ Fig. 6 on PMC ↗

LIMITATIONS & NOTES

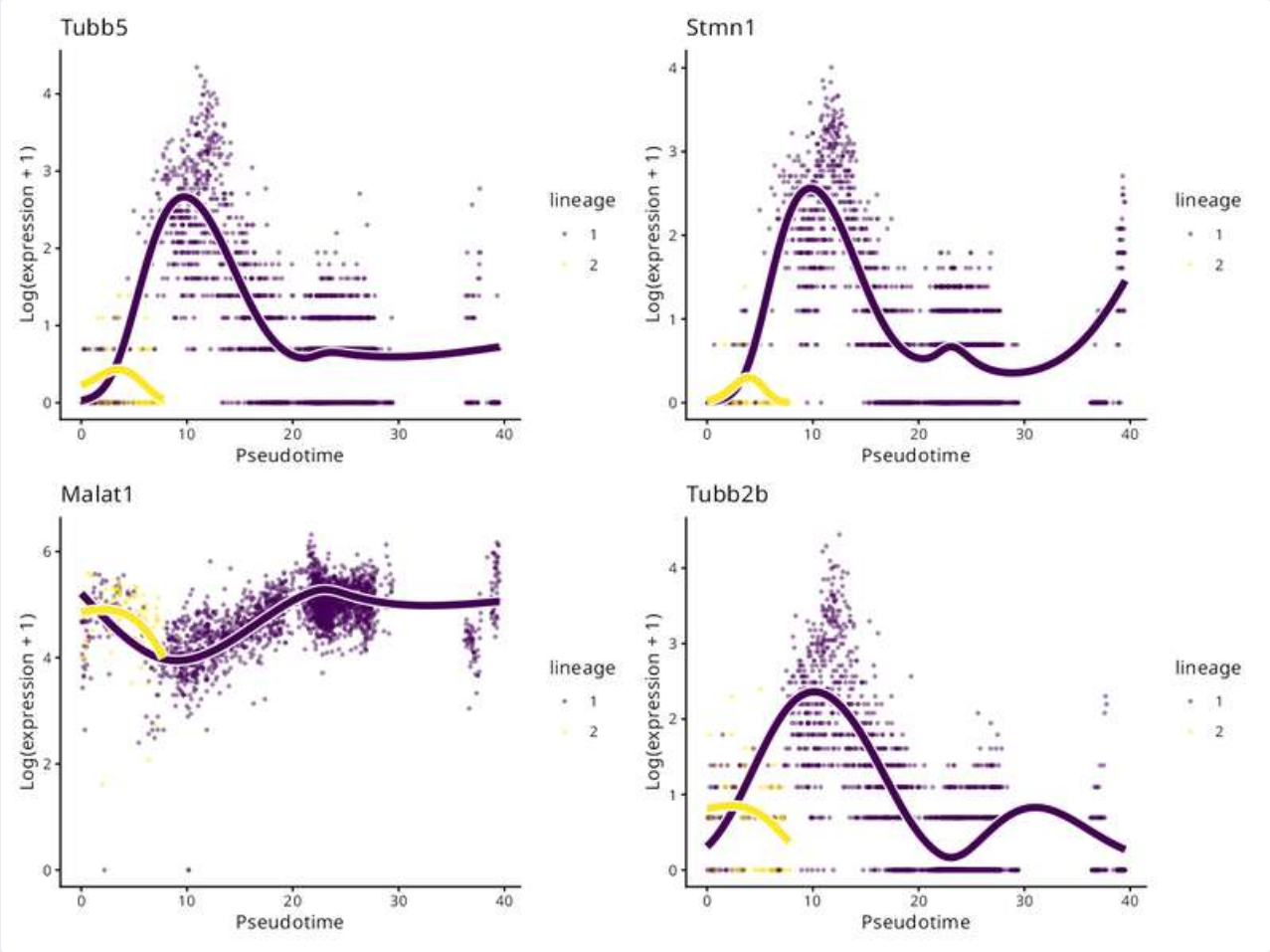
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/tradeSeq/tradeSeq_recreate.R`

✓ re-run end to end · 12 s

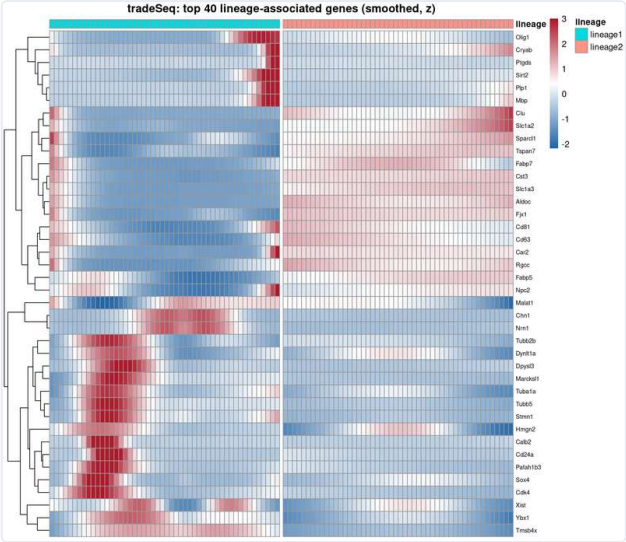
tradeSeq · 3 variations

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



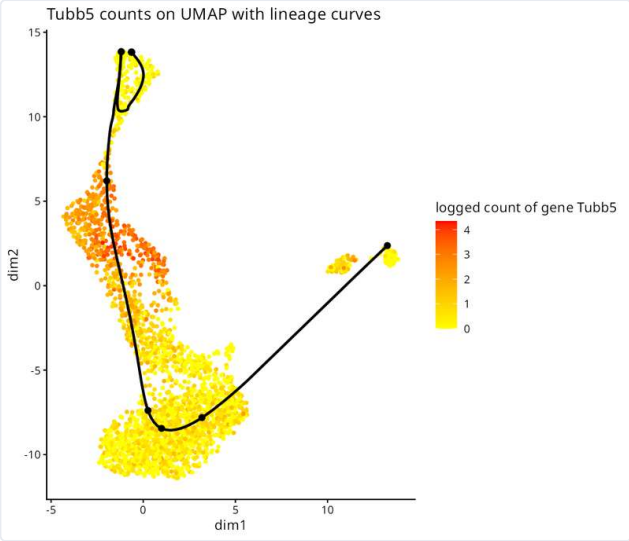
v1 plotSmoothers

TRAJECTORY



v2 dynamic_gene_heatmap

TRAJECTORY



v3 plotGeneCount

TRAJECTORY

LIANA+

#92

OF 100 · PAPERS NAMING IT

R/PYTHON | CELL-CELL COMMUNICATION | CELL-CELL COMMUNICATION

45	0.25%	67	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · LIANA+

Neuro-corpus papers per year (*2026 partial) · niche rank **#23**, growth **3.78×**, visual 4/5, score 0.77. Broad-corpus rate 0.836 → 3.188 per 1k papers (2023 → 2025–26).
GitHub saezlab/liana-py · created — · last push —

BEST FOR

L-R consensus dot plots / tileplots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

LIANA+ consensus ligand-receptor inference on Zeisel cortex/hippocampus (mouse consensus resource)

KEY PLOTTING CALL

RECREATIONS/LIANA_PLUS/LIANA_RECREATE.PY — EXCERPT · ENDS LINE 8

```
ad=load(); ad.obs["celltype"]=ad.obs.celltype.astype(str).astype("category")
li.mt.rank_aggregate(ad,groupby="celltype",resource_name="mouseconsensus",expr_prop=.1,use_raw=False,verbose=False,n_perms=200,seed=1)
p=li.pl.dotplot(adata=ad,colour="magnitude_rank",size="specificity_rank",inverse_size=True,inverse_colour=True,source_labels=["astrocytes_ependymal","microglia","endothelial-mural"],target_labels=["interneurons","pyramidal CA1","pyramidal SS","oligodendrocytes"],top_n=20,orderby="magnitude_rank",orderby_ascending=True,figure_size=(10,7))
p.save("v1_dotplot.png",dpi=130,verbose=False)
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Extended Data Fig. 9 — FIGURE CAPTION

Axonal injury is a targetable driver of glioblastoma progression. *Nature, 2025*

"...- mice. The dashed line represents equal proportions in both genotypes. Cell types with PPearson's chi-squared test <0.05 and relative difference > 10% were considered significantly different (§). e, LIANA ..."

DOI ↗ Extended Data Fig. 9 on PMC ↗

Extended Data Fig. 9 — FIGURE CAPTION

Longitudinal single-cell multiomic atlas of high-risk neuroblastoma reveals chemotherapy-induced tumor microenvironment rewiring. *Nat Genet, 2025*

"...nt macrophage subsets from DX and PTX samples, upregulated post-therapy using logistic regression with no multiple comparison adjustment (Methods) e) Cell-cell interaction analysis using CellChat and LIANA ..."

DOI ↗ Extended Data Fig. 9 on PMC ↗

Figure 3 — TEXT

Tumor heterogeneity and tumor-microglia interactions in primary and recurrent IDH1-mutant gliomas. *Cell Rep Med, 2023*

"...Using LIANA,26 we predicted enriched receptor-ligand interactions between tumor populations (astro-like and OPC-like) and TAM subpopulations (Figures 3F and 3G)."

DOI ↗ Figure 3 on PMC ↗

LIMITATIONS & NOTES

- v2 is a LIANA circle plot because tileplot is broken with current plotnine/mizani.
- Run this first: its tables feed pyCirclize, CellPhoneDB_ktplots and circlize_chord (METHODS §9).

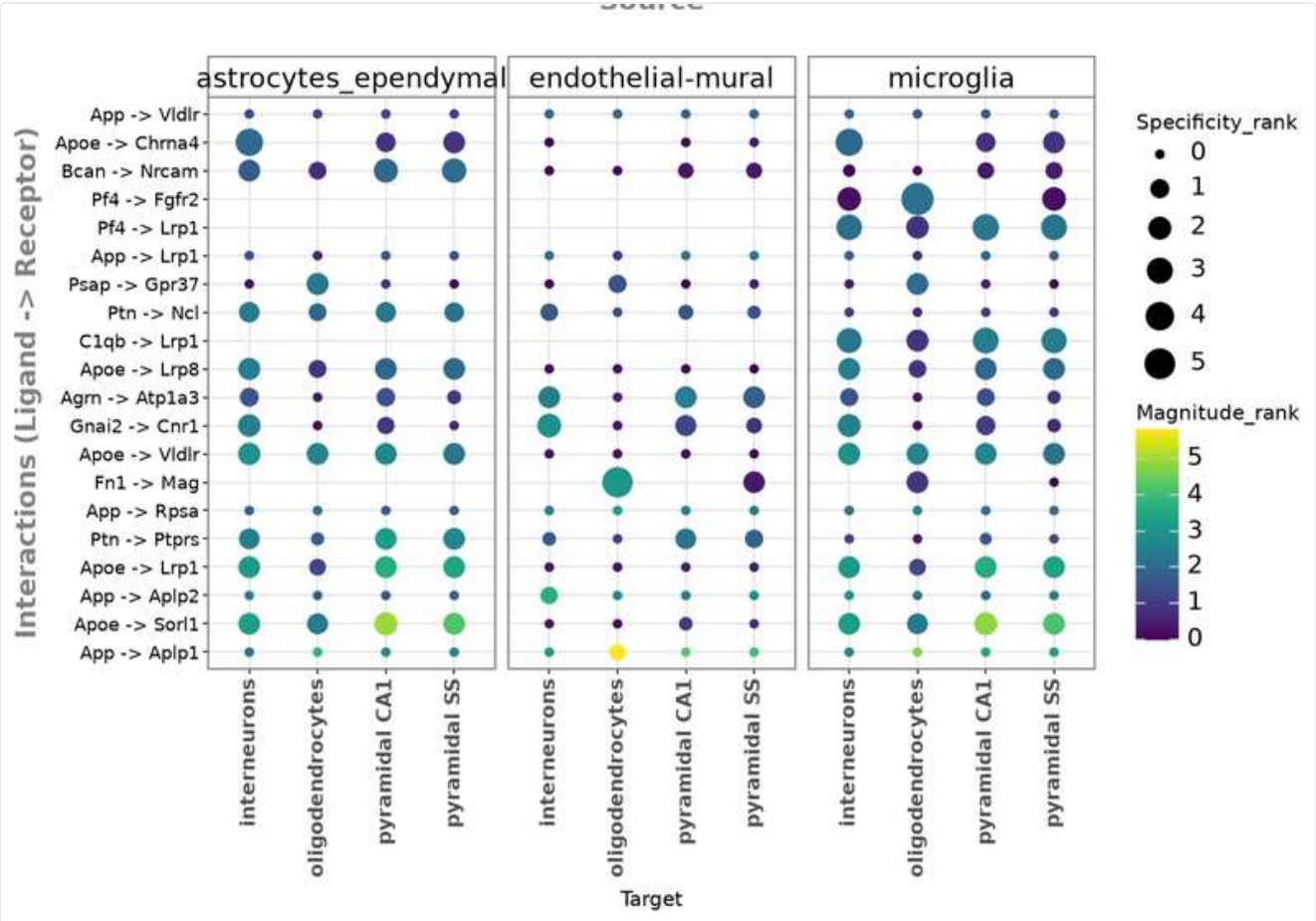
Reproduce: `python3 recreations/LIANA_plus/liana_recreate.py`

√ re-run end to end · 24 s

LIANA+ · 3 variations

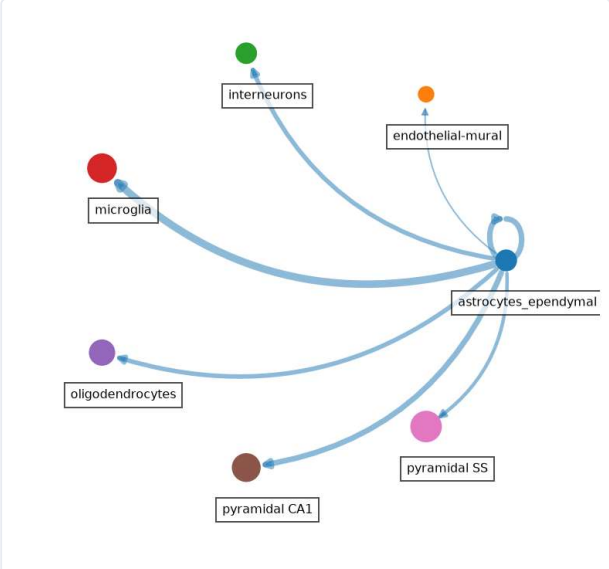
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



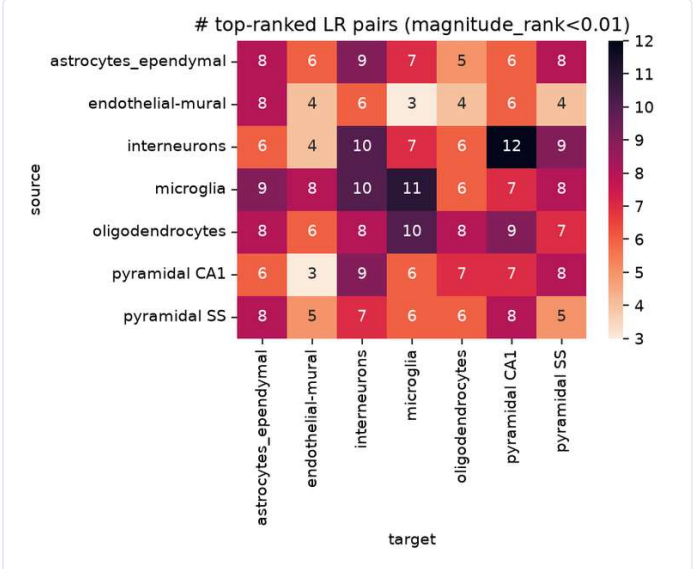
v1 dotplot

CELL–CELL



v2 circle_plot

CELL–CELL



v3 interaction_count_heatmap

CELL–CELL

BayesSpace

#97

OF 100 · PAPERS NAMING IT

R · SPATIAL CLUSTERING · SPATIAL & TISSUE MAPS

41	0.23%	67	analysis+plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · BAYESSPACE

Neuro-corpus papers per year (*2026 partial) · niche rank **#32**, growth **2.19×**, visual 4/5, score 0.671. Broad-corpus rate 1.289 → 2.832 per 1k papers (2023 → 2025–26).
GitHub edward130603/BayesSpace · created 2020-06-23 · last push 2026-01-08

BEST FOR

Enhanced-resolution spatial cluster maps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

BayesSpace on Maynard 2021 human DLPFC (151673): spot clusters, enhanced sub-spot resolution, enhanced gene maps

KEY PLOTTING CALL

RECREATIONS/BAYESSPACE/BAYESSPACE_RECREATE.R — EXCERPT · ENDS LINE 8

```
sce <- spatialPreprocess(sce, platform = "Visium", n.PCs = 7, n.HVGs = 2000, log.normalize = FALSE)
sce <- spatialCluster(sce, q = 7, platform = "Visium", d = 7, init.method = "mclust", model = "t", gamma = 2, nrep = 5000, burn.
in = 500)
pal <- c("#E41A1C","#377EB8","#4DAF4A","#984EA3","#FF7F00","#FFFF33","#A65628")
ggsave("v1_clusterPlot_vs_truth.png", patchwork::wrap_plots(list(clusterPlot(sce, palette = pal, color = NA) + ggtitle("BayesSp
ace (q=7)"), clusterPlot(sce, label = "ground_truth", palette = c(pal, "grey"), color = NA) + ggtitle("Manual layers (Maynard
2021)")), nrow = 1), width = 11, height = 5, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 4 — FIGURE CAPTION

BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. *Nat Genet, 2024*

"...use hippocampus.a, BANKSY separates CA3 neurons from SSC neurons. b–h, Comparisons with nonspatial clustering (λ = 0) and several spatial clustering methods (b, nonspatial; c, MERINGUE; d, Giotto; e ..."

DOI ↗ Fig. 4 on PMC ↗

Fig. 2 — FIGURE CAPTION

Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST. *Nat Commun, 2023*

"...quartiles, and whiskers denote the 1.5× interquartile range. B H&E image and manual annotation from the original study. C Clustering results by nonspatial and spatial methods, Seurat, Giotto, SpaGCN ..."

DOI ↗ Fig. 2 on PMC ↗

Figure 2—figure supplement 5 — FIGURE CAPTION

The gene expression landscape of the human locus coeruleus revealed by single-nucleus and spatially-resolved transcriptomics. *Elife, 2024*

"...ve technical variation in the molecular measurements between the n=8 Visium samples from 4 donors. The integrated measurements were subsequently used as the input for spatially-aware clustering using ..."

DOI ↗ Figure 2—figure supplement 5 on PMC ↗

LIMITATIONS & NOTES

- Enhanced-gene maps use the same per-gene PC regression that enhanceFeatures(nrounds=0) uses; the full sub-spot assay runs out of memory (METHODS §9).

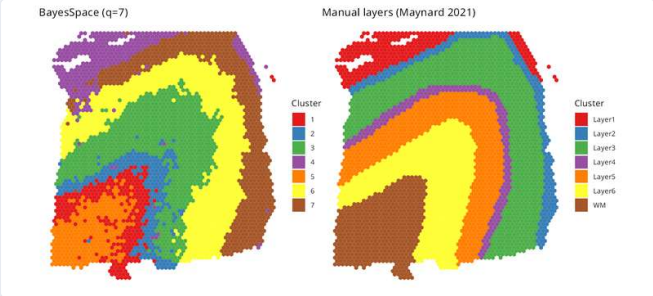
Reproduce: `Rscript recreations/BayesSpace/BayesSpace_recreate.R`

√ re-run end to end · 159 s

BayesSpace · 3 variations

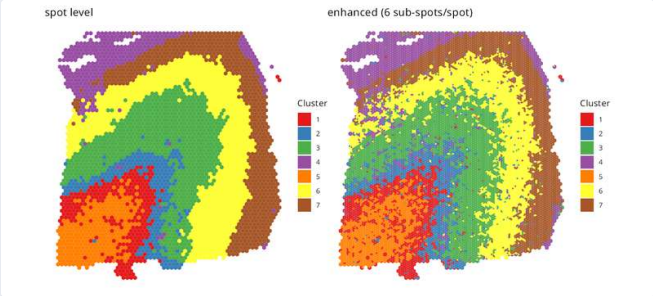
NOVAFLOW RECREATION

Data: Maynard et al. 2021, human DLPFC Visium sample 151673 (STexampleData). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



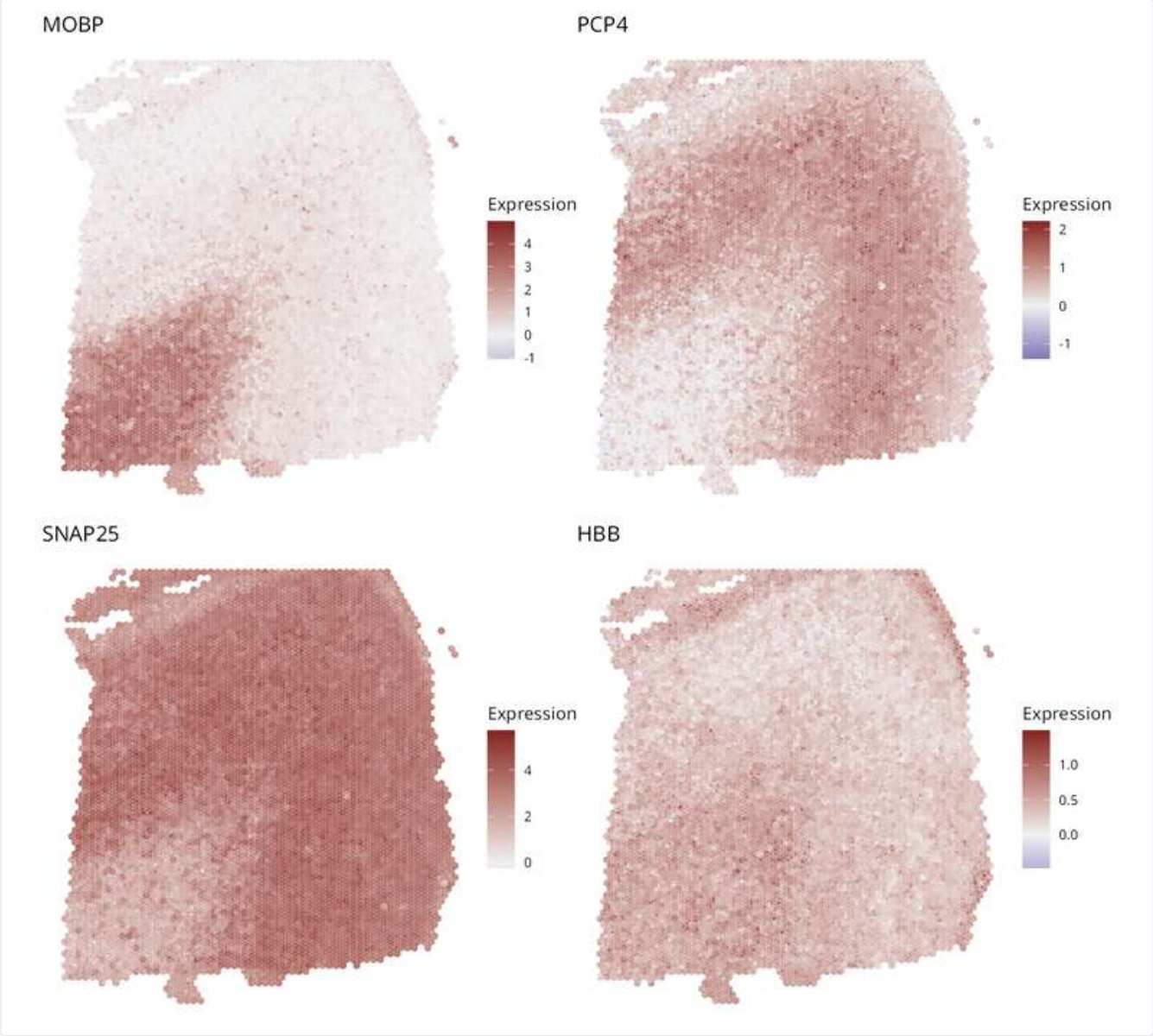
v1 clusterPlot_vs_truth

SPATIAL



v2 enhanced_subspot

SPATIAL



v3 enhanced_genes

SPATIAL

CellRank

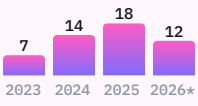
#98

OF 100 · PAPERS NAMING IT

PYTHON | FATE MAPPING | TRAJECTORY, PSEUDOTIME & VELOCITY

40	0.23%	51	analysis+plotting	4
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS



NICHE & COOL · CELLRANK

Neuro-corpus papers per year (*2026 partial) · niche rank **#35**, growth **1.55×**, visual 5/5, score 0.658. Broad-corpus rate 1.428 → 2.223 per 1k papers (2023 → 2025–26).
GitHub theislab/cellrank · created — · last push —

BEST FOR

Fate probabilities, driver genes, circular projections

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

CellRank 2 on dentate gyrus: macrostates, fate probabilities, circular projection, driver-gene trends

KEY PLOTTING CALL

RECREATIONS/CELLRANK/CELLRANK_RECREATE.PY — EXCERPT · ENDS LINE 6

```
a=load(); vk=cr.kernels.VelocityKernel(a).compute_transition_matrix(); ck=cr.kernels.ConnectivityKernel(a).compute_transition_m
atrix(); k=0.8*vk+0.2*ck
g=cr.estimators.GPCCA(k); g.fit(cluster_key="clusters",n_states=[4,10]);
g.plot_macrostates(which="all",basis="umap",legend_loc="right",s=60,show=False);
plt.gcf().savefig("v1_macrostates.png",dpi=150,bbox_inches="tight"); plt.close("all")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 4 — FIGURE CAPTION

Mapping cells through time and space with moscot. *Nature, 2025*

"…f moscot compared with TOME7. c, Mapping heart cells across time points (bottom) and ground-truth annotation of the heart lineage (top). d, Heart-lineage driver genes found by interfacing moscot with CellRank …"

DOI ↗ Fig. 4 on PMC ↗

Extended Data Fig. 2 — FIGURE CAPTION

Inferring and perturbing cell fate regulomes in human brain organoids. *Nature, 2023*

"…ster identities, expression patterns of cluster markers. Volcano plot shows differentially expressed (DE) genes of cluster 5 relative to other clusters. f, UMAP embedding coloured by rank-transformed CellRank …"

DOI ↗ Extended Data Fig. 2 on PMC ↗

Fig. 6 — FIGURE CAPTION

Tracking single-cell evolution using clock-like chromatin accessibility loci. *Nat Biotechnol, 2025*

"…EpiTrace reveals the developmental history during human cortical gyrification.a, UMAP projected cell evolution trajectory built with CellRank by using a hybrid kernel of EpiTrace, CytoTRACE and RNA velocity of …"

DOI ↗ Fig. 6 on PMC ↗

LIMITATIONS & NOTES

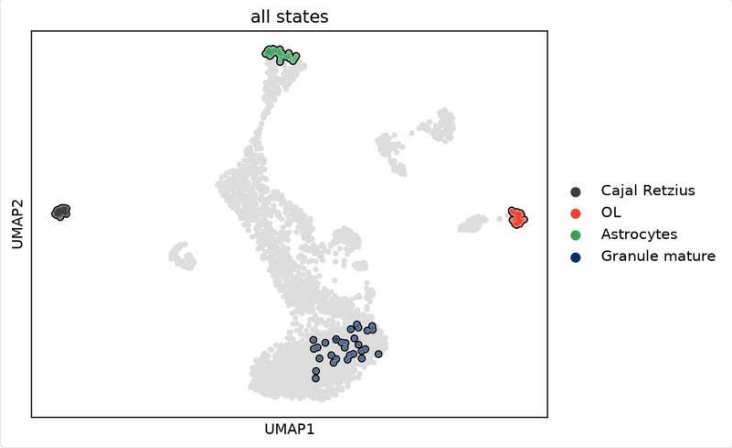
- v4 shows lineage drivers because GAM gene-trend models fail with current pygam/scipy.

Reproduce: `python3 recreations/CellRank/cellrank_recreate.py`

√ re-run end to end · 21 s

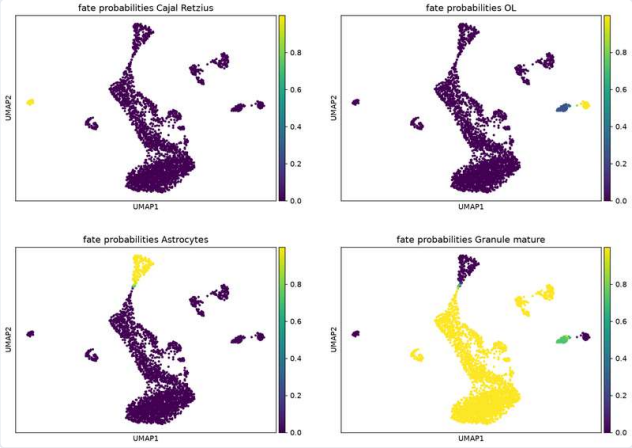
CellRank · 4 variations

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



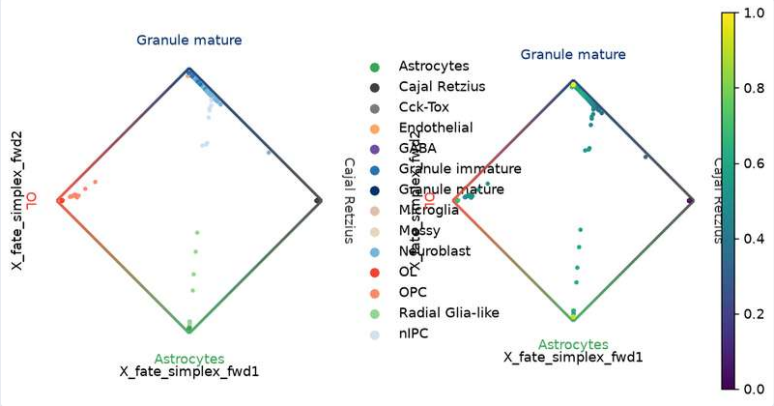
v1 macrostates

TRAJECTORY



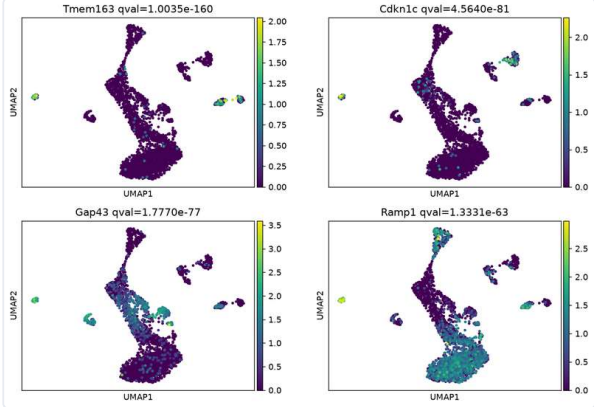
v2 fate_probabilities

TRAJECTORY



v3 circular_projection

TRAJECTORY



v4 lineage_drivers

TRAJECTORY

Nebulosa

#99

OF 100 · PAPERS NAMING IT

R | SINGLE-CELL DENSITY | EMBEDDINGS & FEATURE MAPS

39	0.22%	43	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · NEBULOSA

Neuro-corpus papers per year (*2026 partial) · niche rank **#58**, growth **1.29×**, visual 4/5, score 0.524. Broad-corpus rate 1.811 → 2.337 per 1k papers (2023 → 2025–26).
GitHub powellgenomicslab/Nebulosa · created 2020-07-12 · last push 2026-02-06

BEST FOR

Kernel-density feature plots

KEY PLOTTING CALL

RECREATIONS/NEBULOSA/NEBULOSA_RECREATE.R — EXCERPT · ENDS LINE 4

```
so <- readRDS("../_data/zeisel_seurat.rds")
ggsave("v1_density_single.png", plot_density(so, c("Pvalb","Sst","Vip","Aqp4")) + plot_layout(ncol = 2), width = 9, height = 8,
dpi = 120, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Figure 2 — FIGURE CAPTION

A single-nucleus transcriptomic atlas of medium spiny neurons in the rat nucleus accumbens. *Sci Rep, 2024*

"...Low resolution clustering of MSNs. (a) UMAP dimension reduction plot of MSN clustered at low resolution. Nuclei are colored and labeled by major MSN population. Nebulosa plots visualizing expression on the MSN ..."

DOI ↗ Figure 2 on PMC ↗

Figure 1 — TEXT

Landscape of mast cell populations across organs in mice and humans. *J Exp Med, 2023*

"...We showed these differences by plotting a density plot based on gene-weighted kernel density estimation realized with Nebulosa packages (v1.8.0;"

DOI ↗ Figure 1 on PMC ↗

Fig. 3 — TEXT

BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis. *Nat Genet, 2024*

"...We used Nebulosa (v.1.0.2)70 to visualize the cycling metagene on the UMAP embedding of the BANKSY matrix."

DOI ↗ Fig. 3 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

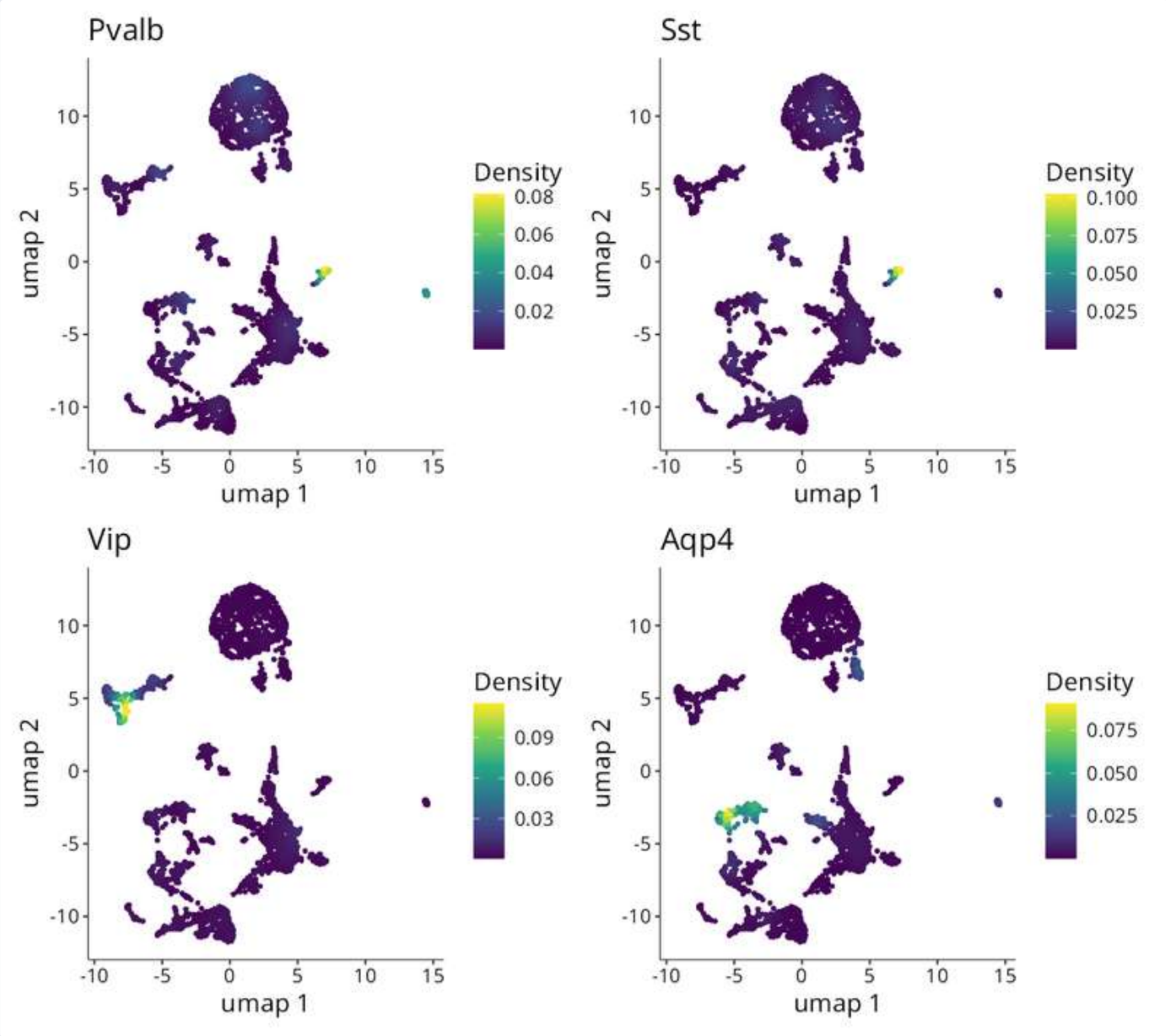
Reproduce: `Rscript recreations/Nebulosa/Nebulosa_recreate.R`

√ re-run end to end · 21 s

Nebulosa · 3 variations

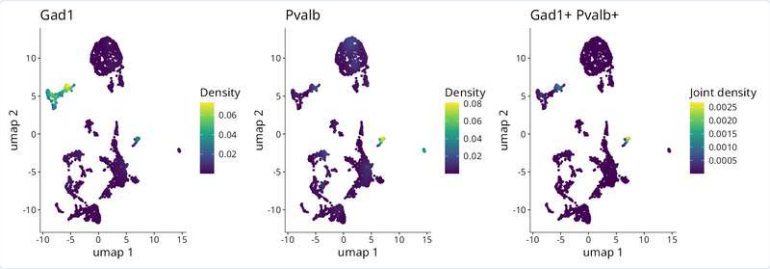
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



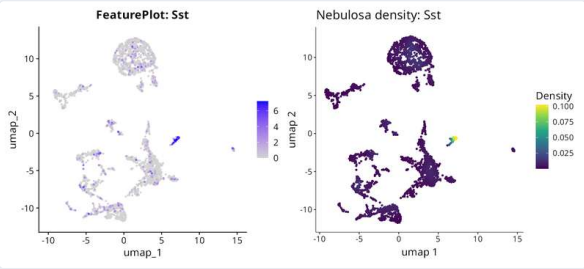
v1 density_single

EMBEDDINGS



v2 joint_density

EMBEDDINGS



v3 density_vs_featureplot

EMBEDDINGS

ggalluvial / ggsankey

#100

OF 100 · PAPERS NAMING IT

R · FLOW/ALLUVIAL · COMPOSITION, DISTRIBUTIONS & LAYOUTS

36	0.2%	37	dedicated plotting	3
papers whose Methods section names it	of the 17,768-paper corpus	papers naming it anywhere in the full text	role	figure variations recreated

POSITION IN TOP 100
BAR HEIGHT = LOG PAPERS

NICHE & COOL · GGSANKEY

Neuro-corpus papers per year (*2026 partial) · niche rank **#11**, growth **10.71×**†, visual 4/5, score 0.845. Broad-corpus rate 0.035 → 0.47 per 1k papers (2023 → 2025–26).
GitHub davidsjoberg/ggsankey · created 2021-03-25 · last push 2024-04-04 · also scored: ggalluvial (#72, growth 0.84×)
† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

Alluvial/Sankey plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Flow plots of Zeisel hierarchy: tissue → level1 class → level2 subclass

KEY PLOTTING CALL

RECREATIONS/GGSANKEY_GGALLUVIAL/SANKEY_RECREATE.R — EXCERPT · ENDS LINE 8

```
md <- readRDS("../_data/zeisel_seurat.rds")@meta.data
top2 <- names(sort(table(md$level2class), decreasing = TRUE))[1:18]; md$sub <- ifelse(md$level2class %in% top2, md$level2class,
"other")
d <- make_long(md, tissue, celltype, sub)
g <- ggplot(d, aes(x = x, next_x = next_x, node = node, next_node = next_node, fill = factor(node), label = node)) +
geom_sankey(flow.alpha = .55, node.color = "grey30") + geom_sankey_label(size = 2.6, color = "black", fill = "white") + theme_s
ankey(base_size = 12) + theme(legend.position = "none") + labs(x = NULL, title = "ggsankey: tissue -> class -> subclass")
ggsave("v1_ggsankey.png", g, width = 10, height = 7, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 2 TEXT
The transcriptomes, connections and development of submucosal neuron classes in the mouse small intestine. *Nat Neurosci, 2025*
"...Finally, the mapping identities were presented as Alluvial plot using the R package ggalluvial (Fig."
DOI ↗ Fig. 2 on PMC ↗

Fig. 4 TEXT
Single-cell transcriptomic analysis reveals diversity within mammalian spinal motor neurons. *Nat Commun, 2023*
"...To intuitively visualize the relationships of LMC subclusters between brachial and lumbar segments, we created an alluvial diagram using the ggalluvial R package ..."
DOI ↗ Fig. 4 on PMC ↗

Fig. 4 TEXT
Microbial signals in primary and metastatic brain tumors. *Nat Med, 2025*
"...Alluvial plots demonstrating the removal of taxa across the different filtration steps were generated using ggalluvial (version 0.12.5)."
DOI ↗ Fig. 4 on PMC ↗

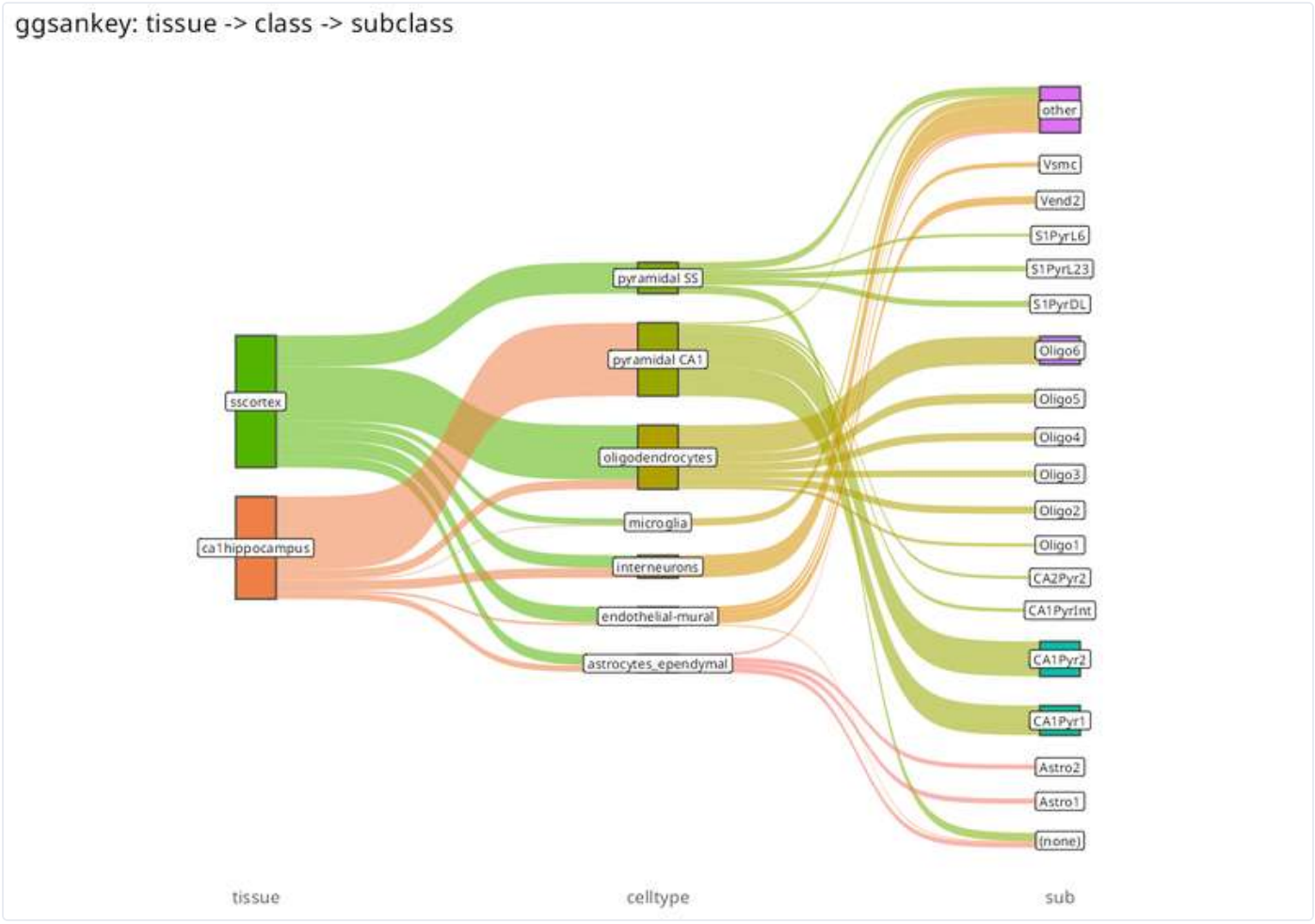
LIMITATIONS & NOTES
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/ggsankey_ggalluvial/sankey_recreate.R` ✓ re-run end to end · 7 s

ggalluvial / ggsankey · 3 variations

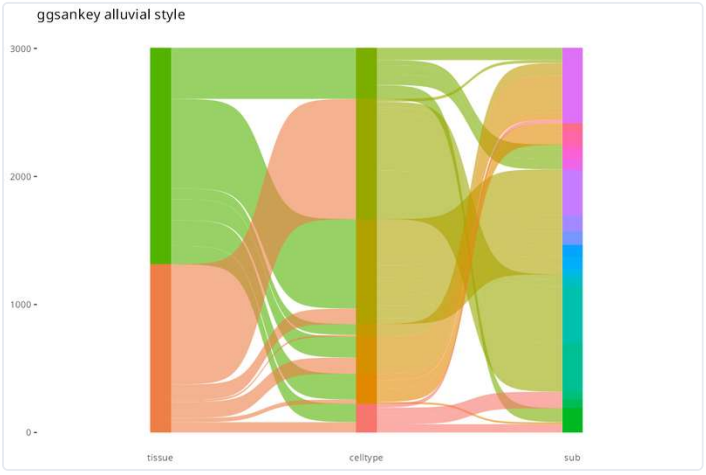
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



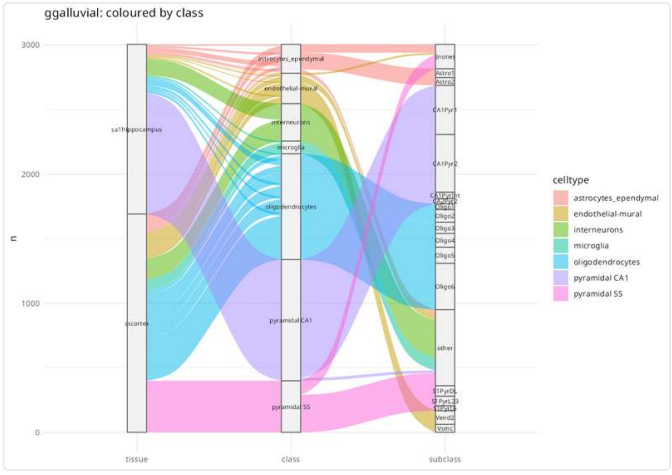
v1 ggsankey

COMPOSITION



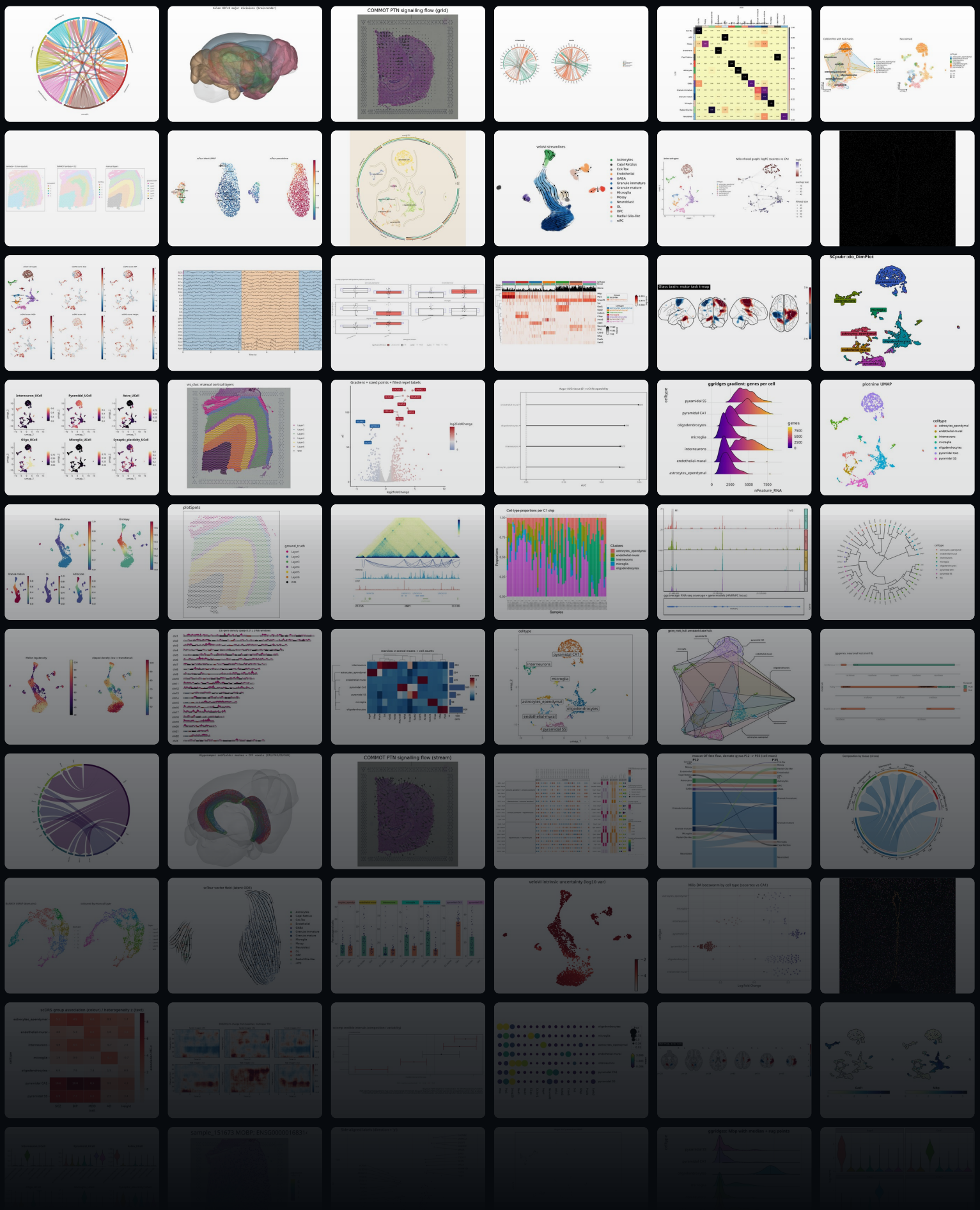
v2 ggsankey_alluvial

COMPOSITION



v3 ggalluvial

COMPOSITION



PART II · 36 PACKAGES · 112 RECREATED FIGURES

novaflow

II

Niche & Cool

Newer or more specialised visualisation tools, ranked by **0.5 × growth percentile + 0.5 × visual score/5**: circos and chord layouts, 3-D atlases, optimal-transport flows, Bayesian composition. This part has the 36 recreations not already in Part I. The other 21 recreated niche tools share Part I spreads, which show their niche statistics.

36
spreads

112
figures

58 / 73
niche tools recreated

25
new tools with low 2023 base (†)

IN THIS PART · RANK · PAGE

1	pyCirclize	146	17	scDRS	170	50	Palantir	194
3	brainrender	148	19	MNE-Python	172	51	ggspavis	196
4	COMMOT	150	20	sccomp	174	55	plotgardener	198
6	MultiNicheNet	152	26	PyComplexHeatmap	176	60	propeller (speckle)	200
7	moscot	154	29	nilearn	178	63	ggcoverage	202
8	scplotter	156	31	SCpubr	180	64	ggtree / ggtreeExtra	204
10	Banksy	158	34	UCell	182	65	Mellon	206
12	scTour	160	37	spatialLIBD	184	66	karyoploteR	208
13	plot1cell	162	41	ggrepel volcano styles	186	67	marsilea	210
14	veloVI	164	42	Augur	188	68	dittoSeq	212
15	miloR	166	47	ggridges	190	70	ggforce	214
16	Datashader	168	48	plotnine	192	71	gggenes	216

PART II · NICHE & COOL · 01 / 36

pyCirclize

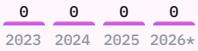
#1

PYTHON · CIRCOS/CHORD · CELL–CELL COMMUNICATION

OF 73 · NICHE & COOL SCORE

20.05×†	5/5	0.98	0	1,127
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · PYCIRCLIZE

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0 → 0.191 per 1k papers (2023 → 2025–26).
GitHub moshi4/pyCirclize · created 2022-12-17 · last push 2026-06-14

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

circos/chord in Python

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

pyCirclize chord + circos ring plots from LIANA+ results on Zeisel (run ../LIANA_plus first)

KEY PLOTTING CALL

RECREATIONS/PYCIRCLIZE/PYCIRCLIZE_RECREATE.PY — EXCERPT · ENDS LINE 6

```
m=pd.read_csv("../LIANA_plus/lr_counts_matrix.csv",index_col=0); idx=sorted(set(m.index)|set(m.columns));
m=m.reindex(index=idx,columns=idx,fill_value=0)
c=Circos.chord_diagram(m,space=3,cmap="tab10",label_kws=dict(size=8,r=110),link_kws=dict(ec="black",lw=.3,direction=1))
c.savefig("v1_chord_interactions.png",dpi=150)
```

IN PUBLISHED PAPERS

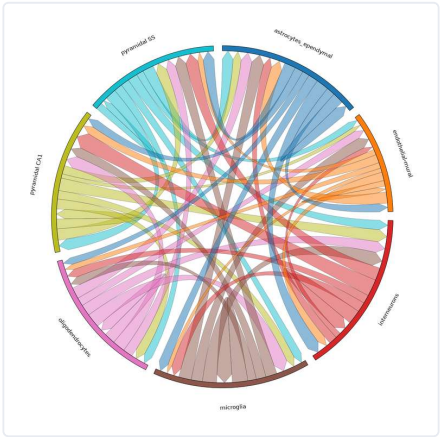
No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/moshi4/pyCirclize.

LIMITATIONS & NOTES

- Reads LIANA+ result tables, so run recreations/LIANA_plus first (METHODS §9).

Reproduce: `python3 recreations/pyCirclize/pycirclize_recreate.py` — √ re-run end to end · 15 s

LEAD FIGURE

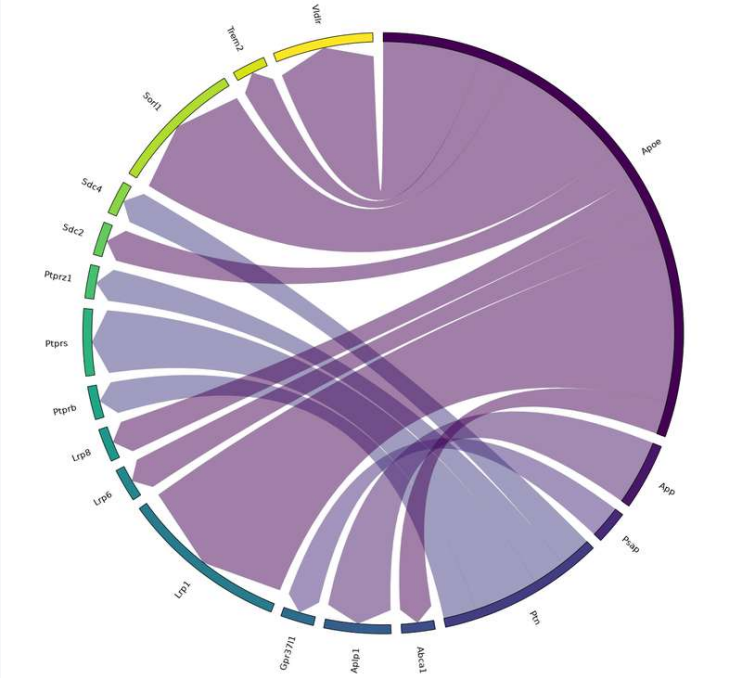


v1 chord_interactions — CELL–CELL

pyCirclize · 3 variations · v1 on facing page

NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v2 `ligand_receptor_chord` ● CELL–CELL

v3 `circos_tracks` ● CELL–CELL

PART II · NICHE & COOL · 02 / 36

brainrender

#3

OF 73 · NICHE & COOL SCORE

PYTHON | 3D BRAIN | BRAIN ATLAS, 3D & NEUROIMAGING

4.81 ×†	5 /5	0.914	4	680
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · BRAINRENDER

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0 → 0.038 per 1k papers (2023 → 2025–26).
GitHub brainglobe/brainrender · created 2019-09-19 · last push 2026-10-13

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

3D Allen CCF renderings

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

brainrender 3D renders of the Allen Mouse CCFv3 (25 um, via BrainGlobe): coloured region meshes, point clouds sampled from real CCF annotation voxels, and a sliced view. brainrender builds the scene (atlas meshes, colours); the final frame is drawn with a plain vedo Plotter without depth peeling, because software-GL offscreen rendering blanks translucent meshes when peeling is on.

KEY PLOTTING CALL

RECREATIONS/BRAINRENDER/BRAINRENDER_RECREATE.PY — EXCERPT · ENDS LINE 28

```
regs={"Isocortex":"#9ecae1","HIP":"#fb6a4a","STR":"#74c476","TH":"#fdae6b","HY":"#9e9ac8","CB":"#fdd0a2","MB":"#a1d99b","OLF":"#fa9fb5"}
# CCF axes: x=AP (anterior->posterior), y=DV (dorsal->ventral), z=ML
s=Scene(atlas_name="allen_mouse_25um")
for r,c in regs.items(): s.add_brain_region(r, color=c, alpha=.45 if r=="Isocortex" else .95)
draw(s, "v1_regions_3quarter.png", "Allen CCFv3 major divisions (brainrender)", azimuth=-35, elevation=25, roll=180, zoom=1.3)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/brainglobe/brainrender.

LIMITATIONS & NOTES

- brainrender builds the scene, but the final frame is drawn with a plain vedo Plotter without depth peeling: software-GL offscreen rendering blanks translucent meshes when peeling is on (METHODS §9).

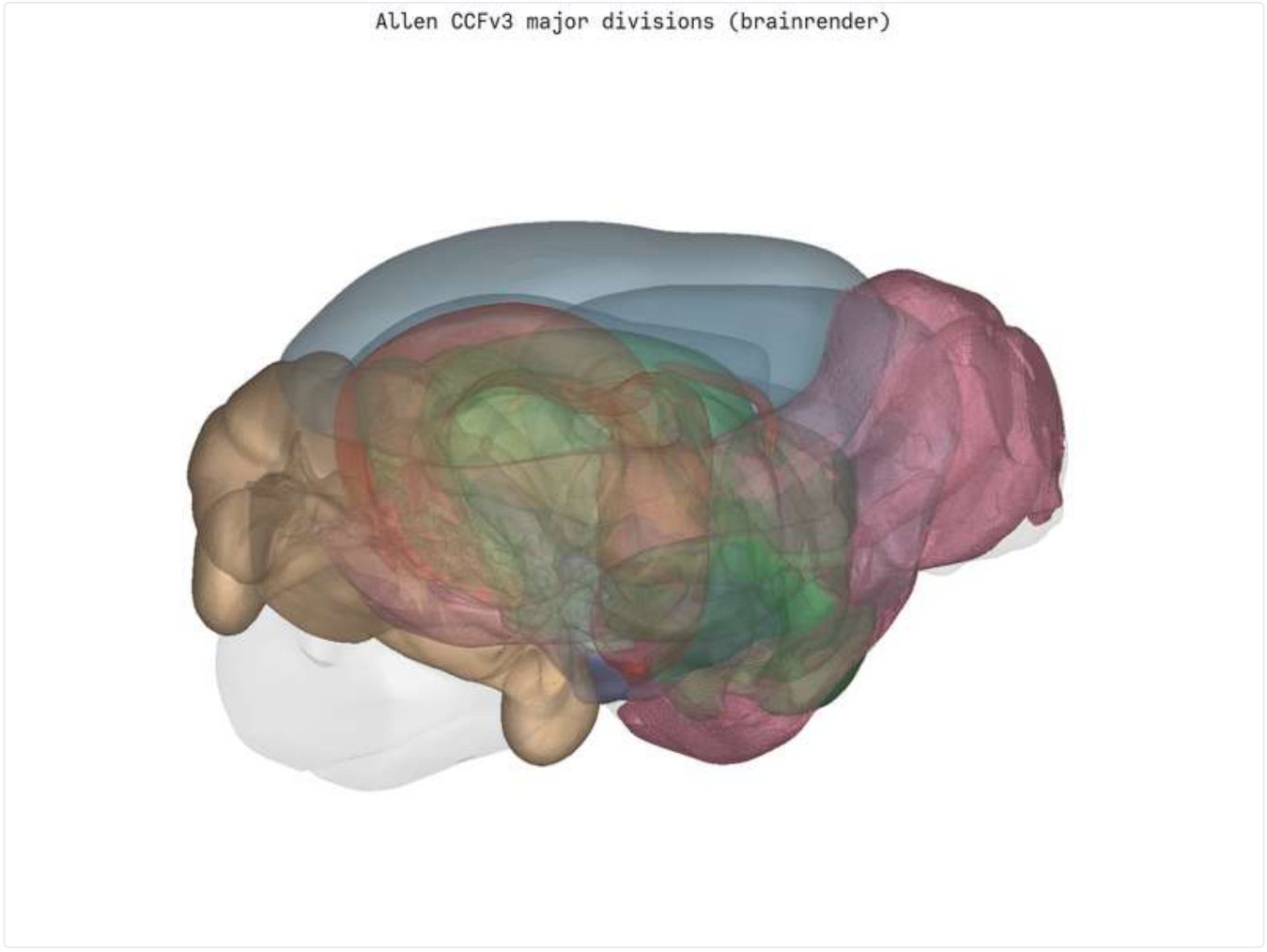
Reproduce: `python3 recreations/brainrender/brainrender_recreate.py`

√ re-run end to end · 29 s

brainrender · 3 variations

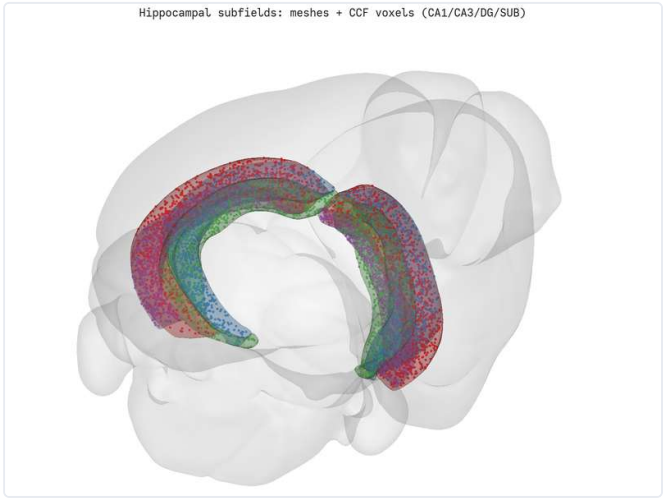
NOVAFLOW RECREATION

Data: Allen Mouse Brain Common Coordinate Framework v3, 25 µm (BrainGlobe atlas API). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



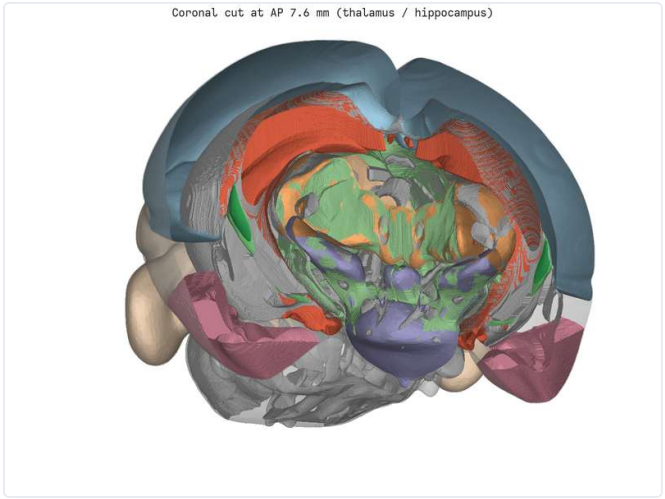
v1 regions_3quarter

BRAIN



v2 hippocampus_points

BRAIN



v3 coronal_cut

BRAIN

PART II · NICHE & COOL · 03 / 36

COMMOT

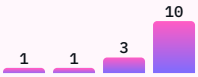
#4

OF 73 · NICHE & COOL SCORE

PYTHON | SPATIAL COMMUNICATION | SPATIAL & TISSUE MAPS

4.67×	5/5	0.904	15	149
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · COMMOT

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.348 → 1.664 per 1k papers (2023 → 2025–26).
GitHub zcang/COMMOT · created 2021-09-03 · last push 2026-06-25

BEST FOR

spatial signalling flow vector fields

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

COMMOT spatial optimal-transport signalling on 10x Visium mouse brain (CellChat mouse secreted LR database)

KEY PLOTTING CALL

RECREATIONS/COMMOT/COMMOT_RECREATE.PY — EXCERPT · ENDS LINE 12

```
for meth,fn in [("grid","v1_flow_grid.png"),("stream","v2_flow_stream.png")]:
    ct.pl.plot_cell_communication(v,database_name="cellchat",pathway_name=pw,plot_method=meth,background_legend=True,scale=.0000
3,ndsize=8,grid_density=.4,summary="sender",background="image",clustering="cluster",cmap="Alphabet",normalize_v=True,normalize_
v_quantile=.995)
    plt.title(f"COMMOT {pw} signalling flow ({meth})"); plt.savefig(fn,dpi=140,bbox_inches="tight"); plt.close("all")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/zcang/COMMOT.

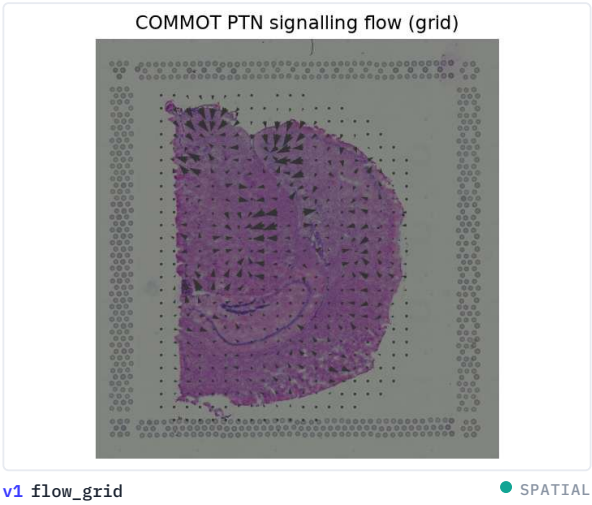
LIMITATIONS & NOTES

- NumPy 2 shim: np.Inf restored because COMMOT still uses it (METHODS §9).

Reproduce: `python3 recreations/COMMOT/commot_recreate.py`

✓ re-run end to end · 1,095 s

LEAD FIGURE



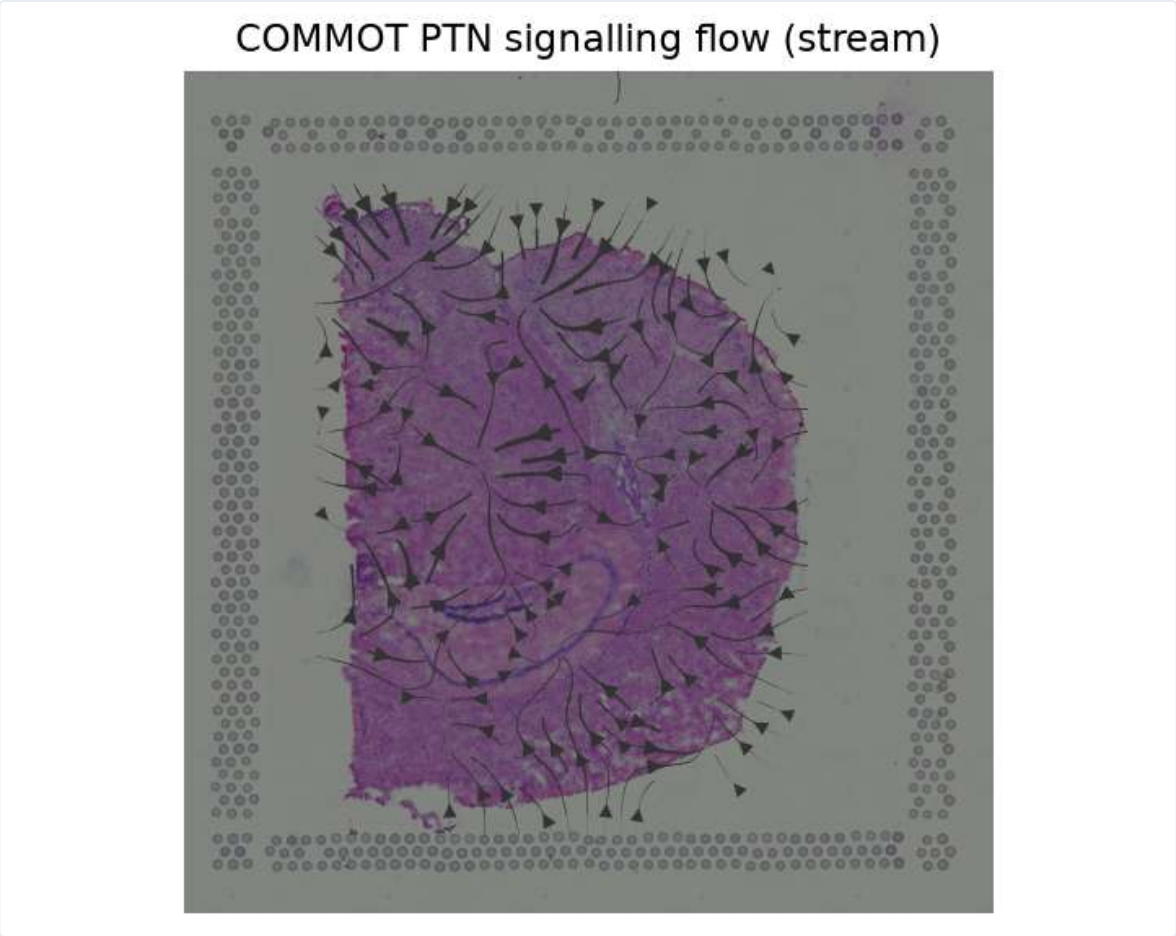
v1 flow_grid

SPATIAL

COMMOT · 3 variations · v1 on facing page

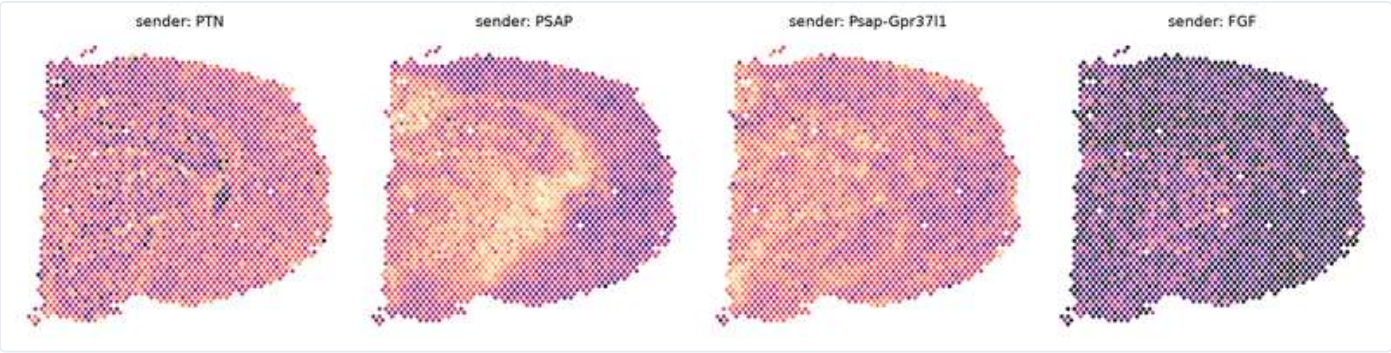
NOVAFLOW RECREATION

Data: 10x Visium mouse coronal brain section (squidpy example) with the CellChat mouse ligand–receptor database. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v2 flow_stream

SPATIAL



v3 sender_maps

SPATIAL

PART II · NICHE & COOL · 04 / 36

MultiNicheNet

#6

OF 73 · NICHE & COOL SCORE

R · LIGAND-TARGET · CELL-CELL COMMUNICATION

89.9×†	4/5	0.893	13	202
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · MULTINICHENET

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0 → 0.889 per 1k papers (2023 → 2025–26).
GitHub saeyslab/multinichenetr · created 2021-02-16 · last push 2025-08-12

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

multi-sample LR circos

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

MultiNicheNet (multi-sample, multi-condition cell-cell communication) on Zeisel 2015: S1 cortex vs CA1, C1 chip as sample (pseudobulk, edgeR), shared non-pyramidal cell types. NicheNet v2 mouse priors (Zenodo 7074291). Group-comparison circos, sample-level LR product + ligand activity bubble plot, ligand-activity/target plot.

KEY PLOTTING CALL

RECREATIONS/MULTINICHENET/MULTINICHENET_RECREATE.R — EXCERPT · ENDS LINE 23

```
top <- get_top_n_lr_pairs(out$prioritization_tables, 50, rank_per_group = FALSE)
cts <- sort(unique(c(top$sender, top$receiver))); cols <- setNames(RColorBrewer::brewer.pal(max(3, length(cts)), "Set2")[seq_along(cts)], cts)
png("v1_circos_%d.png", 1400, 1400, res = 170); circ <- make_circos_group_comparison(top, cols[sort(unique(top$sender))], cols[sort(unique(top$receiver))]); dev.off()
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/saeyslab/multinichenetr.

LIMITATIONS & NOTES

- On Zeisel, MultiNicheNet keeps only 2 receivers (astrocytes, oligodendrocytes) because chips have few cells per type.
- Run single-core: parallel workers exceed typical laptop memory.

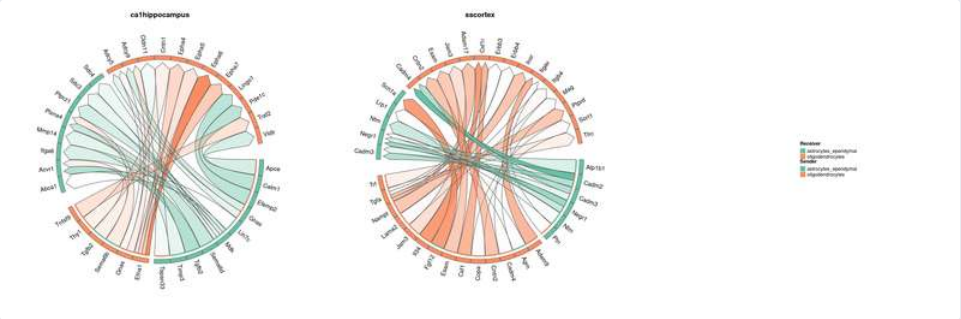
Reproduce: `Rscript recreations/MultiNicheNet/MultiNicheNet_recreate.R`

√ re-run end to end · 922 s

MultiNicheNet · 3 variations

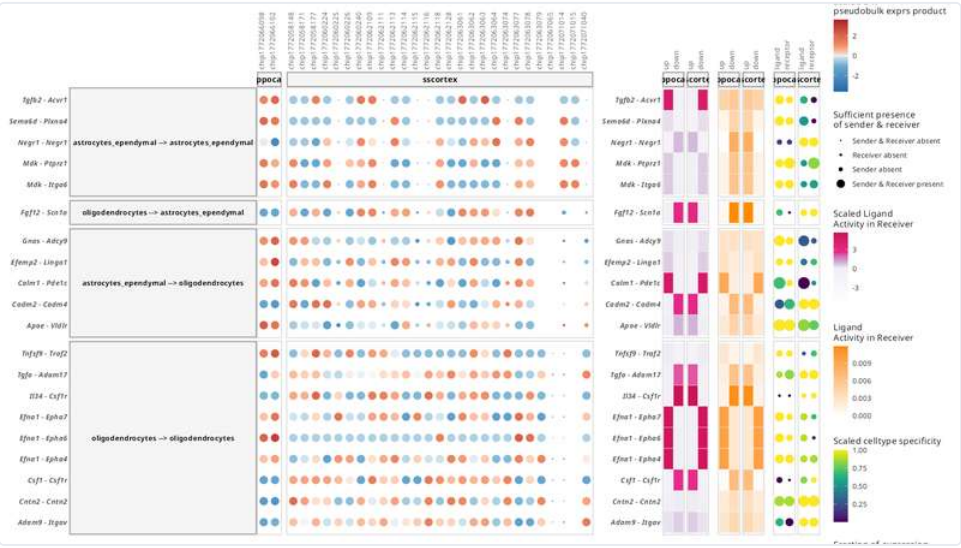
NOVAFLOW RECREATION

Data: Zeisel et al. 2015 scRNA-seq with NicheNet v2 mouse prior networks (Zenodo 7074291). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



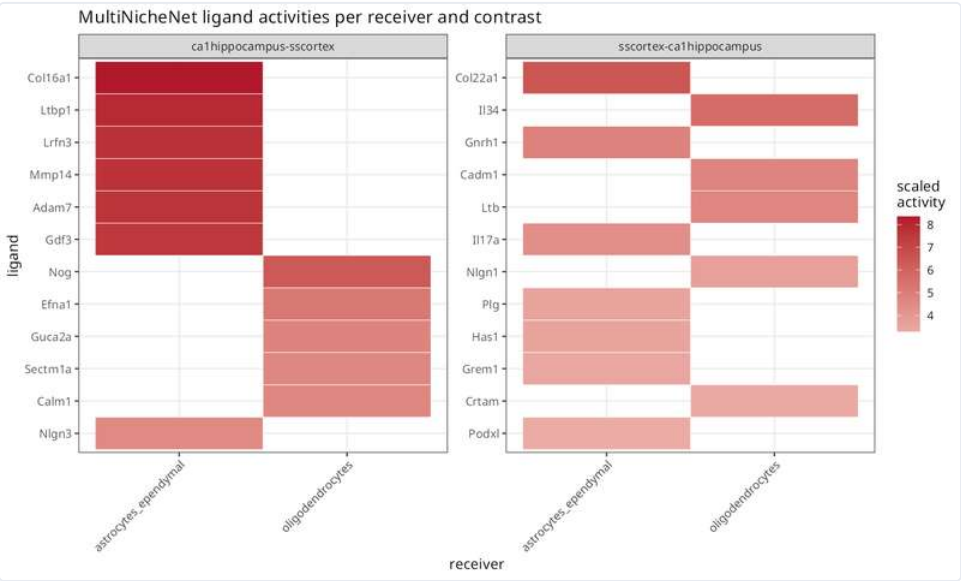
v1 circos_group_comparison

CELL–CELL



v2 sample_lr_activity

CELL–CELL



v3 ligand_activity_tiles

CELL–CELL

moscot

#7

OF 73 · NICHE & COOL SCORE

PYTHON | OPTIMAL TRANSPORT | TRAJECTORY, PSEUDOTIME & VELOCITY

42.91 ×†	4/5	0.887	7	217
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · MOSCOT

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0 → 0.419 per 1k papers (2023 → 2025–26).
GitHub theislab/moscot · created 2021-08-23 · last push 2026-10-14

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

OT cell-transition maps, Sankey

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

moscot optimal-transport TemporalProblem on the La Manno 2018 dentate gyrus (P12 → P35): cell-type transition matrix, Sankey of fate flow, and push-forward of the P12 neuroblast mass onto P35 cells.

KEY PLOTTING CALL

RECREATIONS/MOSCOT/MOSCOT_RECREATE.PY — EXCERPT · ENDS LINE 12

```
tp=mt.problems.TemporalProblem(a).prepare(time_key="day",joint_attr="X_pca").solve(epsilon=1e-2,tau_a=.95,tau_b=.95)
tp.cell_transition(source=12.0,target=35.0,source_groups="clusters",target_groups="clusters",forward=True,key_added="ct")
fig=mtp.cell_transition(tp,key="ct",figsize=(9,8),dpi=130,return_fig=True,cmap="magma_r"); fig.suptitle("moscot: P12 -> P35
cell-type transition probabilities"); fig.savefig("v1_cell_transition.png",bbox_inches="tight",dpi=130)
```

IN PUBLISHED PAPERS

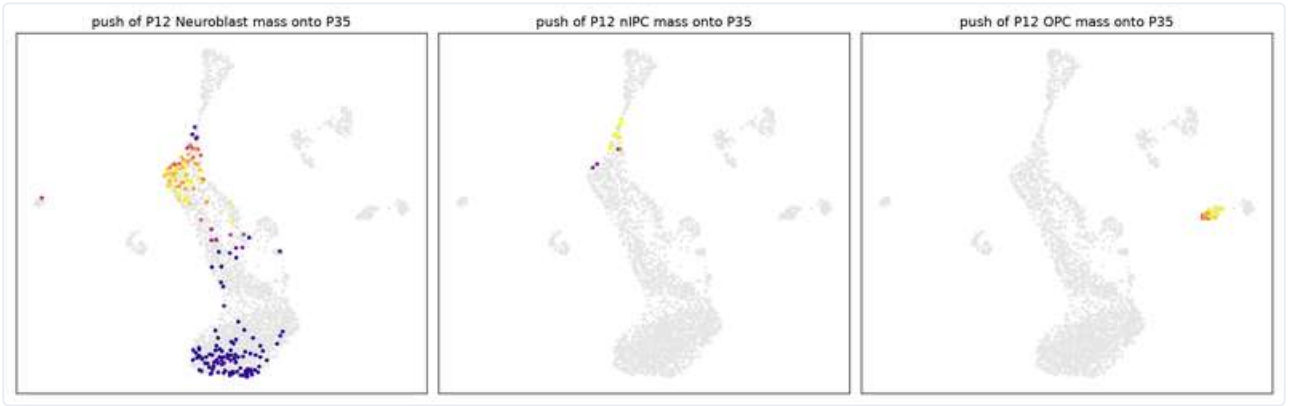
No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/theislab/moscot.

LIMITATIONS & NOTES

- The Sankey is re-drawn from the OT coupling because the built-in Sankey mis-draws (METHODS §9).

Reproduce: `python3 recreations/moscot/moscot_recreate.py` — √ re-run end to end · 14 s

LEAD FIGURE



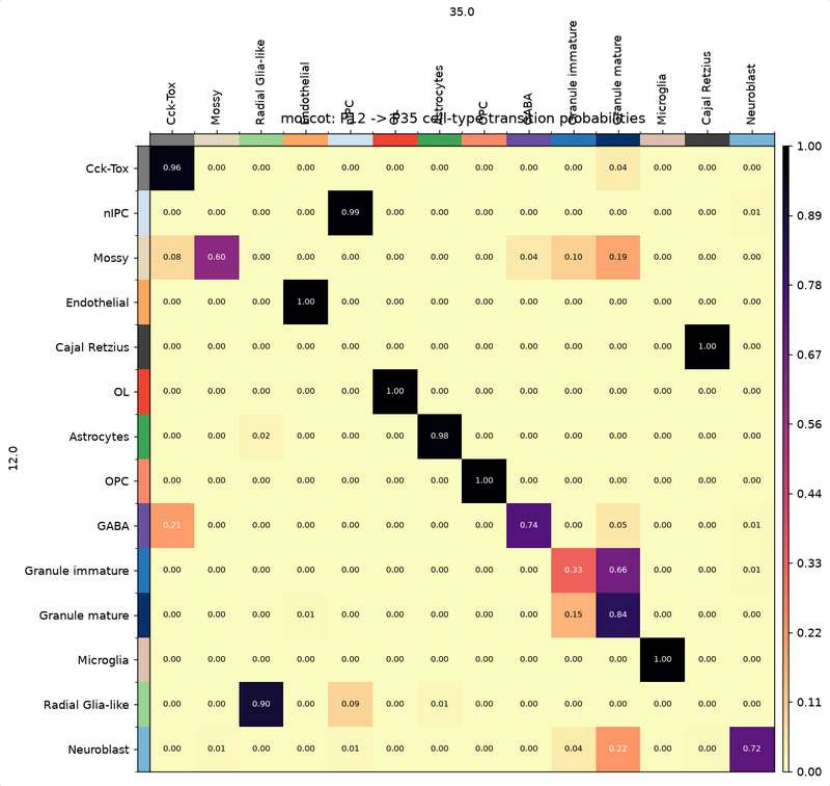
v3 push_umap

TRAJECTORY

moscot · 3 variations · v3 on facing page

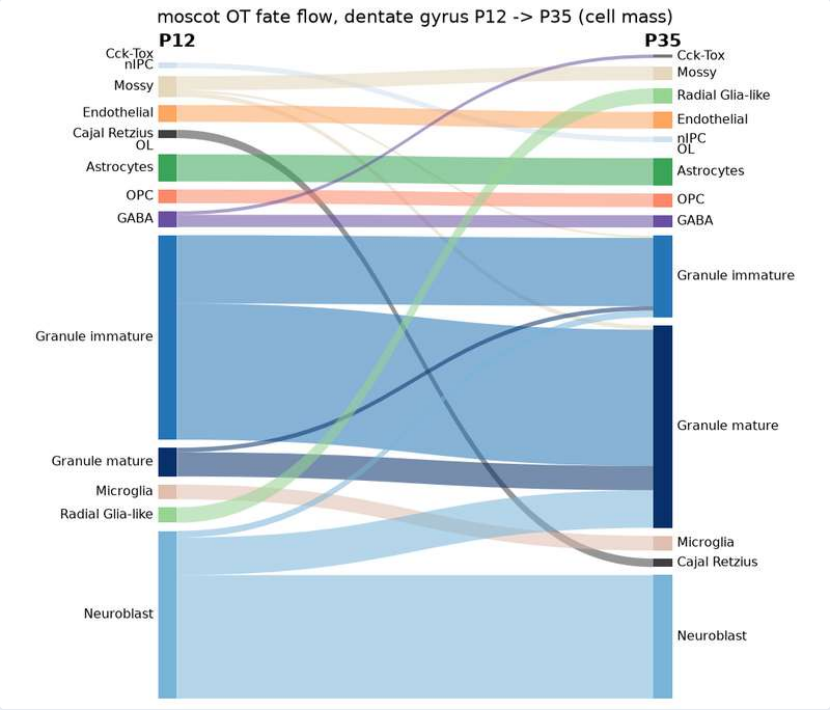
NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 cell_transition

TRAJECTORY



v2 sankey

TRAJECTORY

PART II · NICHE & COOL · 06 / 36

scplotter

#8

OF 73 · NICHE & COOL SCORE

R · SINGLE-CELL PUBLICATION PLOTS · EMBEDDINGS & FEATURE MAPS · COMPOSITION, DISTRIBUTIONS & LAYOUTS

17.51 ×†	4/5	0.872	1	333
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · SCPLOTTER

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0 → 0.165 per 1k papers (2023 → 2025–26).
GitHub pwwang/scplotter · created 2024-09-03 · last push 2026-09-19

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

one-line Seurat publication plots (plotthis backend)

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

scplotter (pwwang; plotthis backend) on Zeisel 2015: hull-marked and hex CellDimPlot, CellStatPlot as circos / radar / ring of cell-type composition by tissue, and FeatureStatPlot ridge/violin of markers. Installed from GitHub (pwwang/plotthis, pwwang/scplotter); not on CRAN for R 4.5.

KEY PLOTTING CALL

RECREATIONS/SCPLOTTER/SCPLOTTER_RECREATE.R — EXCERPT · ENDS LINE 10

```
so <- readRDS("../_data/zeisel_seurat.rds"); so$tissue <- factor(ifelse(so$tissue == "sscortex", "S1 cortex", "CA1"))
so$chip <- paste0("chip", sub("_.*", "", colnames(so))); Idents(so) <- "celltype"
p1 <- CellDimPlot(so, group_by = "celltype", reduction = "umap", theme = "theme_blank", add_mark = TRUE, mark_type = "hull", la
bel = TRUE, label_insitu = TRUE, title = "CellDimPlot with hull marks")
p1b <- CellDimPlot(so, group_by = "celltype", reduction = "umap", theme = "theme_blank", hex = TRUE, hex_bins = 40, title = "hex
-binned")
ggsave("v1_celldimplot.png", p1 + p1b, width = 13, height = 5.5, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/pwwang/scplotter.

LIMITATIONS & NOTES

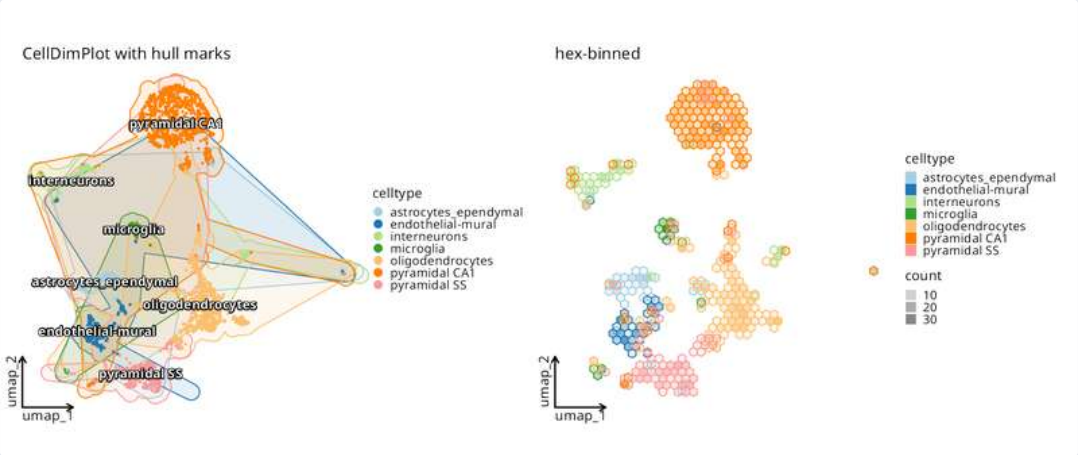
- Installed from GitHub (pwwang/plotthis, pwwang/scplotter); not on CRAN for R 4.5.

Reproduce: `Rscript recreations/scplotter/scplotter_recreate.R`

√ re-run end to end · 24 s

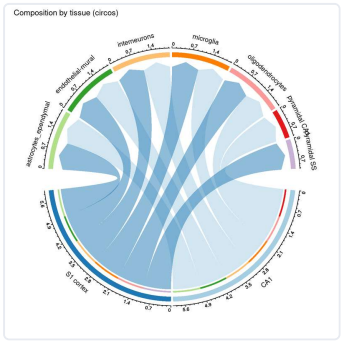
scplotter · 4 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



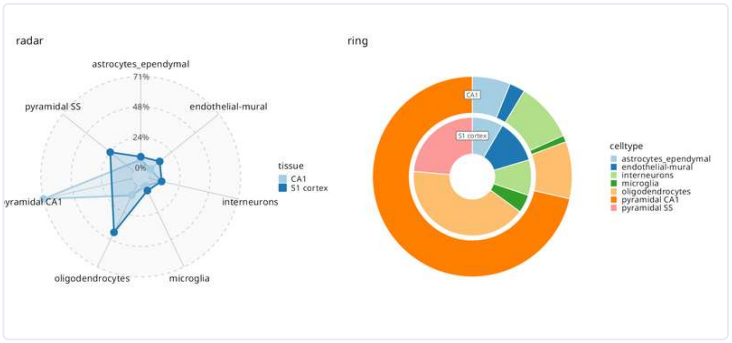
v1 celldimplot

EMBEDDINGS



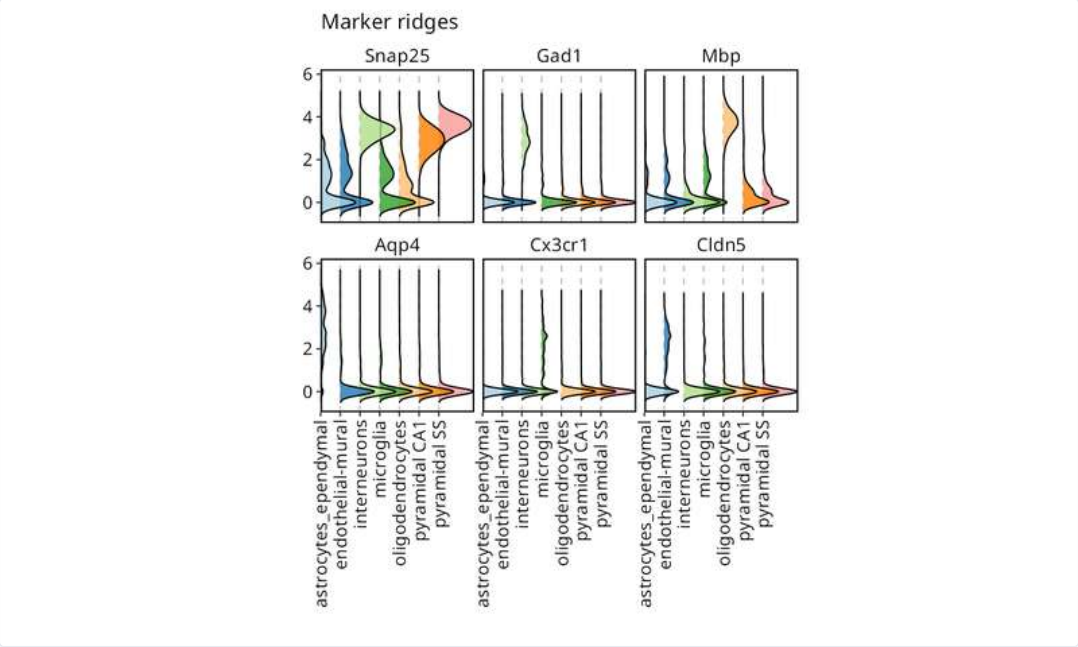
v2 cellstat_circos

COMPOSITION



v3 cellstat_radar_ring

COMPOSITION



v4 featurestat_ridge

COMPOSITION

PART II · NICHE & COOL · 07 / 36

Banksy

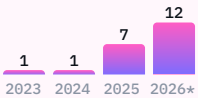
#10

OF 73 · NICHE & COOL SCORE

R/PYTHON | SPATIAL CLUSTERING | SPATIAL & TISSUE MAPS

14.16×†	4/5	0.859	21	167
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · BANKSY

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.07 → 1.118 per 1k papers (2023 → 2025–26).
GitHub prabhakarlab/Banksy · created 2021-05-23 · last push 2026-09-29

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

neighbourhood-augmented domains

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

BANKSY spatial domains (neighbourhood-augmented clustering) on DLPFC 151673

KEY PLOTTING CALL

RECREATIONS/BANKSY/BANKSY_RECREATE.R — EXCERPT · ENDS LINE 11

```
spe <- runBanksyPCA(spe, use_agf = TRUE, lambda = c(0, .2), npcs = 20, seed = 1000)
spe <- clusterBanksy(spe, use_agf = TRUE, lambda = c(0, .2), resolution = .6, seed = 1000)
cn <- clusterNames(spe); spe <- connectClusters(spe)
spe$nonspatial <- factor(colData(spe)[[cn[1]]]); spe$banksy <- factor(colData(spe)[[cn[2]]])
ggsave("v1_banksy_vs_nonspatial.png", patchwork::wrap_plots(list(plotCoords(spe, annotate = "nonspatial", point_size = .6) + gg
title("lambda = 0 (non-spatial)"), plotCoords(spe, annotate = "banksy", point_size = .6) + ggtitle("BANKSY lambda = 0.2"), plot
Coords(spe, annotate = "ground_truth", pal = "libd_layer_colors", point_size = .6) + ggtitle("manual layers")), nrow = 1),
width = 15, height = 5, dpi = 120, bg = "white")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/prabhakarlab/Banksy.

> LIMITATIONS & NOTES
> No package-specific caveat. General caveats apply (p. 8).

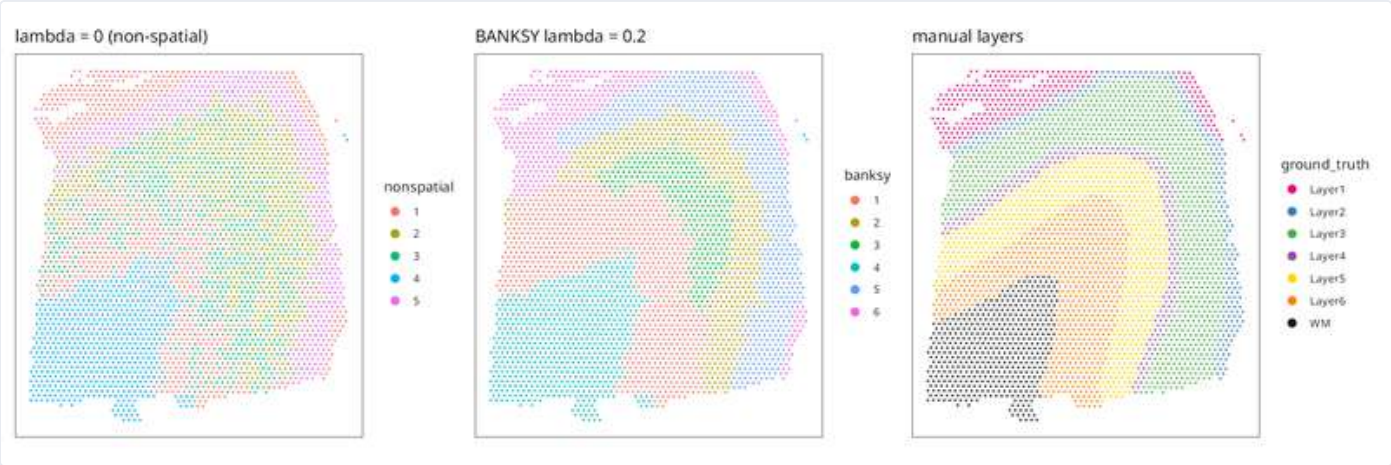
Reproduce: `Rscript recreations/Banksy/Banksy_recreate.R`

√ re-run end to end · 56 s

Banksy · 2 variations

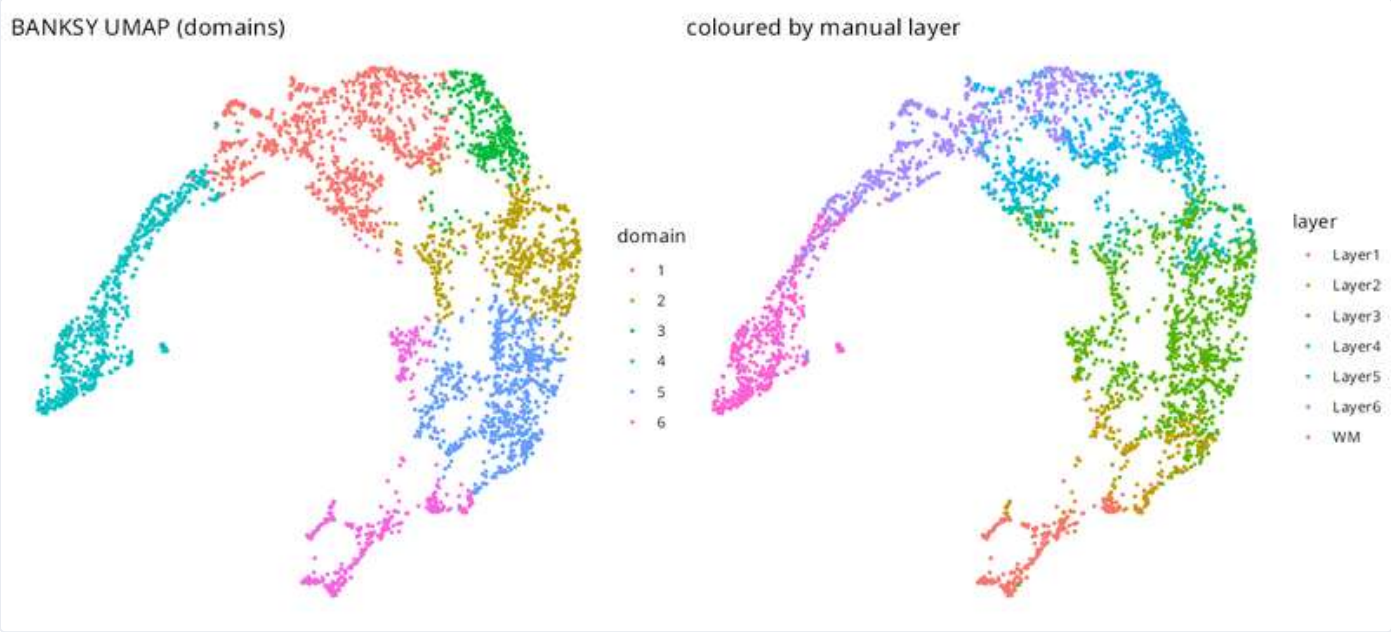
NOVAFLOW RECREATION

Data: Maynard et al. 2021, human DLPFC Visium sample 151673 (STexampleData). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 banksy_vs_nonspatial

SPATIAL



v2 banksy_umap

SPATIAL

scTour

#12

PYTHON | TRAJECTORY | TRAJECTORY, PSEUDOTIME & VELOCITY

OF 73 · NICHE & COOL SCORE

6.52×†	4/5	0.832	7	89
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · SCTOUR

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.104 → 0.737 per 1k papers (2023 → 2025–26).
GitHub LiQian-XC/sctour · created 2022-01-03 · last push 2025-07-20

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

vector field + pseudotime

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

scTour (VAE + neural ODE) on La Manno 2018 dentate gyrus: learned pseudotime, latent-space vector field streamlines, and pseudotime distribution per cell type.

KEY PLOTTING CALL

RECREATIONS/SCTOUR/SCTOUR_RECREATE.PY — EXCERPT · ENDS LINE 18

```
fig,axs=plt.subplots(1,2,figsize=(13,5.5))
sc.pl.umap(a,color="clusters",ax=axs[0],show=False,legend_loc="on data",legend_fontsize=7,frameon=False,title="scTour latent UMAP")
sc.pl.umap(a,color="ptime",ax=axs[1],show=False,cmap="Spectral_r",frameon=False,title="scTour pseudotime")
fig.tight_layout(); fig.savefig("v1_pseudotime_umap.png",dpi=140)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/LiQian-XC/sctour.

LIMITATIONS & NOTES

- Dense X is used because scTour calls the removed scipy .A on sparse views (METHODS §9).

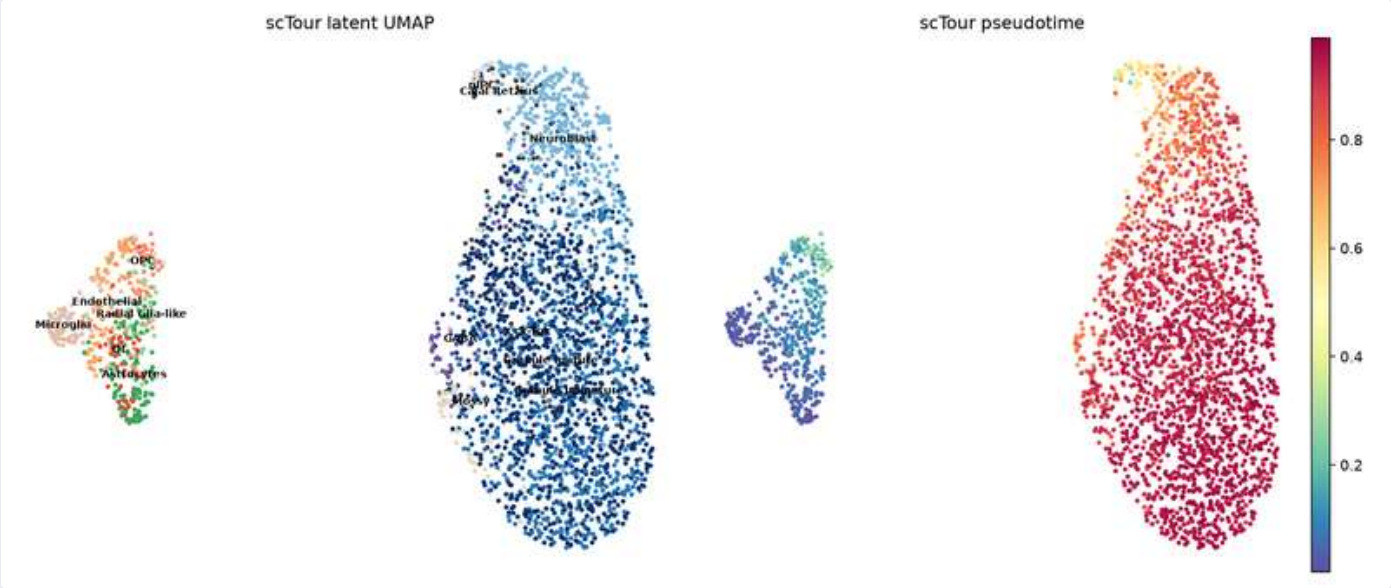
Reproduce: `python3 recreations/scTour/sctour_recreate.py`

√ re-run end to end · 89 s

scTour · 3 variations

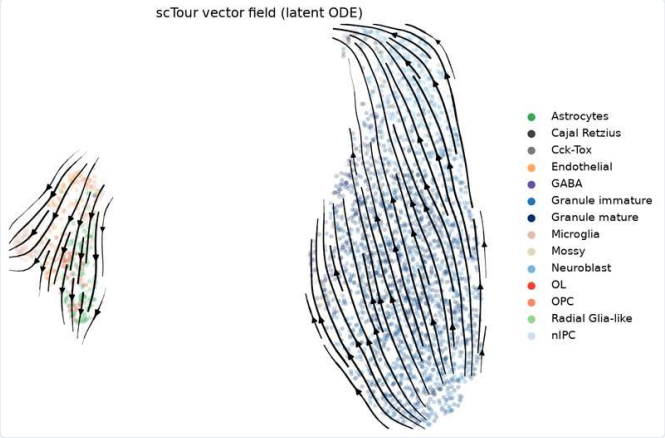
NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



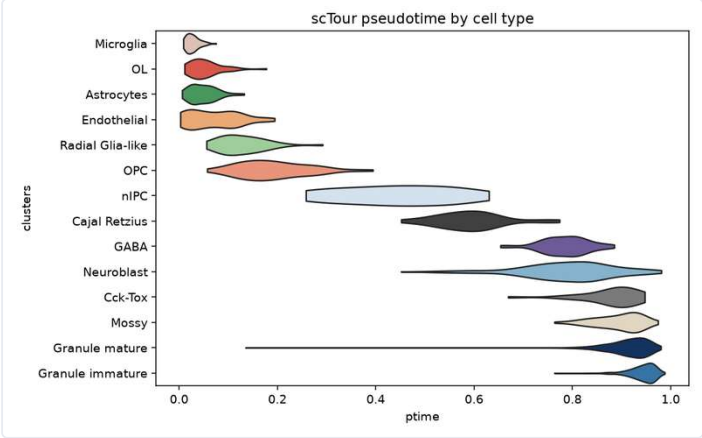
v1 pseudotime_umap

TRAJECTORY



v2 vector_field

TRAJECTORY



v3 ptime_by_celltype

TRAJECTORY

PART II · NICHE & COOL · 09 / 36

plot1cell

#13

OF 73 · NICHE & COOL SCORE

R · SINGLE-CELL PUBLICATION PLOTS · EMBEDDINGS & FEATURE MAPS · COMPOSITION, DISTRIBUTIONS & LAYOUTS · HEATMAPS & DOT MATRICES

2.7 ×†	5 /5	0.829	6	131
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · PLOT1CELL

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.139 → 0.394 per 1k papers (2023 → 2025–26).
GitHub TheHumphreysLab/plot1cell · created 2022-03-22 · last push 2025-09-04

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

circlize 'cell circle' UMAP, multi-group dot plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

plot1cell circular UMAP ("plot_circlize") + tracks, cell-fraction and multi-group dot plots on Zeisel 2015. plot1cell cannot be installed here (hard deps DoubletFinder/loomR/GEOquery/... unavailable for R 4.5), so its plotting functions are sourced verbatim from the GitHub repo (TheHumphreysLab/plot1cell, R/*.R) into src/.

KEY PLOTTING CALL

RECREATIONS/PLOT1CELL/PLOT1CELL_RECREATE.R — EXCERPT · ENDS LINE 16

```
png("v1_circlize_umap.png", 1600, 1600, res = 200)
plot_circlize(cd, do.label = TRUE, pt.size = .5, col.use = cols, bg.color = "#F9F2E4", repel = FALSE, label.cex = .7)
add_track(cd, group = "tissue", colors = c("#1b9e77","#d95f02"), track_num = 2)
add_track(cd, group = "chip", track_num = 3, colors = colorRampPalette(brewer.pal(9, "Greys"))(length(unique(so$chip))))
dev.off()
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/TheHumphreysLab/plot1cell.

LIMITATIONS & NOTES

- plot1cell cannot be installed on R 4.5 (hard deps DoubletFinder/loomR/GEOquery/... unavailable), so its plotting functions are sourced verbatim from the GitHub repo TheHumphreysLab/plot1cell (R/*.R) into src/.

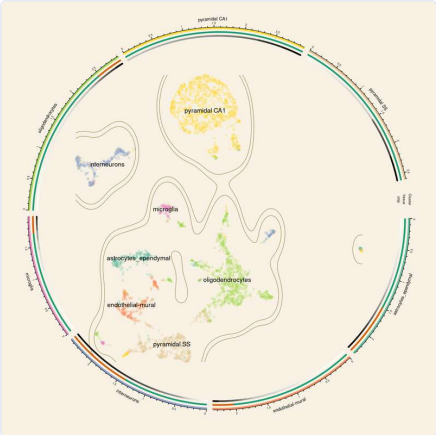
Reproduce: `Rscript recreations/plot1cell/plot1cell_recreate.R`

√ re-run end to end · 13 s

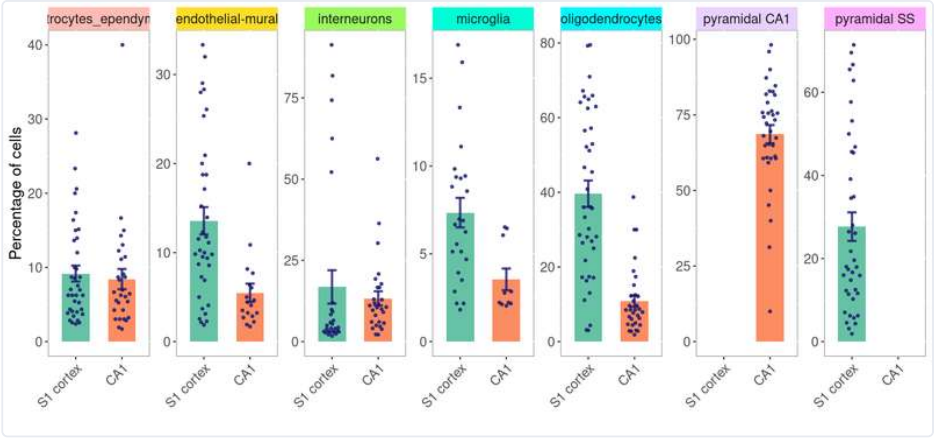
plot1cell · 3 variations

NOVAFLOW RECREATION

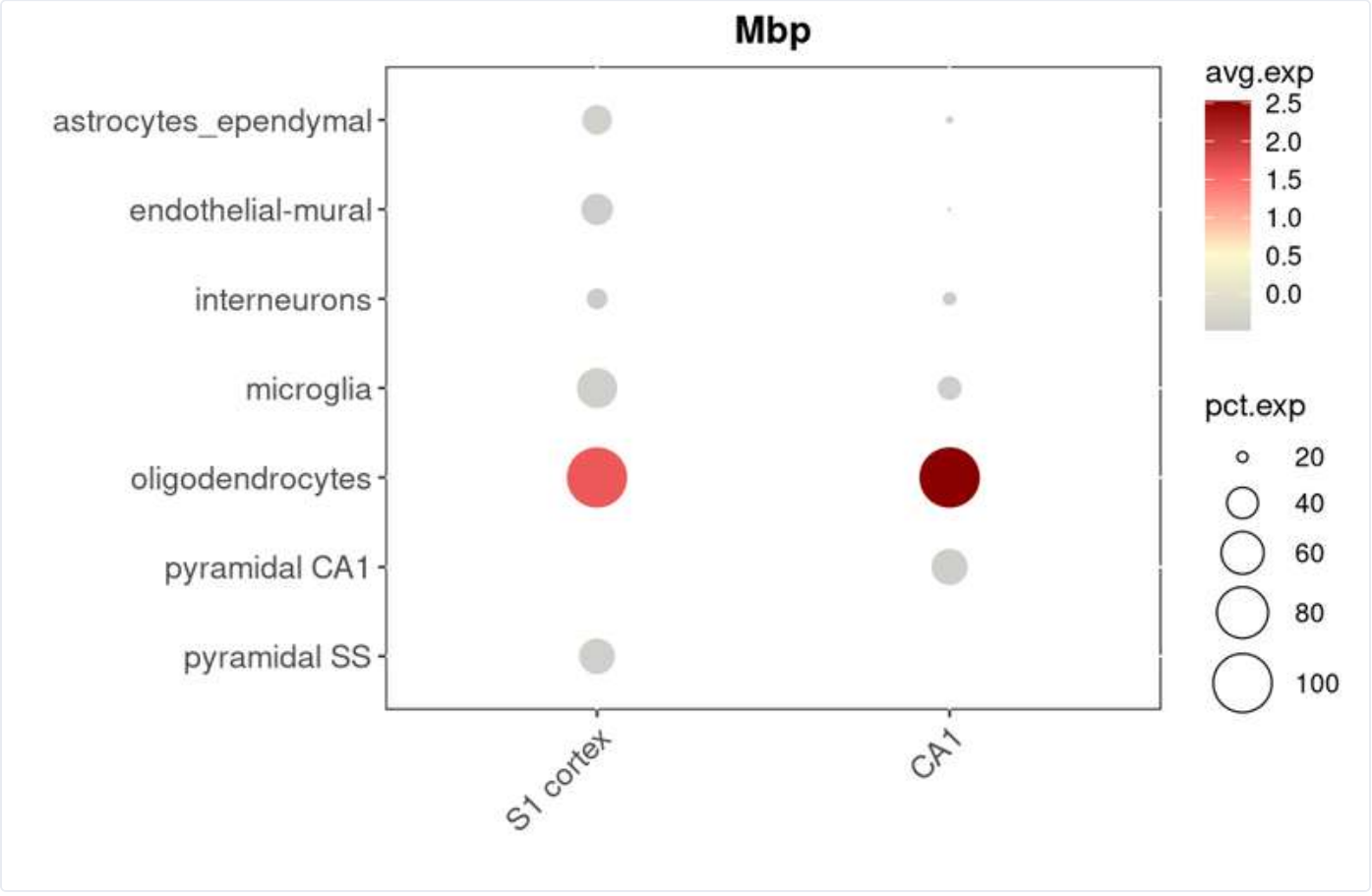
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 circlize_umap ● EMBEDDINGS



v2 cell_fraction ● COMPOSITION



v3 complex_dotplot ● HEATMAPS

veloVI

#14

OF 73 · NICHE & COOL SCORE

PYTHON | RNA VELOCITY | TRAJECTORY, PSEUDOTIME & VELOCITY

5.87 ×†	4/5	0.824	9	45
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · VELOVI

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.07 → 0.457 per 1k papers (2023 → 2025–26).
GitHub YosefLab/velovi · created 2022-03-10 · last push 2026-01-12

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

velocity uncertainty

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

veloVI (variational RNA velocity) on dentate gyrus: streamlines, intrinsic uncertainty, per-cell-type velocity coherence

KEY PLOTTING CALL

RECREATIONS/VELOVI/VELOVI_RECREATE.PY — EXCERPT · ENDS LINE 10

```
scv.settings.figdir="."; a=load(); a=preprocess_data(a)
VELOVI.setup_anndata(a,spliced_layer="Ms",unspliced_layer="Mu"); m=VELOVI(a); m.train(max_epochs=300,accelerator="cpu",enable_p
rogress_bar=False)
lt=m.get_latent_time(n_samples=25); sc_=20/lt.max(0); v=m.get_velocity(n_samples=25,velo_statistic="mean")
a.layers["velocity"]=v/sc_; a.layers["fit_t"]=lt.values*sc_.values[None,:]
scv.tl.velocity_graph(a,n_jobs=4)
scv.pl.velocity_embedding_stream(a,basis="umap",color="clusters",title="veloVI streamlines",legend_loc="right margin",save="v1_
velovi_stream.png",show=False)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/YosefLab/velovi.

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

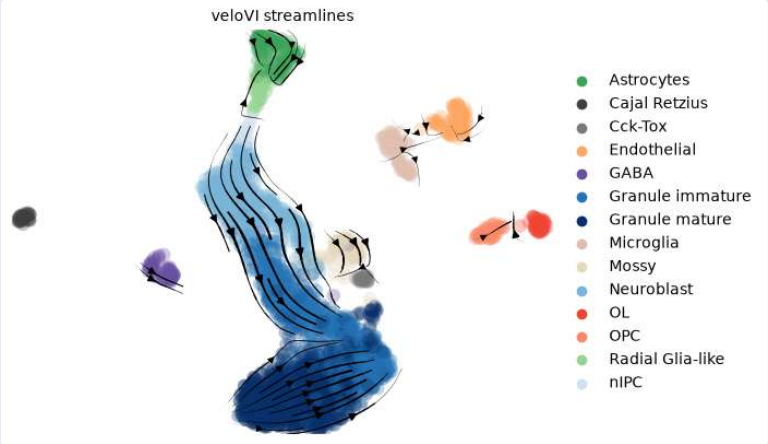
Reproduce: `python3 recreations/veloVI/velovi_recreate.py`

√ re-run end to end · 509 s

veloVI · 3 variations

NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.

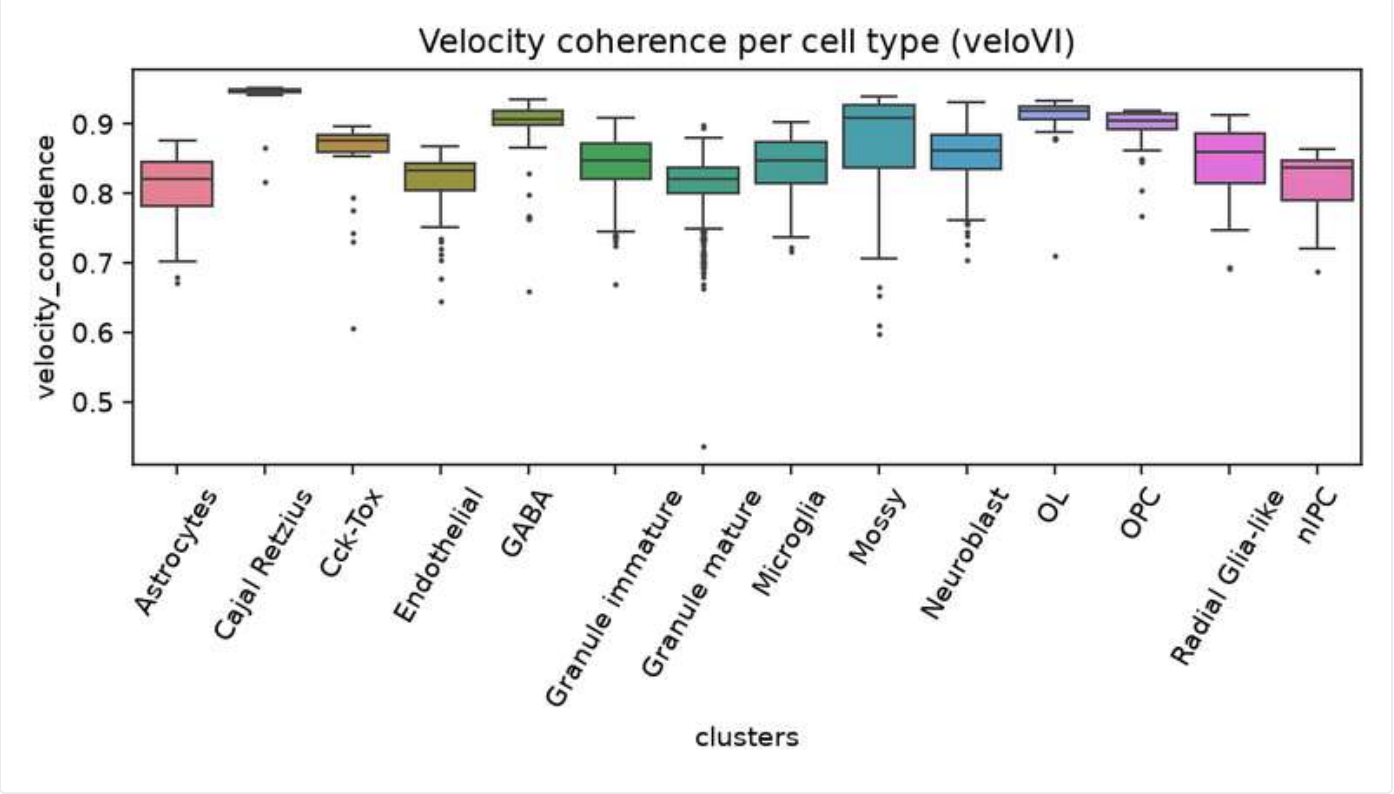


v1 velovi_stream

TRAJECTORY

v2 uncertainty

TRAJECTORY



v3 coherence_by_cluster

TRAJECTORY

miloR

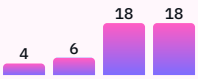
#15

OF 73 · NICHE & COOL SCORE

R/PYTHON | DIFFERENTIAL ABUNDANCE | COMPOSITION, DISTRIBUTIONS & LAYOUTS

2.31×	5/5	0.815	46	450
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · MILOR / MILO

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 1.184 → 2.743 per 1k papers (2023 → 2025–26).
GitHub MarioniLab/miloR · created 2020-07-30 · last push 2026-08-25

BEST FOR

DA beeswarm, nhood graph

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

miloR differential abundance on kNN neighbourhoods, Zeisel 2015: S1 cortex vs CA1 hippocampus, Fluidigm C1 chip (cell-name prefix) as the replicate. Neighbourhood graph, DA beeswarm, nhood size histogram.

KEY PLOTTING CALL

RECREATIONS/MILO/MILOR_RECREATE.R — EXCERPT · ENDS LINE 19

```
m <- calcNhoodDistance(m, d = 20, reduced.dim = "PCA")
da <- testNhoods(m, design = ~ tissue, design.df = design, reduced.dim = "PCA", fdr.weighting = "graph-overlap")
da <- annotateNhoods(m, da, coldata_col = "celltype"); write.csv(da, "milo_da_results.csv", row.names = FALSE)
m <- buildNhoodGraph(m)
p1 <- plotReducedDim(m, "UMAP", colour_by = "celltype", point_size = .6) + guides(colour = guide_legend(override.aes =
list(size = 3))) + ggtitle("Zeisel cell types")
p2 <- plotNhoodGraphDA(m, da, layout = "UMAP", alpha = .1) + ggtitle("Milo nhood graph: logFC sscortex vs CA1")
ggsave("v1_nhood_graph_DA.png", p1 + p2 + plot_layout(widths = c(1, 1.1)), width = 13, height = 5.5, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS · MOST-CITED OF 29 NAMING “MILOR” IN METHODS

94 CITATIONS — **Spatial and single-nucleus transcriptomic analysis of genetic and sporadic forms of Alzheimer's disease.** *Nat Genet, 2024*
DOI ↗ PMC11631771

51 CITATIONS — **Tracking cell-type-specific temporal dynamics in human and mouse brains.** *Cell, 2023*
DOI ↗ PMC10545416

Figure-level matching was run for the Top 100 only, so these papers name the tool but no specific figure is linked.

LIMITATIONS & NOTES

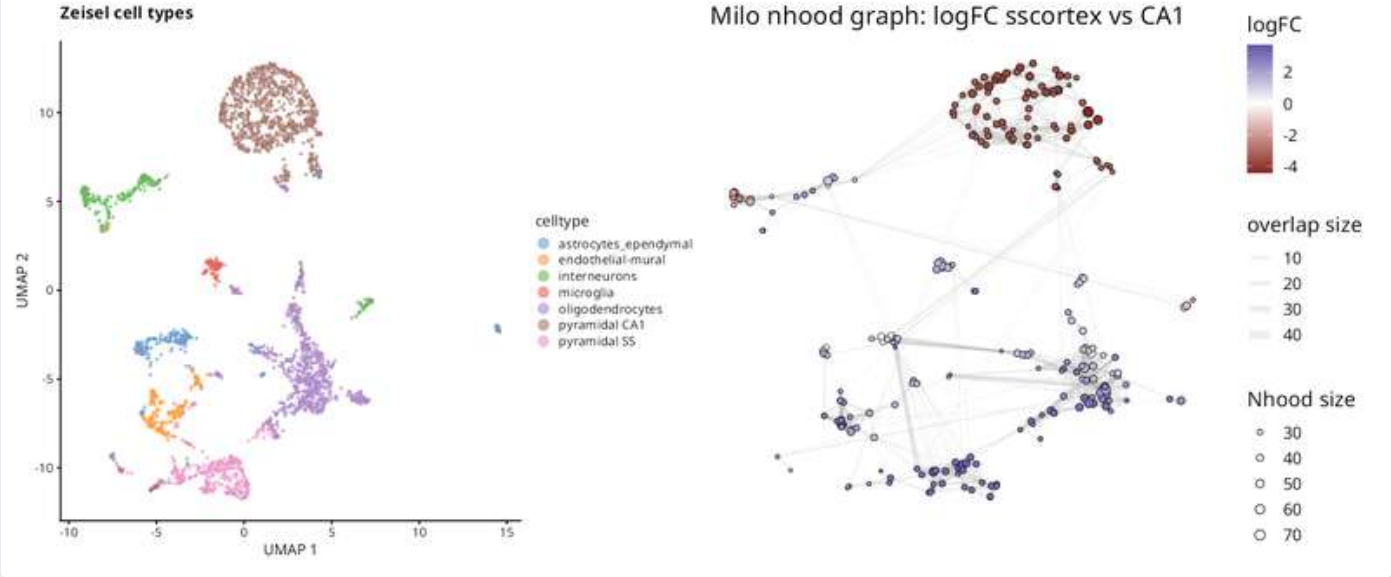
- miloR is built from source with the C++11 flag removed from src/Makevars (METHODS §9).
- The Fluidigm C1 chip ID is used as the replicate (METHODS §8).

Reproduce: `Rscript recreations/Milo/miloR_recreate.R`

✓ re-run end to end · 26 s

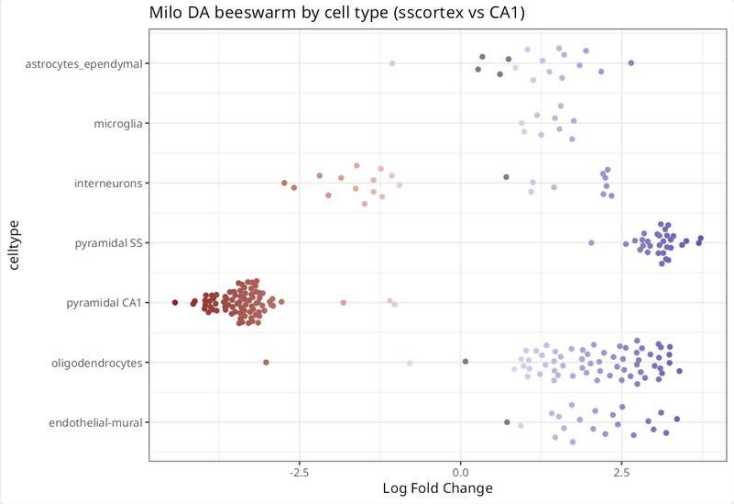
miloR · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



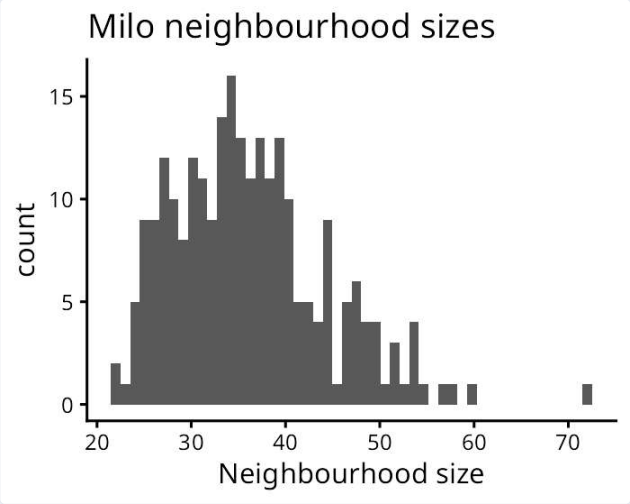
v1 nhood_graph_DA

COMPOSITION



v2 DA_beeswarm

COMPOSITION



v3 nhood_size_hist

COMPOSITION

Datashader

#16

OF 73 · NICHE & COOL SCORE

PYTHON | LARGE DATA | SPATIAL & TISSUE MAPS

4.81 ×†	4/5	0.814	1	3,568
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · DATASHADER

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0 → 0.038 per 1k papers (2023 → 2025–26).
GitHub holoviz/datashader · created 2015-12-23 · last push 2026-10-14

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

millions of points rasterised

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Datashader: all 73,655 MERFISH cells rasterised without overplotting

KEY PLOTTING CALL

RECREATIONS/DATASHADER/DATASHADER_RECREATE.PY — EXCERPT · ENDS LINE 7

```
cv=ds.Canvas(plot_width=900,plot_height=900)
tf.set_background(tf.shade(cv.points(s,"x","y",ds.count()),cmap=cc.fire,how="eq_hist"),"black").to_pil().save("v1_density_fire.png")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/holoviz/datashader.

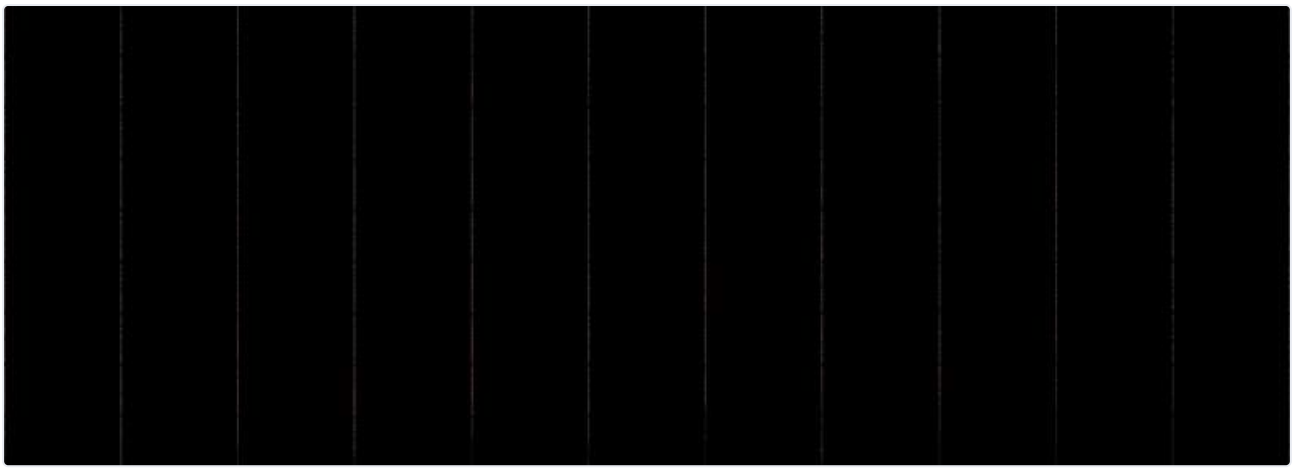
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/Datashader/datashader_recreate.py`

√ re-run end to end · 7 s

LEAD FIGURE



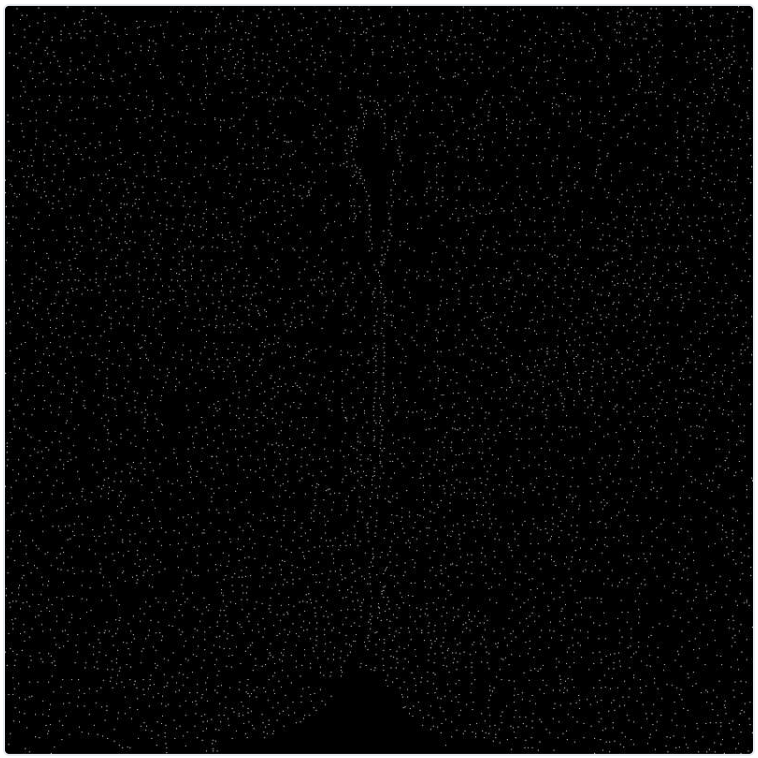
v3 all_slices_strip

SPATIAL

Datashader · 3 variations · v3 on facing page

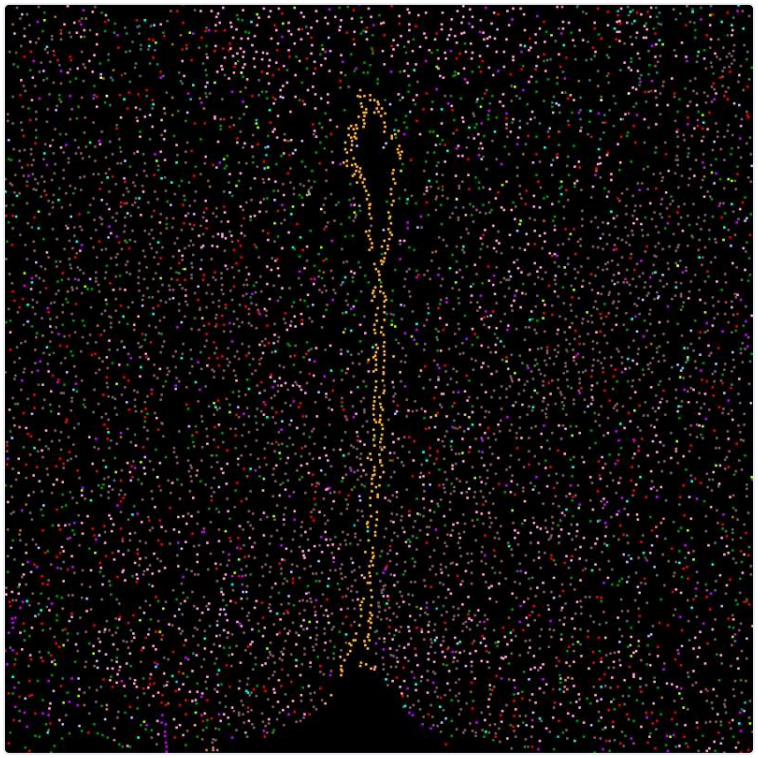
NOVAFLOW RECREATION

Data: Moffitt et al. 2018, mouse hypothalamus MERFISH, all 73,655 cells. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 density_fire

SPATIAL



v2 categorical_shading

SPATIAL

scDRS

#17

OF 73 · NICHE & COOL SCORE

PYTHON | DISEASE RELEVANCE | EMBEDDINGS & FEATURE MAPS | HEATMAPS & DOT MATRICES | COMPOSITION, DISTRIBUTIONS & LAYOUTS

4.63×	4/5	0.798	37	201
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES
SORTED, LOG SCALE

NICHE & COOL · SCDRS

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.209 → 1.003 per 1k papers (2023 → 2025–26).
GitHub martinjzhang/scDRS · created 2020-07-10 · last push 2026-08-27

BEST FOR

disease-score UMAPs, trait x cell-type heatmaps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

scDRS disease-relevance scores on Zeisel 2015 mouse cortex/hippocampus, using the scDRS data-release MAGMA gene sets (top-1000 genes, z-score weights; figshare 19312583, gs_file.zip, 0.9 MB, fetched once into _data/scdrs): schizophrenia, bipolar, MDD, Alzheimer's, and height as a non-brain control. Human→mouse by symbol case mapping (approximation; 1:1 orthologue tables were not used to keep this offline).

KEY PLOTTING CALL

RECREATIONS/SCDRS/SCDRS_RECREATE.PY — EXCERPT · ENDS LINE 25

```
fig,axs=plt.subplots(2,3,figsize=(16,10)); axs=axs.ravel()
sc.pl.umap(a,color="celltype",ax=axs[0],show=False,frameon=False,legend_loc="on data",legend_fontsize=7,title="Zeisel cell types")
for ax,t in zip(axs[1:],gs):
    sc.pl.umap(a,color=f"{t}_norm_score",ax=ax,show=False,frameon=False,cmap="RdBu_r",vcenter=0,vmax=4,vmin=-4,title=f"scDRS score: {t}")
fig.tight_layout(); fig.savefig("v1_umap_scores.png",dpi=130)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/martinjzhang/scDRS.

LIMITATIONS & NOTES

- Human → mouse genes mapped by symbol case: an approximation, since 1:1 orthologue tables were not used, to keep this offline.
- NumPy 2 shim for np.float_ (METHODS §9).

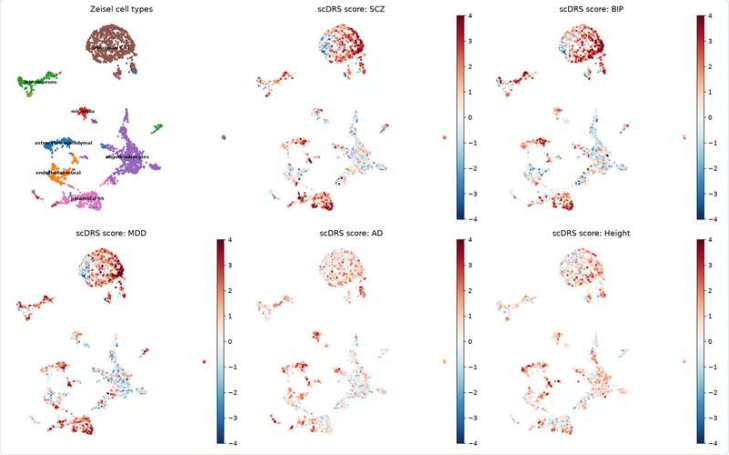
Reproduce: `python3 recreations/scDRS/scdrs_recreate.py`

✓ re-run end to end · 76 s

scDRS · 3 variations

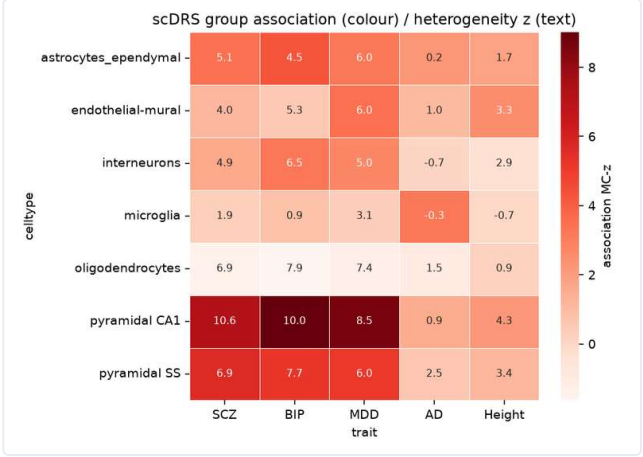
NOVAFLOW RECREATION

Data: Zeisel et al. 2015 scRNA-seq scored with scDRS data-release MAGMA gene sets (figshare 19312583); human→mouse genes mapped by symbol.
Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



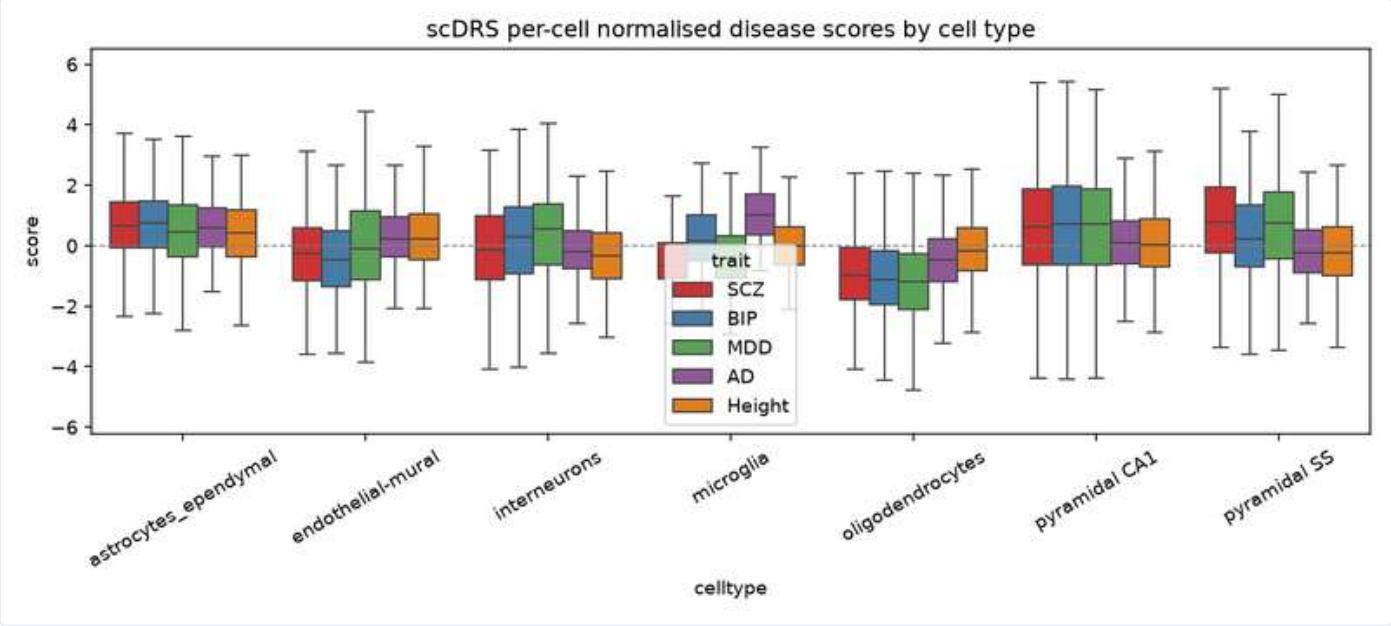
v1 umap_scores

EMBEDDINGS



v2 group_assoc_heatmap

HEATMAPS



v3 score_boxplots

COMPOSITION

MNE-Python

#19

OF 73 · NICHE & COOL SCORE

PYTHON | NEUROPHYSIOLOGY | BRAIN ATLAS, 3D & NEUROIMAGING

2.27×†	5/5	0.798	1	3,549
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · MNE-PYTHON

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0 → 0.013 per 1k papers (2023 → 2025–26).
GitHub mne-tools/mne-python · created 2011-01-28 · last push 2026-10-13

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

cortical surface source maps, topomaps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

MNE-Python on the PhysioNet EEG Motor Movement/Imagery dataset (EEGBCI; subject 1, imagery runs 6/10/14, ~8 MB): raw-trace browser figure, event-related (de)synchronisation time-frequency maps at C3/Cz/C4, and mu/beta-band topomaps for imagined hands vs feet.

KEY PLOTTING CALL

RECREATIONS/MNE/MNE_RECREATE.PY — EXCERPT · ENDS LINE 13

```
fig=raw.plot(duration=10,n_channels=24,scalings=dict(eeg=60e-6),show=False,title="EEGBCI S001 imagery (hands/feet)",show_scroll
bars=False)
fig.set_size_inches(12,7); fig.savefig("v1_raw_browser.png",dpi=120); plt.close(fig)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/mne-tools/mne-python.

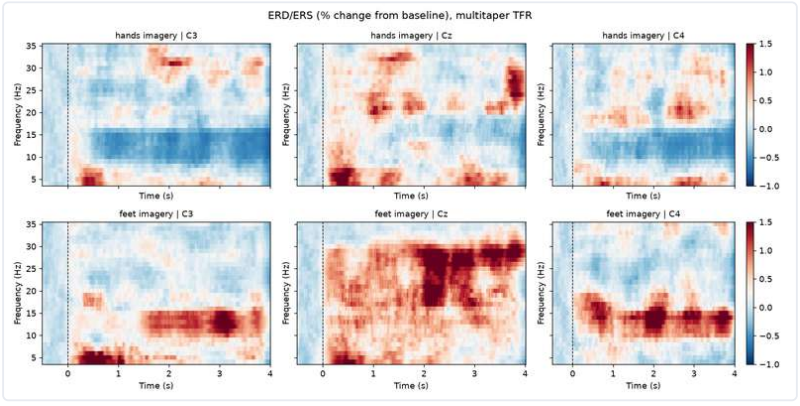
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/MNE/mne_recreate.py`

√ re-run end to end · 4 s

LEAD FIGURE



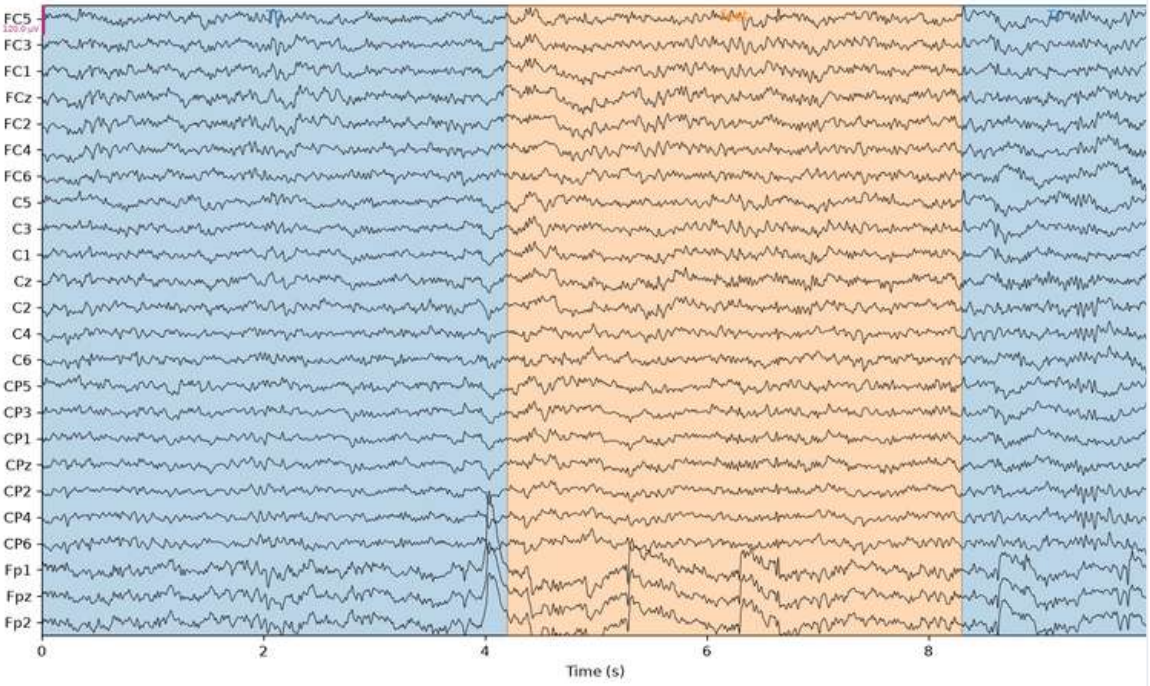
v2 erds_tfr

BRAIN

MNE-Python · 3 variations · v2 on facing page

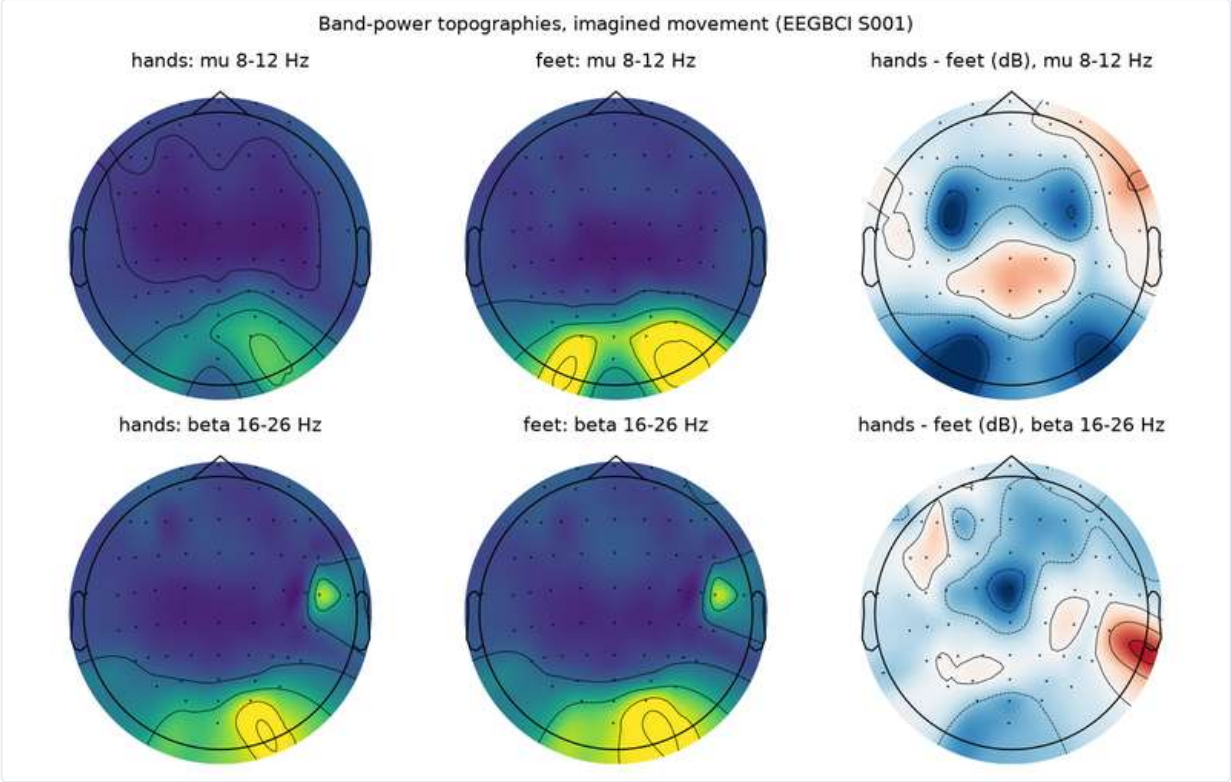
NOVAFLOW RECREATION

Data: PhysioNet EEG Motor Movement/Imagery dataset (EEGBCI), subject 1, imagery runs 6/10/14. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 raw_browser

BRAIN



v3 band_topomaps

BRAIN

sccomp

#20

OF 73 · NICHE & COOL SCORE

R | COMPOSITION | COMPOSITION, DISTRIBUTIONS & LAYOUTS

4.27 ×†	4/5	0.79	10	134
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · SCCOMP

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.07 → 0.33 per 1k papers (2023 → 2025–26).
GitHub MangiolaLaboratory/sccomp · created 2021-02-08 · last push 2026-10-06

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

posterior-predictive composition boxplots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

sccomp Bayesian compositional analysis (CmdStan backend) on Zeisel 2015: cortex vs CA1, C1 chip as sample. Pyramidal subtypes are tissue-exclusive so they are excluded. Posterior boxplots, 1D credible intervals, and the mean-variability 2D plot. Requires cmdstanr + CmdStan (installed with cmdstanr::install_cmdstan()).

KEY PLOTTING CALL

RECREATIONS/SCCOMP/SCCOMP_RECREATE.R — EXCERPT · ENDS LINE 13

```
est <- cnt %>% sccomp_estimate(formula_composition = ~ tissue, sample = "chip", cell_group = "celltype", abundance = "count", c
ores = 2, verbose = FALSE) %>%
  sccomp_remove_outliers(cores = 2, verbose = FALSE) %>% sccomp_test()
ggsave("v1_posterior_boxplot.png", sccomp_boxplot(est, factor = "tissue") + ggtitle("sccomp: proportions with posterior
predictive (cortex vs CA1)"), width = 11, height = 5, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/MangiolaLaboratory/sccomp.

LIMITATIONS & NOTES

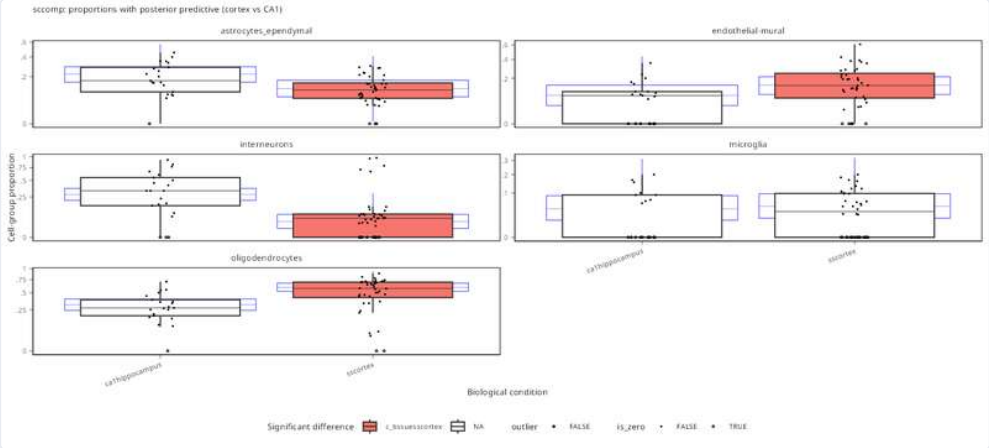
- Requires cmdstanr + CmdStan. Pyramidal subtypes are tissue-exclusive, so they are excluded; the C1 chip is used as the sample.

Reproduce: `Rscript recreations/sccomp/sccomp_recreate.R`

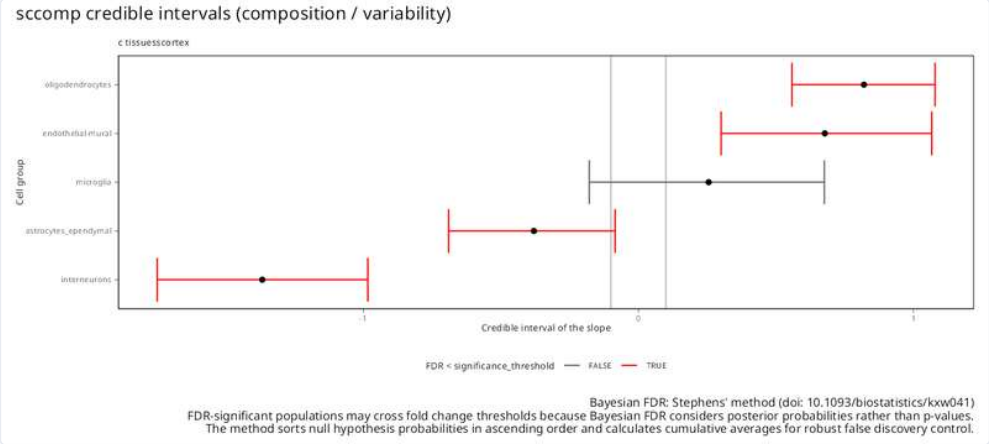
√ re-run end to end · 57 s

sccomp · 3 variations

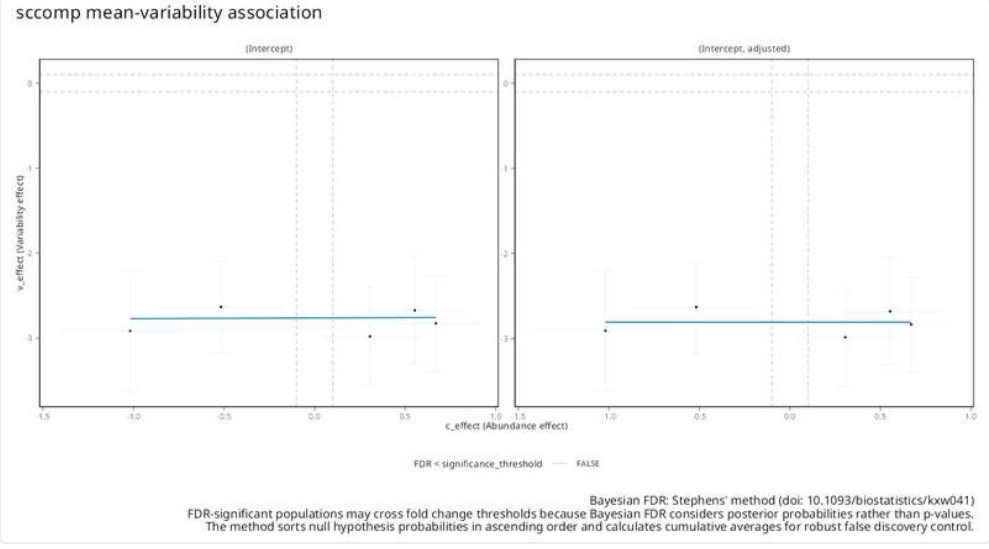
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 `posterior_boxplot` COMPOSITION



v2 `1D_intervals` COMPOSITION



v3 `2D_mean_variability` COMPOSITION

PyComplexHeatmap

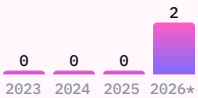
#26

PYTHON · HEATMAP · HEATMAPS & DOT MATRICES

OF 73 · NICHE & COOL SCORE

3.06×†	4/5	0.75	2	407
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · PYCOMPLEXHEATMAP

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.035 → 0.127 per 1k papers (2023 → 2025–26).
GitHub DingWB/PyComplexHeatmap · created 2022-05-05 · last push 2026-09-01

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

ComplexHeatmap-style in Python, dot heatmaps

KEY PLOTTING CALL

RECREATIONS/PYCOMPLEXHEATMAP/PYCH_RECREATE.PY — EXCERPT · ENDS LINE 11

```
X=pd.DataFrame(ad[idx,mk].X.toarray().T,index=mk,columns=ad.obs_names[idx]); meta=ad.obs.iloc[idx][["celltype","tissue"]]
col_ha=HeatmapAnnotation(celltype=anno_simple(meta.celltype,cmap="tab10",legend=True,add_text=False),tissue=anno_simple(meta.ti
ssue,cmap="Set2"),nGene=anno_barplot(ad.obs.iloc[idx].nFeature_RNA,cmap="Greys",height=10),verbose=0,axis=1)
plt.figure(figsize=(9,5)); ClusterMapPlotter(data=X,top_annotation=col_ha,col_split=meta.celltype,col_split_gap=.8,row_cluster=T
rue,col_cluster=False,cmap="Reds",show_rownames=True,label="log-norm",row_dendrogram=True,verbose=0,rasterized=True)
plt.savefig("v1_split_annotated.png",dpi=150,bbox_inches="tight"); plt.close("all")
```

IN PUBLISHED PAPERS

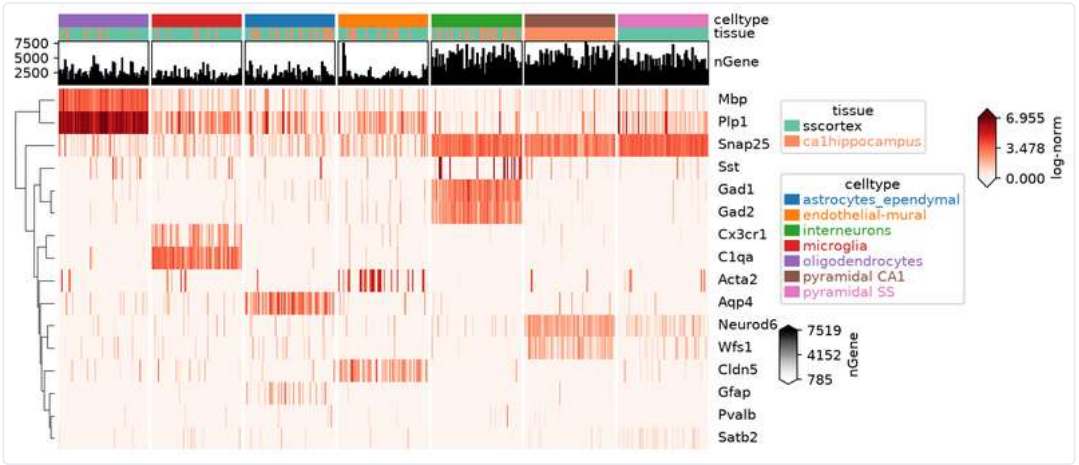
No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/DingWB/PyComplexHeatmap.

LIMITATIONS & NOTES

- Shim: matplotlib.cm.get_cmap restored (removed in matplotlib 3.9+, still called by PyComplexHeatmap) (METHODS §9).

Reproduce: `python3 recreations/PyComplexHeatmap/pych_recreate.py` — √ re-run end to end · 7 s

LEAD FIGURE



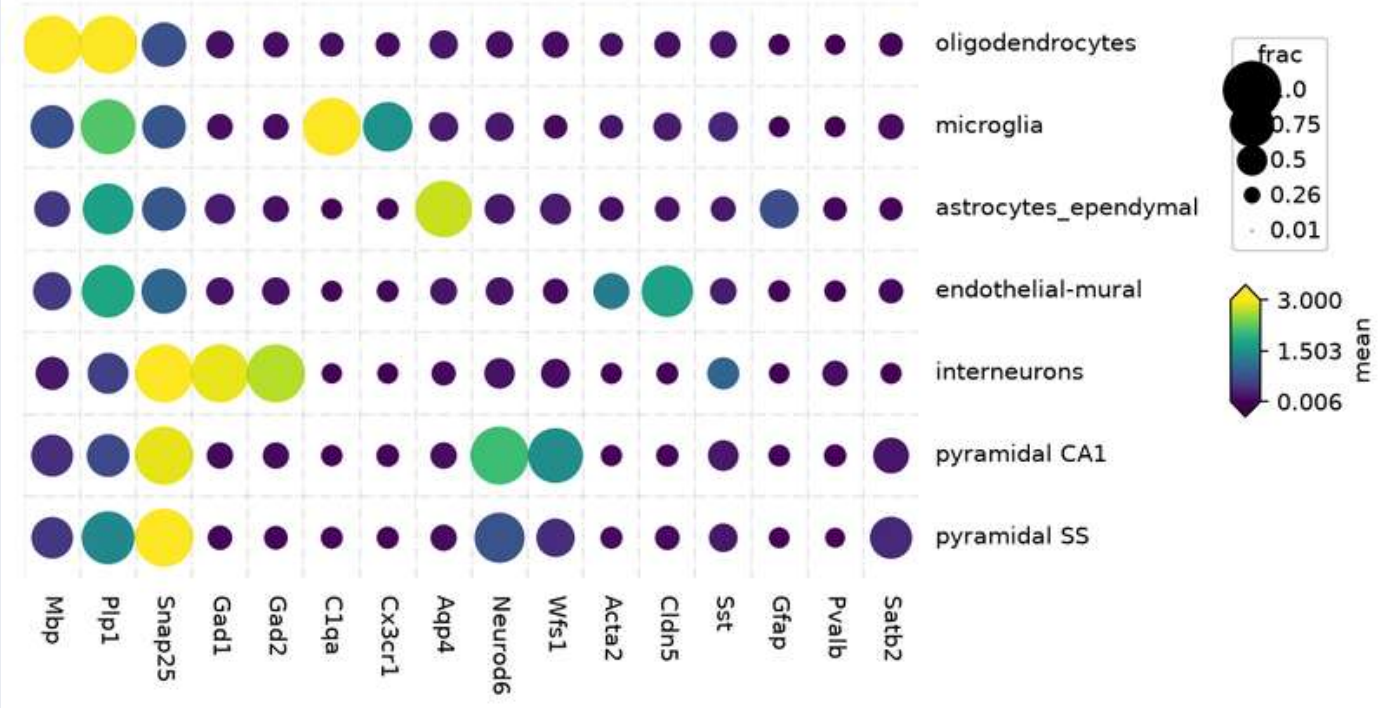
v1 split_annotated

HEATMAPS

PyComplexHeatmap · 2 variations · v1 on facing page

NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v2 dot_clustermap

HEATMAPS

nilearn

#29

PYTHON | NEUROIMAGING | BRAIN ATLAS, 3D & NEUROIMAGING

OF 73 · NICHE & COOL SCORE

1.92×†	5/5	0.733	7	1,444
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · NILEARN

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.035 → 0.076 per 1k papers (2023 → 2025–26). GitHub nilearn/nilearn · created 2011-01-09 · last push 2026-10-14

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

glass brains, surface maps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

nilearn brain visualisations on real public maps (NeuroVault motor-task t-map; Harvard-Oxford atlas; fsaverage surface)

KEY PLOTTING CALL

RECREATIONS/NILEARN/NILEARN_RECREATE.PY — EXCERPT · ENDS LINE 6

```
plotting.plot_glass_brain(mot,threshold=3,colorbar=True,plot_abs=False,display_mode="lyrz",title="Glass brain: motor task t-map").savefig("v1_glass_brain.png",dpi=150)
```

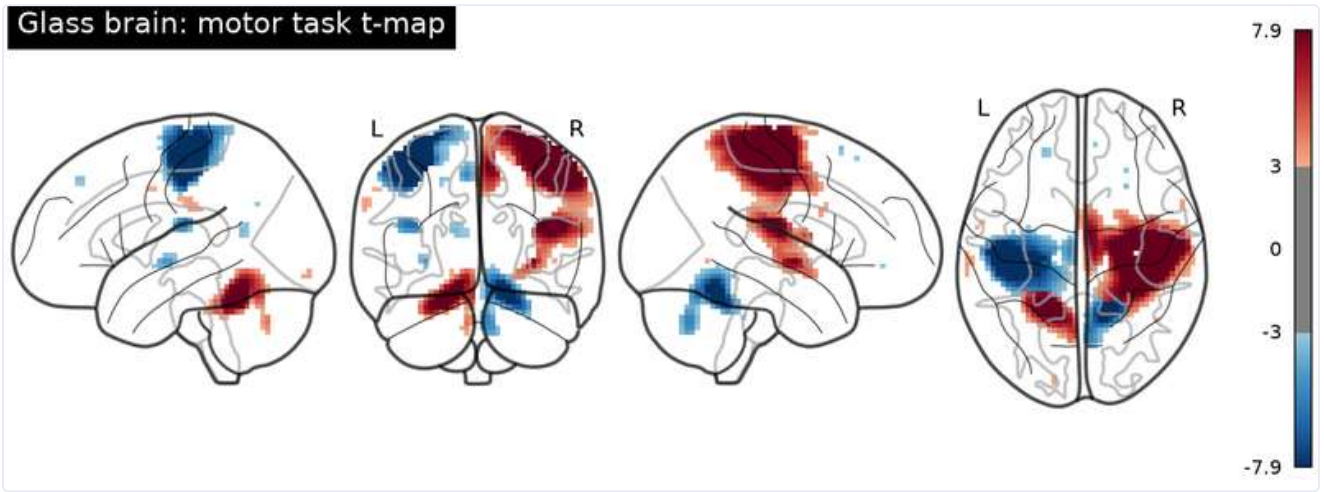
IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/nilearn/nilearn.

> LIMITATIONS & NOTES
>
> No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/nilearn/nilearn_recreate.py` — ✓ re-run end to end · 5 s

LEAD FIGURE



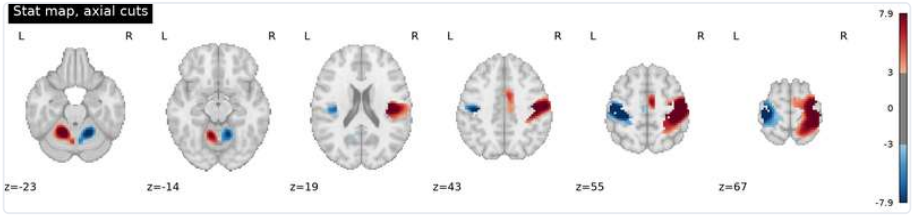
v1 glass_brain

BRAIN

nilearn · 4 variations · v1 on facing page

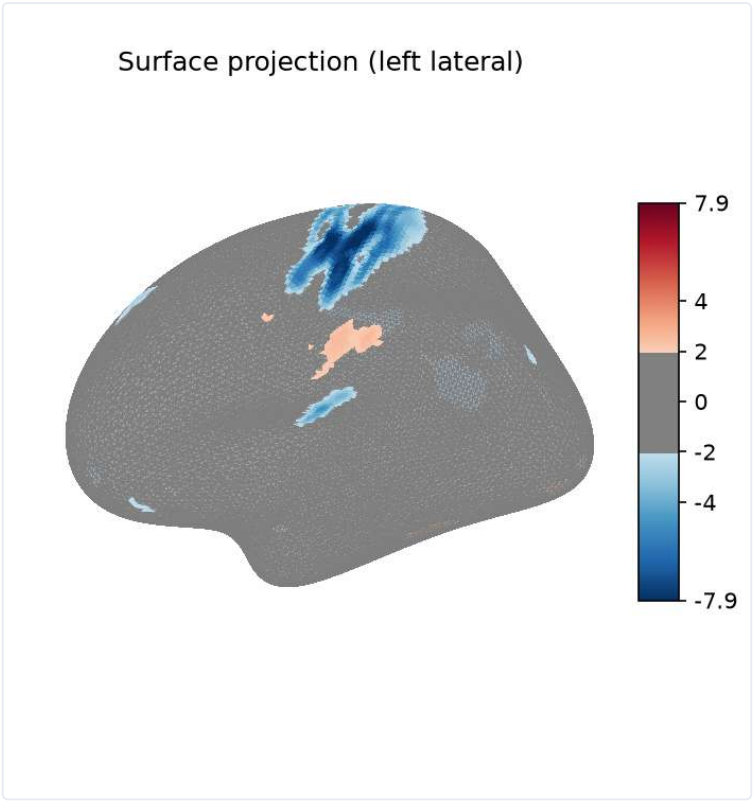
NOVAFLOW RECREATION

Data: NeuroVault motor-task t-map, Harvard-Oxford atlas and fsaverage surface (nilearn datasets). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



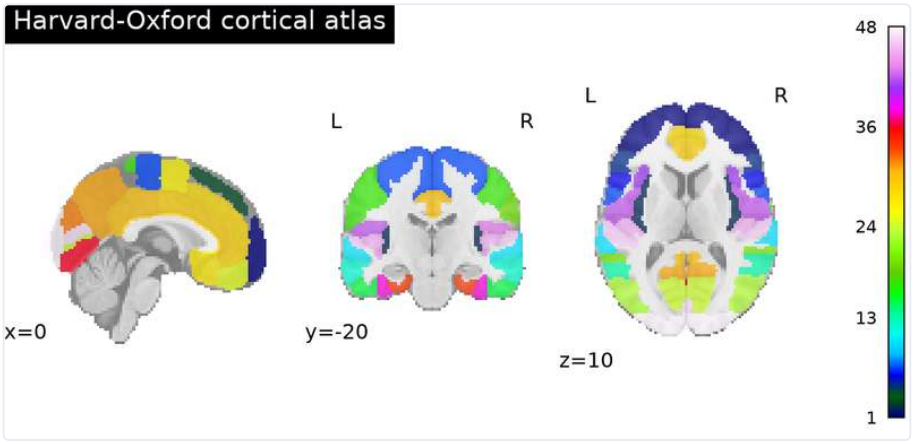
v2 stat_map_cuts

BRAIN



v3 surface_stat_map

BRAIN



v4 atlas_roi

BRAIN

SCpubr

#31

OF 73 · NICHE & COOL SCORE

R · SINGLE-CELL PUBLICATION PLOTS · EMBEDDINGS & FEATURE MAPS · HEATMAPS & DOT MATRICES · CELL-CELL COMMUNICATION · COMPOSITION, DISTRIBUTIONS & LAYOUTS

2.19×	4/5	0.671	14	245
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · SCPUBR

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.418 → 0.927 per 1k papers (2023 → 2025–26).
GitHub enblacar/SCpubr · created 2021-12-21 · last push 2026-03-02

BEST FOR

polished Seurat plots: nebulosa, geyser, alluvial, chord, enrichment

KEY PLOTTING CALL

RECREATIONS/SCPUBR/SCPUBR_RECREATE.R — EXCERPT · ENDS LINE 5

```
sv("v1_do_DimPlot.png", do_DimPlot(so, label = TRUE, repel = TRUE, legend.position = "none", plot.title = "SCpubr::do_DimPlot"))
```

IN PUBLISHED PAPERS

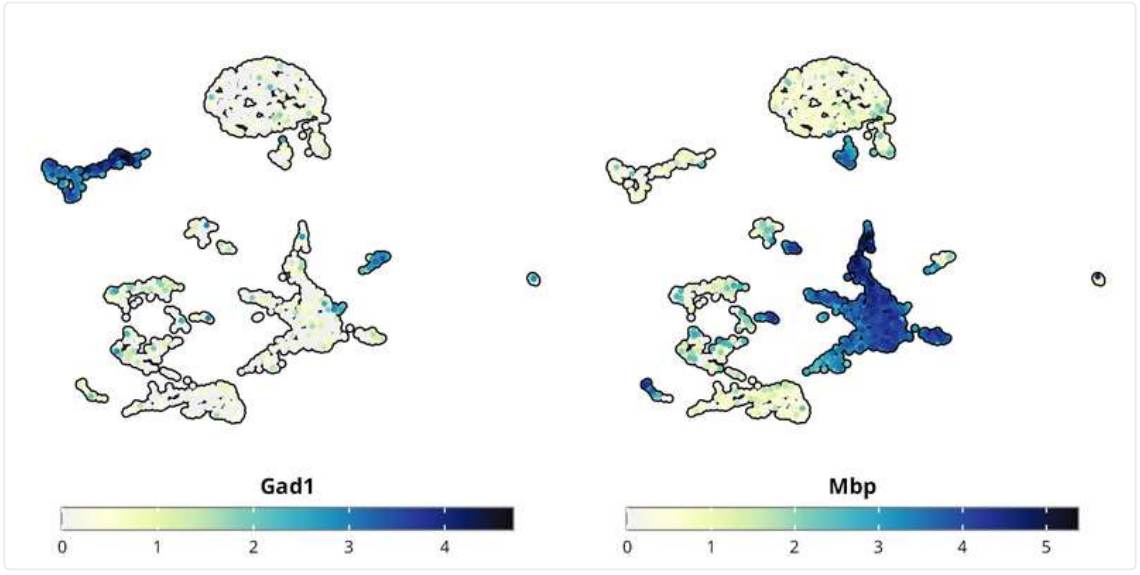
No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/enblacar/SCpubr.

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/SCpubr/SCpubr_recreate.R` — √ re-run end to end · 14 s

LEAD FIGURE



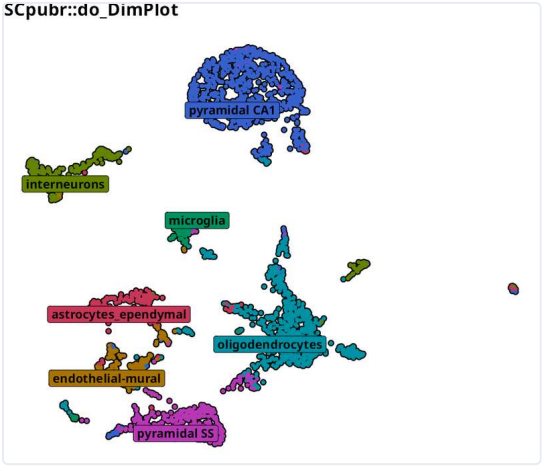
v2 do_NebulosaPlot

EMBEDDINGS

SCpubr · 5 variations · v2 on facing page

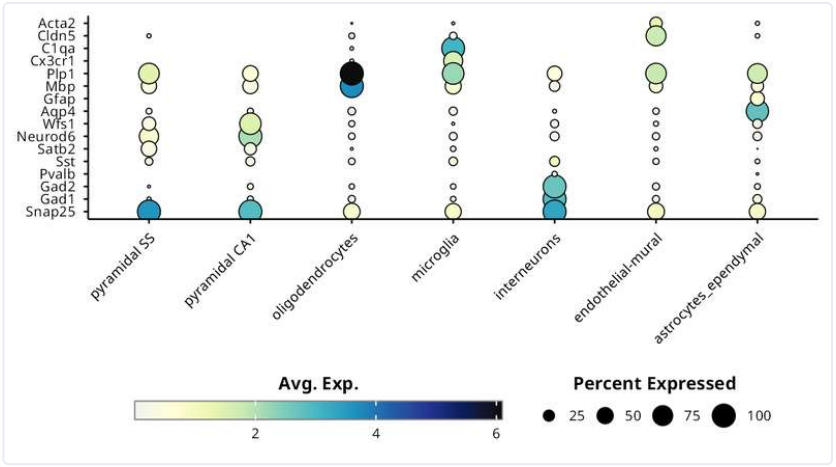
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



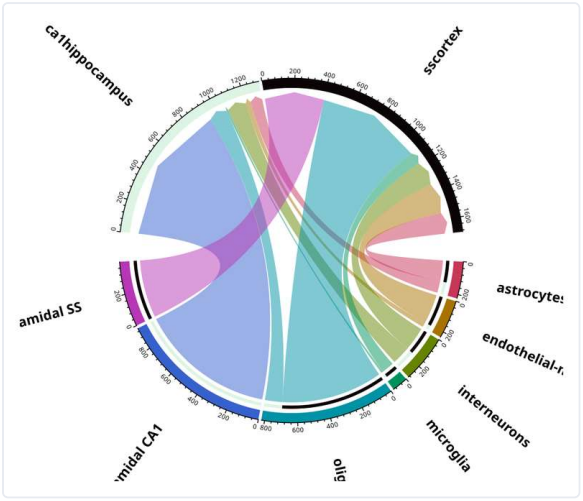
v1 do_DimPlot

EMBEDDINGS



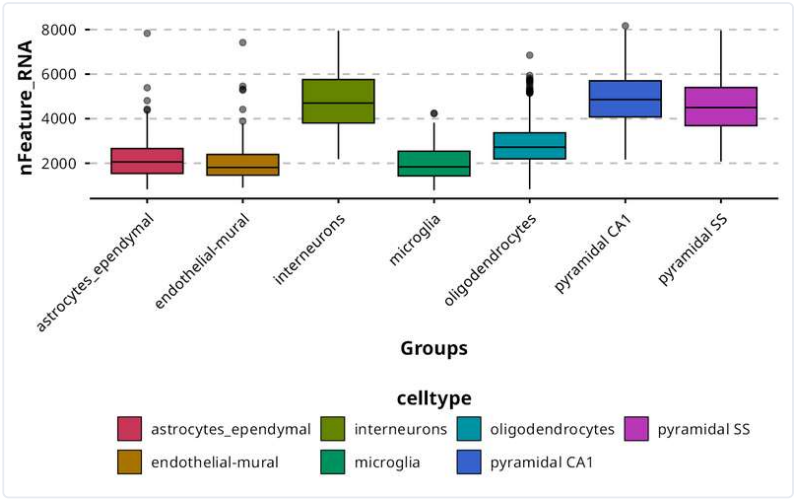
v3 do_DotPlot

HEATMAPS



v4 do_ChordDiagram

CELL–CELL



v5 do_BoxPlot

COMPOSITION

UCell

#34

OF 73 · NICHE & COOL SCORE

R · GENE-SET SCORES · EMBEDDINGS & FEATURE MAPS · COMPOSITION, DISTRIBUTIONS & LAYOUTS · HEATMAPS & DOT MATRICES

3.15×	3/5	0.663	144	213
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · UCELL

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 3.134 → 9.881 per 1k papers (2023 → 2025–26).
GitHub carmonalab/UCell · created 2021-04-06 · last push 2026-09-08

BEST FOR

signature score UMAPs

KEY PLOTTING CALL

RECREATIONS/UCELL/UCELL_RECREATE.R — EXCERPT · ENDS LINE 7

```
so <- AddModuleScore_UCell(so, features = sigs, ncores = 2); sc <- paste0(names(sigs), "_UCell")
ggsave("v1_UCell_featureplots.png", FeaturePlot(so, sc, ncol = 3, order = TRUE) & scale_color_viridis_c(option = "A"), width = 1
2, height = 7, dpi = 110, bg = "white")
```

IN PUBLISHED PAPERS

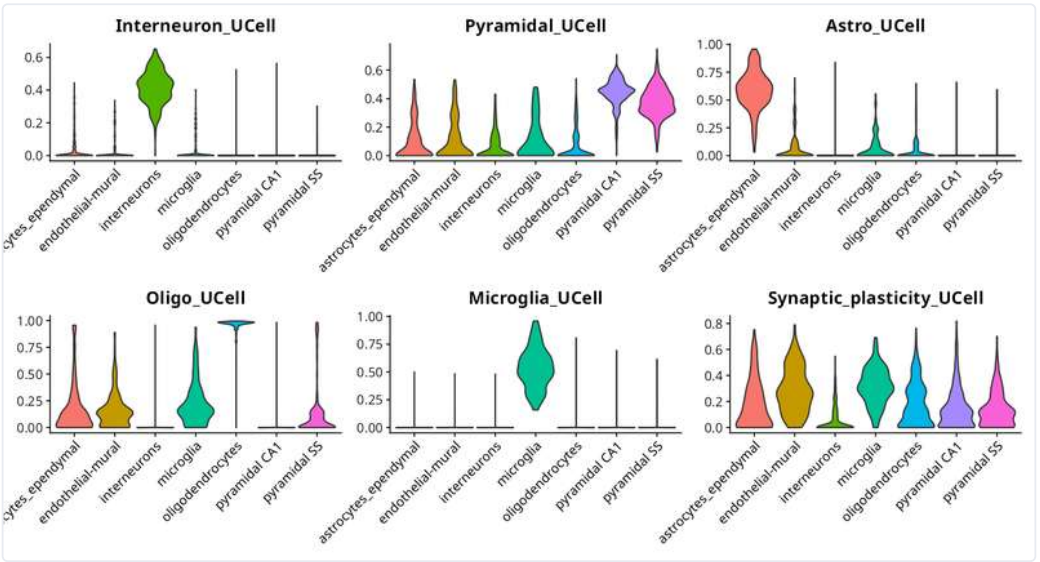
No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/carmonalab/UCell.

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/UCell/UCell_recreate.R` — ✓ re-run end to end · 24 s

LEAD FIGURE

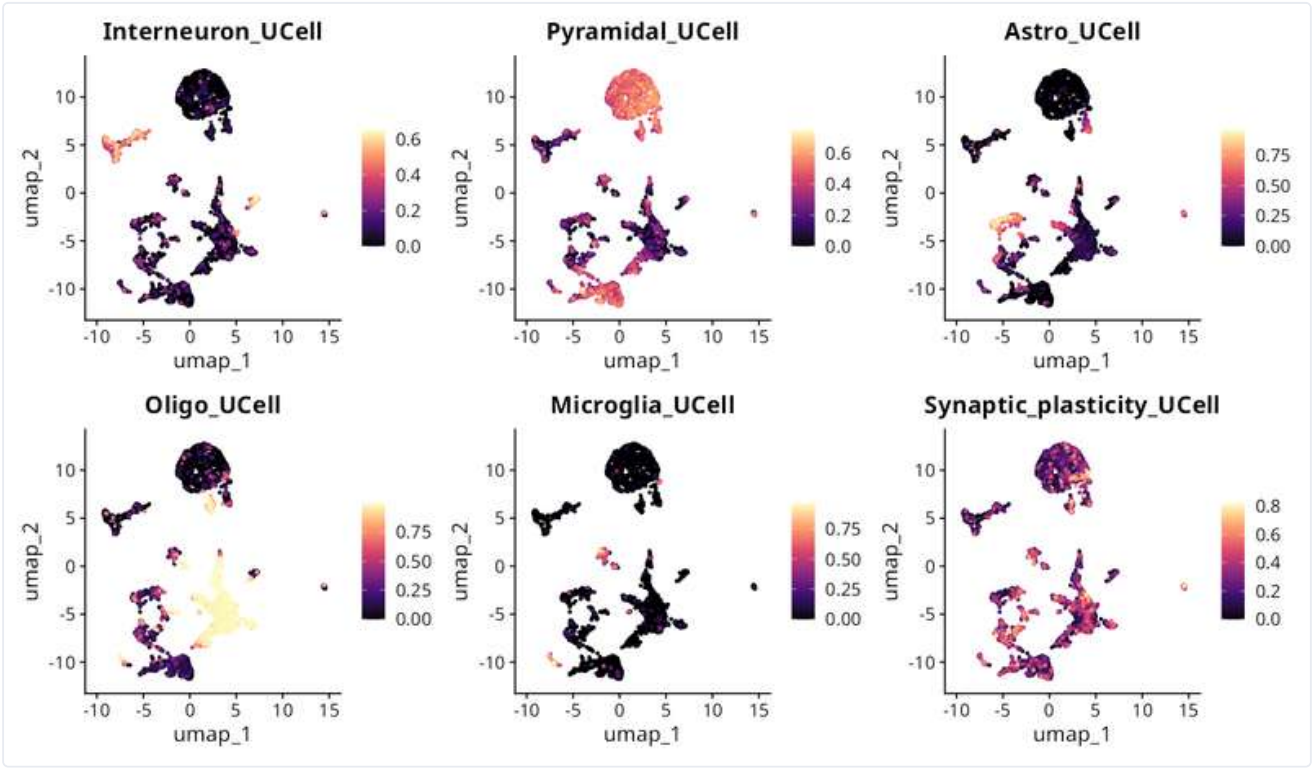


v2 UCell_violins — COMPOSITION

UCell · 3 variations · v2 on facing page

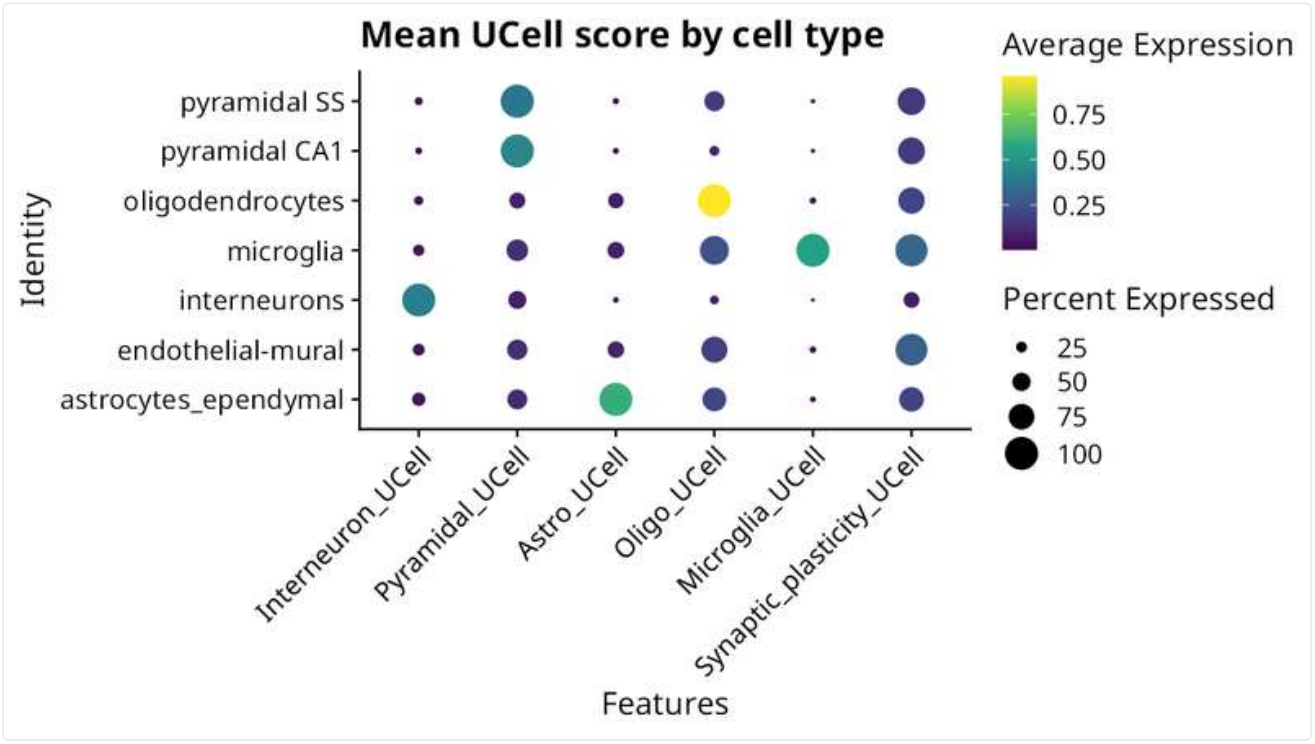
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 UCell_featureplots

EMBEDDINGS



v3 UCell_dotplot

HEATMAPS

spatialLIBD

#37

OF 73 · NICHE & COOL SCORE

R | SPATIAL | SPATIAL & TISSUE MAPS

1.97×	4/5	0.64	73	115
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · SPATIALLIBD

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.975 → 1.93 per 1k papers (2023 → 2025–26).
GitHub LieberInstitute/spatialLIBD · created 2019-12-04 · last push 2026-10-13

BEST FOR

layer-wise Visium plots, DLPFC

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

spatialLIBD look on DLPFC 151673: layer annotation on H&E, gene on H&E, layer-wise marker boxplots

KEY PLOTTING CALL

RECREATIONS/SPATIALLIBD/SPATIALLIBD_RECREATE.R — EXCERPT · ENDS LINE 6

```
spe <- load_dlpfc(); rowData(spe)$gene_search <- paste0(rowData(spe)$gene_name, "; ", rowData(spe)$gene_id); spe$key <- paste0(s
pe$sample_id, "_", colnames(spe))
spe$layer <- factor(spe$ground_truth); cols <- setNames(c("#F0027F","#377EB8","#4DAF4A","#984EA3","#FFD700","#FF7F00","#1A1A1A")
, levels(spe$layer))
ggsave("v1_vis_clus_layers.png", vis_clus(spe, sampleid = "sample_151673", clustervar = "layer", colors = cols, spatial = TRUE,
point_size = 1.6, image_id = "lowres") + ggtitle("vis_clus: manual cortical layers"), width = 7, height = 6.5, dpi = 140, bg = "
white")
```

IN PUBLISHED PAPERS · MOST-CITED OF 35 NAMING "SPATIALLIBD" IN METHODS

537 CITATIONS	**Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST.** *Nat Commun, 2023*	DOI ↗ PMC9977836
227 CITATIONS	**BANKSY unifies cell typing and tissue domain segmentation for scalable spatial omics data analysis.** *Nat Genet, 2024*	DOI ↗ PMC10937399

Figure-level matching was run for the Top 100 only, so these papers name the tool but no specific figure is linked.

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

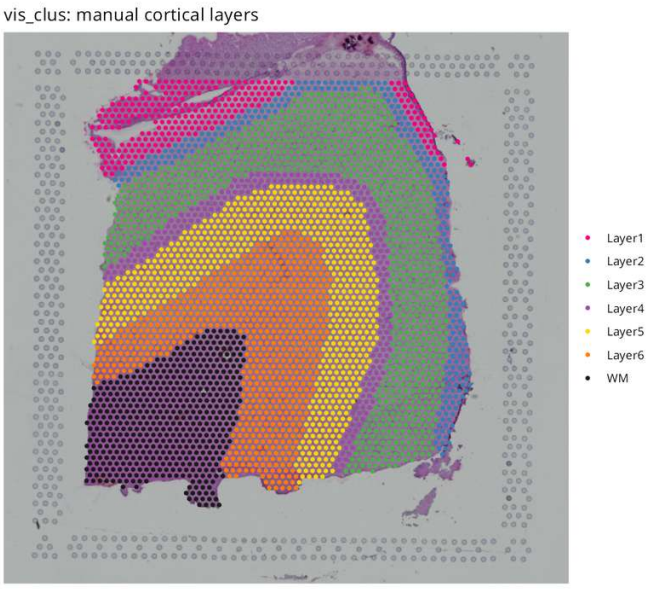
Reproduce: `Rscript recreations/spatialLIBD/spatialLIBD_recreate.R`

√ re-run end to end · 29 s

spatialLIBD · 4 variations

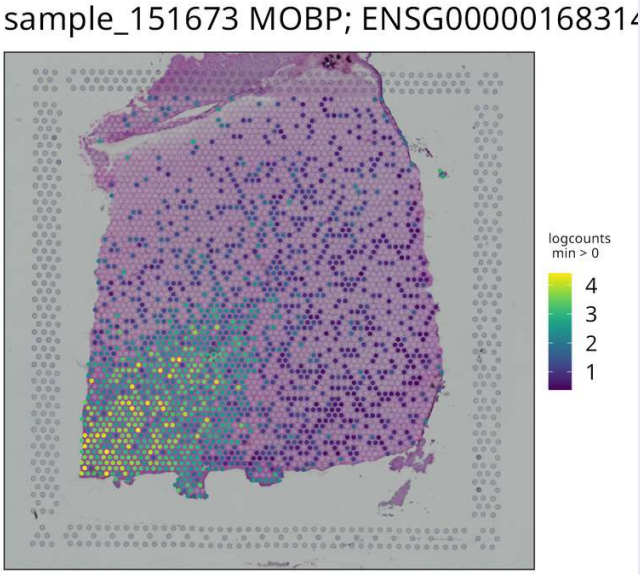
NOVAFLOW RECREATION

Data: Maynard et al. 2021, human DLPFC Visium sample 151673 (STexampleData). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



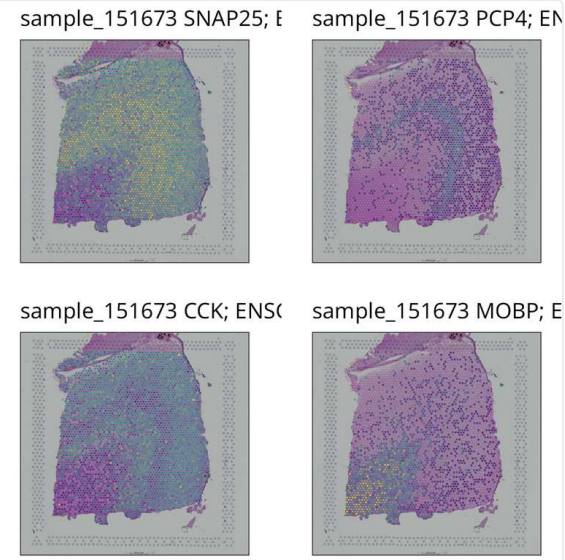
v1 vis_clus_layers

SPATIAL



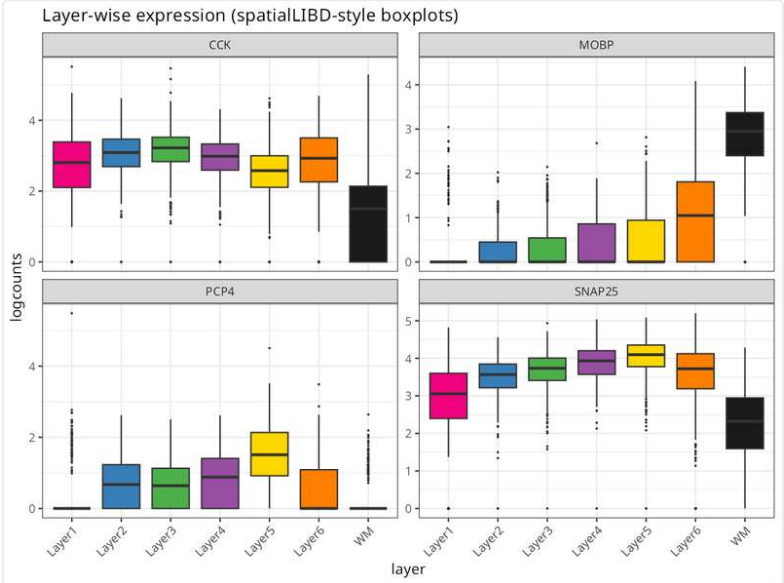
v2 vis_gene_MOBP

SPATIAL



v3 vis_grid_genes

SPATIAL



v4 layer_boxplots

SPATIAL

ggrepel volcano styles

#41

OF 73 · NICHE & COOL SCORE

R VOLCANO DE, QC & SET OVERLAPS

2.3×	3/5	0.608	74	1,262
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · GGREPEL VOLCANO STYLES

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 1.532 → 3.531 per 1k papers (2023 → 2025–26).
GitHub slowkow/ggrepel · created 2016-01-03 · last push 2026-10-09

BEST FOR

label-repelled volcanoes

Also catalogued as: **ggrepel** is Top-100 #64 (92 papers; recreated under ggplot2), examples below are from that entry

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Volcano-plot styles seen in recent neuro papers (ggplot2 + ggrepel), airway DESeq2 results

KEY PLOTTING CALL

RECREATIONS/GGREPEL_VOLCANO_STYLES/VOLCANO_STYLES_RECREATE.R — EXCERPT · ENDS LINE 8

```
lab <- res |> filter(sig) |> group_by(dir) |> slice_min(padj, n = 8)
g <- ggplot(res, aes(log2FoldChange, nl)) + geom_point(data = filter(res, !sig), color = "grey85", size = .6) + geom_point(data
= filter(res, sig), aes(color = log2FoldChange, size = nl), alpha = .8) + scale_color_gradient2(low = "#2166ac", mid =
"grey90", high = "#b2182b") + scale_size(range = c(.4, 3), guide = "none") +
  geom_label_repel(data = lab, aes(label = symbol, fill = dir), color = "white", size = 3, segment.color = "grey40", min.segment
.length = 0, box.padding = .5) + scale_fill_manual(values = c(up = "#b2182b", down = "#2166ac"), guide = "none") +
theme_classic() + labs(title = "Gradient + sized points + filled repel labels")
ggsave("v1_gradient_labels.png", g, width = 7, height = 6, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS · FIGURES BY THE AUTHORS THAT NAME THE PACKAGE

Fig. 3 TEXT

TDP-43 nuclear loss in FTD/ALS causes widespread alternative polyadenylation changes. *Nat Neurosci, 2025*

"...All plots were prepared using tidyplots, ggplot2, ggpubr, patchwork and ggrepel in R."

DOI ↗ Fig. 3 on PMC ↗

Fig. 3 TEXT

A concerted neuron-astrocyte program declines in ageing and schizophrenia. *Nature, 2024*

"...R (v.4.1.3): cluster (v.2.1.2)138, ComplexHeatmap (v.2.10.0)139,140, data.table (v.1.14.8)141, DescTools (v.0.99.48)142, dplyr (v.1.1.2)143, gdata (v.2.19.0)144, ggforce (v.0.4.1)145, ggplot2 (v.3.4.2)146 ..."

DOI ↗ Fig. 3 on PMC ↗

Fig 1 TEXT

Loss of STING in parkin mutant flies suppresses muscle defects and mitochondria damage. *PLoS Genet, 2023*

"...For all plots, ggplot2 v.3.4 and ggrepel v.0.9.2 were used for annotation."

DOI ↗ Fig 1 on PMC ↗

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

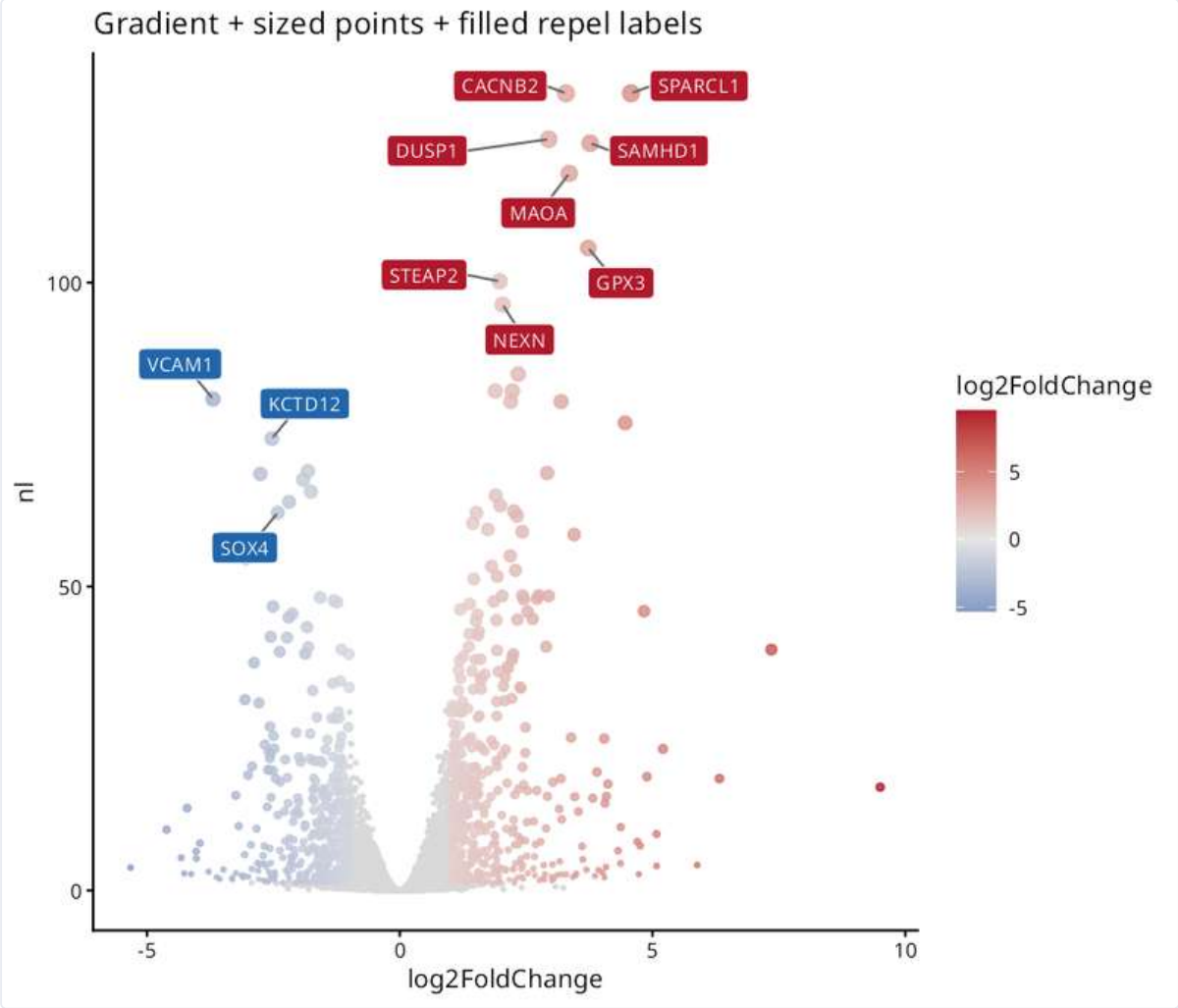
Reproduce: `Rscript recreations/ggrepel_volcano_styles/volcano_styles_recreate.R`

√ re-run end to end · 3 s

ggrepel volcano styles · 3 variations

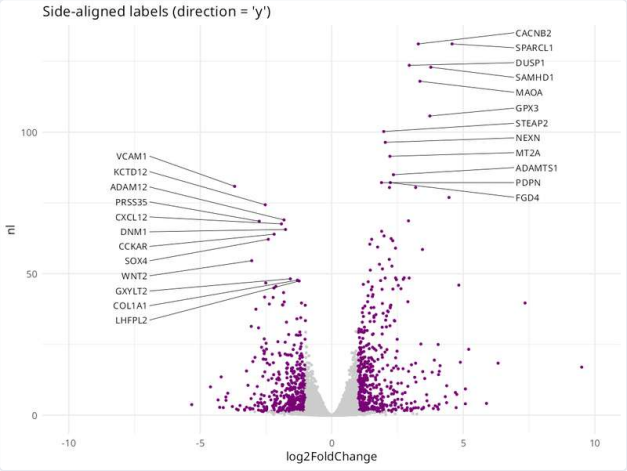
NOVAFLOW RECREATION

Data: airway: human airway smooth-muscle bulk RNA-seq, dexamethasone vs untreated (Himes et al. 2014; Bioconductor *airway*). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



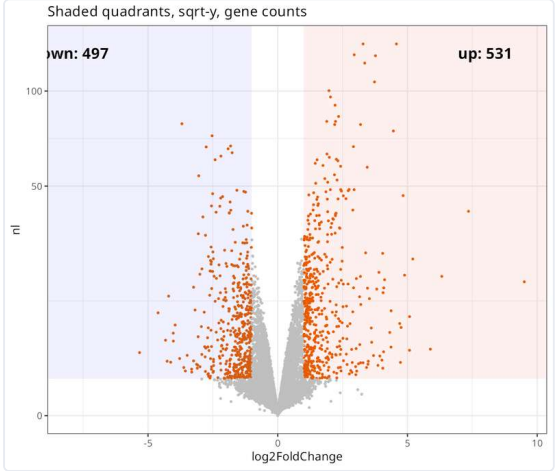
v1 gradient_labels

DE



v2 side_aligned

DE



v3 quadrants_counts

DE

Augur

#42

OF 73 · NICHE & COOL SCORE

R | PERTURBATION PRIORITIZATION | COMPOSITION, DISTRIBUTIONS & LAYOUTS | EMBEDDINGS & FEATURE MAPS

1.86×	4/5	0.606	25	153
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · AUGUR

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.522 → 0.978 per 1k papers (2023 → 2025–26).
GitHub neurorestore/Augur · created 2019-12-20 · last push 2026-03-26

BEST FOR

lollipop/UMAP of cell-type responsiveness

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Augur cell-type prioritisation: which Zeisel cell types best separate tissue of origin (S1 cortex vs CA1)?

KEY PLOTTING CALL

RECREATIONS/AUGUR/AUGUR_RECREATE.R — EXCERPT · ENDS LINE 8

```
au <- calculate_auc(GetAssayData(so, layer = "data"), so@meta.data, cell_type_col = "celltype", label_col = "tissue", n_subsamp
les = 20, subsample_size = 20, n_threads = 2, classifier = "rf")
ggsave("v1_lollipop.png", plot_lollipop(au) + ggtitle("Augur AUC: tissue (S1 vs CA1) separability"), width = 6, height = 4, dpi
= 150, bg = "white")
```

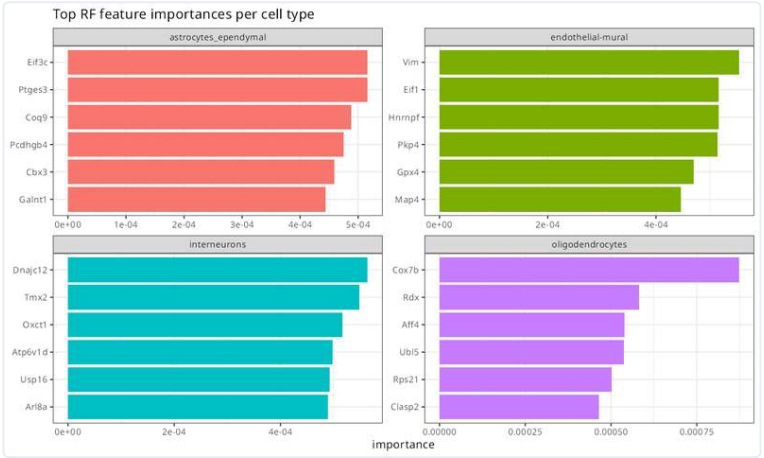
IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/neurorestore/Augur.

> LIMITATIONS & NOTES
> No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/Augur/Augur_recreate.R` — ✓ re-run end to end · 79 s

LEAD FIGURE



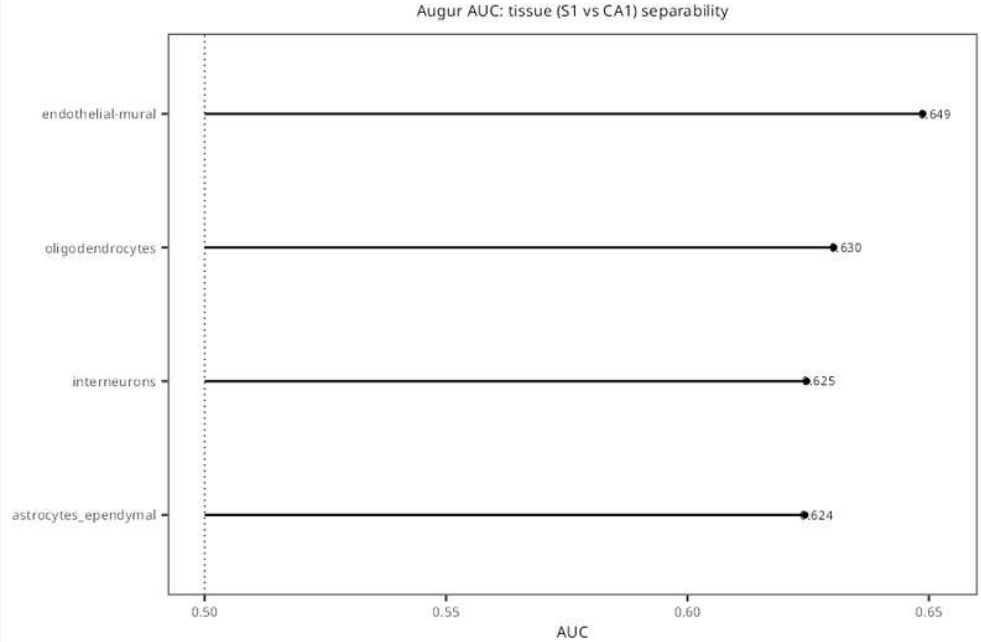
v3 feature_importance

COMPOSITION

Augur · 3 variations · v3 on facing page

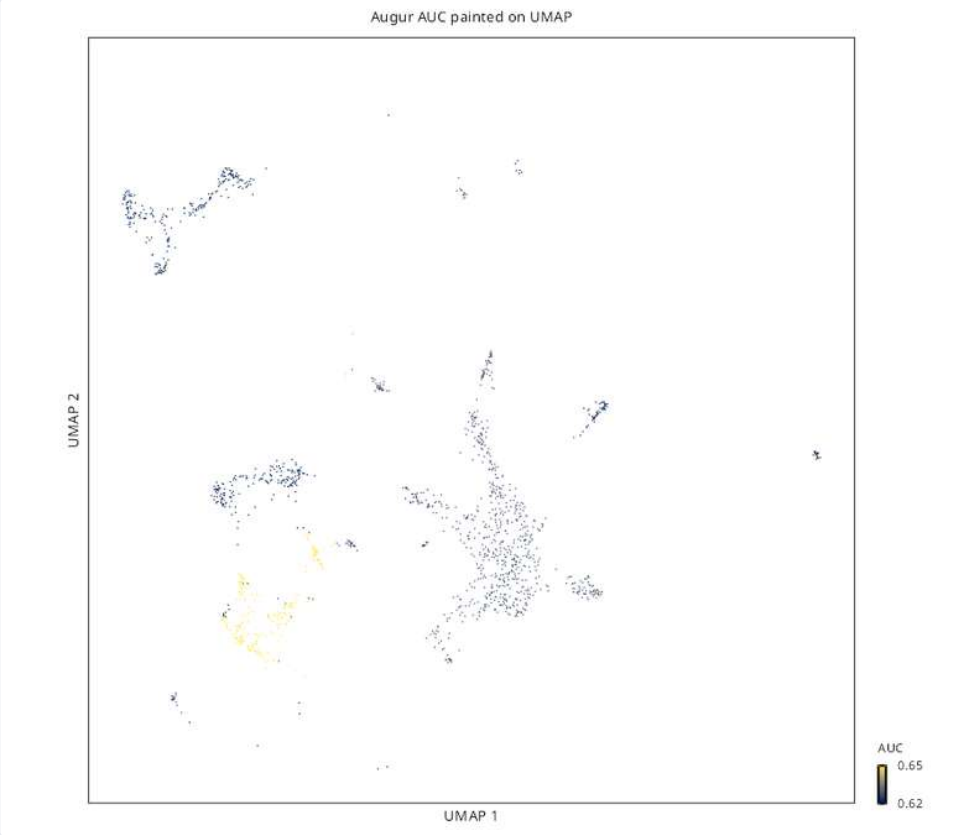
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 lollipop

COMPOSITION



v2 umap_auc

EMBEDDINGS

ggridges

#47

OF 73 · NICHE & COOL SCORE

R | DISTRIBUTION | COMPOSITION, DISTRIBUTIONS & LAYOUTS

1.7×	4/5	0.585	20	425
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · GGRIDGES

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.557 → 0.953 per 1k papers (2023 → 2025–26). GitHub wilkelab/ggridges · created 2017-09-13 · last push 2026-06-18 · also scored: ggdensity (#69, growth 0.51×†)

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

ridgeline plots

KEY PLOTTING CALL

RECREATIONS/GGRIDGES_GGDENSITY/RIDGES_RECREATE.R — EXCERPT · ENDS LINE 5

```
so <- readRDS("../_data/zeisel_seurat.rds"); df <- FetchData(so, c("Snap25","Mbp","celltype","nFeature_RNA","umap_1","umap_2"))
g <- ggplot(df, aes(nFeature_RNA, celltype, fill = after_stat(x))) + geom_density_ridges_gradient(scale = 2.2, rel_min_height
= .01) + scale_fill_viridis_c(option = "C", name = "genes") + theme_ridges() + labs(title = "ggridges gradient: genes per cell")
ggsave("v1_ridges_gradient.png", g, width = 7, height = 5.5, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/wilkelab/ggridges.

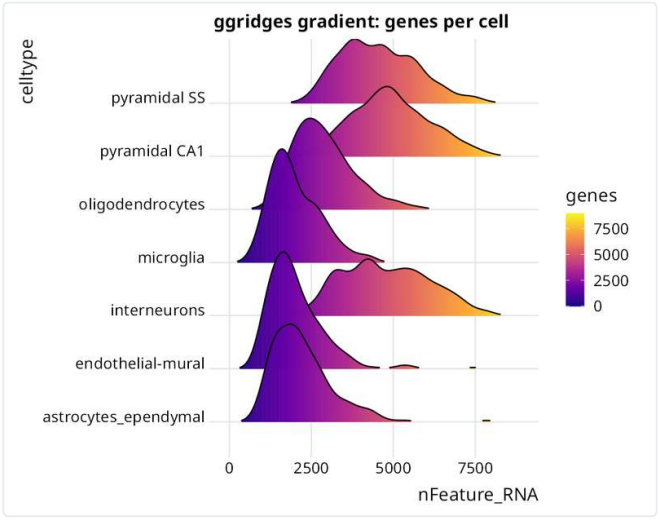
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/ggridges_ggdensity/ridges_recreate.R`

√ re-run end to end · 9 s

LEAD FIGURE

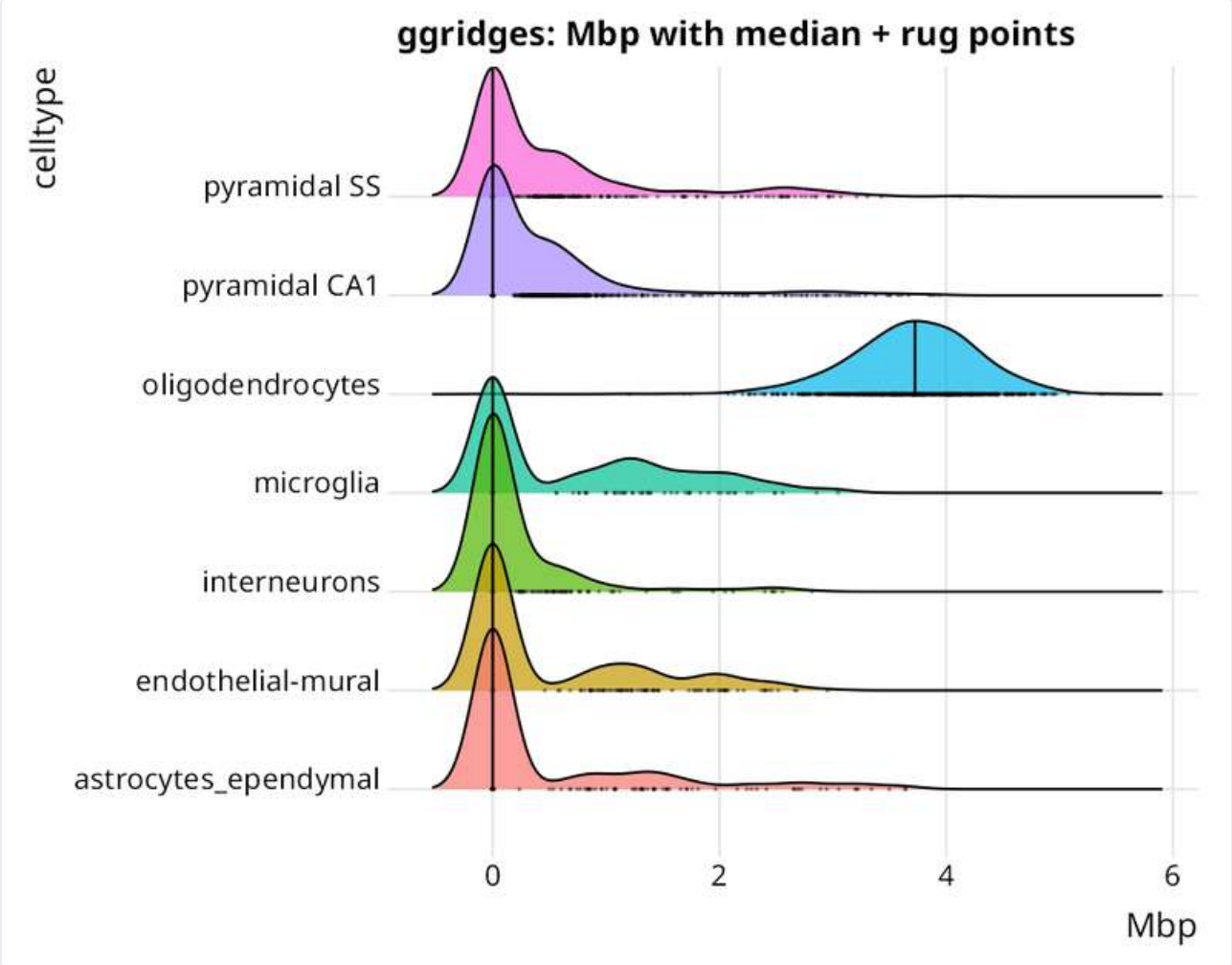


v1 ridges_gradient

COMPOSITION

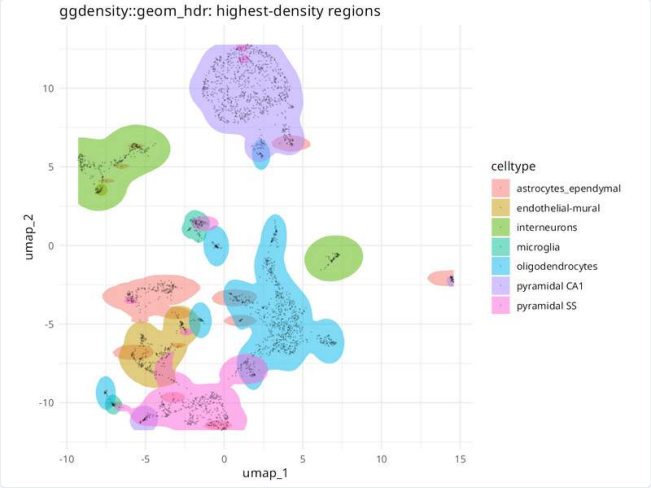
ggridges · 4 variations · v1 on facing page

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



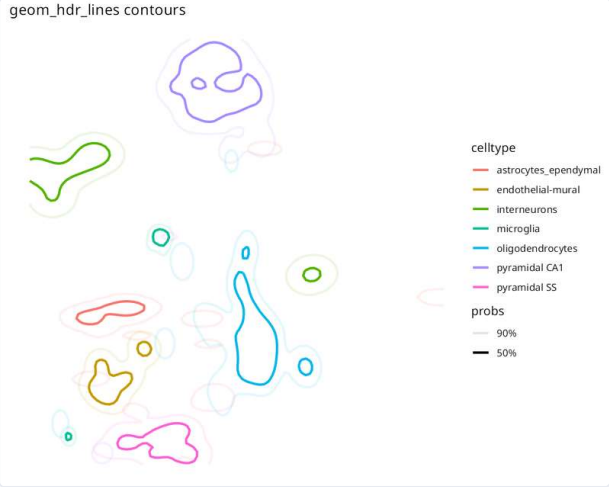
v2 ridges_points

COMPOSITION



v3 hdr_umap

COMPOSITION



v4 hdr_lines

COMPOSITION

plotnine

#48

PYTHON | GENERAL | EMBEDDINGS & FEATURE MAPS | COMPOSITION, DISTRIBUTIONS & LAYOUTS

OF 73 · NICHE & COOL SCORE

2.2×†	3/5	0.581	3	4,771
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · PLOTNINE

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.07 → 0.165 per 1k papers (2023 → 2025–26).
GitHub has2k1/plotnine · created 2017-04-24 · last push 2026-10-08

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

ggplot grammar in Python

KEY PLOTTING CALL

RECREATIONS/PLOTNINE/PLOTNINE_RECREATE.PY — EXCERPT · ENDS LINE 5

```
ad=load(); g=["Gad1","Satb2","Mbp","Aqp4"]; df=pd.DataFrame(ad[:,g].X.toarray(),columns=g);
df=pd.concat([df,ad.obs[["celltype","tissue","UMAP_1","UMAP_2","nFeature_RNA"]].reset_index(drop=True)],axis=1)
p=(ggplot(df,aes("UMAP_1","UMAP_2",color="celltype"))+geom_point(size=.4)+theme_void()+coord_fixed()+ggtitle("plotnine UMAP")+g
uides(color=guide_legend(override_aes={"size":3})))
p.save("v1_umap.png",dpi=150,width=7,height=5,verbose=False)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/has2k1/plotnine.

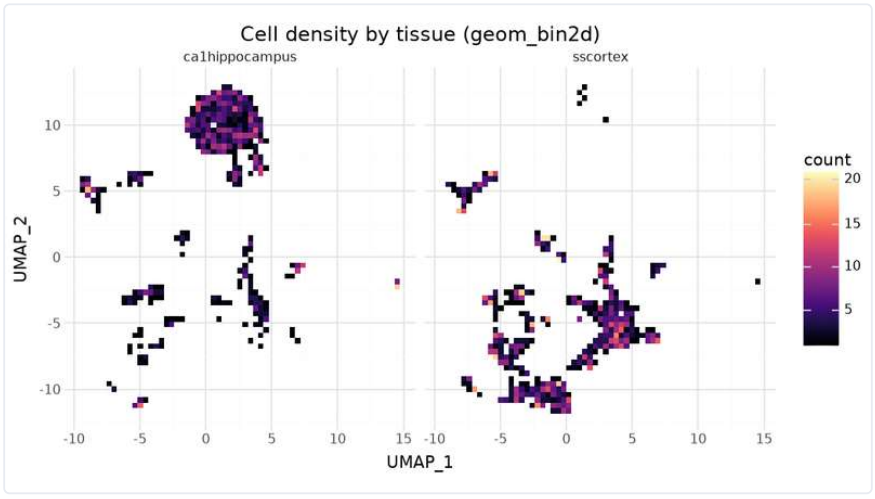
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/plotnine/plotnine_recreate.py`

√ re-run end to end · 7 s

LEAD FIGURE



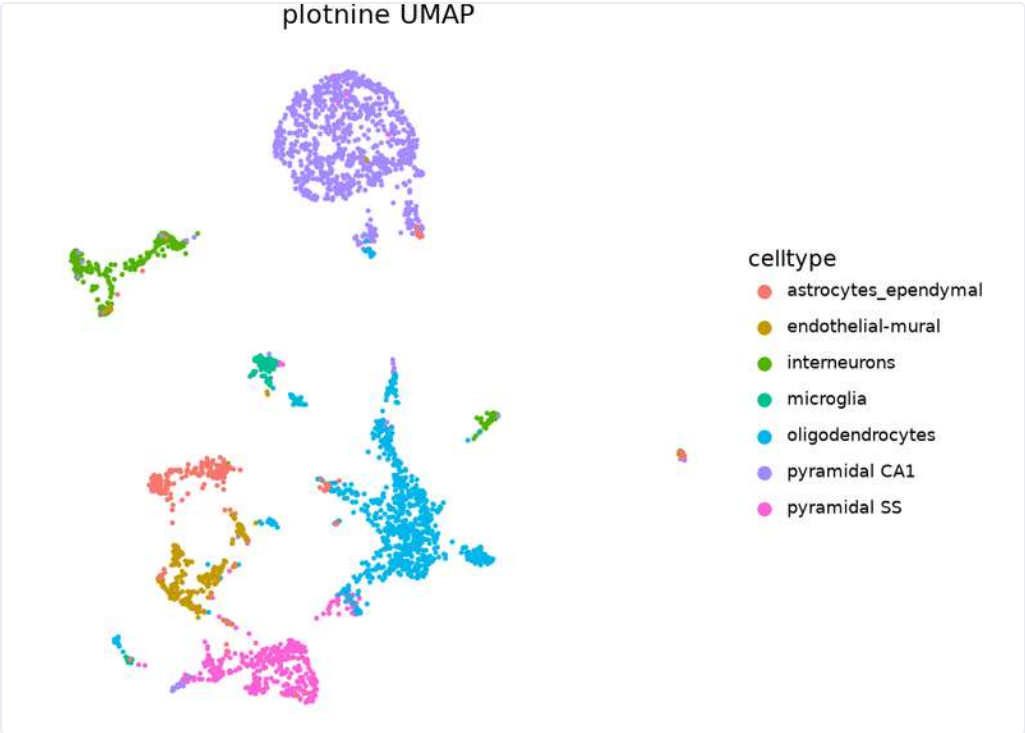
v3 bin2d_facets

COMPOSITION

plotnine · 3 variations · v3 on facing page

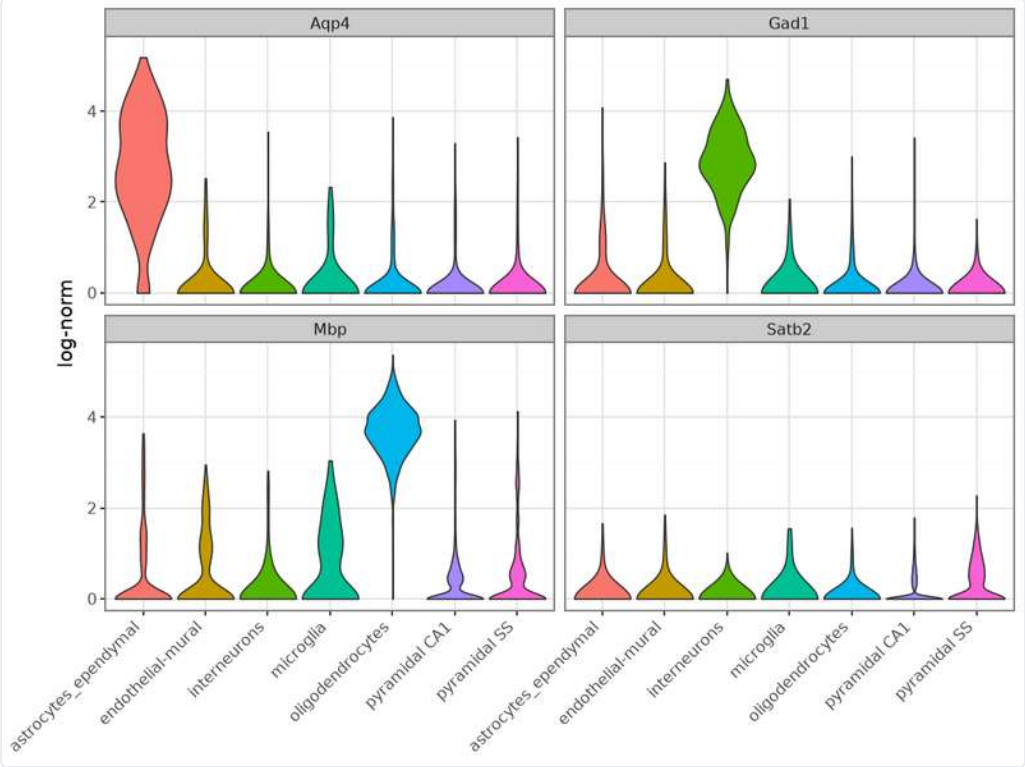
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 umap

EMBEDDINGS



v2 faceted_violins

COMPOSITION

Palantir

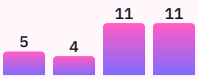
#50

OF 73 · NICHE & COOL SCORE

PYTHON | TRAJECTORY | TRAJECTORY, PSEUDOTIME & VELOCITY

1.62×	4/5	0.571	31	341
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · PALANTIR

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 1.114 → 1.816 per 1k papers (2023 → 2025–26).
GitHub dpeerlab/Palantir · created 2017-11-22 · last push 2026-09-23

BEST FOR

branch probabilities, gene trends

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Palantir on dentate gyrus (diffusion maps → pseudotime, entropy, branch probabilities, gene trends)

KEY PLOTTING CALL

RECREATIONS/PALANTIR/PALANTIR_RECREATE.PY — EXCERPT · ENDS LINE 13

```
palantir.core.run_palantir(a,start,terminal_states=term,num_waypoints=500)
palantir.plot.plot_palantir_results(a,s=3); plt.savefig("v1_palantir_results.png",dpi=130,bbox_inches="tight"); plt.close("all")
```

IN PUBLISHED PAPERS · MOST-CITED OF 24 NAMING "PALANTIR" IN METHODS

258 CITATIONS — **Single-cell brain organoid screening identifies developmental defects in autism.** *Nature, 2023*
DOI ↗ PMC10499611

42 CITATIONS — **Tracking single-cell evolution using clock-like chromatin accessibility loci.** *Nat Biotechnol, 2025*
DOI ↗ PMC12084158

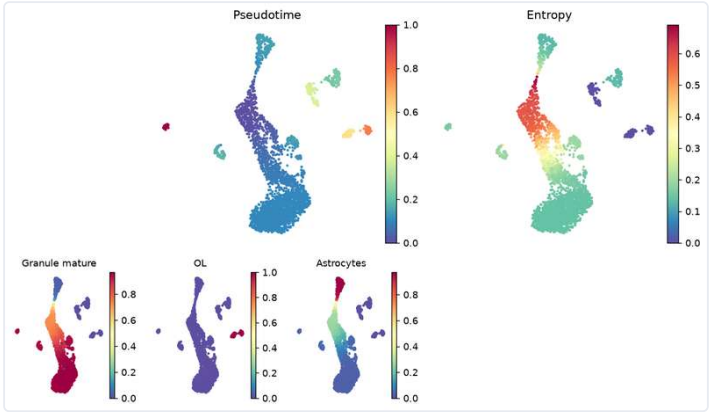
Figure-level matching was run for the Top 100 only, so these papers name the tool but no specific figure is linked.

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/Palantir/palantir_recreate.py` — ✓ re-run end to end · 19 s

LEAD FIGURE

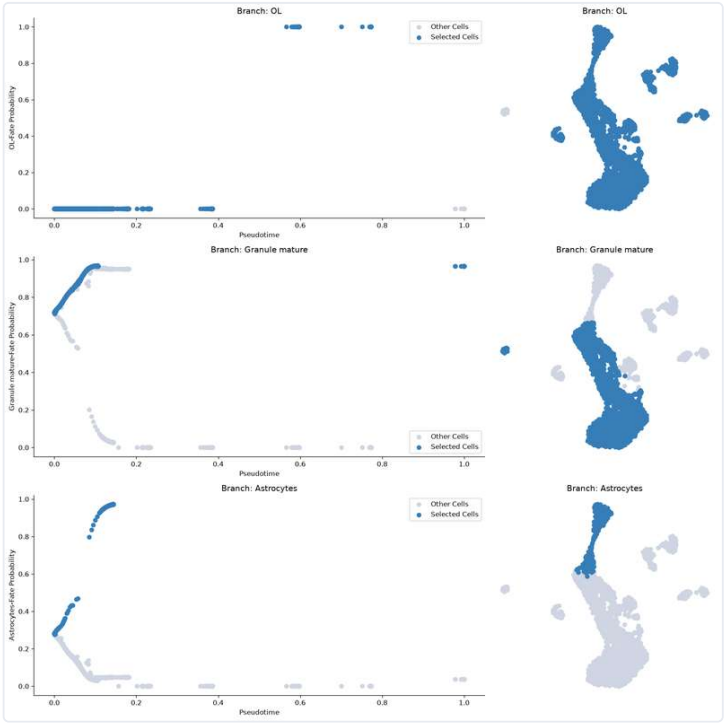


v1 palantir_results — TRAJECTORY

Palantir · 4 variations · v1 on facing page

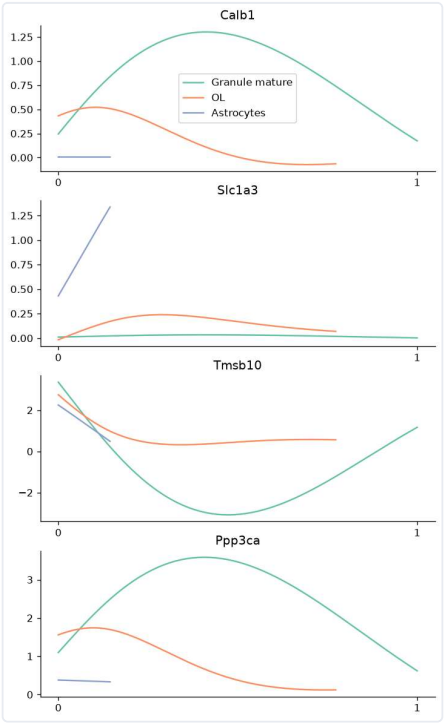
NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



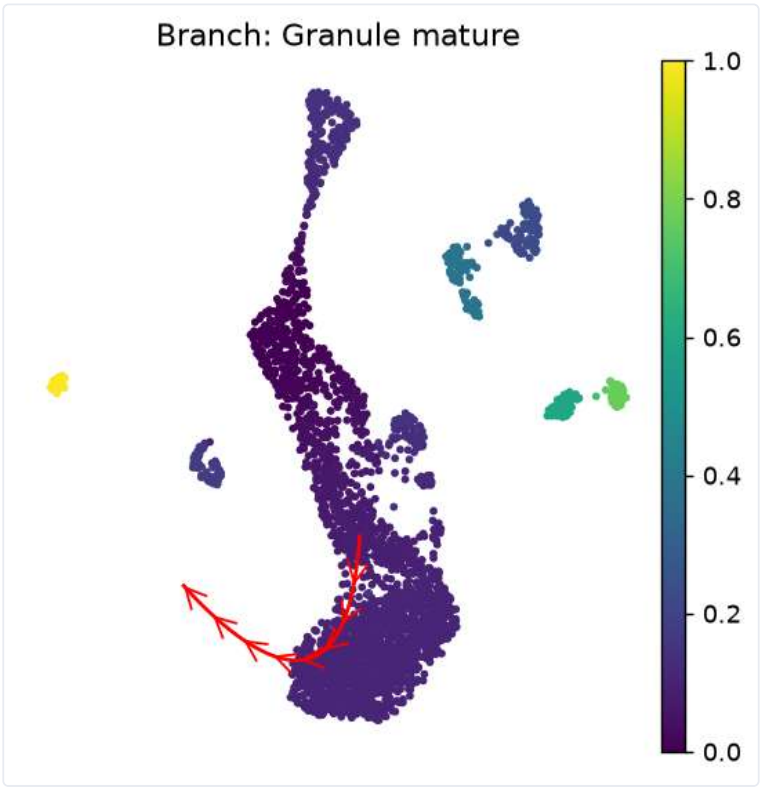
v2 branch_selection

TRAJECTORY



v3 gene_trends

TRAJECTORY



v4 trajectory_arrows

TRAJECTORY

ggspavis

#51

OF 73 · NICHE & COOL SCORE

R SPATIAL SPATIAL & TISSUE MAPS

2.15×	3/5	0.56	9	5
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · GGSPAVIS/SPATIALEXPERIMENT

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.279 → 0.61 per 1k papers (2023 → 2025–26).
GitHub lmweber/ggspavis · created 2020-12-02 · last push 2026-07-14

BEST FOR

spot plots on tissue

KEY PLOTTING CALL

RECREATIONS/GGSPAVIS/GGSPAVIS_RECREATE.R — EXCERPT · ENDS LINE 4

```
spe <- load_dlpfc(); spe$sum <- colSums(counts(spe)); spe$MOBP <- logcounts(spe)["MOBP",]
ggsave("v1_plotSpots_layers.png", plotCoords(spe, annotate = "ground_truth", pal = "libd_layer_colors", point_size = .6) + ggti
tle("plotSpots"), width = 6, height = 5.5, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/lmweber/ggspavis.

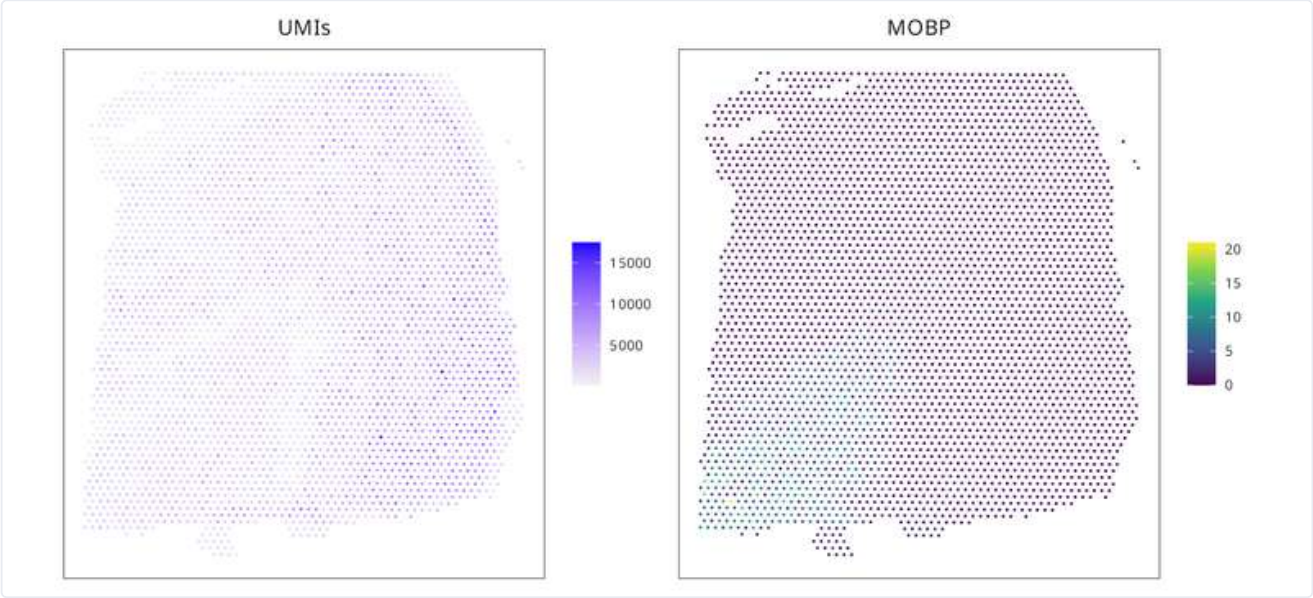
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/ggspavis/ggspavis_recreate.R`

✓ re-run end to end · 18 s

LEAD FIGURE

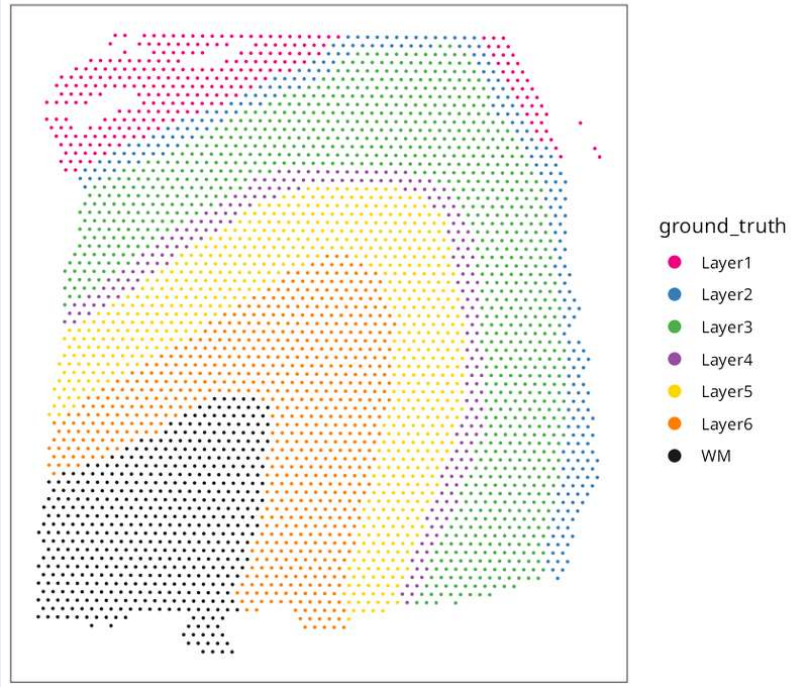


v3 continuous_qc_gene

SPATIAL

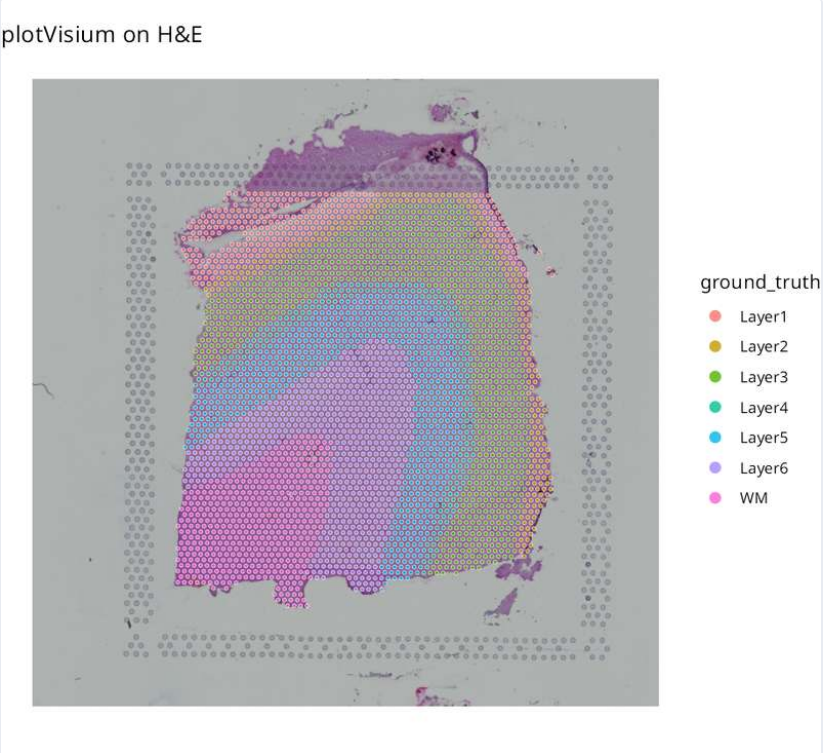
ggspavis · 3 variations · v3 on facing page

Data: Maynard et al. 2021, human DLPFC Visium sample 151673 (STexampleData). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 plotSpots_layers

SPATIAL



v2 plotVisium_HE

SPATIAL

plotgardener

#55

OF 73 · NICHE & COOL SCORE

R | GENOME TRACKS | GENOME TRACKS & GENOMIC SIGNAL

0.93×	5/5	0.544	4	371
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · PLOTGARDENER

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.313 → 0.292 per 1k papers (2023 → 2025–26).
GitHub PhanstielLab/plotgardener · created 2019-07-23 · last push 2026-07-18

BEST FOR

page-layout multi-panel genomic figures, Hi-C

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

plotgardener multi-panel genomic page (plotgardenerData example: IMR90 Hi-C + ChIP + gene track, hg19)

KEY PLOTTING CALL

RECREATIONS/PLOTGARDENER/PLOTGARDENER_RECREATE.R — EXCERPT · ENDS LINE 13

```
pageCreate(width = 7.5, height = 6.5, default.units = "inches", showGuides = FALSE)
params <- pgParams(chrom = "chr21", chromstart = 28000000, chromend = 30300000, assembly = "hg19", x = .5, width = 6.5)
hic <- plotHicTriangle(data = IMR90_HiC_10kb, params = params, zrange = c(0, 70), resolution = 10000, y = .5, height = 2.5,
just = c("left","top"), default.units = "inches")
annoHeatmapLegend(hic, x = 7, y = .5, width = .13, height = 1.2, just = c("right","top"))
plotPairsArches(data = IMR90_DNAloops_pairs, params = params, y = 3.1, height = .5, fill = "#1f4297", linecolor = "#1f4297", fl
ip = TRUE)
plotSignal(data = IMR90_ChIP_H3K27ac_signal, params = params, y = 3.7, height = .6, fill = "#37a7db", linecolor = "#37a7db"); p
lotText("H3K27ac", x = .5, y = 3.7, just = c("left","top"), fontsize = 8)
plotSignal(data = GM12878_ChIP_CTCF_signal, params = params, y = 4.4, height = .6, fill = "#e0644e", linecolor = "#e0644e"); pl
otText("CTCF", x = .5, y = 4.4, just = c("left","top"), fontsize = 8)
plotGenes(params = params, y = 5.1, height = .7); plotGenomeLabel(chrom = "chr21", chromstart = 28000000, chromend = 30300000, a
ssembly = "hg19", x = .5, y = 5.9, length = 6.5, scale = "Mb")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/PhanstielLab/plotgardener.

LIMITATIONS & NOTES

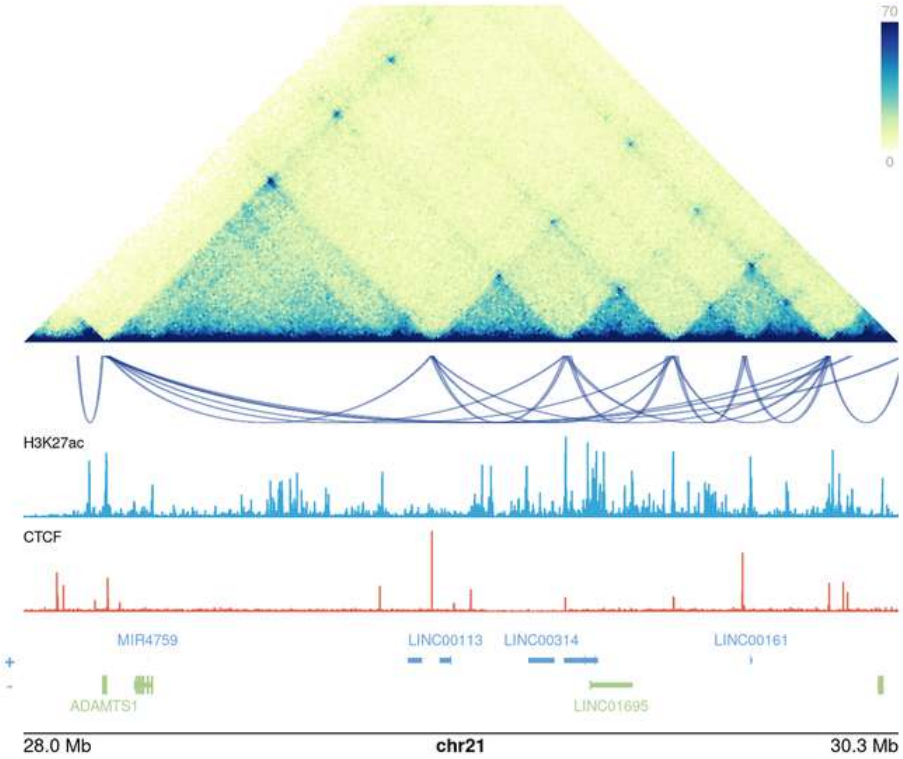
- Non-neuro fallback: plotgardenerData IMR90 example (Hi-C + ChIP + gene track, hg19).

Reproduce: `Rscript recreations/plotgardener/plotgardener_recreate.R`

√ re-run end to end · 14 s

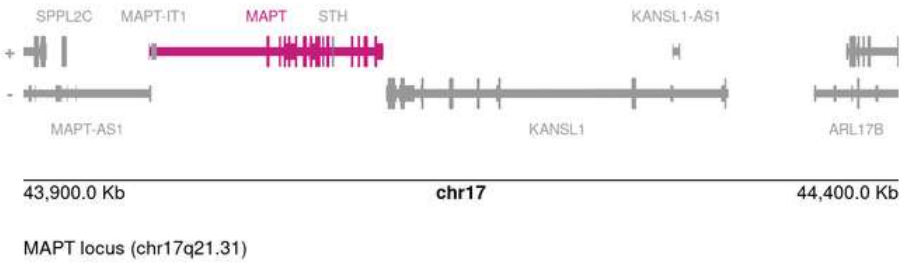
plotgardener · 2 variations

Data: plotgardenerData IMR90 example (Hi-C, ChIP-seq, gene track; hg19); not neuro data. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 page_layout

GENOME



v2 gene_and_ideogram

GENOME

propeller (speckle)

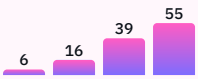
#60

OF 73 · NICHE & COOL SCORE

R | COMPOSITION | COMPOSITION, DISTRIBUTIONS & LAYOUTS

1.89×	3/5	0.512	116	71
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · PROPELLER (SPECKLE)

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 2.299 → 4.356 per 1k papers (2023 → 2025–26).
GitHub phipsonlab/speckle · created 2022-08-19 · last push 2022-12-08

BEST FOR

proportion plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

speckle::propeller differential cell-type composition, Zeisel 2015: S1 cortex vs CA1 hippocampus, using the Fluidigm C1 chip ID (cell-name prefix) as the biological/technical replicate.

KEY PLOTTING CALL

RECREATIONS/SPECKLE_PROPELLER/PROPELLER_RECREATE.R — EXCERPT · ENDS LINE 13

```
res <- propeller(clusters = md$celltype, sample = md$chip, group = md$tissue, transform = "logit")
png("v1_plotCellTypeProps.png", 1300, 900, res = 150); print(plotCellTypeProps(clusters = md$celltype, sample = md$chip) +
theme(axis.text.x = element_text(angle = 90, size = 6)) + ggtitle("Cell-type proportions per C1 chip")); dev.off()
```

IN PUBLISHED PAPERS · MOST-CITED OF 35 NAMING “SPECKLE/PROPELLER” IN METHODS

164 CITATIONS — **Molecular and cellular dynamics of the developing human neocortex.** *Nature, 2025*
DOI ↗ PMC12589127

47 CITATIONS — **Morphodynamics of human early brain organoid development.** *Nature, 2025*
DOI ↗ PMC12390842

Figure-level matching was run for the Top 100 only, so these papers name the tool but no specific figure is linked.

LIMITATIONS & NOTES

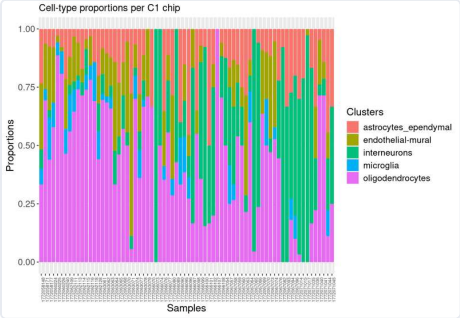
- Shared cell types only, because pyramidal subtypes are tissue-exclusive; the Fluidigm C1 chip ID is used as the replicate.

Reproduce: `Rscript recreations/speckle_propeller/propeller_recreate.R` — √ re-run end to end · 14 s

propeller (speckle) · 3 variations

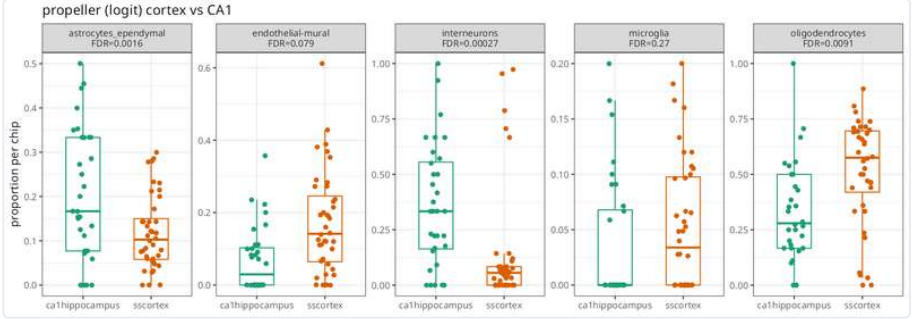
NOVAFLOW RECREATION

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



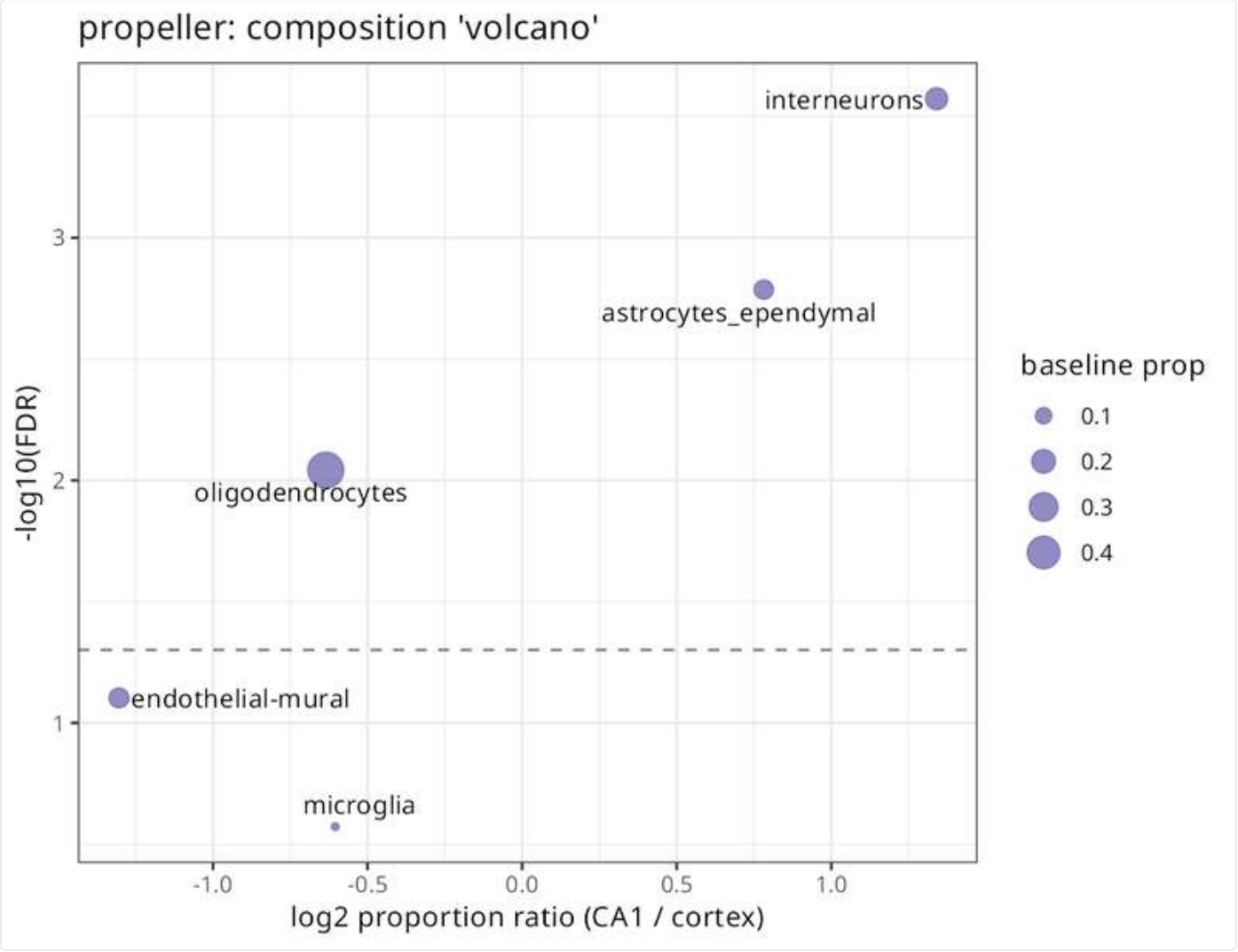
v1 plotCellTypeProps

COMPOSITION



v2 props_by_group

COMPOSITION



v3 composition_volcano

COMPOSITION

ggcoverage

#63

OF 73 · NICHE & COOL SCORE

R | GENOME TRACKS | GENOME TRACKS & GENOMIC SIGNAL

0.97×†	4/5	0.462	3	273
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · GGCOVERAGE

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.104 → 0.102 per 1k papers (2023 → 2025–26). GitHub showteeth/ggcoverage · created 2022-07-18 · last push 2025-01-24

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

ggplot2 coverage tracks + gene/ideogram

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

ggcoverage genome-coverage tracks. NOTE: uses the package's bundled human RNA-seq bigWigs (chr14 HNRNPC locus; ERR127302/3/6/7, KO vs WT) because they are the only coverage files shipped offline - not neuro data.

KEY PLOTTING CALL

RECREATIONS/GGCOVERAGE/GGCOVERAGE_RECREATE.R — EXCERPT · ENDS LINE 10

```
d <- system.file("extdata", package = "ggcoverage"); gtf <- import.gff(file.path(d, "used_hg19.gtf"), format = "gtf")
meta <- read.csv(file.path(d, "RNA-seq", "meta_info.csv"))
trk <- LoadTrackFile(track.folder = file.path(d, "RNA-seq"), format = "bw", region = "chr14:21,677,306-21,737,601", extend = 200
0, meta.info = meta)
cols <- c(KO = "#d7301f", WT = "#225ea8")
base <- ggcoverage(data = trk, plot.type = "facet", group.key = "Type", facet.key = "Type", range.position = "out", mark.region
= data.frame(start = c(21678900, 21732001), end = c(21679900, 21732400), label = c("M1", "M2")), show.mark.label = TRUE)
ggsave("v1_coverage_gene.png", base + geom_gene(gtf.gr = gtf) + ggtitle("ggcoverage: RNA-seq coverage + gene models (HNRNPC
locus)"), width = 10, height = 7, dpi = 130, bg = "white")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/showteeth/ggcoverage.

LIMITATIONS & NOTES

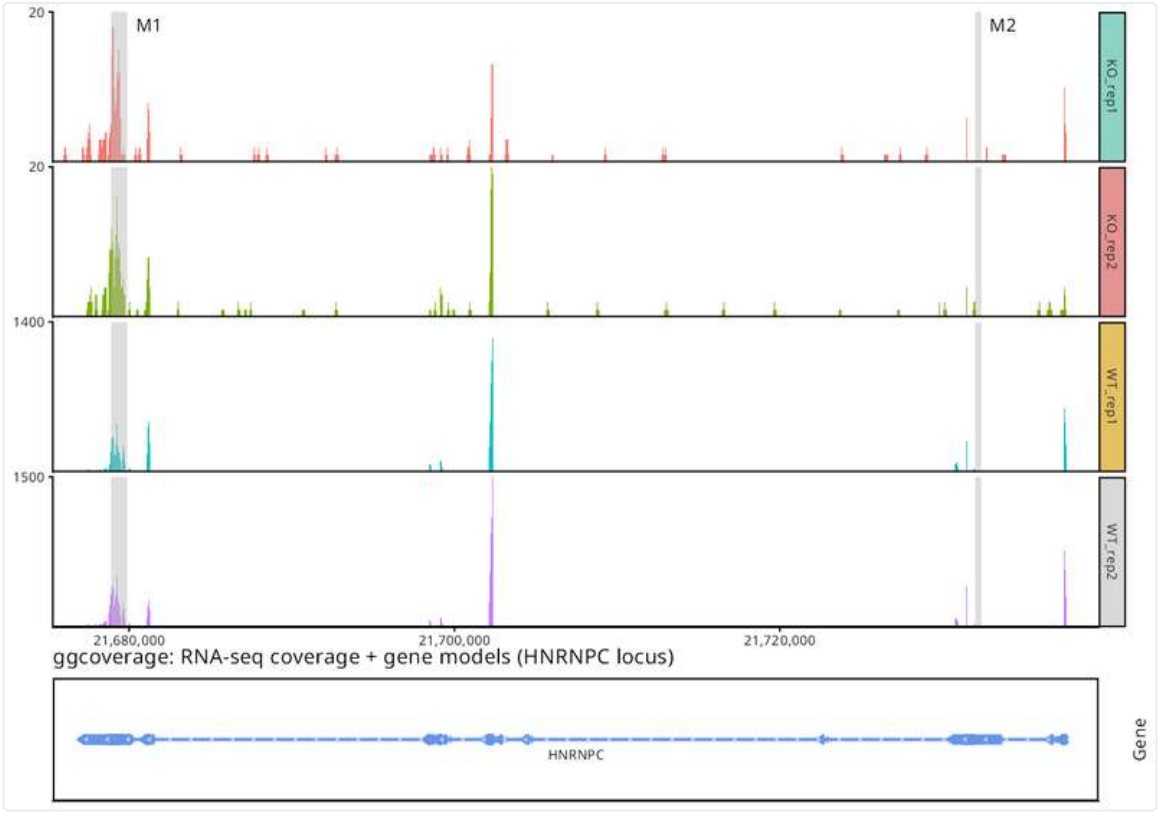
- Non-neuro fallback: the package's bundled human RNA-seq bigWigs (chr14 HNRNPC locus; ERR127302/3/6/7, KO vs WT), the only coverage files shipped offline.

Reproduce: `Rscript recreations/ggcoverage/ggcoverage_recreate.R`

√ re-run end to end · 20 s

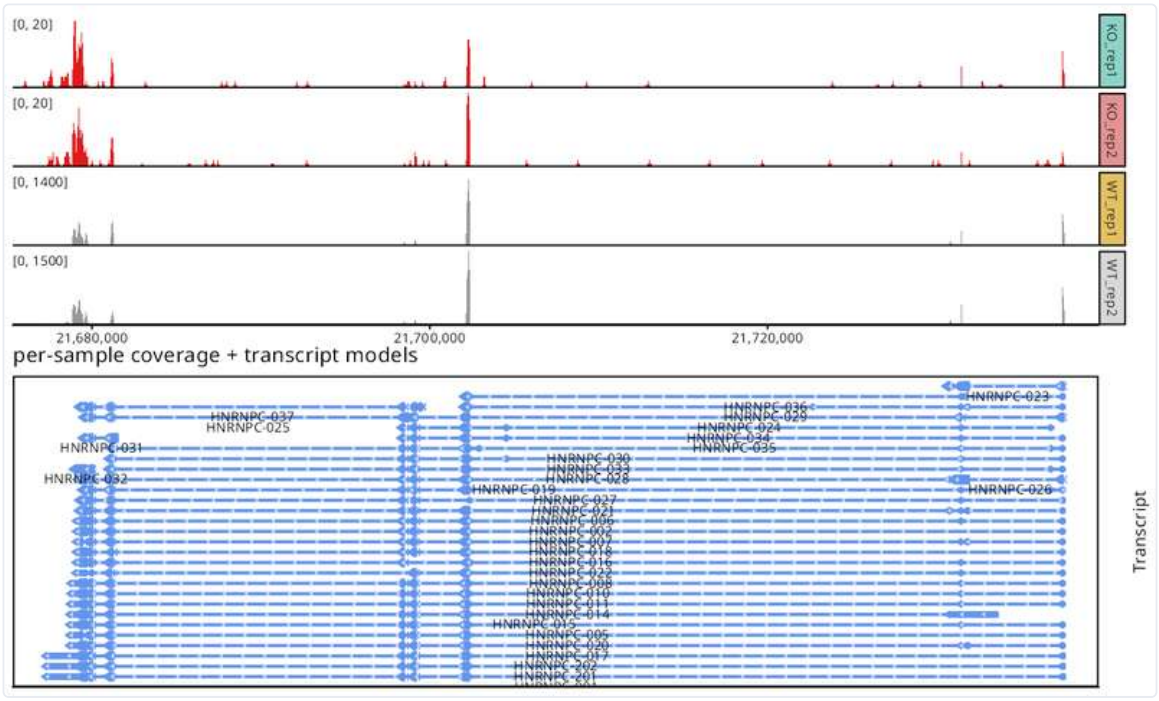
ggcoverage · 2 variations

Data: ggcoverage's bundled human RNA-seq bigWigs (HNRNPC knockdown vs control, chr14); not neuro data. Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 coverage_gene

GENOME



v2 samples_transcripts

GENOME

ggtree / ggtreeExtra

#64

OF 73 · NICHE & COOL SCORE

R TREE NETWORKS, GRAPHS & TREES

0.94×	4/5	0.455	26	99
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · GGTREE/GGTREEEXTRA

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 3.134 → 2.959 per 1k papers (2023 → 2025–26).
GitHub YuLab-SMU/ggtreeExtra · created 2020-05-28 · last push 2026-06-04

BEST FOR

circular trees with heatmap/bar rings

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Circular dendrogram of Zeisel level2 subclasses (mean-expression distance) with ggtreeExtra rings

KEY PLOTTING CALL

RECREATIONS/GGTREE_GGTREEEXTRA/GGTREE_RECREATE.R — EXCERPT · ENDS LINE 9

```
so <- readRDS("../_data/zeisel_seurat.rds"); av <- AverageExpression(so, features = VariableFeatures(so), group.by = "level2cla
ss")$RNA |> as.matrix()
colnames(av) <- gsub("-", "_", colnames(av)); tr <- as.phylo(hclust(as.dist(1 - cor(log1p(av))), "average"))
md <- so@meta.data; md$sub <- gsub("-", "_", md$level2class)
info <- unique(md[, c("sub","celltype")]); info <- info[!duplicated(info$sub),]; n <- as.data.frame(table(md$sub)); colnames(n)
<- c("label","cells"); info <- merge(info, n, by.x = "sub", by.y = "label"); colnames(info)[1] <- "label"
p <- ggtree(tr, layout = "circular") %<+% info + geom_tippoint(aes(color = celltype), size = 2) + geom_tiplab(size = 2, offset
= .02)
ggsave("v1_circular_tree.png", p + theme(legend.position = "right"), width = 8, height = 7, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS · MOST-CITED OF 25 NAMING “GGTREE” IN METHODS

86 CITATIONS — **A comprehensive spatio-cellular map of the human hypothalamus.** *Nature, 2025*
DOI ↗ PMC11922758

76 CITATIONS — **A marmoset brain cell census reveals regional specialization of cellular identities.** *Sci Adv, 2023*
DOI ↗ PMC10569717

Figure-level matching was run for the Top 100 only, so these papers name the tool but no specific figure is linked.

LIMITATIONS & NOTES

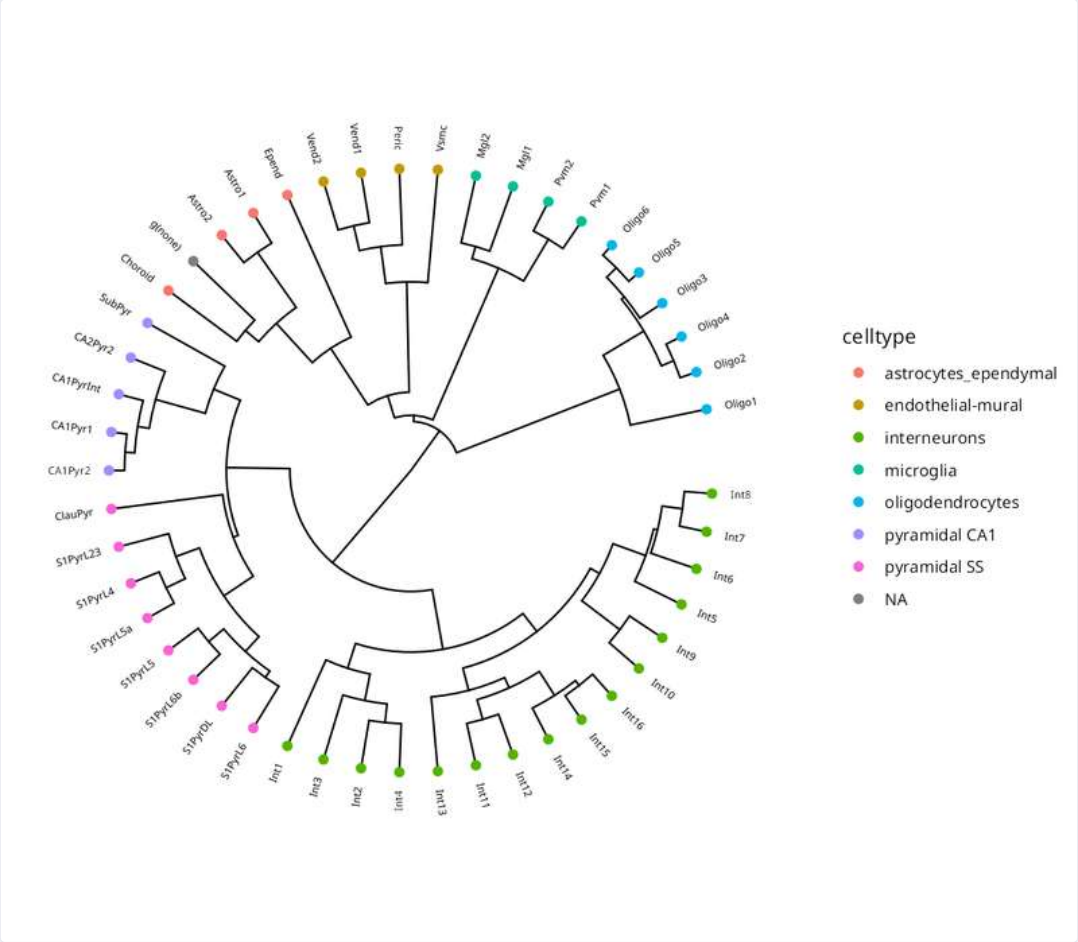
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/ggtree_ggtreeExtra/ggtree_recreate.R`

✓ re-run end to end · 9 s

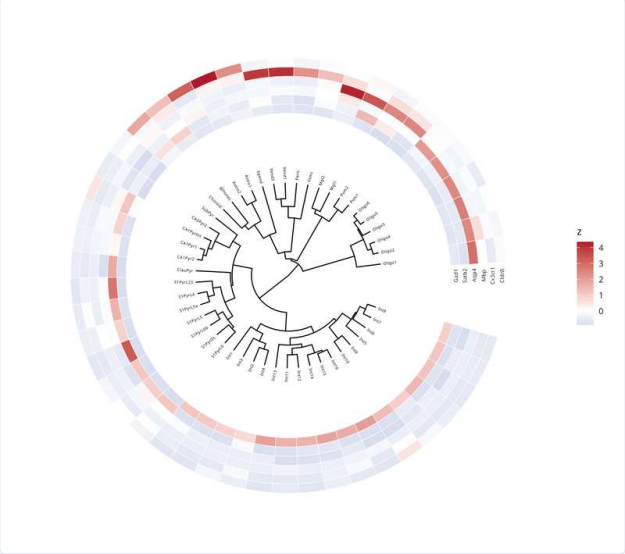
ggtree / ggtreeExtra · 3 variations

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



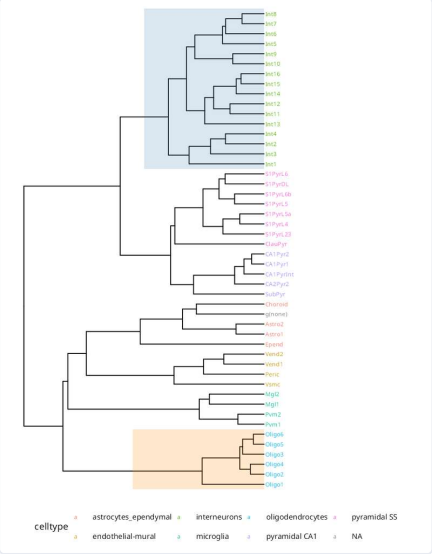
v1 circular_tree

NETWORKS



v2 fan_heatmap_bar_rings

NETWORKS



v3 rect_tree_clade_highlights

NETWORKS

Mellon

#65

OF 73 · NICHE & COOL SCORE

PYTHON | CELL-STATE DENSITY | TRAJECTORY, PSEUDOTIME & VELOCITY

0.93×	4/5	0.444	27	77
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · MELLON

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.836 → 0.775 per 1k papers (2023 → 2025–26).
GitHub settylab/Mellon · created 2022-10-28 · last push 2026-09-24

BEST FOR

cell-state density landscapes

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

Mellon cell-state density (via palantir.utils.run_density) on dentate gyrus

KEY PLOTTING CALL

RECREATIONS/MELLON/MELLON_RECREATE.PY — EXCERPT · ENDS LINE 8

```
a=load(); palantir.utils.run_diffusion_maps(a,n_components=10); palantir.utils.run_density(a)
fig,ax=plt.subplots(1,2,figsize=(11,4.5))
s=ax[0].scatter(*a.obsm["X_umap"].T,c=a.obs["mellon_log_density"],s=6,cmap="magma"); ax[0].set_title("Mellon log-density");
plt.colorbar(s,ax=ax[0]); ax[0].axis("off")
s=ax[1].scatter(*a.obsm["X_umap"].T,c=a.obs["mellon_log_density_clipped"],s=6,cmap="Spectral_r"); ax[1].set_title("clipped
density (low = transitional)"); plt.colorbar(s,ax=ax[1]); ax[1].axis("off")
fig.tight_layout(); fig.savefig("v1_density_umap.png",dpi=140)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/settylab/Mellon.

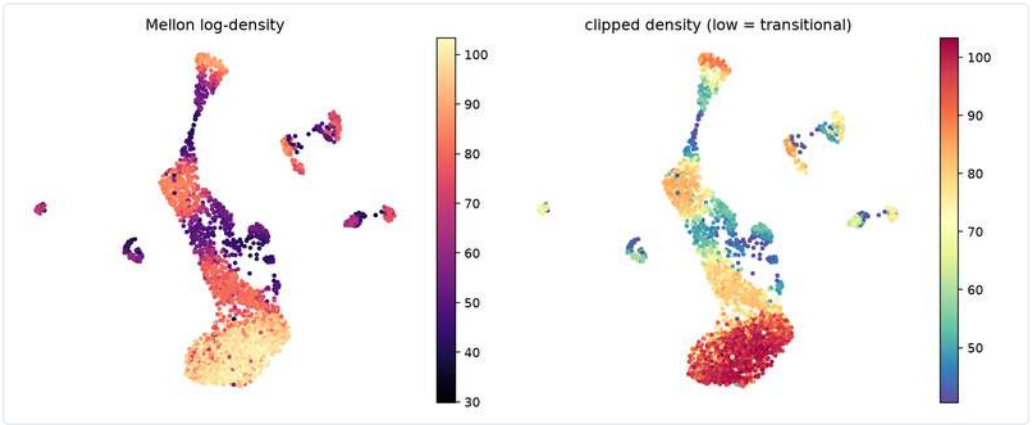
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/Mellon/mellon_recreate.py`

√ re-run end to end · 28 s

LEAD FIGURE



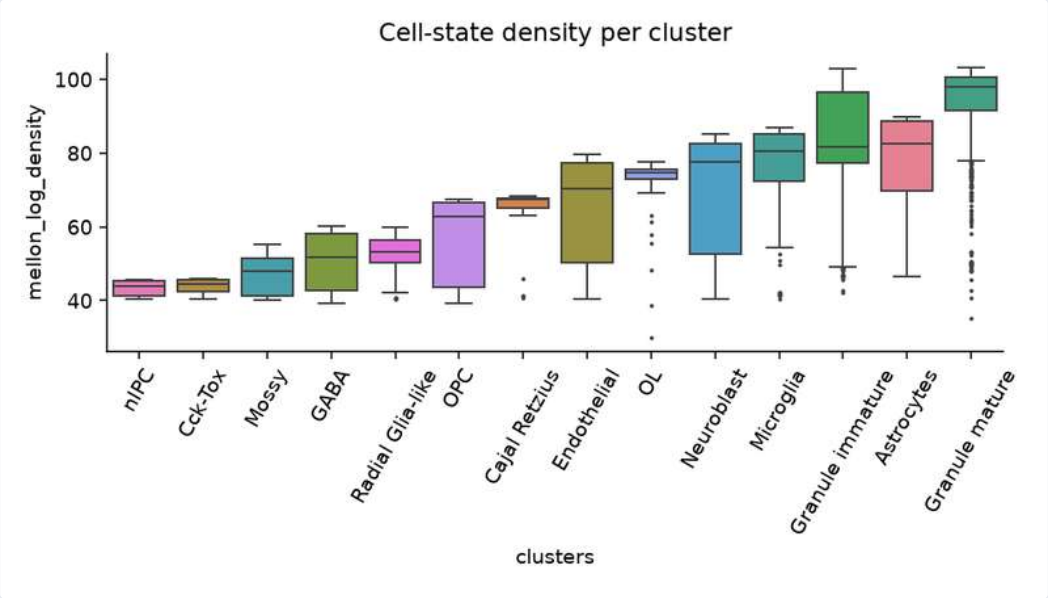
v1 density_umap

TRAJECTORY

Mellon · 3 variations · v1 on facing page

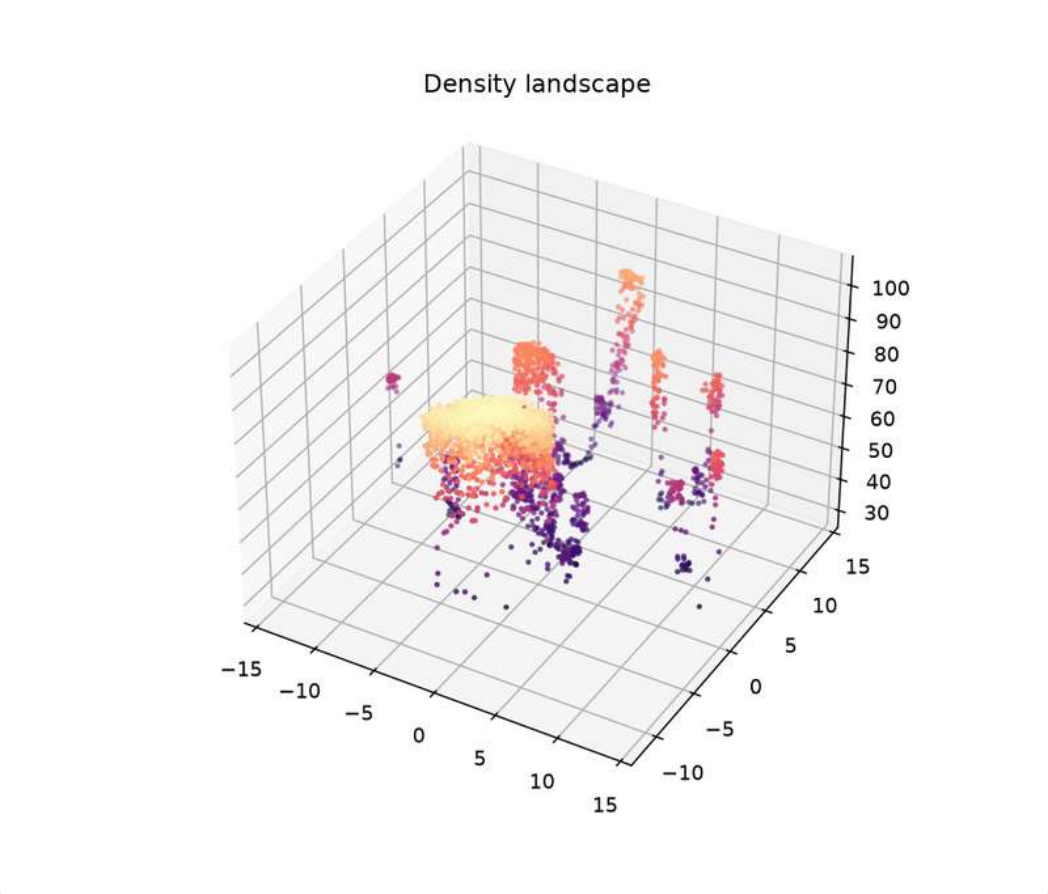
NOVAFLOW RECREATION

Data: La Manno et al. 2018, mouse dentate gyrus scRNA-seq with spliced/unspliced counts (scVelo dataset). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v2 density_by_cluster

TRAJECTORY



v3 density_landscape_3d

TRAJECTORY

karyoploteR

#66

OF 73 · NICHE & COOL SCORE

R | GENOME | GENOME TRACKS & GENOMIC SIGNAL

0.67×	4/5	0.428	13	377
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · KARYOPLOTER

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 1.289 → 0.864 per 1k papers (2023 → 2025–26).
GitHub bernatgel/karyoploteR · created 2016-06-07 · last push 2026-07-06

BEST FOR

whole-karyotype data plots

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

karyoploteR: genome-wide views of airway DE genes (hg19 coordinates via TxDb)

KEY PLOTTING CALL

RECREATIONS/KARYOPLOTER/KARYOPLOTER_RECREATE.R — EXCERPT · ENDS LINE 8

```
res <- as.data.frame(readRDS("../_data/airway_deseq2.rds")$res); res <- res[!is.na(res$padj) & !is.na(res$entrez),]
g <- genes(TxDb.Hsapiens.UCSC.hg19.knownGene); g <- g[g$gene_id %in% res$entrez]; r <- res[match(g$gene_id, res$entrez),];
g$lfc <- r$log2FoldChange; g$padj <- r$padj; g$sym <- r$symbol
g <- keepStandardChromosomes(g, pruning.mode = "coarse"); g <- g[seqnames(g) %in% paste0("chr", c(1:22, "X"))]
sig <- g[g$padj < .01]
png("v1_karyotype_density.png", 1500, 1100, res = 150); kp <- plotKaryotype("hg19", plot.type = 1, chromosomes = paste0("chr",
c(1:22,"X"))); kpPlotDensity(kp, sig, col = "#c51b7d", border = NA, window.size = 2e6); kpAddMainTitle(kp, "DE-gene density
(padj<0.01), 2-Mb windows"); dev.off()
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/bernatgel/karyoploteR.

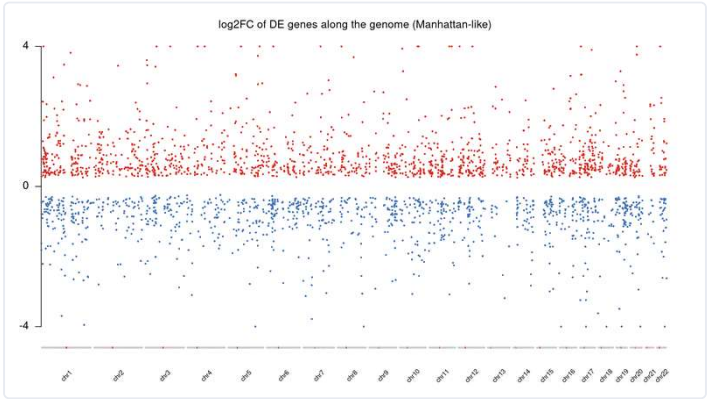
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/karyoploteR/karyoploteR_recreate.R`

√ re-run end to end · 16 s

LEAD FIGURE



v2 lfc_scatter_ideogram

GENOME

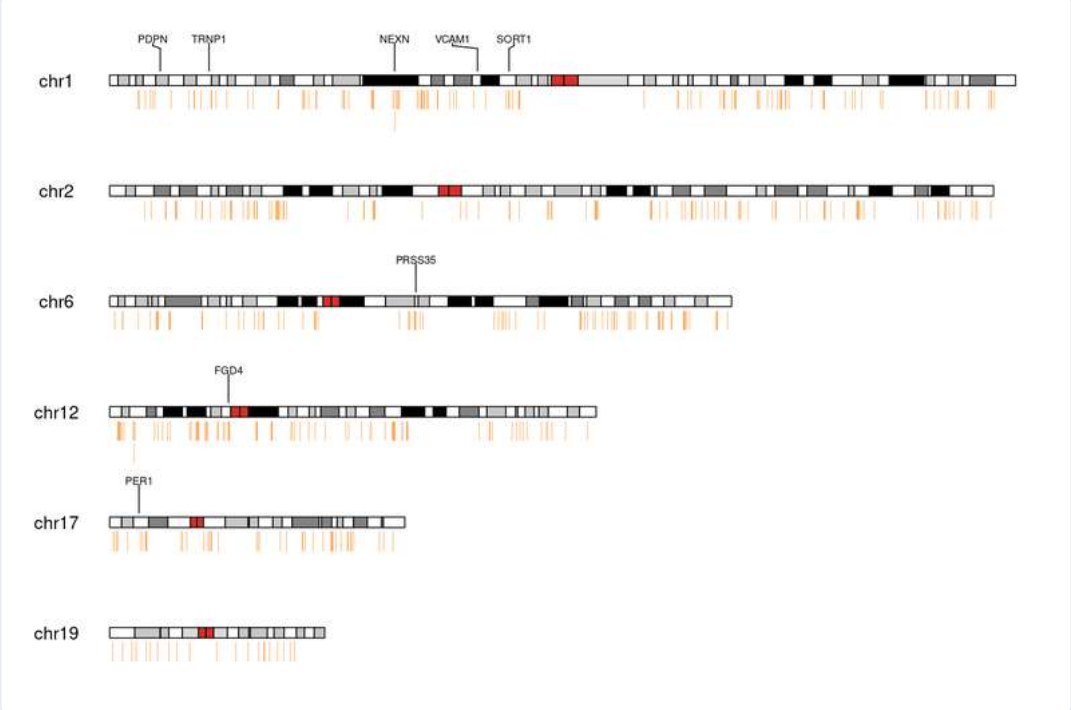
karyoploteR · 3 variations · v2 on facing page

NOVAFLOW RECREATION

Data: airway bulk RNA-seq DE results placed on hg19 coordinates (Himes et al. 2014). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.

v1 karyotype_density

GENOME



v3 markers_chr

GENOME

marsilea

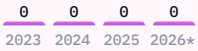
#67

OF 73 · NICHE & COOL SCORE

PYTHON | HEATMAP | HEATMAPS & DOT MATRICES

0.64×†	4/5	0.421	0	324
growth ratio, per-1k rate 2025–26 ÷ 2023 (†low 2023 base)	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · MARSILEA

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.104 → 0.064 per 1k papers (2023 → 2025–26). GitHub Marsilea-viz/marsilea · created 2022-08-22 · last push 2026-09-26

† Growth ratio rests on fewer than 5 broad-corpus mentions in 2023, so it mostly reflects a new release, not a measured trend.

BEST FOR

composable cross-layout heatmaps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

marsilea composable heatmaps on Zeisel cortex/hippocampus

KEY PLOTTING CALL

RECREATIONS/MARSILEA/MARSILEA_RECREATE.PY — EXCERPT · ENDS LINE 9

```
ad=load();
mk=["Snap25","Gad1","Gad2","Pvalb","Sst","Satb2","Neurod6","Wfs1","Aqp4","Gfap","Mbp","Plp1","Cx3cr1","C1qa","Cldn5","Acta2"]
df=pd.DataFrame(ad[:,mk].X.toarray(),columns=mk); df["ct"]=ad.obs.celltype.values
mean=df.groupby("ct",observed=True).mean(); frac=(df[mk]>0).groupby(df.ct,observed=True).mean(); z=(mean-mean.mean())/mean.std()
n=df.ct.value_counts().reindex(mean.index)
h=ma.Heatmap(z.values,cmap="RdBu_r",label="z-score",width=5,height=3.2); h.add_left(mp.Labels(mean.index),pad=.05);
h.add_bottom(mp.Labels(mk,rotation=90)); h.add_right(mp.Numbers(n.values,color="#4C72B0",label="cells"),size=1,pad=.1)
h.add_dendrogram("top",colors="#555"); h.add_dendrogram("left"); h.add_legends(); h.add_title("marsilea: z-scored means + cell
counts"); h.save("v1_heatmap_dendro_bars.png",dpi=150)
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/Marsilea-viz/marsilea.

LIMITATIONS & NOTES

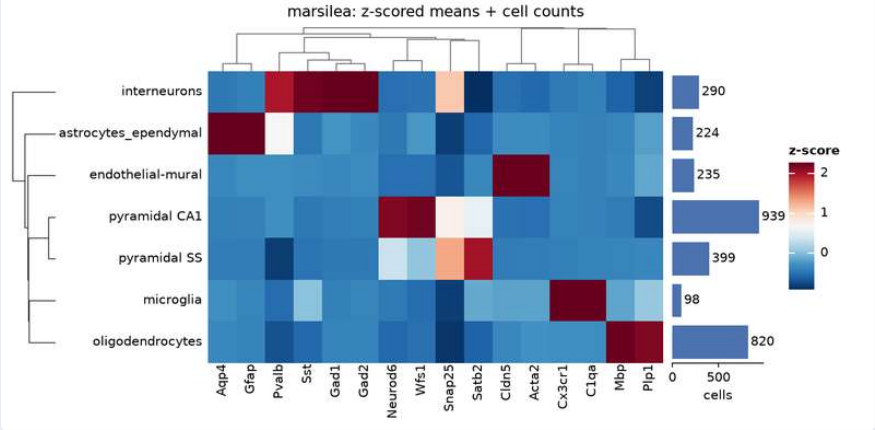
No package-specific caveat. General caveats apply (p. 8).

Reproduce: `python3 recreations/marsilea/marsilea_recreate.py`

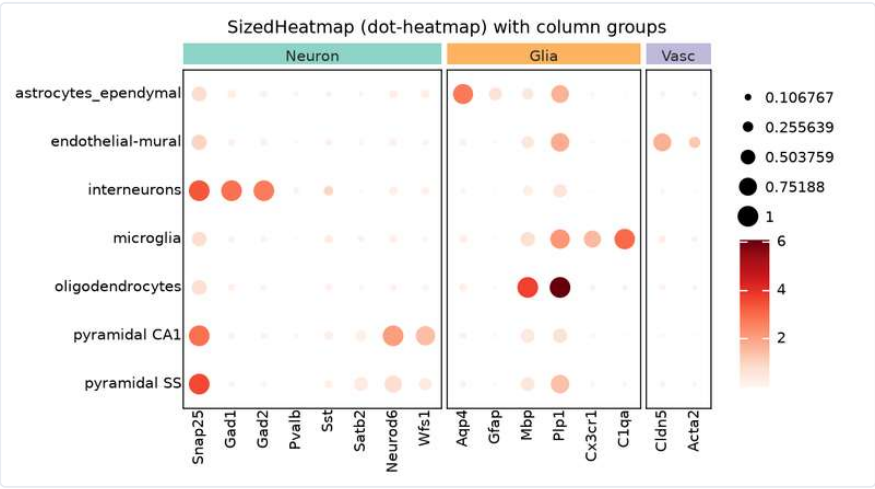
√ re-run end to end · 7 s

marsilea · 3 variations

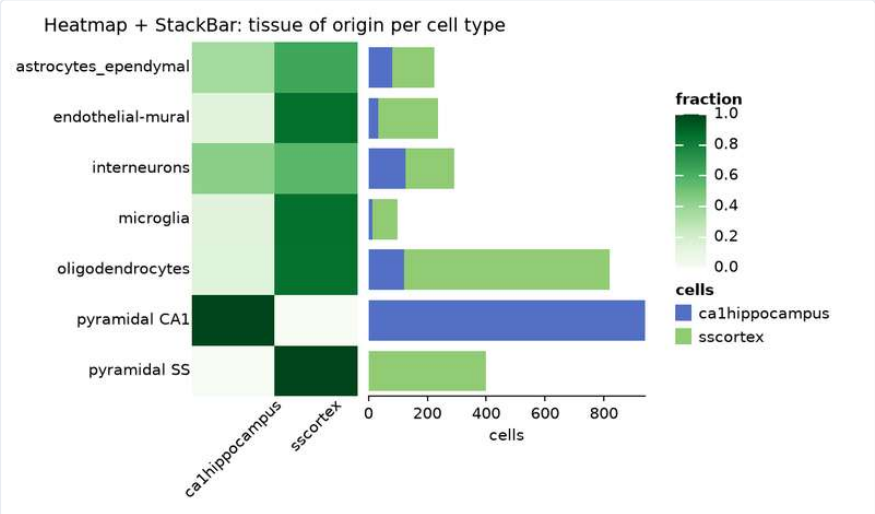
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 `heatmap_dendro_bars` HEATMAPS



v2 `sized_dot_heatmap` HEATMAPS



v3 `stackbar` HEATMAPS

dittoSeq

#68

OF 73 · NICHE & COOL SCORE

R | SINGLE-CELL PUBLICATION PLOTS | EMBEDDINGS & FEATURE MAPS | COMPOSITION, DISTRIBUTIONS & LAYOUTS | HEATMAPS & DOT MATRICES

1.25×	3/5	0.413	27	216
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE

NICHE & COOL · DITTOSEQ

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 1.045 → 1.308 per 1k papers (2023 → 2025–26).
GitHub dtm2451/dittoSeq · created 2019-02-18 · last push 2025-12-10

BEST FOR

colour-blind-safe dimplots, freq bars, hex plots

KEY PLOTTING CALL

RECREATIONS/DITTOSEQ/DITTOSEQ_RECREATE.R — EXCERPT · ENDS LINE 4

```
so <- readRDS("../_data/zeisel_seurat.rds"); mk <-
c("Snap25","Gad1","Gad2","Pvalb","Sst","Satb2","Neurod6","Wfs1","Aqp4","Gfap","Mbp","Plp1","Cx3cr1","C1qa","Cldn5","Acta2")
ggsave("v1_dittoDimPlot_letters.png", dittoDimPlot(so, "celltype", do.label = TRUE, labels.highlight = TRUE, legend.show =
FALSE, do.letter = FALSE, size = .8), width = 6.5, height = 6, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS · MOST-CITED OF 24 NAMING "DITTOSEQ" IN METHODS

120 CITATIONS — **Defining blood-induced microglia functions in neurodegeneration through multiomic profiling.** *Nat Immunol, 2023*
DOI ↗ PMC10307624

23 CITATIONS — **Exercise rejuvenates microglia and reverses T cell accumulation in the aged female mouse brain.** *Aging Cell, 2024*
DOI ↗ PMC11258432

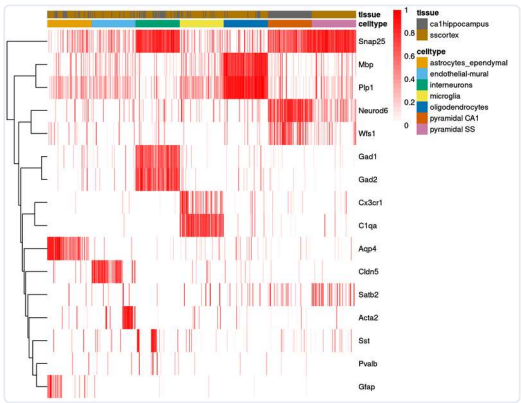
Figure-level matching was run for the Top 100 only, so these papers name the tool but no specific figure is linked.

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/dittoSeq/dittoSeq_recreate.R` — √ re-run end to end · 21 s

LEAD FIGURE

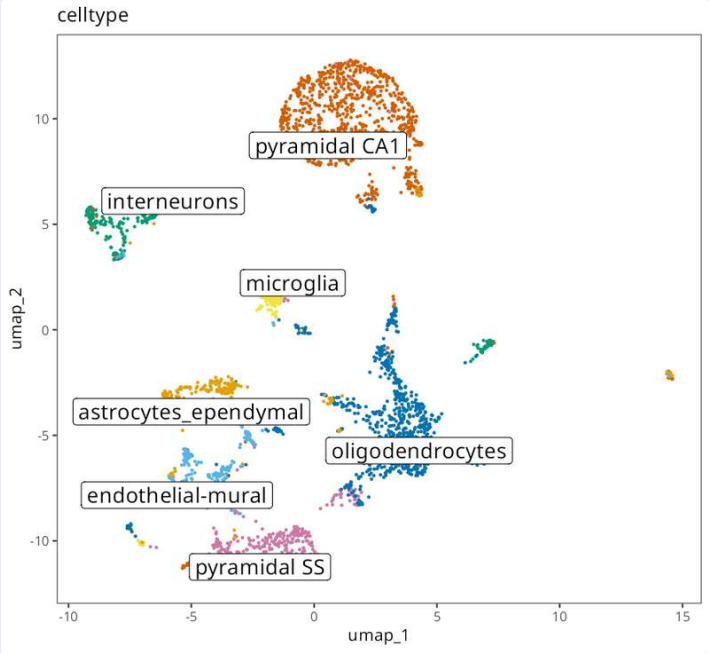


v5 dittoHeatmap — HEATMAPS

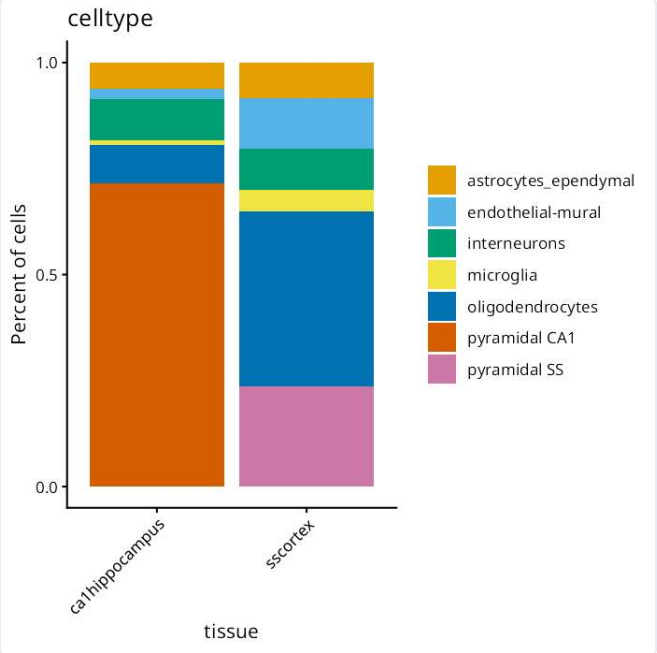
dittoSeq · 5 variations · v5 on facing page

NOVAFLOW RECREATION

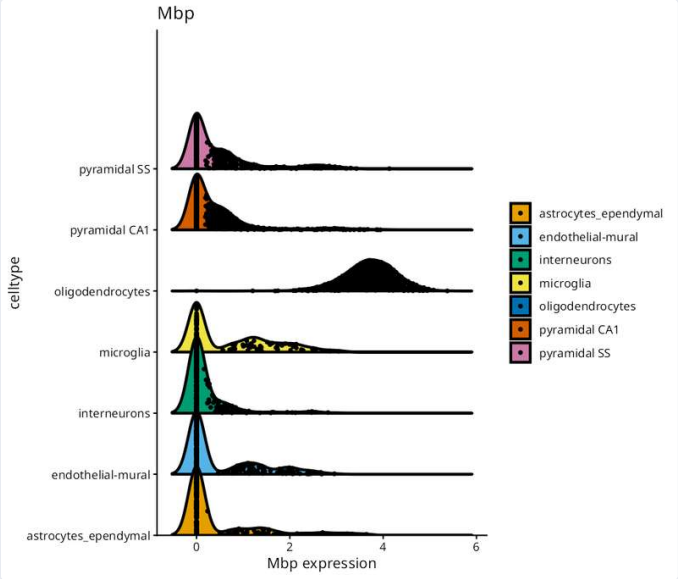
Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



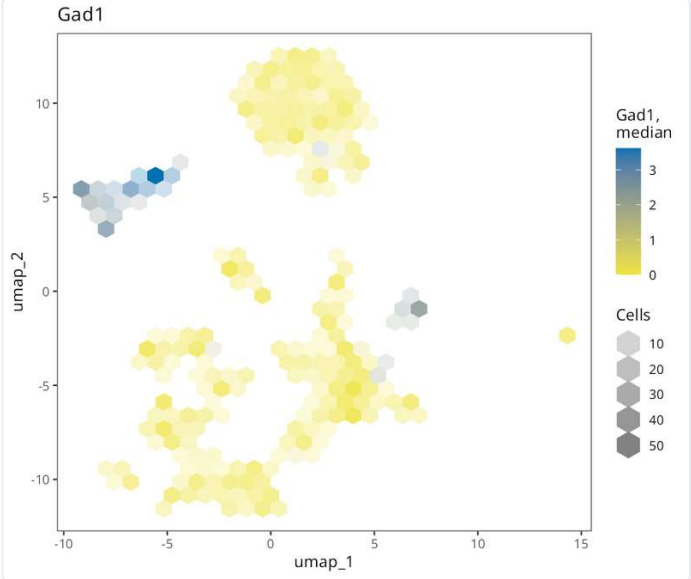
v1 dittoDimPlot_letters — EMBEDDINGS



v2 dittoBarPlot — COMPOSITION



v3 dittoPlot_ridgejitter — COMPOSITION



v4 dittoDimHex — EMBEDDINGS

ggforce

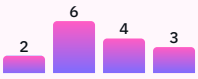
#70

OF 73 · NICHE & COOL SCORE

R | GENERAL | EMBEDDINGS & FEATURE MAPS | COMPOSITION, DISTRIBUTIONS & LAYOUTS

1.14×	3/5	0.386	15	958
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · GGFORCE

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.488 → 0.559 per 1k papers (2023 → 2025–26).
GitHub thomasp85/ggforce · created 2016-01-02 · last push 2025-09-15

BEST FOR

hull marks, zoom facets, parallel sets

KEY PLOTTING CALL

RECREATIONS/GGFORCE/GGFORCE_RECREATE.R — EXCERPT · ENDS LINE 5

```
md <- readRDS("../_data/zeisel_seurat.rds")@meta.data; md$UMAP_1 <- read.csv("../_data/zeisel_obs.csv")$UMAP_1; md$UMAP_2 <- re
ad.csv("../_data/zeisel_obs.csv")$UMAP_2
g <- ggplot(md, aes(UMAP_1, UMAP_2)) + geom_point(aes(color = celltype), size = .4) + geom_mark_hull(aes(fill = celltype, label
= celltype), concavity = 3, expand = unit(2,"mm"), alpha = .12, label.fontsize = 8, con.cap = 0) + theme_void() + theme(legend.
position = "none") + labs(title = "geom_mark_hull: annotated cluster hulls")
ggsave("v1_mark_hull.png", g, width = 8, height = 7, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS

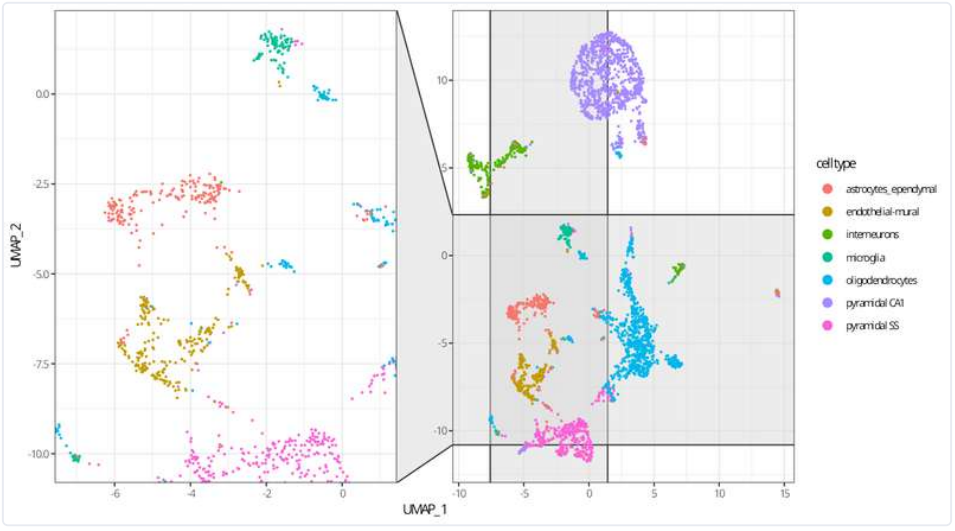
No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/thomasp85/ggforce.

LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/ggforce/ggforce_recreate.R` — √ re-run end to end · 7 s

LEAD FIGURE

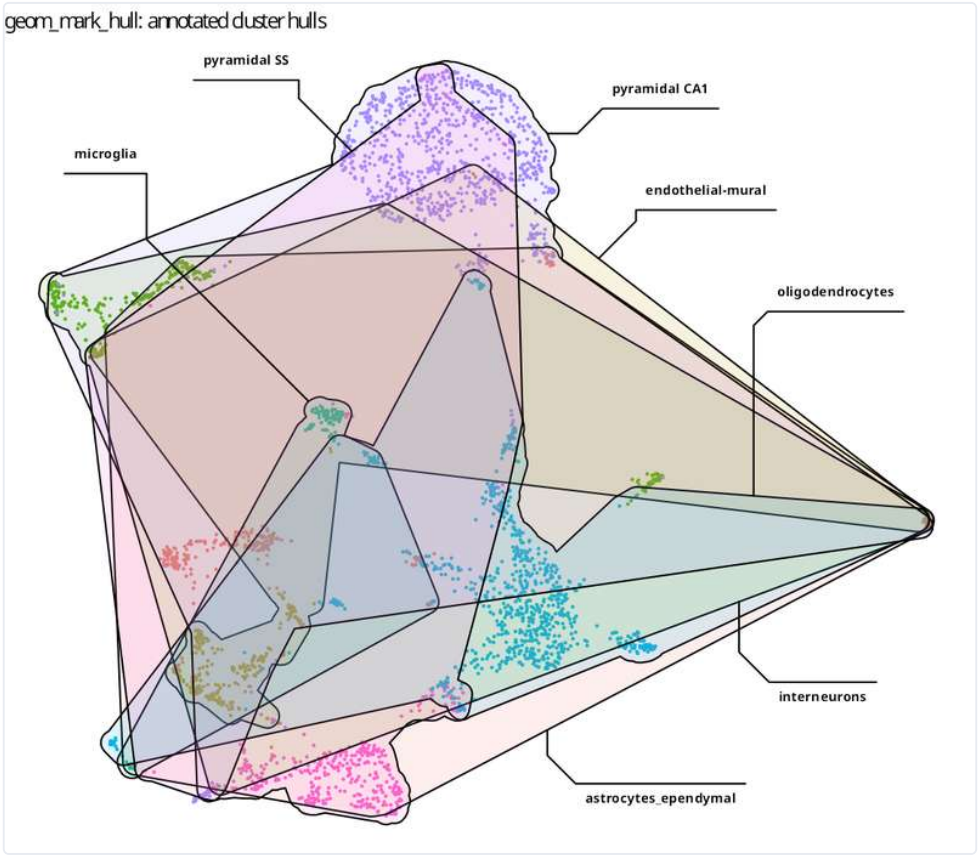


v2 facet_zoom

COMPOSITION

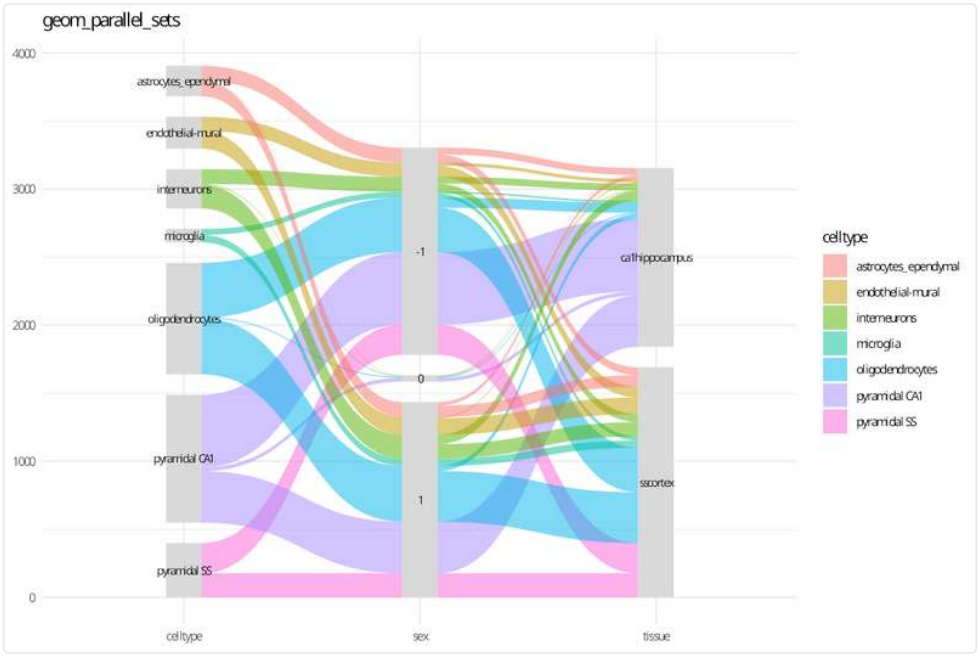
ggforce · 3 variations · v2 on facing page

Data: Zeisel et al. 2015, mouse somatosensory cortex & hippocampus CA1 scRNA-seq (3,005 cells). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v1 mark_hull

EMBEDDINGS



v3 parallel_sets

COMPOSITION

gggenes

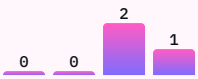
#71

OF 73 · NICHE & COOL SCORE

R | GENOME | GENOME TRACKS & GENOMIC SIGNAL

1.01×	3/5	0.368	3	603
growth ratio, per-1k rate 2025–26 ÷ 2023	visual score (curator judgement)	niche-cool score	neuro-corpus papers naming it in Methods, 2023–26	GitHub stars

GROWTH AMONG 73 CANDIDATES SORTED, LOG SCALE



NICHE & COOL · GGGENES

Neuro-corpus papers per year (*2026 partial). Broad-corpus rate 0.453 → 0.457 per 1k papers (2023 → 2025–26).
GitHub wilkox/gggenes · created 2017-03-02 · last push 2026-07-05

BEST FOR

gene-arrow maps

THIS RECREATION (NOVAFLOW, PUBLIC DATA)

gggenes: gene-arrow maps of the mouse protocadherin-gamma and Hox-adjacent neuronal loci (coordinates from TxDb mm10)

KEY PLOTTING CALL

RECREATIONS/GGGENES/GGGENES_RECREATE.R — EXCERPT · ENDS LINE 9

```
d <- do.call(rbind, lapply(names(reg), function(n) { g <- subsetByOverlaps(gr, reg[[n]]); data.frame(molecule = n, gene = g$sym
bol, start = start(g), end = end(g), strand = as.character(strand(g)), forward = as.character(strand(g)) == "+") }))
d <- d[!is.na(d$gene),]
g <- ggplot(d, aes(xmin = start, xmax = end, y = molecule, fill = forward, label = gene, forward = forward)) + geom_gene_arrow(a
rrowhead_height = unit(3,"mm"), arrowhead_width = unit(1,"mm")) + geom_gene_label(align = "left", min.size = 3) + facet_wrap(~ m
olecule, scales = "free", ncol = 1) + scale_fill_brewer(palette = "Set2") + theme_genes() + labs(title =
"gggenes: neuronal loci (mm10)")
ggsave("v1_gene_arrows.png", g, width = 11, height = 5, dpi = 140, bg = "white")
```

IN PUBLISHED PAPERS

No example paper is listed: figure-level matching covered the Top 100 only, and this tool was not in the corpus package screen. Reference implementation: https://github.com/wilkox/gggenes.

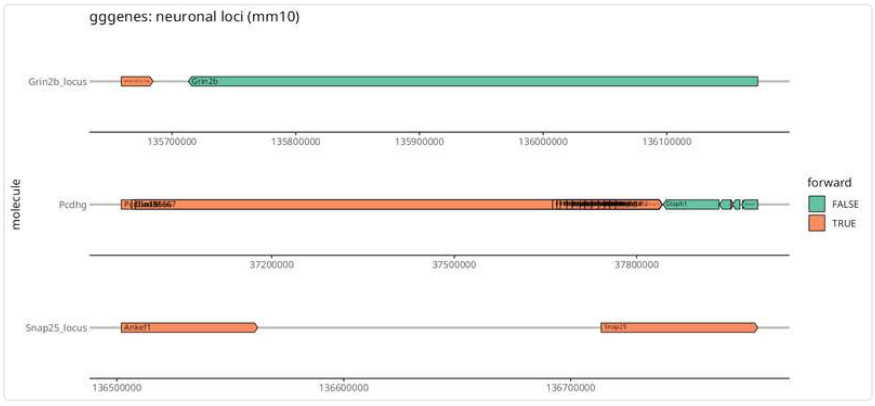
LIMITATIONS & NOTES

No package-specific caveat. General caveats apply (p. 8).

Reproduce: `Rscript recreations/gggenes/gggenes_recreate.R`

✓ re-run end to end · 13 s

LEAD FIGURE



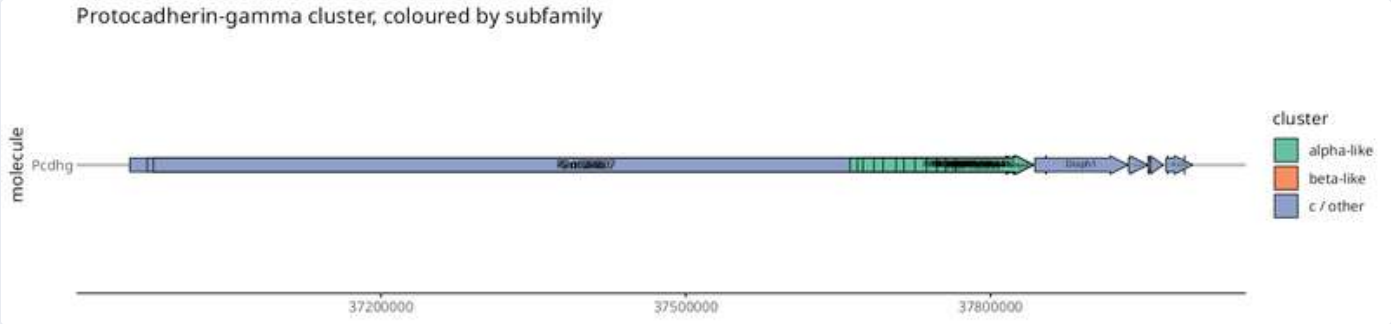
v1 gene_arrows

GENOME

gggenes · 2 variations · v1 on facing page

NOVAFLOW RECREATION

Data: Mouse gene models from TxDb mm10 (protocadherin-γ cluster and a Hox-adjacent locus). Recreated by Novaflow from the script on the facing page; not reproduced from the papers cited there.



v2 `pcdhg_cluster`

GENOME

APPENDIX A

Queued tools and why

Tools that rank in the catalog but are not recreated: **30 mainstream entries** and **15 Niche & Cool tools**, with the reason for each.

Niche & Cool

NICHE #	TOOL	REASON
9	**SCENIC+**	needs cisTarget ranking DBs (>4 GB) + pycisTopic stack
21	**CellOracle**	pip pins old numpy/pandas/anndata; would break shared env
22	**napari-spatialdata**	GUI viewer (Qt); screenshot path not set up
24	**cell2location**	incompatible with installed pyro (deep_getattr removed); needs pinned env
25	**MapMyCells (ABC atlas)**	cell_type_mapper + precomputed ABC stats download (large)
27	**Dynamo**	dynamo-release requires anndata<0.11 + old statsmodels API; tried and removed (broke scanpy/cellrank)
33	**Vitessce**	web/JS viewer, no static plotting API; needs served data + browser
36	**CELLxGENE**	Census query/explorer GUI; streaming data, not a plotting library
38	**scCODA**	needs TensorFlow/TFP (or pertpy, heavy)
39	**stLearn**	pip install failed (old pinned deps)
49	**trackplot**	needs the bwtool binary (not in apt) + neuro bigWigs
52	**SpaGCN**	pip install failed
53	**UniTVelo**	needs TensorFlow
57	**Giotto**	heavy R+python env install, not attempted after reboot
59	**ArchR**	compile failed; also needs BSgenome (~1 GB) + Arrow files

Mainstream Top 100

#	PACKAGE	LANG	CATEGORY	PAPERS	NOTE
8	**Cytoscape**	Desktop	network	1,501	Desktop GUI - queued: style-equivalent via RCy3/py4cytoscape needs a running Cytoscape; igraph/networkx redraws planned
15	**Metascape**	Web	enrichment	707	Web-only (no API) - queued: style-equivalent with clusterProfiler/ComplexHeatmap
16	**Ingenuity Pathway Analysis**	Desktop	pathways	678	Commercial (QIAGEN) - not reproducible openly; queued style-equivalent z-score bar
25	**SCENIC**	R/Python	regulon	385	Queued: needs cisTarget motif databases (~GBs)
26	**QuPath**	Desktop	imaging	380	Desktop GUI image analysis - queued: napari/matplotlib overlays
30	**monocle (v2)**	R	trajectory	310	Queued: legacy Bioconductor monocle (DDRTree)
34	**ShinyGO**	Web	enrichment	263	not yet recreated
35	**ChIPseeker**	R	peak annotation	247	not yet recreated
49	**SynGO**	Web	synaptic enrichment	139	not yet recreated
51	**ClueGO**	Desktop plugin	enrichment network	134	not yet recreated
53	**base R graphics**	R	general	132	not yet recreated
55	**maftools**	R	mutation oncoplot	123	not yet recreated
58	**ArchR**	R	scATAC	107	not yet recreated
62	**factoextra**	R	PCA/clustering	96	not yet recreated
65	**cellxgene**	Web/ Python	interactive	92	not yet recreated
67	**UCSC Cell Browser**	Web	interactive	86	not yet recreated
69	**scater**	R	single-cell QC/ embedding	83	not yet recreated
72	**variancePartition**	R	variance	77	not yet recreated

#	PACKAGE	LANG	CATEGORY	PAPERS	NOTE
73	**cell2location**	Python	spatial deconvolution	77	not yet recreated
76	**RColorBrewer**	R	color palettes	71	not yet recreated
77	**FactoMineR**	R	PCA	70	not yet recreated
78	**shiny**	R	interactive	69	not yet recreated
80	**EnrichmentMap**	Desktop plugin	enrichment network	65	not yet recreated
81	**Seurat spatial (SpatialDimPlot)**	R	spatial	63	not yet recreated
85	**UMAP (umap-learn/ uwot)**	Python/R	embedding	54	not yet recreated
89	**viridis**	R/Python	color palettes	47	not yet recreated
90	**networkx**	Python	network	47	not yet recreated
93	**pertpy**	Python	perturbation/ composition	45	not yet recreated
95	**Morpheus**	Web	heatmap	44	not yet recreated
96	**CellCall/connectome**	R	cell-cell communication	41	not yet recreated

APPENDIX B

Full Top 100 ranked table

Ranked by the number of the 17,768 corpus papers naming each package in Methods. Page numbers link to spreads.

#	PACKAGE	LANG	CATEGORY	PAPERS	%	FULL TEXT	ROLE	STATUS	PAGE
1	**DESeq2 (plotPCA/plotMA)**	R	bulk QC/DE	5,668	31.9	5,871	analysis+plotting	done	20
2	**Seurat**	R	single-cell (embedding/dot/violin/feature)	4,671	26.29	4,846	analysis+plotting	done	22
3	**clusterProfiler**	R	enrichment	3,518	19.8	3,609	analysis+plotting	done	24
4	**ggplot2**	R	general	2,490	14.01	2,546	dedicated plotting	done	26
5	**GSEA (Broad)**	Desktop	enrichment	2,056	11.57	3,310	analysis+plotting	done	28
6	**limma**	R	bulk QC/DE	2,047	11.52	2,116	analysis+plotting	done	30
7	**edgeR**	R	bulk QC/DE	1,892	10.65	1,949	analysis+plotting	done	32
8	**Cytoscape**	Desktop	network	1,501	8.45	1,665	dedicated plotting	queued	
9	**pheatmap**	R	heatmap	1,119	6.3	1,138	dedicated plotting	done	34
10	**WGCNA**	R	network/modules	884	4.98	1,025	analysis+plotting	done	36
11	**CellChat**	R	cell-cell communication	839	4.72	976	analysis+plotting	done	38
12	**STRING**	Web	PPI network	838	4.72	1,142	analysis+plotting	done	40
13	**Enrichr**	Web	enrichment	778	4.38	912	analysis+plotting	done	42
14	**scanpy**	Python	single-cell (embedding/dot/violin/matrix)	763	4.29	789	analysis+plotting	done	44
15	**Metascape**	Web	enrichment	707	3.98	808	analysis+plotting	queued	
16	**Ingenuity Pathway Analysis**	Desktop	pathways	678	3.82	893	analysis+plotting	queued	
17	**ComplexHeatmap**	R	heatmap	567	3.19	578	dedicated plotting	done	46
18	**g:Profiler**	Web/R/Python	enrichment	563	3.17	627	analysis+plotting	done	48
19	**IGV**	Desktop/Web	genome tracks	549	3.09	794	viewer	done	50
20	**fgsea**	R	enrichment	478	2.69	498	analysis+plotting	done	52
21	**monocle3**	R	trajectory	430	2.42	499	analysis+plotting	done	54
22	**deepTools**	Python/CLI	heatmap (genomic)	416	2.34	427	dedicated plotting	done	56
23	**SingleR**	R	annotation (heatmap)	416	2.34	451	analysis+plotting	done	58
24	**AUCell**	R	gene set scores	387	2.18	432	analysis+plotting	done	60
25	**SCENIC**	R/Python	regulon	385	2.17	462	analysis+plotting	queued	
26	**QuPath**	Desktop	imaging	380	2.14	395	viewer	queued	
27	**EnhancedVolcano**	R	volcano	369	2.08	373	dedicated plotting	done	62
28	**UCSC Genome Browser**	Web	genome tracks	363	2.04	509	viewer	done	50
29	**survminer**	R	survival	351	1.98	342	dedicated plotting	done	64
30	**monocle (v2)**	R	trajectory	310	1.74	360	analysis+plotting	queued	
31	**Space Ranger/Loupe**	Desktop	spatial	300	1.69	307	viewer	done	66
32	**enrichplot**	R	enrichment	280	1.58	286	dedicated plotting	done	24
33	**ggpubr**	R	general/statistics	277	1.56	283	dedicated plotting	done	68

#	PACKAGE	LANG	CATEGORY	PAPERS	%	FULL TEXT	ROLE	STATUS	PAGE
34	ShinyGO	Web	enrichment	263	1.48	284	analysis+plotting	queued	
35	ChIPseeker	R	peak annotation	247	1.39	264	analysis+plotting	queued	
36	igraph	R/Python	network	241	1.36	240	analysis+plotting	done	70
37	Signac	R	scATAC (coverage/embedding)	227	1.28	239	analysis+plotting	done	72
38	matplotlib	Python	general	227	1.28	238	dedicated plotting	done	74
39	CopyKAT/inferCNV	R	CNV heatmap	207	1.17	236	analysis+plotting	done	76
40	seaborn	Python	general/statistics	192	1.08	193	dedicated plotting	done	78
41	CellPhoneDB	Python	cell-cell communication	182	1.02	231	analysis+plotting	done	80
42	slingshot	R	trajectory	173	0.97	225	analysis+plotting	done	82
43	Loupe Browser	Desktop	interactive	166	0.93	176	viewer	done	66
44	rrvgo	R	enrichment (GO reduce)	163	0.92	179	dedicated plotting	done	84
45	scVelo	Python	RNA velocity	161	0.91	177	analysis+plotting	done	86
46	VennDiagram	R	set overlap	157	0.88	160	dedicated plotting	done	88
47	gplots (heatmap.2)	R	heatmap	152	0.86	157	dedicated plotting	done	90
48	corrplot	R	correlation	147	0.83	151	dedicated plotting	done	92
49	SynGO	Web	synaptic enrichment	139	0.78	210	analysis+plotting	queued	
50	hdWGCNA	R	network/modules (sc)	134	0.75	147	analysis+plotting	done	94
51	ClueGO	Desktop plugin	enrichment network	134	0.75	152	dedicated plotting	queued	
52	clustree	R	cluster tree	133	0.75	144	dedicated plotting	done	96
53	base R graphics	R	general	132	0.74	134	dedicated plotting	queued	
54	circlize	R	circos/chord	124	0.7	130	dedicated plotting	done	98
55	maftools	R	mutation oncoplot	123	0.69	123	dedicated plotting	queued	
56	velocyto	Python/R	RNA velocity	120	0.68	129	analysis+plotting	done	86
57	UpSetR	R	set overlap	112	0.63	130	dedicated plotting	done	100
58	ArchR	R	scATAC	107	0.6	117	analysis+plotting	queued	
59	RCTD/spacexr	R	spatial deconvolution	105	0.59	130	analysis+plotting	done	102
60	CytoTRACE	R/Python	differentiation potency	103	0.58	131	analysis+plotting	done	104
61	decoupleR/decoupler	R/Python	TF/pathway activity	102	0.57	112	analysis+plotting	done	106
62	factoextra	R	PCA/clustering	96	0.54	101	dedicated plotting	queued	
63	ABC atlas / Allen Brain Map	Web	atlas	96	0.54	169	viewer	done	108
64	ggrepel	R	labels	92	0.52	94	dedicated plotting	done	26
65	cellxgene	Web/Python	interactive	92	0.52	162	viewer	queued	
66	NicheNet	R	ligand-target	90	0.51	122	analysis+plotting	done	110
67	UCSC Cell Browser	Web	interactive	86	0.48	122	viewer	queued	
68	ReactomePA	R	enrichment	84	0.47	87	analysis+plotting	done	112
69	scater	R	single-cell QC/embedding	83	0.47	84	analysis+plotting	queued	

#	PACKAGE	LANG	CATEGORY	PAPERS	%	FULL TEXT	ROLE	STATUS	PAGE
70	**pathview**	R	pathway maps	82	0.46	99	dedicated plotting	done	114
71	**GOplot**	R	enrichment	81	0.46	85	dedicated plotting	done	116
72	**variancePartition**	R	variance	77	0.43	79	analysis+plotting	queued	
73	**cell2location**	Python	spatial deconvolution	77	0.43	107	analysis+plotting	queued	
74	**scCustomize**	R	single-cell	75	0.42	78	dedicated plotting	done	118
75	**PAGA**	Python	trajectory graph	75	0.42	120	analysis+plotting	done	44
76	**RColorBrewer**	R	color palettes	71	0.4	76	dedicated plotting	queued	
77	**FactoMineR**	R	PCA	70	0.39	70	analysis+plotting	queued	
78	**shiny**	R	interactive	69	0.39	117	dashboard	queued	
79	**patchwork**	R	layout	65	0.37	71	dedicated plotting	done	120
80	**EnrichmentMap**	Desktop plugin	enrichment network	65	0.37	72	dedicated plotting	queued	
81	**Seurat spatial (SpatialDimPlot)**	R	spatial	63	0.35	65	analysis+plotting	queued	
82	**ggvenn**	R	set overlap	61	0.34	63	dedicated plotting	done	88
83	**Squidpy**	Python	spatial	60	0.34	74	analysis+plotting	done	122
84	**plotly**	R/Python/JS	interactive	56	0.32	60	dedicated plotting	done	124
85	**UMAP (umap-learn/uwot)**	Python/R	embedding	54	0.3	57	analysis+plotting	queued	
86	**ggraph**	R	network	50	0.28	49	dedicated plotting	done	126
87	**Tangram**	Python	spatial mapping	49	0.28	77	analysis+plotting	done	128
88	**PCAtools**	R	PCA	48	0.27	48	dedicated plotting	done	130
89	**viridis**	R/Python	color palettes	47	0.26	67	dedicated plotting	queued	
90	**networkx**	Python	network	47	0.26	44	analysis+plotting	queued	
91	**tradeSeq**	R	trajectory DE	46	0.26	51	analysis+plotting	done	132
92	**LIANA**	R/Python	cell-cell communication	45	0.25	67	analysis+plotting	done	134
93	**pertpy**	Python	perturbation/composition	45	0.25	53	analysis+plotting	queued	
94	**cowplot**	R	layout	44	0.25	45	dedicated plotting	done	120
95	**Morpheus**	Web	heatmap	44	0.25	102	dedicated plotting	queued	
96	**CellCall/connectome**	R	cell-cell communication	41	0.23	193	analysis+plotting	queued	
97	**BayesSpace**	R	spatial clustering	41	0.23	67	analysis+plotting	done	136
98	**CellRank**	Python	fate mapping	40	0.23	51	analysis+plotting	done	138
99	**Nebulosa**	R	single-cell density	39	0.22	43	dedicated plotting	done	140
100	**ggalluvial**	R	flow/alluvial	36	0.2	37	dedicated plotting	done	142

APPENDIX C

Full Niche & Cool table

All 73 candidates by niche-cool score. Growth = broad-corpus per-1k rate 2025–26 ÷ 2023 (+0.01 smoothing); † = fewer than 5 mentions in 2023. Neuro = neuro-corpus papers naming the tool in Methods, 2023–26.

#	TOOL	LANG	CATEGORY	GROWTH	VISUAL	SCORE	NEURO	STARS	STATUS	PAGE
1	**pyCirclize**	Python	circos/chord	20.05†	5	0.98	0	1,127	done	146
2	**hdWGCNA**	R	sc co-expression	6.66	5	0.938	147	494	done	94
3	**brainrender**	Python	3D brain	4.81†	5	0.914	4	680	done	148
4	**COMMOT**	Python	spatial communication	4.67	5	0.904	15	149	done	150
5	**CytoTRACE2**	R/Python	potency	324.85†	4	0.9	24	269	done	104
6	**MultiNicheNet**	R	ligand-target	89.9†	4	0.893	13	202	done	152
7	**moscot**	Python	optimal transport	42.91†	4	0.887	7	217	done	154
8	**scplotter**	R	single-cell publication plots	17.51†	4	0.872	1	333	done	156
9	SCENIC+	Python	GRN	14.32†	4	0.866	32	267	queued	
10	**Banksy**	R/Python	spatial clustering	14.16†	4	0.859	21	167	done	158
11	**ggsankey**	R	flow	10.71†	4	0.845	8	338	done	142
12	**scTour**	Python	trajectory	6.52†	4	0.832	7	89	done	160
13	**plot1cell**	R	single-cell publication plots	2.7†	5	0.829	6	131	done	162
14	**veloVI**	Python	RNA velocity	5.87†	4	0.824	9	45	done	164
15	**miloR / Milo**	R/Python	differential abundance	2.31	5	0.815	46	450	done	166
16	**Datashader**	Python	large data	4.81†	4	0.814	1	3,568	done	168
17	**scDRS**	Python	disease relevance	4.63	4	0.798	37	201	done	170
18	**BrainGlobe atlas API**	Python	3D brain	2.27†	5	0.798	1	189	done	108
19	**MNE-Python**	Python	neurophysiology	2.27†	5	0.798	1	3,549	done	172
20	**sccomp**	R	composition	4.27†	4	0.79	10	134	done	174
21	CellOracle	Python	GRN perturbation	2.26	5	0.788	29	503	queued	
22	napari-spatialdata	Python	imaging	4.05	4	0.776	36	91	queued	
23	**LIANA+**	Python/R	cell-cell communication	3.78	4	0.77	67	–	done	134
24	cell2location	Python	spatial deconvolution	3.1	4	0.756	107	462	queued	
25	MapMyCells (ABC atlas)	Web/Python	atlas mapping	11.56†	3	0.752	41	58	queued	
26	**PyComplexHeatmap**	Python	heatmap	3.06†	4	0.75	2	407	done	176
27	Dynamo	Python	vector fields	2.05	5	0.746	8	512	queued	
28	**Squidpy**	Python	spatial	2.72	4	0.736	74	608	done	122
29	**nilearn**	Python	neuroimaging	1.92†	5	0.733	7	1,444	done	178
30	**scCustomize**	R	single-cell publication plots	4.24	3	0.684	78	314	done	118
31	**SCpubr**	R	single-cell publication plots	2.19	4	0.671	14	245	done	180
32	**BayesSpace**	R	spatial clustering	2.19	4	0.671	67	189	done	136
33	Vitessce	JS/Python	interactive spatial	1.6†	5	0.664	2	272	queued	
34	**UCell**	R	gene-set scores	3.15	3	0.663	144	213	done	182
35	**CellRank**	Python	fate mapping	1.55	5	0.658	51	–	done	138
36	CELLxGENE	Web/Python	atlas	2.8	3	0.643	162	787	queued	
37	**spatialLIBD**	R	spatial	1.97	4	0.64	73	115	done	184
38	scCODA	Python	composition	2.36	3	0.622	53	203	queued	
39	stLearn	Python	spatial	1.91	4	0.622	54	261	queued	
40	**inferCNV/CopyKAT**	R	CNV	1.91	4	0.622	237	686	done	76
41	**ggrepel volcano styles**	R	volcano	2.3	3	0.608	74	1,262	done	186

#	TOOL	LANG	CATEGORY	GROWTH	VISUAL	SCORE	NEURO	STARS	STATUS	PAGE
42	**Augur**	R	perturbation prioritization	1.86	4	0.606	25	153	done	188
43	**circlize (chord)**	R	circos/chord	1.17	5	0.602	133	1,027	done	98
44	**Tangram**	Python	spatial mapping	1.83	4	0.598	77	373	done	128
45	**scVelo streamlines**	Python	RNA velocity	1.15	5	0.596	177	513	done	86
46	**NicheNet**	R	ligand-target	1.73	4	0.592	122	682	done	110
47	**ggridges**	R	distribution	1.7	4	0.585	20	425	done	190
48	**plotnine**	Python	general	2.2†	3	0.581	3	4,771	done	192
49	**trackplot**	R	genome tracks	1.68†	4	0.578	1	193	queued	
50	**Palantir**	Python	trajectory	1.62	4	0.571	31	341	done	194
51	**ggspavis/SpatialExperiment**	R	spatial	2.15	3	0.56	9	5	done	196
52	**SpaGCN**	Python	spatial clustering	2.11	3	0.554	55	265	queued	
53	**UniTVelo**	Python	RNA velocity	1.44	4	0.55	7	30	queued	
54	**Signac**	R	scATAC	1.43	4	0.544	239	431	done	72
55	**plotgardener**	R	genome tracks	0.93	5	0.544	4	371	done	198
56	**plotly/Dash 3D**	Python/R	interactive 3D	1.36	4	0.537	60	18,841	done	124
57	**Giotto**	R	spatial	1.35	4	0.53	42	—	queued	
58	**Nebulosa**	R	density embedding	1.29	4	0.524	43	116	done	140
59	**ArchR**	R	scATAC	1.25	4	0.513	117	467	queued	
60	**propeller (speckle)**	R	composition	1.89	3	0.512	116	71	done	200
61	**Allen CCF overlays**	Python/MATLAB	atlas registration	1.14	4	0.486	98	126	done	108
62	**CellPhoneDB/ktplots**	Python/R	cell-cell communication	1.1	4	0.476	232	178	done	80
63	**ggcoverage**	R	genome tracks	0.97†	4	0.462	3	273	done	202
64	**ggtree/ggtreeExtra**	R	tree	0.94	4	0.455	26	99	done	204
65	**Mellon**	Python	cell-state density	0.93	4	0.444	27	77	done	206
66	**karyoploteR**	R	genome	0.67	4	0.428	13	377	done	208
67	**marsilea**	Python	heatmap	0.64†	4	0.421	0	324	done	210
68	**dittoSeq**	R	single-cell publication plots	1.25	3	0.413	27	216	done	212
69	**ggdensity**	R	distribution	0.51†	4	0.407	1	234	done	190
70	**ggforce**	R	general	1.14	3	0.386	15	958	done	214
71	**gggenes**	R	genome	1.01	3	0.368	3	603	done	216
72	**ggalluvial**	R	flow	0.84	3	0.334	37	526	done	142
73	**Gviz**	R	genome tracks	0.55	3	0.314	31	94	done	50